

eSDK Huawei Storage Kubernetes CSI Plugins V4.13.0

User Guide

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About This Document

Intended Audience

This document is intended for:

- Technical support engineers
- O&M engineers
- Engineers with basic knowledge of storage and Kubernetes

Symbol Conventions

The symbols that may be found in this document are defined as follows.

Symbol	Description
 DANGER	Indicates a hazard with a high level of risk which, if not avoided, will result in death or serious injury.
 WARNING	Indicates a hazard with a medium level of risk which, if not avoided, could result in death or serious injury.
 CAUTION	Indicates a hazard with a low level of risk which, if not avoided, could result in minor or moderate injury.
 NOTICE	Indicates a potentially hazardous situation which, if not avoided, could result in equipment damage, data loss, performance deterioration, or unanticipated results. NOTICE is used to address practices not related to personal injury.
 NOTE	Supplements the important information in the main text. NOTE is used to address information not related to personal injury, equipment damage, and environment deterioration.

Change History

Issue	Date	Description
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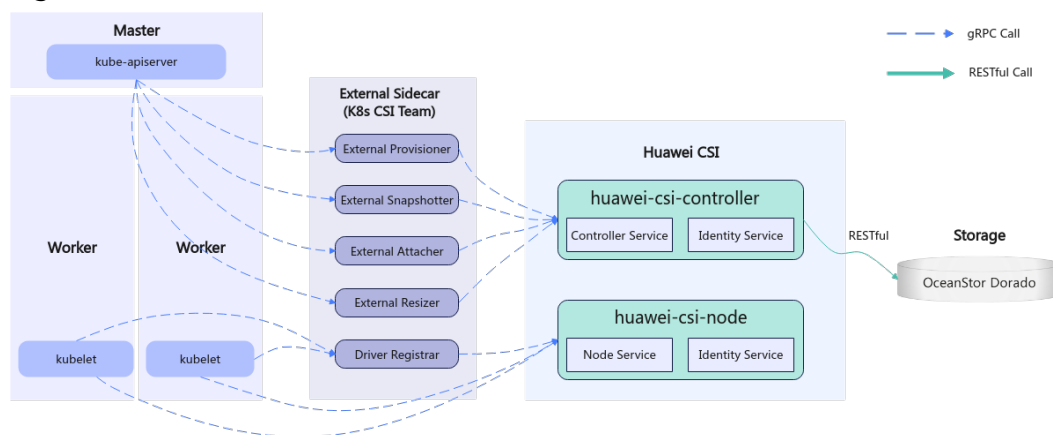
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1 Overview

Container Storage Interface (CSI) is an industry standard protocol used to connect container platforms such as Kubernetes to underlying storage systems. Huawei CSI is a mandatory component that connects Huawei enterprise storage and distributed storage products to Kubernetes clusters, providing persistent storage services for containerized workload.

Kubernetes uses a series of officially maintained sidecar components to register and listen Kubernetes object resources and invoke the CSI driver to trigger physical operations on Huawei storage. For example, when a **Persistent Volume** (PV) is created, the Huawei CSI driver creates a LUN (block storage) or file system on the storage device. **Figure 1-1** shows the overall structure of Kubernetes, Huawei CSI, and Huawei storage.

Figure 1-1 CSI overall architecture



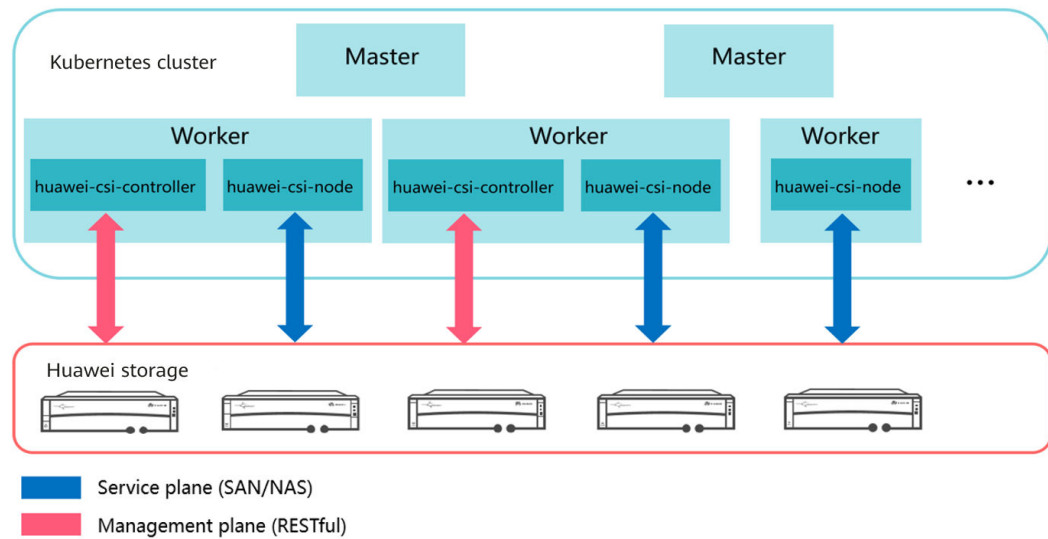
Huawei CSI consists of two components: `huawei-csi-controller` and `huawei-csi-node`.

- `huawei-csi-controller`: one or more Pods (including `Controller Service` and `Identity Service`) running in Deployment mode. It is used to interact with Huawei storage using RESTful. Therefore, the node running the `huawei-csi-controller` component must be connected to the management plane network of the storage.
- `huawei-csi-node`: a Pod (including `Node Service` and `Identity Service`) that runs on Kubernetes worker nodes in DaemonSet mode. It is used to mount

and unmount a LUN/file system provided by Huawei storage on worker nodes. Therefore, the node running the huawei-csi-node component must be connected to the service plane network of the storage.

Figure 1-2 shows the deployment model of Huawei CSI.

Figure 1-2 CSI deployment model



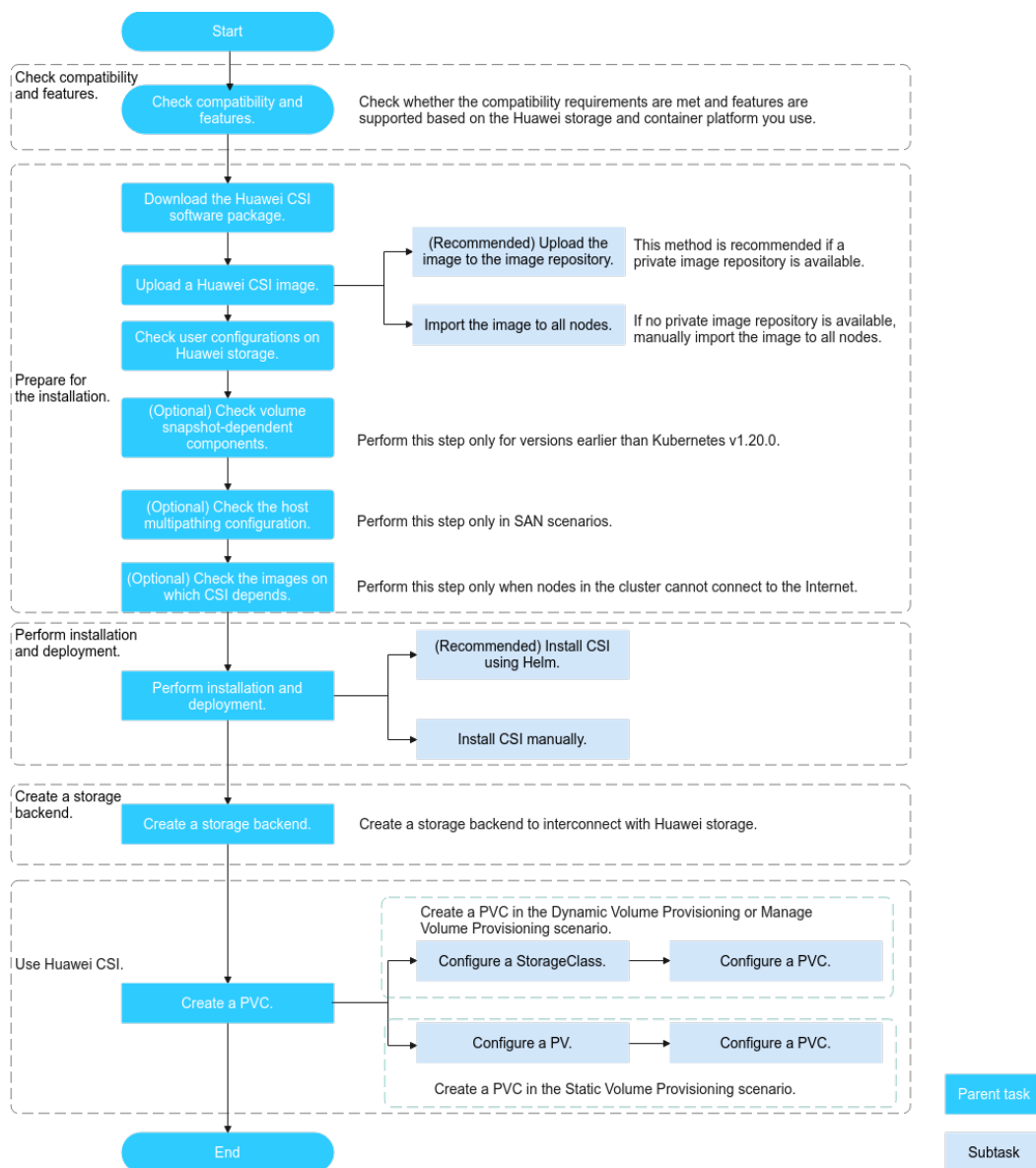
This document describes how to install, deploy, and use the Huawei CSI V4.13.0 plug-in.

2 Quick Start

This chapter describes how to quickly install and use Huawei CSI to manage Persistent Volume Claims (PVCs).

Huawei CSI Use Process

Figure 2-1 CSI installation and use process



Compatibility and Features

Before using this plug-in, learn about its compatibility with Huawei storage, container platforms, and host operating systems (OSs), as well as supported features. For details, see [3 Compatibility and Features](#).

Installation Preparations

Before installing Huawei CSI, you need to prepare the configurations of related environments such as container platforms and hosts. For details, see [4.1.1 Installation Preparations](#).

Installation and Deployment

Huawei CSI provides two installation modes: installation using Helm and manual installation, which are suitable for different container platforms such as Kubernetes and OpenShift. For details, see [4 Installation and Deployment](#).

Creating a Storage Backend

Before using Huawei CSI, you need to create storage backend resources. For details, see [5.1.1 Configuring the Storage Backend](#).

Using Huawei CSI

Now, you can use Huawei CSI to manage PVCs. For details, see [5.2.1 Configuring a StorageClass](#) and [5.3.1 Configuring PVs](#).

3 Compatibility and Features

- 3.1 Flash Storage (OceanStor Dorado/OceanStor V5/OceanStor V6 and Later)
- 3.2 Flash Storage (OceanStor A600/A800)
- 3.3 iMaster DME (Interconnected with an OceanStor A800 Cluster)
- 3.4 Mass Storage (OceanStor Pacific/FusionStorage Block Series)
- 3.5 Mass Storage (OceanDisk Series)

3.1 Flash Storage (OceanStor Dorado/OceanStor V5/OceanStor V6 and Later)

3.1.1 File Service

3.1.1.1 Compatibility

3.1.1.1.1 Storage

Compatibility with Huawei CSI Storage

Table 3-1 Supported Huawei storage products

Storage Product	Version
OceanStor V5	V500R007 and V500R007 Kunpeng
OceanStor	6.1.3, 6.1.5, 6.1.6, 6.1.7, 6.1.8, V700R001C00, V700R001C10, V700R001C20, and V700R001C30
OceanStor Dorado	6.1.0, 6.1.2, 6.1.3, 6.1.5, 6.1.6, 6.1.7, 6.1.8, V700R001C00, V700R001C10, V700R001C20, and V700R001C30

Table 3-2 Huawei storage versions supported by Huawei enterprise storage dtrees

Storage Product	Version
OceanStor Dorado	6.1.0, 6.1.2, 6.1.3, 6.1.5, 6.1.6, 6.1.7, 6.1.8, V700R001C00, V700R001C10, V700R001C20, and V700R001C30

3.1.1.1.2 Container Platform

Huawei CSI Platform Compatibility

Table 3-3 Supported container management platforms

Container Management Platform	Version
Kubernetes	1.16 to 1.36
Red Hat OpenShift Container Platform	4.12 to 4.22
Tanzu Kubernetes Grid Integrated	TKGI 1.17 to 1.19
CCE Agile	22.3.2 and 23.12.1
CCE	22.9.5

3.1.1.1.3 OS

Huawei CSI OS Compatibility

Table 3-4 Supported host OSs

OS Name	OS Version
CentOS x86_64	7.6, 7.7, 7.9, 8.2, 8.4
CentOS Arm	7.6
Rocky Linux x86_64	8.6
SUSE 15 x86_64	SP2 and SP3
Red Hat CoreOS x86_64	4.12 to 4.20
Ubuntu x86_64	18.04, 20.04, 22.04, and 24.04
Ubuntu Arm	22.04 and 24.04
Kylin x86_64	7.6, V10 SP1, V10 SP2, and V10 SP3

OS Name	OS Version
Kylin Arm	V10 SP1, V10 SP2, and V10 SP3
Debian x86_64	9, 11, and 12
EulerOS x86_64	V2R9, V2R10, V2R11, and V2R12
EulerOS Arm	V2R10 and V2R12
UOS x86_64	V20
BC-Linux Arm	21.10
OpenEuler x86_64	22.03 LTS SP1 and 24.03 LTS SP1
Red Hat Enterprise Linux x86_64	8.6, 8.7, 8.8, 8.9, 8.10, and 9.4
Anolis OS x86_64	8.9

3.1.1.2 Features

3.1.1.2.1 Huawei CSI

Table 3-5 Features supported by Huawei storage and constraints

Feature	OceanStor V5	OceanStor	OceanStor Dorado
Static Provisioning	NFS 3	• NFS 3/4.0/4.1/4.2 • NFS over RDMA is supported only in 6.1.7 and later versions. • NFS 4.2 is supported only in 6.1.8 and later versions.	• NFS 3/4.0/4.1/4.2/NFS+ • NFS 4.1 is supported only in 6.1.3 and later versions. • NFS over RDMA is supported only in 6.1.7 and later versions. • NFS 4.2 is supported only in 6.1.8 and later versions. • NFS+ is supported only in 6.1.7 and later versions.
Dynamic Provisioning			
Manage Provisioning			
Expand Persistent Volume	Volumes created in Dynamic Provisioning or Manage Provisioning mode are supported.		

Feature	OceanStor V5	OceanStor	OceanStor Dorado	
Create VolumeSnapshot	Volumes created in Dynamic Provisioning or Manage Provisioning mode are supported.			
Delete VolumeSnapshot	Supported			
Restore VolumeSnapshot	Supported	Supported only in 6.1.5 and later versions		
Clone Persistent Volume	Non-HyperMetro volumes created in Dynamic Provisioning or Manage Provisioning mode are supported.	Only 6.1.5 and later versions support volumes created in Dynamic Provisioning or Manage Provisioning mode.		
Raw Block Volume	Not supported			
Topology	Supported			
Generic Ephemeral Volumes	Supported			
Access Mode	RWO, ROX, RWX, and RWOP. RWOP is supported only in Kubernetes 1.22 or later.			
QoS	Supported only by system users	Supported		
Application type	Not supported	Supported		
Volume HyperMetro	Not supported	Only the HyperMetro active-active (AA) mode is supported.		
Storage multi-tenant	Supported	Supported in 6.1.3 and later versions		

Table 3-6 Features supported by Huawei enterprise storage dtrees

Feature	Supported
Static Provisioning	√
Dynamic Provisioning	√
Manage Provisioning	√
Expand Persistent Volume	√
Access Mode	√ (RWX/RWO/ROX/RWOP: Kubernetes 1.22 or later supports RWOP.)
Multi-tenant	√
Create VolumeSnapshot	X
Delete VolumeSnapshot	X
Restore VolumeSnapshot	X
Clone Persistent Volume	X
QoS	X
Volume HyperMetro	X
Application type	X

3.1.2 Block Service

3.1.2.1 Compatibility

3.1.2.1.1 Storage

Compatibility with Huawei CSI Storage

Table 3-7 Supported Huawei storage products

Storage Product	Version
OceanStor V5	V500R007 and V500R007 Kunpeng
OceanStor	6.1.3, 6.1.5, 6.1.6, 6.1.7, 6.1.8, V700R001C00, V700R001C10, V700R001C20, and V700R001C30
OceanStor Dorado	6.1.0, 6.1.2, 6.1.3, 6.1.5, 6.1.6, 6.1.7, 6.1.8, V700R001C00, V700R001C10, V700R001C20, and V700R001C30

3.1.2.1.2 Container Platform

Huawei CSI Platform Compatibility

Table 3-8 Supported container management platforms

Container Management Platform	Version
Kubernetes	1.16 to 1.36
Red Hat OpenShift Container Platform	4.12 to 4.22
Tanzu Kubernetes Grid Integrated	TKGI 1.17 to 1.19
CCE Agile	22.3.2 and 23.12.1
CCE	22.9.5

3.1.2.1.3 OS

Huawei CSI OS Compatibility

Table 3-9 Supported host OSs

OS Name	OS Version	Native DM-Multipath Version	Huawei UltraPath Version	Native NVMe-Native Version
CentOS x86_64	7.6, 7.7, and 7.9	Delivered with the OS, supporting FC/iSCSI	UltraPath 31.1.0, supporting FC/iSCSI	Not supported
CentOS x86_64	8.2 and 8.4	Delivered with the OS, supporting FC/iSCSI	UltraPath 31.1.0, supporting FC/iSCSI UltraPath-NVMe 31.1.RC8, supporting NVMe over RoCE/NVMe over FC	Not supported
CentOS Arm	7.6	Delivered with the OS, supporting FC/iSCSI	Not supported	Not supported
Rocky Linux x86_64	8.6	Delivered with the OS, supporting FC/iSCSI	UltraPath-NVMe 31.2.1, supporting NVMe over RoCE	Not supported

OS Name	OS Version	Native DM-Multipath Version	Huawei UltraPath Version	Native NVMe-Native Version
	9.2	Delivered with the OS, supporting FC	Not supported	Not supported
SUSE 15 x86_64	SP2	Delivered with the OS, supporting FC/iSCSI	UltraPath 31.1.0, supporting FC/iSCSI UltraPath 31.3.0, supporting iSCSI UltraPath-NVMe 31.1.RC8, supporting NVMe over RoCE/NVMe over FC	Not supported
	SP3	Delivered with the OS, supporting FC/iSCSI	UltraPath 31.1.0, supporting FC/iSCSI UltraPath-NVMe 31.1.RC8, supporting NVMe over RoCE/NVMe over FC	Not supported
	SP6	Delivered with the OS, supporting FC/iSCSI	UltraPath 31.1.0, supporting FC/iSCSI UltraPath-NVMe 32.4.0, supporting NVMe over RoCE/ NVMe over FC	Not supported
	SP7	Delivered with the OS, supporting FC/iSCSI	Not supported	Not supported
Red Hat CoreOS x86_64	4.12 to 4.18, and 4.20	Delivered with the OS, supporting FC/iSCSI	Not supported	Not supported
	4.19	Delivered with the OS, supporting FC/iSCSI	Not supported	Delivered with the OS, supporting NVMe over TCP

OS Name	OS Version	Native DM-Multipath Version	Huawei UltraPath Version	Native NVMe-Native Version
Ubuntu x86_64	18.04, 20.04, and 22.04	Delivered with the OS, supporting FC/iSCSI	Not supported	Not supported
	24.04	Not supported	Not supported	Delivered with the OS, supporting NVMe over TCP
Ubuntu Arm	22.04	Delivered with the OS, supporting FC/iSCSI	Not supported	Not supported
	24.04	Not supported	Not supported	Delivered with the OS, supporting NVMe over TCP
Kylin x86_64	7.6, V10 SP2, and V10 SP3	Delivered with the OS, supporting FC/iSCSI	UltraPath 31.2.0, supporting FC/iSCSI	Not supported
	V10 SP1	Delivered with the OS, supporting FC/iSCSI	UltraPath 31.2.0, supporting FC/iSCSI UltraPath 31.3.0, supporting iSCSI	Not supported
Kylin Arm	V10 SP1, V10 SP2, and V10 SP3	Delivered with the OS, supporting FC/iSCSI	UltraPath 31.3.0, supporting iSCSI	Not supported
Debian x86_64	9 and 11	Delivered with the OS, supporting FC/iSCSI	Not supported	Not supported
	12	Delivered with the OS, supporting FC/iSCSI	Not supported	Delivered with the OS, supporting NVMe over TCP

OS Name	OS Version	Native DM-Multipath Version	Huawei UltraPath Version	Native NVMe-Native Version
EulerOS x86_64	V2R9, V2R10, V2R11, and V2R12	Delivered with the OS, supporting FC/iSCSI	Not supported	Not supported
EulerOS Arm	V2R10 and V2R12	Delivered with the OS, supporting FC/iSCSI	Not supported	Not supported
UOS x86_64	V20	Delivered with the OS, supporting FC/iSCSI	Not supported	Not supported
UOS ARM	V20	Not supported	Ultrapath-NVMe 31.05.039, supporting NVMe over RoCE	Not supported
BC-Linux Arm	21.10	Delivered with the OS, supporting FC/iSCSI	Not supported	Not supported
OpenEuler x86_64	22.03 LTS SP1	Delivered with the OS, supporting iSCSI	Not supported	Not supported
Red Hat Enterprise Linux x86_64	8.6, 8.7, 8.8, 8.9, and 9.4	Delivered with the OS, supporting FC/iSCSI	Not supported	Not supported
	8.10	Delivered with the OS, supporting FC/iSCSI	Ultrapath-NVMe 31.08.013, supporting NVMe over FC	Not supported
	9.6	Delivered with the OS, supporting FC	Not supported	Not supported
AlmaLinux x86_64	8.6	Not supported	Ultrapath-NVMe 31.05.039, supporting NVMe over RoCE	Not supported

OS Name	OS Version	Native DM-Multipath Version	Huawei UltraPath Version	Native NVMe-Native Version
	8.8	Not supported	Ultrapath-NVMe 31.05.025, supporting NVMe over RoCE	Not supported

3.1.2.2 Features

3.1.2.2.1 Huawei CSI

Table 3-10 Features supported by Huawei storage and constraints

Feature	OceanStor V5	OceanStor	OceanStor Dorado
Static Provisioning	FC/iSCSI	FC/iSCSI/NVMe over RoCE/NVMe over FC/ NVMe over TCP	FC/iSCSI/NVMe over RoCE/NVMe over FC/ NVMe over TCP
Dynamic Provisioning			
Manage Provisioning			
Expand Persistent Volume	• Volumes created in Dynamic Provisioning or Manage Provisioning mode are supported. • The provisioned PVC whose volumeType is lun and accessModes is ReadOnlyMany does not support capacity expansion.		
Create VolumeSnaps hot	Volumes created in Dynamic Provisioning or Manage Provisioning mode are supported.		
Delete VolumeSnaps hot	Supported		
Restore VolumeSnaps hot	Supported		
Clone Persistent Volume	Volumes created in Dynamic Provisioning or Manage Provisioning mode are supported.		
Raw Block Volume	Supported		

Feature	OceanStor V5	OceanStor	OceanStor Dorado
Topology	Supported		
Generic Ephemeral Volumes	Supported		
Access Mode	• RWO/ROX/RWOP: supported. RWOP is supported only in Kubernetes 1.22 or later. • RWX: supported only by Raw Block volumes		
QoS	Supported. Only system users can configure QoS.		
Application type	Not supported	Supported	
Volume HyperMetro	Not supported	Supported (FC/iSCSI+DM-Multipath)	
Multiple vStores	Not supported. Only single-vStore interconnection is supported.		

 NOTE

- When using the iSCSI protocol to connect the CSI to a storage system with multiple controller enclosures, you need to log in to the storage background and run the **change target_name display_mode mode=Full** command to ensure that the target IQN obtained by the CSI from the storage system is the same as the actual IQN.
- If a container platform is deployed on a virtualization platform, the iSCSI protocol is recommended when the CSI is connected to SAN storage.
- If the FC, NVMe over FC, or NVMe over RoCE protocol is required, the virtualization platform needs to be configured. In this case, the customer's virtualization team needs to provide technical support.
- When NVMe over RoCE or NVMe over FC is used, the supported nvme-cli tool version is 1.9 or later. When NVMe over TCP is used, the supported nvme-cli tool version is 2.0 or later. The query command is **nvme version**.
- A single cluster can connect to only one storage vStore. Initiators are associated with a single vStore on the storage system. If multiple vStores are interconnected, some vStores cannot obtain the initiator permission. As a result, storage resources fail to be mapped and mounted.

3.2 Flash Storage (OceanStor A600/A800)

3.2.1 File Service

3.2.1.1 Compatibility

3.2.1.1.1 Storage

Compatibility with Huawei CSI Storage

Table 3-11 Supported Huawei storage products

Storage Product	Version
OceanStor A series	V700R001C10, V700R001C20, and V700R001C30

3.2.1.1.2 Container Platform

Huawei CSI Platform Compatibility

Table 3-12 Supported container management platforms

Container Management Platform	Version
Kubernetes	1.16 to 1.36
Red Hat OpenShift Container Platform	4.19, 4.21, and 4.22 (only the NFS protocol is supported)

3.2.1.1.3 OS

Huawei CSI OS Compatibility

Table 3-13 Supported host OSs

OS Name	OS Version
EulerOS Arm	V2R12, V2R13, and V2R15
Kylin Arm	V10 SP3
BCLinux Arm	21.10U4

3.2.1.2 Features

3.2.1.2.1 Huawei CSI

Table 3-14 Features and constraints supported by Huawei A series storage

Feature	OceanStor A Series
Static Provisioning	- DataTurbo, NFS 3/4.0/4.1/4.2, NFS over RDMA - Only local file systems of storage with a single zone are supported.
Dynamic Provisioning	
Manage Provisioning	
Expand Persistent Volume	Volumes created in Dynamic Provisioning or Manage Provisioning mode are supported.
Create VolumeSnapshot	Not supported
Delete VolumeSnapshot	Not supported
Restore VolumeSnapshot	Not supported
Clone Persistent Volume	Not supported
Raw Block Volume	Not supported
Topology	Supported
Generic Ephemeral Volumes	Supported
Access Mode	RWO, ROX, RWX, and RWOP. RWOP is supported only in Kubernetes 1.22 or later.
QoS	Not supported
Application type	Supported
Volume HyperMetro	Not supported
Multiple vStores	Not supported
KV cache	Supported in V700R001C20 and later versions

Table 3-15 Features supported by Huawei A series storage dtrees

Feature	Supported or Not
Static Provisioning	√
Dynamic Provisioning	√
Manage Provisioning	√
Expand Persistent Volume	√
Access Mode	√ (RWX/RWO/ROX/RWOP: Kubernetes 1.22 or later supports RWOP.)

Feature	Supported or Not
Multiple vStores	X
Create VolumeSnapshot	X
Delete VolumeSnapshot	X
Restore VolumeSnapshot	X
Clone Persistent Volume	X
QoS	X
Volume HyperMetro	X
Application type	X

3.3 iMaster DME (Interconnected with an OceanStor A800 Cluster)

3.3.1 File Service

3.3.1.1 Compatibility

3.3.1.1.1 Storage

Compatibility with Huawei CSI Storage

Table 3-16 Supported Huawei storage products

Storage Product	Version
iMaster DME	24.0.0, 24.1.0, 25.1.0

Table 3-17 iMaster DME versions supported by iMaster DME dtrees

Storage Product	Version
iMaster DME	25.1.0

3.3.1.1.2 Container Platform

Huawei CSI Platform Compatibility

Table 3-18 Supported container management platforms

Container Management Platform	Version
Kubernetes	1.16 to 1.36

3.3.1.1.3 OS

Huawei CSI OS Compatibility

Table 3-19 Supported host OSs

OS Name	OS Version
OpenEuler x86_64	22.03 LTS SP4
EulerOS Arm	V2R15

3.3.1.2 Features

3.3.1.2.1 Huawei CSI

Table 3-20 Features supported by Huawei storage and constraints

Feature	OceanStor A800 Cluster on iMaster DME
Static Provisioning	- DataTurbo/NFS 3 - Global file systems of the OceanStor A800 cluster are supported.
Dynamic Provisioning	
Manage Provisioning	
Expand Persistent Volume	Volumes created in Dynamic Provisioning or Manage Provisioning mode are supported.
Create VolumeSnapshot	Not supported
Delete VolumeSnapshot	Not supported
Restore VolumeSnapshot	Not supported
Clone Persistent Volume	Not supported
Raw Block Volume	Not supported
Topology	Supported

Feature	OceanStor A800 Cluster on iMaster DME
Generic Ephemeral Volumes	Supported
Access Mode	RWO, ROX, RWX, and RWOP. RWOP is supported only in Kubernetes 1.22 or later.
QoS	Not supported
Application type	Not supported
Volume HyperMetro	Not supported
Multiple vStores	Not supported
KV cache	Supported in 25.1.0 and later versions

Table 3-21 Features supported by Huawei iMaster DME dtrees

Feature	Supported or Not
Static Provisioning	√
Dynamic Provisioning	√
Manage Provisioning	√
Expand Persistent Volume	√
Access Mode	√ (RWX/RWO/ROX/RWOP: Kubernetes 1.22 or later supports RWOP.)
Multiple vStores	X
Create VolumeSnapshot	X
Delete VolumeSnapshot	X
Restore VolumeSnapshot	X
Clone Persistent Volume	X
QoS	X
Volume HyperMetro	X
Application type	X

3.4 Mass Storage (OceanStor Pacific/FusionStorage Block Series)

3.4.1 File Service

3.4.1.1 Compatibility

3.4.1.1.1 Storage

Compatibility with Huawei CSI Storage

Table 3-22 Supported Huawei storage products

Storage Product	Version
OceanStor Pacific series	8.1.0, 8.1.1, 8.1.2, 8.1.3, 8.1.5, 8.2.0, 8.2.1, V800R001C10, V800R001C20, and V800R001C30

Table 3-23 Huawei storage versions supported by Huawei distributed storage dtrees

Storage Product	Version
OceanStor Pacific series	8.1.0, 8.1.1, 8.1.2, 8.1.3, 8.1.5, 8.2.0, 8.2.1, V800R001C10, V800R001C20, and V800R001C30

3.4.1.1.2 Container Platform

Huawei CSI Platform Compatibility

Table 3-24 Supported container management platforms

Container Management Platform	Version
Kubernetes	1.16 to 1.36
Red Hat OpenShift Container Platform	4.12 to 4.22
Tanzu Kubernetes Grid Integrated	TKGI 1.17 to 1.19
CCE Agile	22.3.2 and 23.12.1
CCE	22.9.5

3.4.1.1.3 OS

Table 3-25 Supported host OSs

OS Name	OS Version
CentOS x86_64	7.6, 7.7, 7.9, 8.2, and 8.4
CentOS Arm	7.6
Rocky Linux x86_64	8.6
SUSE 15 x86_64	SP2 and SP3
Red Hat CoreOS x86_64	4.12 to 4.20
Ubuntu x86_64	18.04, 20.04, 22.04, and 24.04
Ubuntu Arm	22.04 and 24.04
Kylin x86_64	7.6, V10 SP1, V10 SP2, and V10 SP3
Kylin Arm	V10 SP1, V10 SP2, and V10 SP3
Debian x86_64	9, 11, and 12
EulerOS x86_64	V2R9, V2R10, V2R11, and V2R12
EulerOS Arm	V2R10 and V2R12
UOS x86_64	V20
BC-Linux Arm	21.10
Anolis OS x86_64	8.8
OpenEuler x86_64	22.03 LTS SP1
Red Hat Enterprise Linux x86_64	8.6, 8.7, 8.8, 8.9, 8.10, and 9.4

3.4.1.2 Features

3.4.1.2.1 Huawei CSI

Table 3-26 Features supported by Huawei storage and constraints

Feature	OceanStor Pacific Series
Static Provisioning	• DPC/NFS 3/4.1 • DPC is supported only in 8.1.2 and later versions. • NFS 4.1 is supported by 8.1.2 and later versions. • Only OceanStor Pacific series 8.2.0 and later versions support NFS over RDMA. When NFS over RDMA is used, only NFS 3 is supported.
Dynamic Provisioning	
Manage Provisioning	
Expand Persistent Volume	Volumes created in Dynamic Provisioning or Manage Provisioning mode are supported.
Create VolumeSnapshot	Not supported
Delete VolumeSnapshot	Not supported
Restore VolumeSnapshot	Not supported
Clone Persistent Volume	Not supported
Raw Block Volume	Not supported
Topology	Supported
Generic Ephemeral Inline Volumes	Supported
Access Mode	RWO, ROX, RWX, and RWOP. RWOP is supported only in Kubernetes 1.22 or later.
QoS	Supported
Soft and hard quotas	Supported
Storage multi-tenant	Supported

Table 3-27 Features supported by Huawei distributed storage dtrees

Feature	Supported
Static Provisioning	√
Dynamic Provisioning	√
Manage Provisioning	√
Expand Persistent Volume	√
Access Mode	√ (RWX/RWO/ROX/RWOP: Kubernetes 1.22 or later supports RWOP.)

Feature	Supported
Multi-tenant	√
Create VolumeSnapshot	X
Delete VolumeSnapshot	X
Restore VolumeSnapshot	X
Clone Persistent Volume	X
QoS	X
Volume HyperMetro	X
Application type	X

3.4.2 Block Service

3.4.2.1 Compatibility

3.4.2.1.1 Storage

Compatibility with Huawei CSI Storage

Table 3-28 Supported Huawei storage products

Storage Product	Version
FusionStorage block	8.0.1
OceanStor Pacific series	8.1.0, 8.1.1, 8.1.2, 8.1.3, 8.1.5, 8.2.0, 8.2.1, V800R001C10, V800R001C20, and V800R001C30

3.4.2.1.2 Container Platform

Huawei CSI Platform Compatibility

Table 3-29 Supported container management platforms

Container Management Platform	Version
Kubernetes	1.16 to 1.36
Red Hat OpenShift Container Platform	4.12 to 4.22
Tanzu Kubernetes Grid Integrated	TKGI 1.17 to 1.19

Container Management Platform	Version
CCE Agile	22.3.2 and 23.12.1
CCE	22.9.5

3.4.2.1.3 OS

Huawei CSI OS Compatibility

Table 3-30 Supported host OSs

OS Name	OS Version	Native DM-Multipath Version	Huawei UltraPath Version
CentOS x86_64	7.6, 7.7, 7.9, 8.2, and 8.4	Delivered with the OS, supporting iSCSI	UltraPath 31.1.0, supporting iSCSI
CentOS Arm	7.6	Delivered with the OS, supporting iSCSI	Not supported
Rocky Linux x86_64	8.6	Delivered with the OS, supporting iSCSI	Not supported
SUSE 15 x86_64	SP2	Delivered with the OS, supporting iSCSI	UltraPath 31.1.0, supporting iSCSI UltraPath 31.3.0, supporting iSCSI
	SP3	Delivered with the OS, supporting iSCSI	UltraPath 31.1.0, supporting iSCSI
	SP7	Delivered with the OS, supporting iSCSI	Not supported
Red Hat CoreOS x86_64	4.12 to 4.20	Delivered with the OS, supporting iSCSI	Not supported

OS Name	OS Version	Native DM-Multipath Version	Huawei UltraPath Version
Ubuntu x86_64	18.04, 20.04, and 22.04	Delivered with the OS, supporting iSCSI	Not supported
	24.04 (Only OceanStor Pacific storage is supported.)	Delivered with the OS, supporting iSCSI	Not supported
Ubuntu Arm	22.04	Delivered with the OS, supporting iSCSI	Not supported
	24.04 (Only OceanStor Pacific storage is supported.)	Delivered with the OS, supporting iSCSI	Not supported
Kylin x86_64	7.6, V10 SP2, and V10 SP3	Delivered with the OS, supporting iSCSI	UltraPath 31.2.0, supporting iSCSI
	V10 SP1	Delivered with the OS, supporting iSCSI	UltraPath 31.2.0, supporting iSCSI UltraPath 31.3.0, supporting iSCSI
Kylin Arm	V10 SP1 and V10 SP2	Delivered with the OS, supporting iSCSI	Not supported
	V10 SP3	Delivered with the OS, supporting iSCSI	UltraPath 31.3.0, supporting iSCSI
Debian x86_64	9, 11, and 12	Delivered with the OS, supporting iSCSI	Not supported
EulerOS x86_64	V2R9, V2R10, V2R11, and V2R12	Delivered with the OS, supporting iSCSI	Not supported

OS Name	OS Version	Native DM-Multipath Version	Huawei UltraPath Version
EulerOS Arm	V2R10 and V2R12	Delivered with the OS, supporting iSCSI	Not supported
UOS x86_64	V20	Delivered with the OS, supporting iSCSI	Not supported
BC-Linux Arm	21.10	Delivered with the OS, supporting iSCSI	Not supported
Anolis OS x86_64 (Only OceanStor Pacific storage is supported.)	8.8	Delivered with the OS, supporting iSCSI	Not supported
OpenEuler x86_64	22.03 LTS SP1	Delivered with the OS, supporting iSCSI	Not supported
Red Hat Enterprise Linux x86_64	8.6, 8.7, 8.8, 8.9, 8.10, and 9.4	Delivered with the OS, supporting iSCSI	Not supported

3.4.2.2 Features

3.4.2.2.1 Huawei CSI

Feature	FusionStorage Block	OceanStor Pacific Series
Static Provisioning	iSCSI/SCSI	iSCSI/SCSI
Dynamic Provisioning		
Manage Provisioning		
Expand Persistent Volume	• Volumes created in Dynamic Provisioning or Manage Provisioning mode are supported. • The provisioned PVC whose volumeType is lun and accessModes is ReadOnlyMany does not support capacity expansion.	

Feature	FusionStorage Block	OceanStor Pacific Series
Create VolumeSnapshot	Volumes created in Dynamic Provisioning or Manage Provisioning mode are supported.	
Delete VolumeSnapshot	Supported	
Restore VolumeSnapshot	Supported	
Clone Persistent Volume	Volumes created in Dynamic Provisioning or Manage Provisioning mode are supported.	
Raw Block Volume	Supported	
Topology	Supported	
Generic Ephemeral Inline Volumes	Supported	
Access Mode	RWO/ROX/RWOP: supported by all types of volumes. RWOP is supported only by Kubernetes 1.22 and later versions. RWX: supported only by Raw Block volumes.	
QoS	Supported	
Soft and hard quotas	Not supported	
Storage multi-tenant	Not supported	

3.5 Mass Storage (OceanDisk Series)

3.5.1 File Service

3.5.1.1 Compatibility

3.5.1.1.1 Storage

Compatibility with Huawei CSI Storage

Table 3-31 Supported Huawei storage products

Storage Product	Version
OceanDisk 1500T (NAS)	V700R001C01, V700R001C30

3.5.1.1.2 Container Platform

Huawei CSI Platform Compatibility

Table 3-32 Supported container management platforms

Container Management Platform	Version
Kubernetes	1.16 to 1.36
Red Hat OpenShift Container Platform	4.12 to 4.22
Tanzu Kubernetes Grid Integrated	TKGI 1.17 to 1.19
CCE Agile	22.3.2 and 23.12.1
CCE	22.9.5

3.5.1.1.3 OS

Table 3-33 Supported host OSs

OS Name	OS Version
CentOS x86_64	7.6, 7.7, 7.9, 8.2, 8.4
CentOS Arm	7.6
Rocky Linux x86_64	8.6
SUSE 15 x86_64	SP2 and SP3
Red Hat CoreOS x86_64	4.12 to 4.20
Ubuntu x86_64	18.04, 20.04, 22.04, and 24.04
Ubuntu Arm	22.04 and 24.04
Kylin x86_64	7.6, V10 SP1, V10 SP2, and V10 SP3
Kylin Arm	V10 SP1, V10 SP2, and V10 SP3
Debian x86_64	9, 11, and 12
EulerOS x86_64	V2R9, V2R10, V2R11, and V2R12
EulerOS Arm	V2R10 and V2R12
UOS x86_64	V20
BC-Linux Arm	21.10
OpenEuler x86_64	22.03 LTS SP1 and 24.03 LTS SP1
Red Hat Enterprise Linux x86_64	8.6, 8.7, 8.8, 8.9, 8.10, and 9.4

OS Name	OS Version
Anolis OS x86_64	8.9

3.5.1.2 Features

3.5.1.2.1 Huawei CSI

Table 3-34 Features supported by Huawei storage and constraints

Feature	OceanDisk 1500T
Static Provisioning	NFS 3/4.0
Dynamic Provisioning	
Manage Provisioning	
Expand Persistent Volume	Volumes created in Dynamic Provisioning or Manage Provisioning mode are supported.
Create VolumeSnapshot	Not supported
Delete VolumeSnapshot	Not supported
Restore VolumeSnapshot	Not supported
Clone Persistent Volume	Not supported
Raw Block Volume	Not supported
Topology	Supported
Generic Ephemeral Inline Volumes	Supported
Access Mode	RWO, ROX, RWX, and RWOP. RWOP is supported only in Kubernetes 1.22 or later.
QoS	Supported
Application type	Supported
Storage multi-tenant	Supported

3.5.2 Block Service

3.5.2.1 Compatibility

3.5.2.1.1 Storage

Compatibility with Huawei CSI Storage

Table 3-35 Supported Huawei storage products

Storage Product	Version
OceanDisk 1500/1600 series	1.5.0, V700R001C10, V700R001C11, V700R001C30

3.5.2.1.2 Container Platform

Huawei CSI Platform Compatibility

Table 3-36 Supported container management platforms

Container Management Platform	Version
Kubernetes	1.16 to 1.36
Red Hat OpenShift Container Platform	4.12 to 4.22
Tanzu Kubernetes Grid Integrated	TKGI 1.17 to 1.19
CCE Agile	22.3.2 and 23.12.1
CCE	22.9.5

3.5.2.1.3 OS

Huawei CSI OS Compatibility

Table 3-37 Supported host OSs

OS Name	OS Version	Native DM-Multipath Version	Huawei UltraPath Version
CentOS x86_64	7.6, 7.7, and 7.9	Delivered with the OS, supporting FC/iSCSI	UltraPath 31.1.0, supporting FC/iSCSI
CentOS x86_64	8.2 and 8.4	Delivered with the OS, supporting FC/iSCSI	UltraPath 31.1.0, supporting FC/iSCSI UltraPath-NVMe 31.1.RC8, supporting NVMe over RoCE

OS Name	OS Version	Native DM-Multipath Version	Huawei UltraPath Version
CentOS ARM	7.6	Delivered with the OS, supporting FC/iSCSI	Not supported
Rocky Linux x86_64	8.6	Delivered with the OS, supporting FC/iSCSI	UltraPath-NVMe 31.2.1, supporting NVMe over RoCE
	9.2	Delivered with the OS, supporting FC	Not supported
SUSE 15 x86_64	SP2	Delivered with the OS, supporting FC/iSCSI	UltraPath 31.1.0, supporting FC/iSCSI UltraPath 31.3.0, supporting iSCSI UltraPath-NVMe 31.1.RC8, supporting NVMe over RoCE
	SP3	Delivered with the OS, supporting FC/iSCSI	UltraPath 31.1.0, supporting FC/iSCSI UltraPath-NVMe 31.1.RC8, supporting NVMe over RoCE
	SP6	Delivered with the OS, supporting FC/iSCSI	UltraPath 31.1.0, supporting FC/iSCSI UltraPath-NVMe 32.4.0, supporting NVMe over RoCE
	SP7	Delivered with the OS, supporting FC/iSCSI	Not supported
Red Hat CoreOS x86_64	4.12 to 4.20	Delivered with the OS, supporting FC/iSCSI	Not supported
Ubuntu x86_64	18.04, 20.04, and 22.04	Delivered with the OS, supporting FC/iSCSI	Not supported

OS Name	OS Version	Native DM-Multipath Version	Huawei UltraPath Version
Ubuntu ARM	22.04	Delivered with the OS, supporting FC/iSCSI	Not supported
Kylin x86_64	7.6, V10 SP2, and V10 SP3	Delivered with the OS, supporting FC/iSCSI	UltraPath 31.2.0, supporting FC/iSCSI
	V10 SP1	Delivered with the OS, supporting FC/iSCSI	UltraPath 31.2.0, supporting FC/iSCSI UltraPath 31.3.0, supporting iSCSI
Kylin ARM	V10 SP1, V10 SP2, and V10 SP3	Delivered with the OS, supporting FC/iSCSI	UltraPath 31.3.0, supporting iSCSI
Debian x86_64	9, 11, and 12	Delivered with the OS, supporting FC/iSCSI	Not supported
EulerOS x86_64	V2R9, V2R10, V2R11, and V2R12	Delivered with the OS, supporting FC/iSCSI	Not supported
EulerOS ARM	V2R10 and V2R12	Delivered with the OS, supporting FC/iSCSI	Not supported
UOS x86_64	V20	Delivered with the OS, supporting FC/iSCSI	Not supported
UOS ARM	V20	Not supported	Ultrath-NVMe 31.05.039, supporting NVMe over RoCE
BC-Linux ARM	21.10	Delivered with the OS, supporting FC/iSCSI	Not supported

OS Name	OS Version	Native DM-Multipath Version	Huawei UltraPath Version
OpenEuler x86_64	22.03 LTS SP1	Delivered with the OS, supporting iSCSI	Not supported
Red Hat Enterprise Linux x86_64	8.6, 8.7, 8.8, 8.9, 8.10, and 9.4	Delivered with the OS, supporting FC/iSCSI	Not supported
	9.6	Delivered with the OS, supporting FC	Not supported
AlmaLinux x86_64	8.6	Not supported	Ultrath-NVMe 31.05.039, supporting NVMe over RoCE
	8.8	Not supported	Ultrath-NVMe 31.05.025, supporting NVMe over RoCE

3.5.2.2 Features

3.5.2.2.1 Huawei CSI

Feature	OceanDisk 1500/1600
Static Provisioning	SAN: FC/iSCSI/NVMe over RoCE
Dynamic Provisioning	
Manage Provisioning	
Expand Persistent Volume	Volumes created in Dynamic Provisioning or Manage Provisioning mode are supported.
Create VolumeSnapshot	Not supported
Delete VolumeSnapshot	Not supported
Restore VolumeSnapshot	Not supported
Clone Persistent Volume	Not supported

Feature	OceanDisk 1500/1600
Raw Block Volume	Supported
Topology	Supported
Generic Ephemeral Inline Volumes	Supported
Access Mode	RWO/ROX/RWOP: supported by all types of volumes. RWOP is supported only by Kubernetes 1.22 and later versions. RWX: supported only by Raw Block volumes.
QoS	Supported
Application type	Supported
Storage multi-tenant	Not supported

 NOTE

- When using the iSCSI protocol to connect the CSI to a storage system with multiple controller enclosures, you need to log in to the storage background and run the **change target_name display_mode mode=Full** command to ensure that the target IQN obtained by the CSI from the storage system is the same as the actual IQN.
- If a container platform is deployed on a virtualization platform, the iSCSI protocol is recommended when the CSI is connected to SAN storage.
- If the FC, NVMe over FC, or NVMe over RoCE protocol is required, the virtualization platform needs to be configured. In this case, the customer's virtualization team needs to provide technical support.
- When NVMe over RoCE or NVMe over FC is used, the supported nvme-cli tool version is 1.9 or later. When NVMe over TCP is used, the supported nvme-cli tool version is 2.0 or later. The query command is **nvme version**.
- A single cluster can connect to only one storage vStore. Initiators are associated with a single vStore on the storage system. If multiple vStores are interconnected, some vStores cannot obtain the initiator permission. As a result, storage resources fail to be mapped and mounted.

4 Installation and Deployment

4.1 CSI

4.1 CSI

4.1.1 Installation Preparations

This chapter describes the preparations for the installation.

Prerequisites

- The container management platform has been deployed and is running properly. The container platform compatibility requirements in [3 Compatibility and Features](#) are met.
- (Mandatory for enterprise storage) Initial configuration for interconnecting with Huawei enterprise storage has been completed, including storage pool division and port configuration. The version of the storage product meets the requirements in [3 Compatibility and Features](#).
- (Mandatory for distributed storage) Initial configuration for interconnecting with Huawei distributed storage has been completed, including storage pool division and port configuration. The version of the storage product meets the requirements in [3 Compatibility and Features](#).
- The connectivity between Huawei storage and the container platform host has been configured. For example, the worker node running huawei-csi-controller communicates properly with the management IP address of the storage device to be connected, and the worker node running huawei-csi-node communicates properly with the service IP address of the storage device to be connected. In iSCSI scenarios, the **ping** command can be used to verify the connectivity.
- Ensure that the language of the operating system is English.
- Ensure that storage resource names, such as storage pool names and tenant names, are in English.

4.1.1.1 Downloading the Huawei CSI Software Package

This section describes how to download the software package and the component structure of the software package.

Step 1 Open a browser and enter https://github.com/Huawei/eSDK_K8S_Plugin/releases in the address box.

Step 2 Download the software package of the 4.13.0 version based on the CPU architecture.

 NOTE

- Software package naming rule: Plug-in name (**eSDK_Storage_CSI**) + Version number + CPU architecture
- CSI supports the following CPU architectures: x86, Arm, and PPC64LE.

Step 3 Decompress the downloaded software package. The following table shows the component structure of the software package.

Table 4-1 Component description

Component	Description
image/huawei-csi-v4.13.0- <i>arch</i> .tar	huawei-csi-driver image. <i>arch</i> indicates the CPU architecture.
image/storage-backend-controller-v4.13.0- <i>arch</i> .tar	Back-end management controller image. <i>arch</i> indicates the CPU architecture.
image/storage-backend-sidecar-v4.13.0- <i>arch</i> .tar	Back-end management sidecar image. <i>arch</i> indicates the CPU architecture.
image/huawei-csi-extender-v4.13.0- <i>arch</i> .tar	huawei-csi-extender image. <i>arch</i> indicates the CPU architecture.
bin/	Binary file used by an image provided by Huawei.
bin/oceanctl	Command line tool provided by Huawei, which can be used to manage storage backends.
helm/	Helm project used to deploy Huawei CSI.
manual/	Used to manually install and deploy Huawei CSI.
examples/	.yaml sample file used during CSI use.
examples/backend	.yaml sample file used to create a storage backend.

----End

4.1.1.2 Uploading a Huawei CSI Image

Huawei provides the **huawei-csi** image for users. For details about how to obtain the image file, see [4.1.1.1 Downloading the Huawei CSI Software Package](#).

To use the CSI image on the container management platform, you need to import the CSI image to the cluster in advance:

- Method 1: (Recommended) Use Docker to upload the CSI image to the image repository. For details, see [Uploading an Image to the Image Repository](#).
- Method 2: Manually import the CSI image to all nodes where Huawei CSI needs to be deployed. For details, see [Uploading an Image to a Local Node](#).

Uploading an Image to the Image Repository

The installation of Huawei CSI depends on the following image files provided by Huawei. Import and upload the image files in sequence. For details about how to obtain the image files, see [4.1.1.1 Downloading the Huawei CSI Software Package](#).

- huawei-csi-v4.13.0-arch.tar
- storage-backend-controller-v4.13.0-arch.tar
- storage-backend-sidecar-v4.13.0-arch.tar
- huawei-csi-extender-v4.13.0-arch.tar

Prerequisites

A Linux host with Docker installed is available, and the host can access the image repository.

Procedure

Step 1 Run the following command to import the CSI image to the current node. *arch* indicates the CPU architecture.

```
docker load -i huawei-csi-v4.13.0-<arch>.tar
```

Step 2 Run the following command to add the image repository address to the image tag. **repo.huawei.com** indicates the image repository address.

```
docker tag huawei-csi:4.13.0 <repo.huawei.com>/huawei-csi:4.13.0
```

Step 3 Run the following command to upload the CSI image to the image repository. **repo.huawei.com** indicates the image repository address.

```
docker push <repo.huawei.com>/huawei-csi:4.13.0
```

----End

NOTE

- Run the corresponding CLI command to load and push the image based on the container runtime environment (for example, Docker or Containerd).
- For details about how to import and upload images to the CCE or CCE Agile platform, see the user manual of the platform.

Uploading an Image to a Local Node

If the image has been uploaded to the image repository, skip this section.

Prerequisites

- The node has the corresponding Huawei CSI image file. For details about how to obtain the image file, see [4.1.1.1 Downloading the Huawei CSI Software Package](#).

- Docker or another container engine has been installed on the node.

Procedure

- Step 1** Use a remote access tool, such as PuTTY, to log in to the node where the image is to be imported through the management IP address.
- Step 2** Copy the **image** directory in the Kubernetes CSI component package to any directory on the current node.
- Step 3** Run the **cd image** command to go to the **image** working directory. For details about the tool path, see [Table 4-1](#).
- Step 4** Run the following commands in sequence to import all Huawei CSI images in the **image** directory to the local node. In the commands, *name* indicates the name of a .tar image package.
- Run the following command using the Docker container engine:
`docker load -i <name>.tar`
 - Run the following command using the containerd container engine:
`ctr -n k8s.io image import <name>.tar`
 - Run the following command using the Podman container engine:
`podman load -i <name>.tar`

NOTE

If another container engine is installed on the node, use the image import command for the corresponding container engine.

----End

4.1.1.3 Checking User Configurations on Huawei Storage

After Huawei storage is connected to the container platform, Huawei CSI needs to manage storage resources on Huawei storage based on service requirements, such as creating and mapping volumes. In this case, Huawei CSI needs to use the users created on Huawei storage to communicate with Huawei storage. The following table lists the user information required for different storage devices.

Table 4-2 User requirements for connecting storage to CSI

Storage Type	User Type	Role	Level	Type
OceanStor V5	System user	Administrator	Administrator	Local user/ LDAP user ²
	vStore user	vStore administrator	Administrator	Local user/ LDAP user ²
OceanStor	System user	Administrator/ User-defined role ¹	N/A	Local user/ LDAP user ²

Storage Type	User Type	Role	Level	Type
OceanStor Dorado	System user	Administrator/ User-defined role ¹	N/A	Local user/ LDAP user ²
	vStore user	vStore administrator	N/A	Local user/ LDAP user ²
OceanStor A series	System user	Administrator	N/A	Local user/ LDAP user ²
OceanStor Pacific series	System user	Administrator	N/A	Local user
OceanDisk	System user	Administrator	N/A	Local user

 NOTE

- If a user-defined role is used, you need to configure permissions for the role. For details about how to configure the minimum permissions, see [10.2 Configuring Custom Permissions](#).
- If the LDAP authentication mode is used, you need to set the **authenticationMode** parameter when creating a backend. For details, see the parameter description in [5.1.1 Configuring the Storage Backend](#).
- Based on the principle of minimum and system security requirements, users under the super administrator role are not recommended.

4.1.1.4 Checking Volume Snapshot-Dependent Components

This section describes how to check the volume snapshot-dependent components in the cluster.

NOTICE

If Huawei CSI is deployed in a version earlier than Kubernetes v1.20, perform the following steps:

Step 1 Run the following command to check the Kubernetes version. The following uses Kubernetes v1.16.0 as an example.

```
kubectl get node
```

The following is an example of the command output.

NAME	STATUS	ROLES	AGE	VERSION
test-master	Ready	master	311d	v1.16.0
test-node	Ready	<none>	311d	v1.16.0

Step 2 Copy the **helm** directory in the Kubernetes CSI component package to any directory on the master node. For details about the Helm tool path, see [Table 4-1](#).

Step 3 Go to the **helm/esdk/crds/snapshot-crds** directory and run the following command to delete the snapshot CRD installation file.

```
rm -rf ./huawei-csi-snapshot-crd-v1.yaml
```

----End

4.1.1.5 Checking the Host Multipathing Configuration

If you plan to use the FC/iSCSI/NVMe over RoCE/NVMe over FC protocol to access Huawei storage in a container environment, you are advised to use host multipathing software to enhance the link redundancy and performance of the host and storage. If you do not want to use the software, skip this section.

For details about the OSs and multipathing software that can be interconnected with Huawei CSI software, see the OS compatibility in [3 Compatibility and Features](#).

NOTE

- If you want to use the FC/iSCSI protocol to connect to Huawei storage, you are advised to use native DM-Multipath provided by the OS.
- If you want to use the NVMe over RoCE/NVMe over FC protocol to connect to Huawei storage, you are advised to use Huawei-developed UltraPath-NVMe.
- If you want to use the SCSI protocol to connect to Huawei storage, disable DM-Multipath provided by the OS.

Prerequisites

Multipathing software has been correctly installed on a host.

- If you use native DM-Multipath provided by the OS, contact your host or OS provider to obtain the documents and software packages required for the installation.
- If you use Huawei-developed UltraPath or UltraPath-NVMe, contact Huawei engineers to obtain the UltraPath or UltraPath-NVMe documents and software packages. For details about the software package version, see the OS compatibility in [3 Compatibility and Features](#).

Procedure

- If you use the iSCSI/FC protocol to connect to Huawei enterprise storage, configure and check host multipathing by referring to [OceanStor Dorado and OceanStor 6.x and V700R001 Host Connectivity Guide for Red Hat](#).
- If you use the NVMe over RoCE/NVMe over FC/NVMe over TCP protocol to connect to Huawei enterprise storage, configure and check host multipathing by referring to [OceanStor Dorado and OceanStor 6.x and V700R001 Host Connectivity Guide for Connecting to Linux Hosts Using NoF](#).
- If you use iSCSI to connect to Huawei scale-out storage, configure and check host multipathing by referring to Configuring Multipathing for an Application Server in [FusionStorage 8.0.1 Block Storage Basic Service Configuration Guide](#)

- If you use the native multipathing software provided by the OS, check whether the **/etc/multipath.conf** file contains the following configuration item.

```
defaults {  
    user_friendly_names yes  
    find_multipaths no  
}
```

If the configuration item does not exist, add it to the beginning of the **/etc/multipath.conf** file.

NOTE

For details about the functions of the **user_friendly_names** and **find_multipaths** parameters, see [dm_multipath/config_file_defaults](#).

4.1.1.6 Checking the Status of Host-Dependent Software

This section describes how to check whether the status of host-dependent software on worker nodes in a cluster is normal. In this example, the host OS is CentOS 7.9 x86_64.

- Check the status of the iSCSI client.
`systemctl status iscsi iscsid`
- Check the status of the NFS client.
`systemctl status rpcbind`
- Check the status of DM-Multipath.
`systemctl status multipathd.socket multipathd`
- Check the status of UltraPath.
`systemctl status nxup`
- Check the status of UltraPath-NVMe.
`systemctl status upudev upService_plus`
- Check the status of the NVMe-Native multipathing software. If **Y** is displayed in the command output, the status is enabled.
`cat /sys/module/nvme_core/parameters/multipath`
- Check the DataTurbo client status. For details, see [AI Storage Kit 25.x.x DTFS User Guide](#).
- Check the status of the NFS+ client. For details, see [NFS+ Client 1.x User Guide](#).

4.1.1.7 Checking the Images on Which CSI Depends

The installation of Huawei CSI depends on the images listed in the following table. If all worker nodes in the cluster have been connected to the Internet and can pull images online, you can skip this section. If nodes in the cluster cannot connect to the Internet, download the corresponding image file based on the Kubernetes version and upload it to the image repository or import it to all worker nodes in the Kubernetes cluster.

The huawei-csi-controller service depends on the following sidecar images: livenessprobe, csi-provisioner, csi-attacher, csi-resizer, csi-snapshotter, snapshot-controller, health-monitor-controller, storage-backend-controller, storage-backend-sidecar, huawei-csi-driver, and huawei-csi-extender. The huawei-csi-node service depends on the following sidecar images: livenessprobe, csi-node-driver-registrar, and huawei-csi-driver.

For details about the functions and details of each image, see the following table.

Table 4-3 Images on which Huawei CSI depends

Container Name	Container Image	K8s Version Requirements	Feature Description
livenessprobe	registry.k8s.io/sig-storage/livenessprobe:v2.15.0	v1.16+	This image is provided by the Kubernetes community, used to monitor the health status of CSI and report it to Kubernetes so that Kubernetes can automatically detect CSI program problems and restart the Pod to rectify the problems.
csi-resizer	registry.k8s.io/sig-storage/csi-resizer:v1.13.2	v1.16+	This image is provided by the Kubernetes community, used to call CSI to provide more storage space for a PVC when expanding the capacity of the PVC.
csi-node-driver-registrar	registry.k8s.io/sig-storage/csi-node-driver-registrar:v2.13.0	v1.16+	This image is provided by the Kubernetes community, used to obtain CSI information and register a node with kubelet using the plug-in registration mechanism of kubelet so that Kubernetes can detect the connection between the node and Huawei storage.
csi-snapshotter	registry.k8s.io/sig-storage/csi-snapshotter:v8.2.0	v1.25+	This image is provided by the Kubernetes community, used to call CSI to create or delete a snapshot on the storage system when creating or deleting a VolumeSnapshot.
	registry.k8s.io/sig-storage/csi-snapshotter:v6.3.0	v1.20-v1.24	
snapshot-controller	registry.k8s.io/sig-storage/snapshot-controller:v8.2.0	v1.25+	This image is provided by the Kubernetes community, used to listen to the VolumeSnapshot and VolumeSnapshotContent objects in the Kubernetes API and trigger csi-snapshotter to create a snapshot on the storage system when creating or deleting a VolumeSnapshot.
	registry.k8s.io/sig-storage/snapshot-controller:v6.3.0	v1.20-v1.24	

Container Name	Container Image	K8s Version Requirements	Feature Description
csi-provisioner	registry.k8s.io/sig-storage/csi-provisioner:v5.2.0	v1.20+	This image is provided by the Kubernetes community, used to create or delete PVCs. • Calls the huawei-csi-controller service to create a LUN or file system on the storage system as a PV when creating a PVC. • Calls the huawei-csi-controller service to delete the LUN or file system corresponding to the PV when deleting a PVC.
	registry.k8s.io/sig-storage/csi-provisioner:v3.0.0	v1.17-v1.19	
	quay.io/k8scsi/csi-provisioner:v1.4.0	v1.16.x	
csi-attacher	registry.k8s.io/sig-storage/csi-attacher:v4.8.0	v1.17+	Calls the huawei-csi-controller service to perform the "Publish/Unpublish Volume" operation when creating or deleting a Pod.
	quay.io/k8scsi/csi-attacher:v1.2.1	v1.16.x	
storage-backend-controller	storage-backend-controller:4.13.0	v1.16+	This image is provided by Huawei CSI software package, used to manage storageBackendClaim resources.
storage-backend-sidecar	storage-backend-sidecar:4.13.0	v1.16+	This image is provided by Huawei CSI software package, used to manage storageBackendContent resources.
huawei-csi-driver	huawei-csi:4.13.0	v1.16+	This image is provided by Huawei CSI software package, used to provide all features supported by Huawei CSI.
huawei-csi-extender	huawei-csi-extender:4.13.0	v1.16+	This image is provided by Huawei CSI software package, used to provide extended features of Huawei CSI.
health-monitor-controller	registry.k8s.io/sig-storage/csi-external-health-monitor-controller:v0.17.0	v1.16+	This image is provided by the Kubernetes community, and used to check the health status of volumes.

 NOTE

If the cluster is not connected to the Internet, manually download the container images and upload them to the cluster. For details, see [7.3 Downloading a Container Image](#).

4.1.2 Installation

This section describes how to install Huawei CSI.

 NOTE

In the current version, resource requests and limits are added to Huawei CSI. For details, see [10.3 Huawei CSI Resource Management](#).

Prerequisites

- Operations described in [4.1.1 Installation Preparations](#) have been completed.
- All worker nodes of the cluster communicate properly with the service network of the storage device to be connected. In iSCSI scenarios, the **ping** command can be used to verify the connectivity.
- Software clients required by the corresponding protocol, such as iSCSI and NFS clients, have been installed on all worker nodes of the cluster.

4.1.2.1 Installation Using Helm

This section describes how to install and deploy Huawei CSI using Helm. The Helm software version must be 3.0 or later.

Helm Installation Description

NOTICE

- Huawei CSI can be installed as the root user or a non-root user. When installing Huawei CSI as a non-root user, ensure that the current user can access the API Server of the Kubernetes cluster. For details about how to configure access to the Kubernetes cluster as a non-root user, see [7.7 Configuring Access to the Kubernetes Cluster as a Non-root User](#).
 - Huawei CSI must be run as the root user.
-

Helm is a software package management tool in the Kubernetes ecosystem. Similar to Ubuntu APT, CentOS YUM, or Python pip, Helm manages Kubernetes application resources.

You can use Helm to package, distribute, install, upgrade, and roll back Kubernetes applications in a unified manner.

- For details about how to obtain and install Helm, see <https://helm.sh/docs/intro/install/>.
- For details about the mapping between Helm and Kubernetes versions, see https://helm.sh/docs/topics/version_skew/.

When installing `huawei-csi-controller`, Helm deploys the following components in the workloads of the Deployment type in the specified namespace:

- `huawei-csi-driver`: Huawei CSI driver.
- `storage-backend-controller`: Huawei backend management controller, used to manage `storageBackendClaim` resources.
- `storage-backend-sidecar`: used to manage `storageBackendContent` resources.
- `Kubernetes External Provisioner`: used to provide or delete volumes.
- `Kubernetes External Attacher`: used to attach or detach volumes.
- `Kubernetes External Resizer`: used to expand the capacity of volumes.
- `Kubernetes External liveness-probe`: used to determine the health status of a Pod.
- (Optional) `huawei-csi-extender`: used to support changing persistent volumes.
- (Optional) `Kubernetes External Snapshotter`: used to provide snapshot support (installed as CRD).
- (Optional) `Kubernetes External Snapshot Controller`: used to control volume snapshots.
- (Optional) `Kubernetes External Health Monitor Controller`: used to monitor and control volume health.

When installing `huawei-csi-node`, Helm deploys the following components in the workloads of the DaemonSet type in the specified namespace:

- `huawei-csi-driver`: Huawei CSI driver.
- `Kubernetes Node Registrar`: used to process driver registration.
- `liveness-probe`: used to determine the health status of a Pod.

4.1.2.1.1 Installing Huawei CSI on Kubernetes, OpenShift, and Tanzu

Installation Procedure

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the cluster through the management IP address.
- Step 2** Copy the `helm` directory in the Kubernetes CSI component package to any directory on the master node. For details about the Helm tool path, see [Table 4-1](#).
- Step 3** Go to the `helm/esdk` working directory.
- ```
cd helm/esdk
```
- Step 4** Prepare the `values.yaml` file. Huawei CSI provides the `values.yaml` template file in the `helm/esdk` directory of the software package. You can also modify parameters according to [4.1.2.1.3 Parameters in the values.yaml File of Helm](#) to customize Huawei CSI.
- Step 5** Perform the following configuration before the installation:
- If the container platform is Kubernetes, skip this step.
  - If the container platform is OpenShift, perform the configuration in [Installation and Configuration on the OpenShift Platform](#).
  - If the container platform is Tanzu, perform the configuration in [Installation and Configuration on the Tanzu Platform](#).

**Step 6** Run the following command to update the storage backend CRD.

```
kubectl apply -f ./crds/backend/
```

**Step 7** (Optional) Check snapshot-dependent components by following the instructions provided in [4.1.1.4 Checking Volume Snapshot-Dependent Components](#). After confirming that the components are correct, run the following command to update the snapshot CRD. If **controller.snapshot.enabled** is set to **false** or the Kubernetes version is earlier than v1.20, you can skip this step. For details, see [Table 4-5](#).

```
kubectl apply -f ./crds/snapshot-crd/ --validate=false
```

**Step 8** Run the following command to install Huawei CSI. In the preceding command, *helm-huawei-csi* indicates the custom Helm chart name, *.* indicates that the Helm project in the current directory is used, and *huawei-csi* indicates the custom Helm chart namespace.

```
helm install helm-huawei-csi ./ -n huawei-csi --create-namespace
```

The following is an example of the command output.

```
NAME: helm-huawei-csi
LAST DEPLOYED: Wed Jun 8 11:50:28 2022
NAMESPACE: huawei-csi
STATUS: deployed
REVISION: 1
TEST SUITE: None
```

**Step 9** After the huawei-csi service is deployed, run the following command to check whether the service is started.

```
kubectl get pod -n huawei-csi
```

The following is an example of the command output. If the Pod status is **Running**, the installation is successful.

| NAME                                   | READY | STATUS  | RESTARTS | AGE |
|----------------------------------------|-------|---------|----------|-----|
| huawei-csi-controller-6dfcc4b79f-9vjtq | 9/9   | Running | 0        | 24m |
| huawei-csi-controller-6dfcc4b79f-csphc | 9/9   | Running | 0        | 24m |
| huawei-csi-node-g6f4k                  | 3/3   | Running | 0        | 20m |
| huawei-csi-node-tqs87                  | 3/3   | Running | 0        | 20m |

----End

## Installation and Configuration on the OpenShift Platform

For the OpenShift platform, perform the following steps to create the **SecurityContextConstraints** resource.

**Step 1** Run the following command to edit the **helm\_scc.yaml** file.

```
vi helm_scc.yaml
```

**Step 2** Modify the **helm\_scc.yaml** file. In the following commands, *huawei-csi* indicates the created namespace. Replace it based on the actual situation.

```
apiVersion: security.openshift.io/v1
kind: SecurityContextConstraints
metadata:
 name: helm-scc
allowHostDirVolumePlugin: true
allowHostIPC: true
allowHostNetwork: true
allowHostPID: true
allowHostPorts: true
allowPrivilegeEscalation: true
allowPrivilegedContainer: true
```

```
defaultAddCapabilities:
- SYS_ADMIN
runAsUser:
 type: RunAsAny
seLinuxContext:
 type: RunAsAny
fsGroup:
 type: RunAsAny
users:
- system:serviceaccount:huawei-csi:huawei-csi-controller
- system:serviceaccount:huawei-csi:huawei-csi-node
```

**Step 3** Run the following command to create a **SecurityContextConstraints** file.

```
oc create -f helm_scc.yaml
```

----End

## Installation and Configuration on the Tanzu Platform

On the Tanzu platform, run the following command to configure the **kubelet** installation directory.

**Step 1** Go to the **helm/esdk** directory in the installation package, and run the following command to open the configuration file. For details about the installation package directory, see [Table 4-1](#).

```
vi values.yaml
```

**Step 2** Modify the **kubeletConfigDir** parameter as follows and save the settings.

```
Specify kubelet config dir path.
kubernetes and openshift is usually /var/lib/kubelet
Tanzu is usually /var/vcap/data/kubelet
CCE is usually /mnt/paas/kubernetes/kubelet
kubeletConfigDir: /var/vcap/data/kubelet
```

----End

For TKGI 1.16 or earlier of the Tanzu platform, run the following commands to configure the RBAC permission.

**Step 1** Run the following command to create a file named **rbac.yaml**.

```
vi rbac.yaml
```

**Step 2** Copy the following content to the **rbac.yaml** file, save the file, and exit.

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
 name: huawei-csi-psp-role
rules:
- apiGroups: ['policy']
 resources: ['podsecuritypolicies']
 verbs: ['use']

apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
 name: huawei-csi-psp-role-cfg
roleRef:
 kind: ClusterRole
 name: huawei-csi-psp-role
apiGroup: rbac.authorization.k8s.io
subjects:
- kind: Group
 apiGroup: rbac.authorization.k8s.io
```

```
name: system:serviceaccounts:huawei-csi
- kind: Group
 apiGroup: rbac.authorization.k8s.io
 name: system:serviceaccounts:default
```

**Step 3** Run the following command to create the RBAC permission.

```
kubectrl create -f rbac.yaml
```

----End

#### 4.1.2.1.2 Installing Huawei CSI on the CCE or CCE Agile Platform

This section describes how to install Huawei CSI on the CCE or CCE Agile platform.

### Creating a Helm Installation Package

The CCE or CCE Agile platform cannot directly install Huawei CSI using Helm. You need to manually create a Helm installation package and upload it to the chart list on the platform for installation.

**Step 1** Use a remote access tool, such as PuTTY, to log in to any node where Helm is deployed through the management IP address.

**Step 2** Copy the **helm** directory in the Huawei CSI component package to any directory on the node. For details about the Helm tool path, see [Table 4-1](#).

**Step 3** Go to the **helm** working directory.

```
cd helm/
```

**Step 4** Modify the **kubeletConfigDir** and **csiDriver.driverName** parameters in the **helm/esdk/values.yaml** file.

```
vi ./esdk/values.yaml
```

Modify the following parameters:

```
Specify kubelet config dir path.
kubernetes and openshift is usually /var/lib/kubelet
Tanzu is usually /var/vcap/data/kubelet
CCE is usually /mnt/paas/kubernetes/kubelet
kubeletConfigDir: /mnt/paas/kubernetes/kubelet

The CSI driver parameter configuration
csiDriver:
 # Driver name, it is strongly recommended not to modify this parameter
 # The CCE platform needs to modify this parameter, e.g. csi.oceanstor.com
 driverName: csi.oceanstor.com
```

**Step 5** Run the following command to create a Helm installation package. This command will generate the installation package to the current path.

```
helm package ./esdk/ -d ./
```

----End

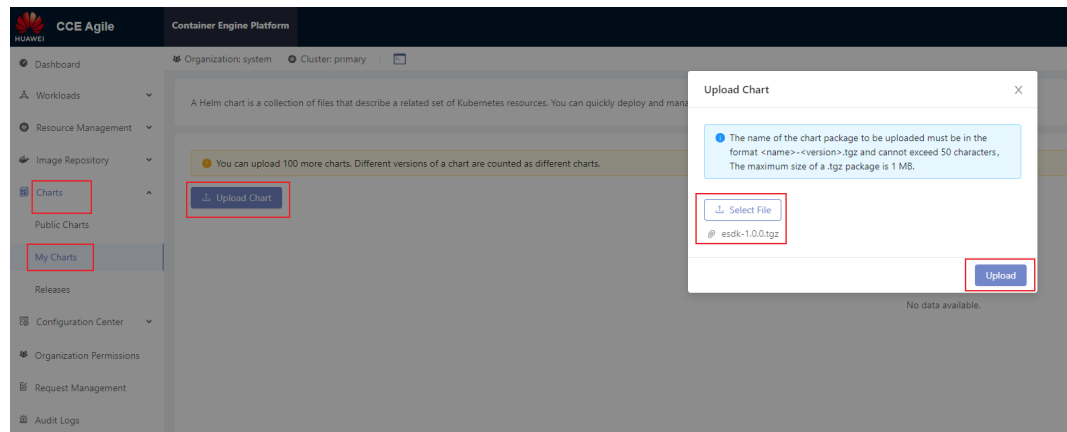
### Installing Huawei CSI

**Step 1** Use a remote access tool, such as PuTTY, to log in to any master node where the CCE Agile platform is deployed through the management IP address.

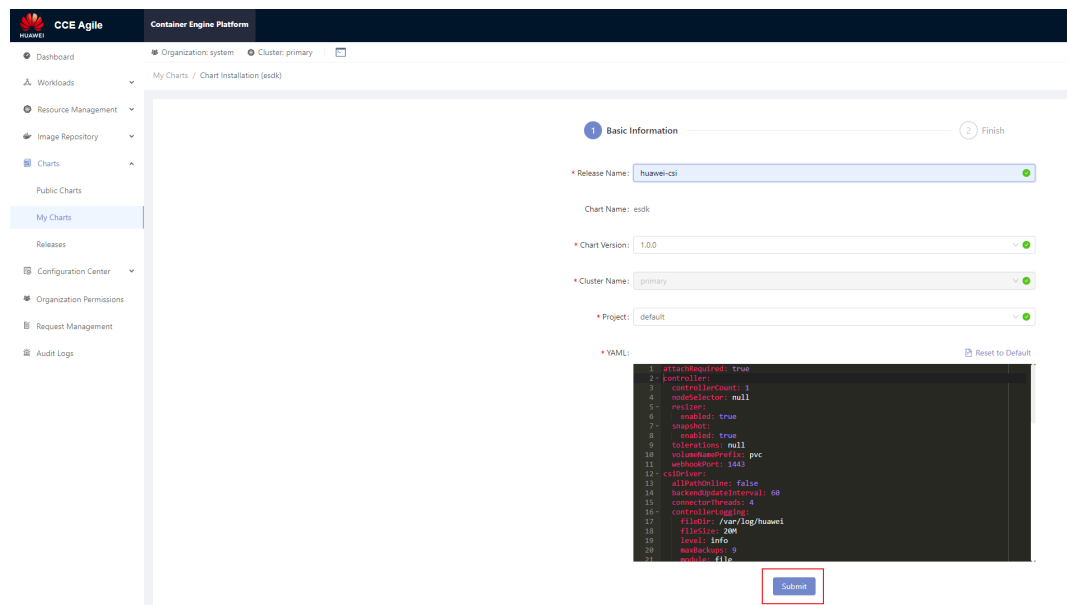
**Step 2** Run the following command to create a namespace for deploying Huawei CSI. *huawei-csi* indicates the custom namespace.

```
kubectrl create namespace huawei-csi
```

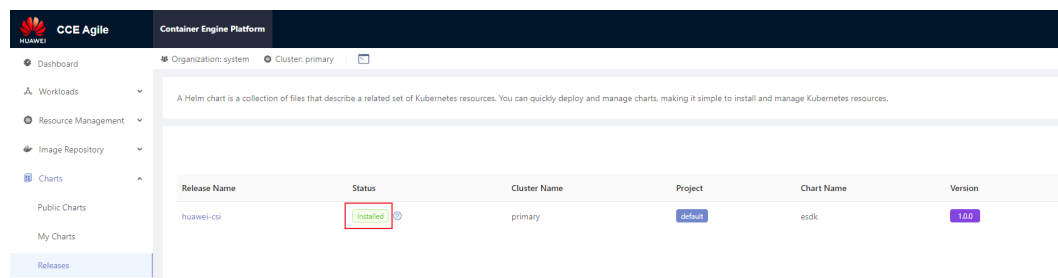
- Step 3** Export the Helm installation package. For details, see [Creating a Helm Installation Package](#).
- Step 4** On the home page, choose **Charts > My Charts > Upload Chart**. The **Upload Chart** dialog box is displayed. Import the exported Helm installation package to the CCE Agile platform.



- Step 5** After the installation package is uploaded, choose **Charts > My Charts**. On the **My Charts** page that is displayed, choose **Install > Submit**. The chart release name can be customized.



- Step 6** On the home page, choose **Charts > Releases** and select the project specified during installation (for example, **default** in the following figure). After the installation is successful, **Installed** is displayed in the **Status** column.



----End

#### 4.1.2.1.3 Parameters in the values.yaml File of Helm

When using Helm to install CSI, you need to prepare the **values.yaml** file of the Helm project based on the features required during deployment. Huawei CSI provides the **values.yaml** template file in the **helm/esdk** directory of the software package.

This section describes the configuration items in the **values.yaml** file and backend configuration examples in typical scenarios.

### images Parameters

The images parameters in the **values.yaml** file are used to configure the component image information on which Huawei CSI depends during running. Set the following parameters:

**Table 4-4** images parameters

| Parameter                        | Description                                  | Mandatory | Default Value                                      |
|----------------------------------|----------------------------------------------|-----------|----------------------------------------------------|
| images.huawei CSIService         | huawei-csi image.                            | Yes       | huawei-csi:4.13.0                                  |
| images.storage BackendSidecar    | Huawei back-end management sidecar image.    | Yes       | storage-backend-sidecar:4.13.0                     |
| images.storage BackendController | Huawei back-end management controller image. | Yes       | storage-backend-controller:4.13.0                  |
| images.huawei CSIExtender        | huawei-csi-extender image.                   | No        | huawei-csi-extender:4.13.0                         |
| images.sidecar.livenessProbe     | <b>livenessprobe</b> sidecar image.          | Yes       | registry.k8s.io/sig-storage/livenessprobe:v2.15.0  |
| images.sidecar.provisioner       | <b>csi-provisioner</b> sidecar image.        | Yes       | registry.k8s.io/sig-storage/csi-provisioner:v5.2.0 |



| Parameter                              | Description                                              | Mandatory | Default Value                                                              |
|----------------------------------------|----------------------------------------------------------|-----------|----------------------------------------------------------------------------|
| images.sidecar.attacher                | <a href="#">csi-attacher</a> sidecar image.              | Yes       | registry.k8s.io/sig-storage/csi-attacher:v4.8.0                            |
| images.sidecar.resizer                 | <a href="#">csi-resizer</a> sidecar image.               | Yes       | registry.k8s.io/sig-storage/csi-resizer:v1.13.2                            |
| images.sidecar.snapshotter             | <a href="#">csi-snapshotter</a> sidecar image.           | Yes       | registry.k8s.io/sig-storage/csi-snapshotter:v8.2.0                         |
| images.sidecar.snapshotController      | <a href="#">snapshot-controller</a> sidecar image.       | Yes       | registry.k8s.io/sig-storage/snapshot-controller:v8.2.0                     |
| images.sidecar.registrar               | <a href="#">csi-node-driver-registrar</a> sidecar image. | Yes       | registry.k8s.io/sig-storage/csi-node-driver-registrar:v2.13.0              |
| images.sidecar.healthMonitorController | <a href="#">health-monitor-controller</a> sidecar image. | No        | registry.k8s.io/sig-storage/csi-external-health-monitor-controller:v0.17.0 |

#### NOTICE

- For details about the values of **huaweiCSIService**, **storageBackendSidecar**, **storageBackendController**, and **huaweiCSIExtender**, see [4.1.1.2 Uploading a Huawei CSI Image](#). Use the name and version of the finally generated image.
- For details about other sidecar image parameters, see [4.1.1.7 Checking the Images on Which CSI Depends](#). Use the name and version of the finally uploaded image.

## controller Parameters

The controller parameters are used to configure the huawei-csi-controller component.

**Table 4-5** controller parameters

| Parameter                  | Description                                       | Mandatory | Default Value | Remarks                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------|---------------------------------------------------|-----------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| controller.controllerCount | Number of huawei-csi-controller component copies. | Yes       | 1             | <p>If the Kubernetes version is earlier than v1.17, the huawei-csi-controller component can be deployed only in single-copy mode because the csi-provisioner sidecar image provided by the Kubernetes community does not support the <b>--leader-election</b> parameter.</p> <p>Therefore, if the Kubernetes version is earlier than v1.17, this parameter can only be set to <b>1</b>.</p> |

| Parameter                   | Description                                                                                                                                                                                                                                                                         | Mandatory | Default Value | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| controller.volumeNamePrefix | <p>PV name prefix. The default value is <b>pvc</b>, that is, the name of a created PV is <b>pvc-<i>&lt;uuid&gt;</i></b>. The prefix must comply with the naming rules of <a href="#">DNS Subdomain Names</a>, and the total length of the PV name cannot exceed 253 characters.</p> | No        | pvc           | <p>The corresponding provisioner parameter name is <b>--volume-name-prefix</b>.</p> <p>It is recommended that the prefix contain no more than 20 characters.</p> <p>For details, see <a href="#">Configuring the PV Name Prefix</a>.</p> <ul style="list-style-type: none"><li>• If the connected backend is OceanStor V5 SAN storage, it is recommended that the prefix contain a maximum of 5 characters.</li><li>• If the connected backend is OceanStor V5 NAS storage, the prefix can contain only lowercase letters, hyphens (-), and digits.</li><li>• If the connected backend is OceanStor Dorado, OceanStor, OceanDisk, or OceanStor A series storage, the prefix can contain only lowercase letters, hyphens (-), and digits.</li><li>• If the connected backend is OceanStor Pacific series</li></ul> |

| Parameter                 | Description                             | Mandatory | Default Value                                                                                                                     | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------|-----------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           |                                         |           |                                                                                                                                   | <p>storage, the prefix can contain a maximum of 58 characters, including only letters, digits, underscores (_), hyphens (-), and periods (.).</p> <ul style="list-style-type: none"><li>• If the connected backend is FusionStorage Block, the prefix can contain a maximum of 58 characters, including only letters, digits, underscores (_), and hyphens (-).</li></ul>                                                                                                                                                    |
| controller.hostNamePrefix | Prefix of a storage host resource name. | No        | <ul style="list-style-type: none"><li>• Mass storage: The default prefix is empty.</li><li>• For other storage: "k8s_".</li></ul> | <ul style="list-style-type: none"><li>• The prefix characters must comply with the rules for the host name of the connected storage system.</li><li>• The total length of the prefix and the original host name must comply with the host name length limit of the connected storage system.</li></ul> <p><b>NOTICE</b><br/>This parameter needs to be configured when CSI is installed for the first time. Do not modify this parameter during subsequent upgrade or redeployment. Otherwise, volume mounting may fail.</p> |

| Parameter                       | Description                                                                                                                                 | Mandatory | Default Value | Remarks                                                                                                                     |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| controller.webhookPort          | Port used by the webhook service.                                                                                                           | Yes       | 4433          | If a port conflict occurs, change the port number to an idle one.                                                           |
| controller.snapshot.enabled     | Whether to enable the snapshot feature.                                                                                                     | Yes       | true          | If you want to use snapshot-related functions, enable this feature.<br><br>The Kubernetes version must be later than v1.20. |
| controller.resizer.enabled      | Whether to enable the capacity expansion feature.                                                                                           | Yes       | true          | The Kubernetes version must be later than v1.16.                                                                            |
| controller.resizer.kubeApiQps   | Number of requests that can be sent by the client to kube-apiserver per second.                                                             | No        | 5             |                                                                                                                             |
| controller.resizer.kubeApiBurst | Maximum number of requests that can be sent by the client in a short period.                                                                | No        | 10            |                                                                                                                             |
| controller.nodeSelector         | Node selector of huawei-csi-controller. After this parameter is set, huawei-csi-controller will be scheduled only to a node with the label. | No        | -             | For details about the node selector, see <a href="#">Assign Pods to Nodes</a> .                                             |
| controller.tolerations          | Taint toleration of huawei-csi-controller. After this parameter is set, huawei-csi-controller can tolerate taints on a node.                | No        | -             | For details about taints and tolerations, see <a href="#">Taints and Tolerations</a> .                                      |

| Parameter                                          | Description                                                                                                                                           | Mandatory | Default Value | Remarks                                                                                                        |
|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|----------------------------------------------------------------------------------------------------------------|
| controller.affinity                                | Node affinity of huawei-csi-controller. After this parameter is set, huawei-csi-controller will be preferentially scheduled to a node with the label. | No        | -             | For details about node affinity, see <a href="#">Assigning Pods to Nodes</a> .                                 |
| controller.livenessProbePort                       | Liveness probe port of huawei-csi-controller, used for health check.                                                                                  | Yes       | 9808          | If a port conflict occurs, change the port number to an idle one.                                              |
| controller.csiExtender.volumeModify.enabled        | Whether to enable the PVC change feature.                                                                                                             | No        | false         | If you want to use PVC change-related functions, enable this feature.                                          |
| controller.csiExtender.volumeModify.retryBaseDelay | Minimum retry interval when a PVC change fails to be created.                                                                                         | No        | 5s            | The default value is recommended.                                                                              |
| controller.csiExtender.volumeModify.retryMaxDelay  | Maximum retry interval when a PVC change fails to be created.                                                                                         | No        | 5m            | The default value is recommended.                                                                              |
| controller.csiExtender.volumeModify.reconcileDelay | Interval for reconciling VolumeModifyClaim objects.                                                                                                   | No        | 1s            | The default value is recommended.                                                                              |
| controller.csiExtender.volumeModify.workerThreads  | Number of worker threads for processing PVC change tasks.                                                                                             | No        | 10            | Value range: (0, 100]                                                                                          |
| controller.exportCsiService.enabled                | Whether to run CSI services on the Service of the Kubernetes cluster.                                                                                 | No        | false         | After this function is enabled, other services in the Kubernetes cluster can access CSI services through gRPC. |

| Parameter                                | Description                                                                     | Mandatory | Default Value  | Remarks                                                                                              |
|------------------------------------------|---------------------------------------------------------------------------------|-----------|----------------|------------------------------------------------------------------------------------------------------|
| controller.exportCsiService.port         | Port used when CSI services run on the Service of the Kubernetes cluster.       | No        | 9090           | If a port conflict occurs, change the port number to an idle one.                                    |
| controller.exportCsiService.audience     | Audience of the CSI service token                                               | No        | csi.huawei.com |                                                                                                      |
| controller.healthMonitor.enabled         | Whether to enable the volume monitoring service.                                | No        | false          | After this function is enabled, volume monitoring features can be used.                              |
| controller.healthMonitor.monitorInterval | Monitoring polling interval.                                                    | No        | 60s            | Set the volume monitoring polling interval.                                                          |
| controller.healthMonitor.workerThreads   | Number of monitoring PV worker threads.                                         | No        | 10             | Set the number of monitoring PV worker threads.                                                      |
| controller.provisioner.kubeAPIQps        | Number of requests that can be sent by the client to kube-apiserver per second. | No        | 5              |                                                                                                      |
| controller.provisioner.kubeAPIBurst      | Maximum number of requests that can be sent by the client in a short period.    | No        | 10             |                                                                                                      |
| controller.provisioner.topology          | Whether to enable the topology feature gate.                                    | No        | false          |                                                                                                      |
| controller.provisioner.strictTopology    | Whether to use the strict topology mode.                                        | No        | false          | This parameter takes effect only when <b>controller.provisioner.topology</b> is set to <b>true</b> . |

| Parameter                                        | Description                                                                     | Mandatory | Default Value | Remarks |
|--------------------------------------------------|---------------------------------------------------------------------------------|-----------|---------------|---------|
| controller.attacher.kubeApiQps                   | Number of requests that can be sent by the client to kube-apiserver per second. | No        | 5             |         |
| controller.attacher.kubeApiBurst                 | Maximum number of requests that can be sent by the client in a short period.    | No        | 10            |         |
| controller.snapshotter.kubeApiQps                | Number of requests that can be sent by the client to kube-apiserver per second. | No        | 5             |         |
| controller.snapshotter.kubeApiBurst              | Maximum number of requests that can be sent by the client in a short period.    | No        | 10            |         |
| controller.storageBackendController.kubeApiQps   | Number of requests that can be sent by the client to kube-apiserver per second. | No        | 5             |         |
| controller.storageBackendController.kubeApiBurst | Maximum number of requests that can be sent by the client in a short period.    | No        | 10            |         |
| controller.storageBackendSched.kubeApiQps        | Number of requests that can be sent by the client to kube-apiserver per second. | No        | 5             |         |



| Parameter                                   | Description                                                                     | Mandatory | Default Value | Remarks |
|---------------------------------------------|---------------------------------------------------------------------------------|-----------|---------------|---------|
| controller.storageBackendSicar.kubeApiBurst | Maximum number of requests that can be sent by the client in a short period.    | No        | 10            |         |
| controller.huaweiCsiExtender.kubeApiQps     | Number of requests that can be sent by the client to kube-apiserver per second. | No        | 5             |         |
| controller.huaweiCsiExtender.kubeApiBurst   | Maximum number of requests that can be sent by the client in a short period.    | No        | 10            |         |
| controller.huaweiCsiDriver.kubeApiQps       | Number of requests that can be sent by the client to kube-apiserver per second. | No        | 5             |         |
| controller.huaweiCsiDriver.kubeApiBurst     | Maximum number of requests that can be sent by the client in a short period.    | No        | 10            |         |

 **NOTE**

If **controller.snapshot.enabled** is set to **true**, you need to install the volume snapshot CRD resource in the **helm/crd/snapshot-crds** directory.

## node Parameters

The node parameters are used to configure the huawei-csi-node component.

**Table 4-6** node parameters

| Parameter              | Description                                                                                                                                                                                                                                                         | Mandatory | Default Value                                                                                                                                                                                                                                                                                                              | Remarks                                                                                |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| node.maxVolumesPerNode | Maximum number of volumes provisioned by Huawei CSI that can be used by a node. If this parameter is not specified or is set to <b>0</b> , the number is unlimited.<br><br>If <b>nodeName</b> is specified during Pod creation, this configuration will be ignored. | No        | 100                                                                                                                                                                                                                                                                                                                        | For details, see <a href="#">Volume Limits</a> .                                       |
| node.nodeSelector      | Node selector of huawei-csi-node. After this parameter is set, huawei-csi-node will be scheduled only to a node with the label.                                                                                                                                     | No        | -                                                                                                                                                                                                                                                                                                                          | For details about the node selector, see <a href="#">Assign Pods to Nodes</a> .        |
| node.tolerations       | Taint toleration of huawei-csi-node. After this parameter is set, huawei-csi-node can tolerate taints on a node.                                                                                                                                                    | No        | <ul style="list-style-type: none"><li>- key: "node.kubernetes.io/memory-pressure" operator: "Exists" effect: "NoExecute"</li><li>- key: "node.kubernetes.io/disk-pressure" operator: "Exists" effect: "NoExecute"</li><li>- key: "node.kubernetes.io/network-unavailable" operator: "Exists" effect: "NoExecute"</li></ul> | For details about taints and tolerations, see <a href="#">Taints and Tolerations</a> . |

| Parameter                         | Description                                                                                                                               | Mandatory | Default Value | Remarks                                                                                                                                                                                                                     |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| node.affinity                     | Node affinity of huawei-csi-node. After this parameter is set, huawei-csi-node will be preferentially scheduled to a node with the label. | No        | -             | For details about node affinity, see <a href="#">Assigning Pods to Nodes</a> .                                                                                                                                              |
| node.livenessProbePort            | Liveness probe port of huawei-csi-node, used for health check.                                                                            | Yes       | 9800          | If a port conflict occurs, change the port number to an idle one.                                                                                                                                                           |
| node.kubeletVolumeDevicesDirName  | Name of the directory where a block device is mounted to kubelet.                                                                         | No        | volumeDevices | After a block device is successfully mounted, the directory structure of the mount path is as follows:<br><code>/var/lib/kubelet/plugins/kubernetes.io/csi/{kubeletVolumeDevicesDirName}/publish/{specName}/{podUID}</code> |
| node.huaweiCsiDriver.kubeAPIQps   | Number of requests that can be sent by the client to kube-apiserver per second.                                                           | No        | 5             |                                                                                                                                                                                                                             |
| node.huaweiCsiDriver.kubeAPIBurst | Maximum number of requests that can be sent by the client in a short period.                                                              | No        | 10            |                                                                                                                                                                                                                             |

## csiDriver Parameters

The csiDriver parameters include the basic configurations for running Huawei CSI, such as Huawei driver name and multipathing type.

**Table 4-7** csiDriver parameters

| Parameter                    | Description                                                                                                      | Mandatory | Default Value  | Remarks                                                                                                                                                                                                                                                        |
|------------------------------|------------------------------------------------------------------------------------------------------------------|-----------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| csiDriver.driverName         | Registered driver name.                                                                                          | Yes       | csi.huawei.com | <ul style="list-style-type: none"><li>• Use the default value.</li><li>• For the CCE Agile platform, modify this field. For example, <b>csi.oceanstor.com</b>.</li></ul>                                                                                       |
| csiDriver.endpoint           | Communication endpoint.                                                                                          | Yes       | /csi/csi.sock  | Use the default value.                                                                                                                                                                                                                                         |
| csiDriver.connectorThreads   | Maximum number of disks that can be concurrently scanned/detached. The value is an integer ranging from 1 to 10. | Yes       | 4              | A larger value indicates that more concurrent disk scanning and detaching operations are performed on a single node at the same time. When DM-Multipath is used, a large number of concurrent requests may cause unknown problems and affect the overall time. |
| csiDriver.volumeUseMultipath | Whether to use multipathing software. The value is a Boolean value.                                              | Yes       | true           | It is strongly recommended that multipathing software be enabled to enhance the redundancy and performance of storage links.                                                                                                                                   |

| Parameter                   | Description                                                                                                                                                                                                                                              | Mandatory                                                        | Default Value            | Remarks                                                                                                                                                               |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| csiDriver.scsiMultipathType | Multipathing software used when the storage protocol is <b>fc</b> or <b>iscsi</b> . The following parameter values can be configured: <ul style="list-style-type: none"><li>DM-multipath</li><li>HW-UltraPath</li><li><b>HW-UltraPath-NVMe</b></li></ul> | Mandatory when <b>volumeUseMultipath</b> is set to <b>true</b> . | DM-multipath             | The <b>DM-multipath</b> value is recommended.                                                                                                                         |
| csiDriver.nvmeMultipathType | Multipathing software used when the storage protocol is NVMe. The following parameter values can be configured: <ul style="list-style-type: none"><li><b>HW-UltraPath-NVMe</b></li><li><b>NVMe-Native</b></li></ul>                                      | Mandatory when <b>volumeUseMultipath</b> is set to <b>true</b> . | <b>HW-UltraPath-NVMe</b> | <ul style="list-style-type: none"><li>roce-nvme and fc-nvme support only <b>HW-UltraPath-NVMe</b>.</li><li>The tcp-nvme protocol supports only NVMe-Native.</li></ul> |
| csiDriver.scanVolumeTimeout | Timeout interval for waiting for multipathing aggregation when DM-Multipath is used on the host. The value ranges from 1 to 600 seconds.                                                                                                                 | Yes                                                              | 3                        | -                                                                                                                                                                     |

| Parameter                    | Description                                                                                    | Mandatory | Default Value | Remarks                                                                                                                                                                                                                                                           |
|------------------------------|------------------------------------------------------------------------------------------------|-----------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| csiDriver.execCommandTimeout | Timeout interval for running commands on the host.                                             | Yes       | 30            | In scenarios such as mounting and capacity expansion, the CSI plug-in needs to run some host commands, for example, running the <b>mount</b> command to mount a file system. This parameter is used to control the timeout interval for running a single command. |
| csiDriver.enableRoCEConnect  | Whether to enable CSI automatic disk scanning when the roce-nvme or tcp-nvme protocol is used. | No        | true          | If an external tool is used to establish an NVMe_over_RoCE or NVMe_over_TCP connection, set this parameter to <b>false</b> . For example, if the SNSD automatic connection function is enabled on storage devices, you can set this parameter to <b>false</b> .   |

| Parameter                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Mandatory                                                                                                                                                              | Default Value | Remarks |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------|
| csiDriver.allPathOnline         | <p>Whether to check that the number of paths aggregated by DM-Multipath or NVMe-Native is equal to the actual number of online paths. The following parameter values can be configured:</p> <ul style="list-style-type: none"><li>• <b>true:</b> The drive letter mounting condition is met only when the number of paths aggregated by multipathing software is equal to the actual number of online paths.</li><li>• <b>false:</b> By default, the number of paths aggregated by multipathing software is not checked. As long as virtual drive letters are generated upon aggregation, the drive letter mounting condition is met.</li></ul> | This parameter is mandatory when <b>csiDriver.scsiMultipathType</b> is set to <b>DM-multipath</b> or <b>csiDriver.nvmeMultipathType</b> is set to <b>NVMe-Native</b> . | false         | -       |
| csiDriver.backendUpdateInterval | Interval for updating backend capabilities. The value ranges from 60 to 600 seconds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Yes                                                                                                                                                                    | 60            | -       |

| Parameter                           | Description                                                                                                                                                                                          | Mandatory | Default Value   | Remarks                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| csiDriver.controllerLogging.module  | Record type of the controller log. The following parameter values can be configured: <ul style="list-style-type: none"><li>file</li><li>console</li></ul>                                            | Yes       | file            | When the value is <b>file</b> , logs are retained in the specified directory of the node. When the Pod where CSI is located is destroyed, logs are still retained.<br><br>When the value is <b>console</b> , logs are retained in the temporary space of the Pod where CSI is located. When the Pod where CSI is located is destroyed, the logs are also destroyed. |
| csiDriver.controllerLogging.level   | Output level of the controller log. The following parameter values can be configured: <ul style="list-style-type: none"><li>debug</li><li>info</li><li>warning</li><li>error</li><li>fatal</li></ul> | Yes       | info            | -                                                                                                                                                                                                                                                                                                                                                                   |
| csiDriver.controllerLogging.fileDir | Directory of the controller log in <b>file</b> output mode.                                                                                                                                          | Yes       | /var/log/huawei | <b>NOTICE</b><br>The custom log directory must be in the <b>/var/log/</b> directory. Otherwise, log files cannot be persisted.<br><br>Ensure that the directory has sufficient space for storing logs. It is recommended that the space be greater than or equal to 200 MB.                                                                                         |



| Parameter                              | Description                                                                                                                                                                                              | Mandatory | Default Value | Remarks                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| csiDriver.controllerLogging.fileSize   | Size of a single controller log file in <b>file</b> output mode.                                                                                                                                         | Yes       | 20M           | -                                                                                                                                                                                                                                                                                                                                                                   |
| csiDriver.controllerLogging.maxBackups | Maximum number of controller log file backups in <b>file</b> output mode.                                                                                                                                | Yes       | 9             | -                                                                                                                                                                                                                                                                                                                                                                   |
| csiDriver.nodeLogging.module           | Record type of the node log. The following parameter values can be configured: <ul style="list-style-type: none"><li>• file</li><li>• console</li></ul>                                                  | Yes       | file          | When the value is <b>file</b> , logs are retained in the specified directory of the node. When the Pod where CSI is located is destroyed, logs are still retained.<br><br>When the value is <b>console</b> , logs are retained in the temporary space of the Pod where CSI is located. When the Pod where CSI is located is destroyed, the logs are also destroyed. |
| csiDriver.nodeLogging.level            | Output level of the node log. The following parameter values can be configured: <ul style="list-style-type: none"><li>• debug</li><li>• info</li><li>• warning</li><li>• error</li><li>• fatal</li></ul> | Yes       | info          | -                                                                                                                                                                                                                                                                                                                                                                   |

| Parameter                        | Description                                                         | Mandatory | Default Value   | Remarks                                                                                                                                                                                                                                                                     |
|----------------------------------|---------------------------------------------------------------------|-----------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| csiDriver.nodeLogging.fileDir    | Directory of the node log in <b>file</b> output mode.               | Yes       | /var/log/huawei | <b>NOTICE</b><br>The custom log directory must be in the <b>/var/log/</b> directory. Otherwise, log files cannot be persisted.<br><br>Ensure that the directory has sufficient space for storing logs. It is recommended that the space be greater than or equal to 200 MB. |
| csiDriver.nodeLogging.fileSize   | Size of a single node log file in <b>file</b> output mode.          | Yes       | 20M             | -                                                                                                                                                                                                                                                                           |
| csiDriver.nodeLogging.maxBackups | Maximum number of node log file backups in <b>file</b> output mode. | Yes       | 9               | -                                                                                                                                                                                                                                                                           |
| csiDriver.reportNodeIP           | Whether to report the node IP address.                              | No        | false           | After this item is enabled, the CSI node will automatically report the node IP address.                                                                                                                                                                                     |

| Parameter                     | Description                               | Mandatory | Default Value | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------|-------------------------------------------|-----------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| csiDriver.enablePerNodeSecret | Whether to create a Secret for each node. | No        | false         | <p>By default, the information required by CSI on all nodes is saved to a Secret. After this function is enabled, an independent Secret is created for each node to save information.</p> <p><b>NOTICE</b><br/>After this function is enabled, the following conditions must be met:</p> <ul style="list-style-type: none"><li>• The K8s node name must meet the K8s naming rules in <a href="#">Names</a>.</li><li>• The storage backend whose name is the same as that of the K8s node cannot exist in the cluster.</li></ul> |

**NOTICE**

If Huawei CSI has been deployed in your container environment, ensure that the value of **csiDriver.driverName** is the same as that configured during previous deployment. Otherwise, existing volumes or snapshots provisioned by Huawei CSI in the system cannot be managed by the newly deployed Huawei CSI.

## Other Parameters

Other parameters include some features of the CSI plug-in or the policies for obtaining images.

**Table 4-8** Other parameters

| Parameter              | Description                                                                                                                                                                                      | Mandatory | Default Value    | Remarks                                                                                                                                                                                                                                                                                  |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| kubernetes.namespace   | Kubernetes namespace where Huawei CSI is running, which can be customized. The name must consist of lowercase letters, digits, and hyphens (-), for example, <b>my-name</b> and <b>123-abc</b> . | No        | huawei-csi       | -                                                                                                                                                                                                                                                                                        |
| kubeletConfigDir       | Working directory of kubelet.                                                                                                                                                                    | Yes       | /var/lib/kubelet | <ul style="list-style-type: none"><li>• Use the default value.</li><li>• For the Tanzu platform, change the value of this field to <b>/var/vcap/data/kubelet</b>.</li><li>• For the CCE Agile platform, change the value of this field to <b>/mnt/paas/kubernetes/kubelet</b>.</li></ul> |
| sidecarImagePullPolicy | Pull policy of the sidecar image.                                                                                                                                                                | Yes       | IfNotPresent     | -                                                                                                                                                                                                                                                                                        |
| huaweiImagePullPolicy  | Pull policy of the huawei-csi image.                                                                                                                                                             | Yes       | IfNotPresent     | -                                                                                                                                                                                                                                                                                        |
| imagePullSecrets       | Used by the Kubernetes cluster to pass the identity authentication of an image registry to pull private images.                                                                                  | No        | -                | For details, see <a href="#">Pull an Image from a Private Registry</a> .                                                                                                                                                                                                                 |

| Parameter                      | Description                                                                                                                                                                                                                                                     | Mandatory | Default Value | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CSIDriverObject.isCreate       | Whether to create the CSIDriver object.                                                                                                                                                                                                                         | Yes       | false         | The <b>CSIDriver</b> feature is a GA version in Kubernetes v1.18. Therefore, to use this feature, the Kubernetes version must be later than v1.18. If the Kubernetes version is earlier than v1.18, set this parameter to <b>false</b> .                                                                                                                                                                                                          |
| CSIDriverObject.attachRequired | Whether the CSI plug-in skips the attach operation. The following parameter values can be configured: <ul style="list-style-type: none"><li>• <b>true</b>: The attach operation is required.</li><li>• <b>false</b>: The attach operation is skipped.</li></ul> | Yes       | true          | The <b>attachRequired</b> parameter can be configured in Kubernetes v1.18. If <b>CSIDriverObject.isCreate</b> is set to <b>true</b> and <b>attachRequired</b> is set to <b>false</b> , the huawei-csi plug-in will not deploy the csi-attacher sidecar. <ul style="list-style-type: none"><li>• If NAS storage is used, this parameter can be set to <b>false</b>.</li><li>• If SAN storage is used, set this parameter to <b>true</b>.</li></ul> |

| Parameter                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Mandatory | Default Value | Remarks                                                                                                                                                                                                                                                                                                                        |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CSIDriverObject.fsGroupPolicy | <p>Whether the ownership and permissions of a basic volume can be changed before the volume is mounted. The following parameter values can be configured:</p> <ul style="list-style-type: none"><li>• <b>"ReadWriteOnceWithFSType"</b>: The volume ownership and permission can be changed only when <b>fsType</b> is specified and <b>accessModes</b> of the volume contains <b>ReadWriteOnce</b>.</li><li>• <b>"File"</b>: Kubernetes can use <b>fsGroup</b> to change the permissions and ownership of a volume to match <b>fsGroup</b> requested by a user in the Pod security policy, regardless of <b>fsGroup</b> or <b>accessModes</b>.</li><li>• <b>"None"</b>: A volume is mounted without any change.</li><li>• <b>"null"</b>: The <b>fsGroupPolicy</b> parameter is not set.</li></ul> | No        | null          | <p>The <b>fsGroupPolicy</b> parameter can be configured in Kubernetes v1.20, and takes effect only when <b>CSIDriverObject.isCreate</b> is set to <b>true</b>.</p> <p>This feature is a Beta version in Kubernetes v1.20 but a GA version in Kubernetes v1.23. Therefore, the Kubernetes version must be later than v1.20.</p> |

| Parameter                    | Description                                        | Mandatory | Default Value | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------|----------------------------------------------------|-----------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| leaderElection.leaseDuration | Leader duration.                                   | No        | 15s           | This parameter takes effect only in the multi-controller scenario.                                                                                                                                                                                                                                                                                                                                                                                       |
| leaderElection.renewDeadline | Time for the leader to be re-elected.              | No        | 10s           | This parameter takes effect only in the multi-controller scenario.                                                                                                                                                                                                                                                                                                                                                                                       |
| leaderElection.retryPeriod   | Leader election retry time.                        | No        | 5s            | This parameter takes effect only in the multi-controller scenario.                                                                                                                                                                                                                                                                                                                                                                                       |
| service.ipFamilyPolicy       | IP protocol stack selection policy of a service.   | No        | SingleStack   | The value can be: <ul style="list-style-type: none"><li>• <b>SingleStack</b>: The service uses only one IP address family (IPv4 or IPv6).</li><li>• <b>PreferDualStack</b>: The service preferentially uses dual stack. If the cluster does not support dual stack, single stack is used.</li><li>• <b>RequireDualStack</b>: The service must use dual stack. If the cluster does not support dual stack, the service will fail to be created.</li></ul> |
| service.ipFamilies           | List of IP protocol stacks supported by a service. | No        | IPv4          | The value can be: <ul style="list-style-type: none"><li>• IPv4</li><li>• IPv6</li></ul>                                                                                                                                                                                                                                                                                                                                                                  |

| Parameter | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Mandatory | Default Value                                                                                                      | Remarks                                                                                                                                                                                                              |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| resources | <p>You can allocate container resources related to huawei-csi-controller and huawei-csi-node:</p> <p><b>resources.</b> &lt;component&gt;. &lt;container-name&gt;</p> <p>&lt;component&gt; supports the following values:</p> <ul style="list-style-type: none"><li>• <b>node:</b> configured when the sidecar in the huawei-csi-node component is configured.</li><li>• <b>controller:</b> configured when the sidecar in the huawei-csi-controller component is configured.</li></ul> <p><b>controller.</b> &lt;container-name&gt; supports the following values:</p> <ul style="list-style-type: none"><li>• livenessProbe</li><li>• csiProvisioner</li><li>• csiAttacher</li><li>• csiResizer</li><li>• csiSnapshotter</li><li>• snapshotController</li><li>• storageBackendController</li><li>• storageBackendSidecar</li><li>• huaweiCsiExtender</li></ul> | No        | For the default values of different container resources, see <a href="#">10.3 Huawei CSI Resource Management</a> . | <p>The following uses livenessProbe of huawei-csi-controller as an example:</p> <pre>resources:   controller:     limits:       cpu: 100m       memory: 128Mi     requests:       cpu: 10m       memory: 128Mi</pre> |



| Parameter | Description                                                                                                                                                                                                                                            | Mandatory | Default Value | Remarks |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|---------|
|           | <ul style="list-style-type: none"><li>huaweiCsiDriver</li></ul> <b>node.&lt;container-name&gt;</b> supports the following values: <ul style="list-style-type: none"><li>huaweiCsiDriver</li><li>livenessProbe</li><li>csiNodeDriverRegistrar</li></ul> |           |               |         |

**NOTICE**

Ensure that the namespace entered in **kubernetes.namespace** exists on Kubernetes. If the namespace does not exist, run the following command to create it. In this example, the namespace for running Huawei CSI is **huawei-csi**.

```
kubectrl create namespace huawei-csi
```

### 4.1.2.2 Manual Installation

This section describes how to manually install Huawei CSI.

 **NOTE**

Currently, only the Kubernetes platform supports manual installation of Huawei CSI.

### Procedure

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the cluster through the management IP address.
- Step 2** Copy the **manual** directory in the Kubernetes CSI component package to any directory on the master node.
- Step 3** Run the following command to create a namespace. **huawei-csi** is used as an example namespace.
- ```
kubectrl create ns huawei-csi
```
- Step 4** Go to the **manual/esdk** working directory. For details about the path, see [Table 4-1](#).
- ```
cd manual/esdk
```
- Step 5** Run the following command to update the storage backend CRD.
- ```
kubectrl apply -f ./crds/backend/
```
- Step 6** (Optional) Check snapshot-dependent components by following the instructions provided in [4.1.1.4 Checking Volume Snapshot-Dependent Components](#). After confirming that the components are correct, run the following command to

update the snapshot CRD. If the Kubernetes version is earlier than v1.20, skip this step.

```
kubectrl apply -f ./crds/snapshot-crd/ --validate=false
```

Step 7 (Optional) Run the following command to install CSIDriver. If the CSIDriver feature is not used, you can skip this step. For details, see the [CSIDriver](#) feature.

```
kubectrl apply -f ./deploy/csidriver.yaml
```

Step 8 Run the following command to install the huawei-csi-controller service.

NOTE

If the Kubernetes version is earlier than v1.20, modify the `./deploy/huawei-csi-controller.yaml` file as follows:

- If the Kubernetes version is earlier than v1.20, the snapshot feature is not supported. In this case, delete the snapshot-related container configurations items **csi-snapshotter** and **snapshot-controller**.

If the Kubernetes version is earlier than v1.17, modify the `./deploy/huawei-csi-controller.yaml` file as follows:

- If the Kubernetes version is earlier than v1.17, the snapshot feature is not supported. In this case, delete the snapshot-related container configurations items **csi-snapshotter** and **snapshot-controller**.
- If the Kubernetes version is earlier than v1.17, the csi-provisioner sidecar image provided by the Kubernetes community does not support the **--leader-election** parameter. Therefore, the **leader-election** parameter of the csi-provisioner container is deleted and only single-copy deployment is supported.
- Modify the dependent image version based on the version requirements in [4.1.1.7 Checking the Images on Which CSI Depends](#).

```
kubectrl apply -f ./deploy/huawei-csi-controller.yaml
```

Step 9 Run the following command to install the huawei-csi-node service.

```
kubectrl apply -f ./deploy/huawei-csi-node.yaml
```

Step 10 Run the following command to check whether the services are started.

```
kubectrl get pod -n huawei-csi
```

The following is an example of the command output. If the Pod status is **Running**, the installation is successful.

NAME	READY	STATUS	RESTARTS	AGE
huawei-csi-controller-68745d489c-v5xkj	9/9	Running	0	13m
huawei-csi-node-4hbqp	3/3	Running	0	13m
huawei-csi-node-f7dkf	3/3	Running	0	13m
huawei-csi-node-xrntc	3/3	Running	0	13m

----End

NOTE

In the multi-copy controller deployment scenario, you can modify the **spec.replica** field of the Deployment resource in the `./deploy/huawei-csi-controller.yaml` file to specify the number of copies. After the modification, run the **kubectrl apply -f ./deploy/huawei-csi-controller.yaml** command to make the configuration take effect.

4.1.3 Uninstallation

This section describes how to uninstall Huawei CSI. The uninstallation method varies according to the installation mode.

NOTICE

If you do not uninstall Huawei CSI for the purpose of an upgrade, ensure that all resources (such as PV, PVC, snapshot, and storage backend resources) provisioned by Huawei CSI have been cleared on your container platform before uninstalling Huawei CSI. Otherwise, once you uninstall Huawei CSI, these resources cannot be automatically scheduled, managed, or cleared.

4.1.3.1 Uninstallation Using Helm

4.1.3.1.1 Uninstalling Huawei CSI on Kubernetes, OpenShift, and Tanzu

This section describes how to uninstall Huawei CSI on the Kubernetes, OpenShift, and Tanzu platforms.

Procedure

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
- Step 2** Run the following command to uninstall Huawei CSI. In the command, *helm-huawei-csi* indicates the custom Helm chart name and *huawei-csi* indicates the namespace where the Helm chart resides. This command will uninstall the huawei-csi-controller, huawei-csi-node, and RBAC resources of Huawei CSI.

```
helm uninstall helm-huawei-csi -n huawei-csi
```

After the uninstallation command is executed, you need to check whether the uninstallation is successful. In the preceding command, *huawei-csi* indicates the namespace where the chart is located.

```
helm list -n huawei-csi
```

The following is an example of the command output. If the command output is empty, the service is successfully uninstalled.

```
NAME NAMESPACE REVISION UPDATED STATUS CHART APP VERSION
```

- Step 3** Uninstall the huawei-csi-host-info object. For details, see [Uninstalling the huawei-csi-host-info Object](#).
- Step 4** Uninstall the webhook resource. For details, see [Uninstalling a Webhook Resource](#).
- Step 5** (Optional) Uninstall the snapshot-dependent component service. For details, see [Uninstalling the Snapshot-Dependent Component Service](#).
- Step 6** (Optional) Uninstall the Lease resource. For details, see [Uninstalling a Lease Resource](#).
- Step 7** (Optional) Run the following command to delete the namespace where Huawei CSI is located. The default namespace **huawei-csi** is used as an example.

```
kubectl delete ns huawei-csi
```

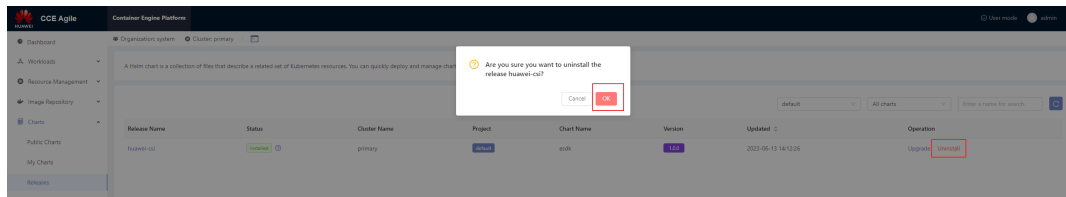
----End

4.1.3.1.2 Uninstalling Huawei CSI on CCE or CCE Agile

This section describes how to uninstall Huawei CSI on the CCE or CCE Agile platform. The following uses CCE Agile v22.3.2 as an example.

Procedure

- Step 1** Log in to the CCE Agile platform.
- Step 2** On the home page, choose **Charts > Releases**. The **Releases** page is displayed.
- Step 3** Select a Huawei CSI release and click **Uninstall**. In the displayed dialog box, click **OK**.



- Step 4** Uninstall the huawei-csi-host-info object. For details, see [Uninstalling the huawei-csi-host-info Object](#).
- Step 5** Uninstall the webhook resource. For details, see [Uninstalling a Webhook Resource](#).
- Step 6** (Optional) Uninstall the snapshot-dependent component service. For details, see [Uninstalling the Snapshot-Dependent Component Service](#).
- Step 7** (Optional) Run the following command to delete the namespace where Huawei CSI is located. **huawei-csi** is used as an example namespace.

```
kubectl delete ns huawei-csi
```

----End

4.1.3.1.3 Uninstalling CSI-Dependent Component Services

This section describes how to uninstall the CSI-dependent component services.

Uninstalling the huawei-csi-host-info Object

Secret object **huawei-csi-host-info** stores the initiator information about each node in the cluster, for example, iSCSI initiators. When you run the **helm uninstall** command, the resource will not be uninstalled. To uninstall the resource, perform the following steps:

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
- Step 2** Run the following command to check whether the Secret object exists. **huawei-csi** is the namespace where the Secret object is located.

```
kubectl get secret -n huawei-csi -l secret-provisioner=csi.huawei.com
```
- Step 3** If the Secret exists in [Step 2](#), run the following command to delete the Secret:

```
kubectl delete secret -n huawei-csi -l secret-provisioner=csi.huawei.com
```

- Step 4** Run the following command to check whether the Secret object exists. **huawei-csi** is the namespace where the Secret object is located.

```
kubectl get secret huawei-csi-host-info -n huawei-csi
```

- Step 5** If the Secret exists in [Step 4](#), run the following command to uninstall the Secret object. **huawei-csi-host-info** is the name of the Secret object.

```
kubectl delete secret huawei-csi-host-info -n huawei-csi
```

----End

Uninstalling a Webhook Resource

The webhook resource named **storage-backend-controller.xuanwu.huawei.io** is used to verify the backend key information and connectivity with the storage. When you run the **helm uninstall** command, the resource will not be uninstalled. To uninstall the resource, perform the following steps:

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
- Step 2** Run the following command to query the webhook-dependent component service.

```
kubectl get validatingwebhookconfigurations.admissionregistration.k8s.io storage-backend-controller.xuanwu.huawei.io
```

The following is an example of the command output.

NAME	WEBHOOKS	AGE
storage-backend-controller.xuanwu.huawei.io	1	12d

- Step 3** Run the following command to uninstall the webhook-dependent component service.

```
kubectl delete validatingwebhookconfigurations.admissionregistration.k8s.io storage-backend-controller.xuanwu.huawei.io
```

- Step 4** Run the following command to check whether the service is successfully uninstalled. If the command output is empty, the uninstallation is successful.

```
kubectl get validatingwebhookconfigurations.admissionregistration.k8s.io storage-backend-controller.xuanwu.huawei.io
```

----End

Uninstalling the Snapshot-Dependent Component Service

NOTICE

- Do not uninstall the snapshot-dependent component service when snapshots exist. Otherwise, Kubernetes will automatically delete all user snapshots and they cannot be restored. For details, see [Delete a CustomResourceDefinition](#).
- Do not uninstall the snapshot-dependent component service during the CSI upgrade.

Scenario Description

- Currently, Huawei CSI uses the snapshot feature.
- Currently, only Huawei CSI is available in the Kubernetes cluster, and Huawei CSI is no longer used.

- Before the uninstallation, ensure that no VolumeSnapshot resource managed by Huawei CSI exists in the Kubernetes cluster.

Procedure

Step 1 Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.

Step 2 Run the following command to uninstall the snapshot-dependent component service.

```
kubectl delete crd volumesnapshotclasses.snapshot.storage.k8s.io  
volumesnapshotcontents.snapshot.storage.k8s.io volumesnapshots.snapshot.storage.k8s.io
```

Step 3 Run the following command to check whether the service is successfully uninstalled. If the command output is empty, the uninstallation is successful.

```
kubectl get crd | grep snapshot.storage.k8s.io
```

----End

Uninstalling a Lease Resource

When huawei-csi-controller is deployed in multi-copy mode, a Lease resource is generated to store the current Holder information. When you run the **helm uninstall** command, the resource will not be uninstalled. To uninstall the resource, perform the following steps:

Step 1 Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.

Step 2 Run the following command to query the Lease information.

```
kubectl get lease -n huawei-csi
```

The following is an example of the command output.

NAME	HOLDER	AGE
csi-huawei-com	node-1	24d
external-attacher-leader-csi-huawei-com	node-1	24d
external-resizer-csi-huawei-com	node-1	24d
external-snapshotter-leader-csi-huawei-com	node-1	24d
snapshot-controller-leader	node-1	24d
storage-backend-controller	node-1	24d
huawei-csi-extender	node-1	24d

Step 3 Run the following command to uninstall the Lease resource.

```
kubectl delete lease -n huawei-csi csi-huawei-com external-attacher-leader-csi-huawei-com external-resizer-csi-huawei-com external-snapshotter-leader-csi-huawei-com snapshot-controller-leader storage-backend-controller huawei-csi-extender
```

Step 4 Run the following command to check whether the uninstallation is successful.

```
kubectl get lease -n huawei-csi
```

The following is an example of the command output. If the command output is empty, the uninstallation is successful.

```
No resources found in huawei-csi namespace.
```

----End

4.1.3.2 Manual Uninstallation

This section describes how to manually uninstall Huawei CSI.

NOTICE

If you do not uninstall Huawei CSI for the purpose of an upgrade, ensure that all resources (such as PV, PVC, snapshot, and storage backend resources) provisioned by Huawei CSI have been cleared on your container platform before uninstalling Huawei CSI. Otherwise, once you uninstall Huawei CSI, these resources cannot be automatically scheduled, managed, or cleared.

Uninstalling the huawei-csi-node Service

Step 1 Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.

Step 2 Run the following command to uninstall the huawei-csi-node service. Replace *huawei-csi* with the namespace where Huawei CSI is located.

```
kubectl delete daemonset huawei-csi-node -n huawei-csi
```

Step 3 Run the following command to check whether the service is successfully uninstalled. If **NotFound** is displayed, the service is successfully uninstalled.

```
kubectl get daemonset huawei-csi-node -n huawei-csi
```

----End

Uninstalling the huawei-csi-controller Service

Step 1 Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.

Step 2 Run the following command to uninstall the huawei-csi-controller service. Replace *huawei-csi* with the namespace where Huawei CSI is located.

```
kubectl delete deployment huawei-csi-controller -n huawei-csi
```

Step 3 Run the following command to check whether the service is successfully uninstalled. If **NotFound** is displayed, the service is successfully uninstalled.

```
kubectl get deployment huawei-csi-controller -n huawei-csi
```

----End

Uninstalling the csidriver Object

If [the CSIDriver feature is not used during installation](#), you can skip the following steps.

Step 1 Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.

Step 2 Run the following command to uninstall the csidriver object.

```
kubectl delete csidriver csi.huawei.com
```

Step 3 Run the following command to check whether the service is successfully uninstalled. If **NotFound** is displayed, the service is successfully uninstalled.

```
kubectl get csidriver csi.huawei.com
```

----End

Deleting the RBAC Permission

Step 1 Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.

Step 2 Delete the RBAC permission.

```
kubectl -n huawei-csi -l provisioner=csi.huawei.com delete  
ServiceAccount,Service,role,rolebinding,ClusterRole,ClusterRoleBinding
```

----End

Uninstalling Other Resources

- Uninstall the huawei-csi-host-info object. For details, see [Uninstalling the huawei-csi-host-info Object](#).
- Uninstall the webhook resource. For details, see [Uninstalling a Webhook Resource](#).
- (Optional) Uninstall the snapshot-dependent component service. For details, see [Uninstalling the Snapshot-Dependent Component Service](#).
- (Optional) Uninstall the Lease resource. For details, see [Uninstalling a Lease Resource](#).
- (Optional) Run the following command to delete the namespace where Huawei CSI is located. The default namespace **huawei-csi** is used as an example.

```
kubectl delete ns huawei-csi
```

4.1.4 Upgrade

This section describes how to upgrade Huawei CSI.

NOTE

In the current version, resource requests and limits are added to Huawei CSI. For details, see [10.3 Huawei CSI Resource Management](#).

4.1.4.1 Upgrade Using Helm

This section describes how to upgrade Huawei CSI.

- To upgrade Huawei CSI from 2.x to 4.13.0, uninstall it by referring to the user guide of the earlier version and install Huawei CSI by referring to [4.1.2.1 Installation Using Helm](#).
- To upgrade Huawei CSI from 2.x or 3.x to 4.13.0, see [4.1.4.1.1 Upgrading from 2.x or 3.x to 4.x](#).
- To upgrade Huawei CSI from 4.x to 4.13.0, see [4.1.4.1.2 Upgrading Huawei CSI on Kubernetes, OpenShift, and Tanzu](#).

NOTICE

- Some CSI 2.x versions are unavailable now. If the upgrade fails, CSI may fail to be rolled back to a version which is unavailable now.
- During the upgrade or rollback, you cannot use Huawei CSI to create new resources or mount or unmount an existing PVC.
- During the upgrade or rollback, do not uninstall the snapshot-dependent component service.
- During the upgrade or rollback, the existing resources such as PVCs, snapshots, and Pods will run properly and will not affect your service access.

4.1.4.1.1 Upgrading from 2.x or 3.x to 4.x

Backing Up Storage Backend Configurations

If you have evaluated the risks mentioned in the preceding notice and need to upgrade CSI from 2.x or 3.x to 4.13.0, perform the following steps to back up storage backend configurations:

Step 1 Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.

Step 2 Run the following command to back up the backend information to the **configmap.json** file. For the OpenShift platform, replace **kubect** with **oc**.

```
kubect get cm huawei-csi-configmap -n huawei-csi -o json > configmap.json
```

----End

Upgrading Huawei CSI

Perform the upgrade according to the procedure described in [Upgrading Huawei CSI](#).

Configuring the Storage Backend

Step 1 Configure the storage backend based on the backend information backed up in [Backing Up Storage Backend Configurations](#) according to [5.1 Storage Backend Management](#). The command for configuring the storage backend based on the backup backend information is as follows:

```
oceanctl create backend -f configmap.json -i json -n huawei-csi
```

----End

4.1.4.1.2 Upgrading Huawei CSI on Kubernetes, OpenShift, and Tanzu

Prerequisites

- Huawei CSI of an earlier version is installed using Helm.
- A Huawei CSI image of a new version has been created and uploaded to the image repository or imported to all nodes by following the instructions provided in [4.1.1.2 Uploading a Huawei CSI Image](#).

Upgrading Huawei CSI

If CSI of an earlier version is deployed using Helm, perform the following steps to upgrade Huawei CSI.

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
- Step 2** Copy the CSI component package of the target version to any directory on the master node.
- Step 3** Go to the **helm/esdk** working directory. For the directory path, see [Table 4-1](#).

```
cd helm/esdk
```
- Step 4** Run the **kubectl apply -f ./crds/backend/** command to update the storage backend CRD.

```
kubectl apply -f ./crds/backend/
```
- Step 5** (Optional) Check snapshot-dependent components by following the instructions provided in [4.1.1.4 Checking Volume Snapshot-Dependent Components](#). After confirming that the components are correct, run the following command to update the snapshot CRD. If **controller.snapshot.enabled** is set to **false** or the Kubernetes version is earlier than v1.20, you can skip this step. For details, see [Table 4-5](#).

```
kubectl apply -f ./crds/snapshot-crd/ --validate=false
```
- Step 6** Run the following command to obtain the original service configuration file. **helm-huawei-csi** indicates the Helm chart name specified during the installation of the earlier version, and **huawei-csi** indicates the Helm chart namespace specified during the installation of the earlier version.

```
helm get values helm-huawei-csi -n huawei-csi -a > ./update-values.yaml
```
- Step 7** Run the **vi update-values.yaml** command to open the file obtained in [Step 6](#), modify the **images** configuration items, and update the image to the latest version. For details about the parameters to be modified, see [Table 4-9](#).

Table 4-9 images configuration items

Container Name	Description	K8s Version Requirements	New Value
storage-backend-controller	huawei-csi image.	v1.16+	storage-backend-controller:4.13.0
storage-backend-sidecar	Image used by Huawei backends to manage storageBackendContent resources.	v1.16+	storage-backend-sidecar:4.13.0
huawei-csi-driver	Image used by Huawei backends to manage storageBackendClaim resources.	v1.16+	huawei-csi:4.13.0

Container Name	Description	K8s Version Requirements	New Value
huawei-csi-extender	huawei-csi-extender image.	v1.16+	huawei-csi-extender:4.13.0
images.sidecar.livenessProbe	livenessprobe sidecar image.	v1.16+	registry.k8s.io/sig-storage/livenessprobe:v2.15.0
images.sidecar.resizer	csi-resizer sidecar image.	v1.16+	registry.k8s.io/sig-storage/csi-resizer:v1.13.2
images.sidecar.registrar	csi-node-driver-registrar sidecar image.	v1.16+	registry.k8s.io/sig-storage/csi-node-driver-registrar:v2.13.0
images.sidecar.snapshotter	csi-snapshotter sidecar image.	v1.25+	registry.k8s.io/sig-storage/csi-snapshotter:v8.2.0
		v1.20-v1.24	registry.k8s.io/sig-storage/csi-snapshotter:v6.3.0
images.sidecar.snapshotController	snapshot-controller sidecar image.	v1.25+	registry.k8s.io/sig-storage/snapshot-controller:v8.2.0
		v1.20-v1.24	registry.k8s.io/sig-storage/snapshot-controller:v6.3.0
images.sidecar.provisioner	csi-provisioner sidecar image.	v1.20+	registry.k8s.io/sig-storage/csi-provisioner:v5.2.0
		v1.17-v1.19	registry.k8s.io/sig-storage/csi-provisioner:v3.0.0
		v1.16.x	quay.io/k8scsi/csi-provisioner:v1.4.0
images.sidecar.attacher	csi-attacher sidecar image.	v1.17+	registry.k8s.io/sig-storage/csi-attacher:v4.8.0
		v1.16.x	quay.io/k8scsi/csi-attacher:v1.2.1

Container Name	Description	K8s Version Requirements	New Value
images.sidecar.healthMonitorController	health-monitor-controller sidecar image.	v.1.16.x	registry.k8s.io/sig-storage/csi-external-health-monitor-controller:v0.17.0

Step 8 (Optional) If you need to update configuration items or add configuration information during the upgrade, modify the configuration information in the **update-values.yaml** file by referring to [4.1.2.1.3 Parameters in the values.yaml File of Helm](#).

 NOTE

During the upgrade, if the **update-values.yaml** and **values.yaml** configuration files contain the same configuration item, the configuration in the **update-values.yaml** file takes effect preferentially.

Step 9 Run the following command to upgrade Huawei CSI. In the following command, **helm-huawei-csi** indicates the specified Helm chart name, **huawei-csi** indicates the specified Helm chart namespace, and **update-values.yaml** indicates the file obtained in [Step 6](#).

```
helm upgrade helm-huawei-csi ./ -n huawei-csi -f ./values.yaml -f ./update-values.yaml
```

Step 10 After the huawei-csi service is deployed, run the following command to check whether the service is started.

```
kubectl get pod -n huawei-csi
```

The following is an example of the command output. If the Pod status is **Running**, the service is started successfully.

NAME	READY	STATUS	RESTARTS	AGE
huawei-csi-controller-6dfcc4b79f-9vjtq	9/9	Running	0	24m
huawei-csi-controller-6dfcc4b79f-cspnc	9/9	Running	0	24m
huawei-csi-node-g6f4k	3/3	Running	0	20m
huawei-csi-node-tqs87	3/3	Running	0	20m

----End

4.1.4.1.3 Upgrading Huawei CSI on CCE or CCE Agile

Prerequisites

You have downloaded the CSI software package of a new version.

Procedure

Step 1 Uninstall CSI. For details, see [4.1.3.1.2 Uninstalling Huawei CSI on CCE or CCE Agile](#).

Step 2 Install CSI of the new version. For details, see [4.1.2.1.2 Installing Huawei CSI on the CCE or CCE Agile Platform](#).

----End

4.1.4.2 Manual Upgrade

This section describes how to manually upgrade Huawei CSI.

During the upgrade or rollback, the existing resources such as PVCs, snapshots, and Pods will run properly and will not affect your service access.

NOTICE

- Some CSI 2.x versions are unavailable now. If the upgrade fails, CSI may fail to be rolled back to a version which is unavailable now.
- During the upgrade or rollback, you cannot use Huawei CSI to create new resources or mount or unmount an existing PVC.
- During the upgrade or rollback, do not uninstall the snapshot-dependent component service.

Upgrading CSI from 2.x or 3.x to 4.13.0

To upgrade CSI from 2.x or 3.x to 4.13.0, perform the following operations:

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
- Step 2** Run the following command to back up the backend information to the **configmap.json** file. For the OpenShift platform, replace **kubectrl** with **oc**.
- ```
kubectrl get cm huawei-csi-configmap -n huawei-csi -o json > configmap.json
```
- Step 3** Uninstall CSI. For details, see [4.1.3.2 Manual Uninstallation](#).
- Step 4** Install CSI of the current version. For details, see [4.1.2.2 Manual Installation](#).
- Step 5** Install the backend information backed up in [Step 2](#) according to [5.1 Storage Backend Management](#). The command for configuring the storage backend based on the backup backend information is as follows:
- ```
oceanctl create backend -f configmap.json -i json -n huawei-csi
```

----End

Upgrading CSI from 4.x to 4.13.0

To upgrade CSI from 4.x to 4.13.0, perform the following operations:

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
- Step 2** Uninstall CSI. For details, see [4.1.3.2 Manual Uninstallation](#).
- Step 3** Install CSI of the current version. For details, see [4.1.2.2 Manual Installation](#).

----End

4.1.5 Rollback

4.1.5.1 Rollback Using Helm

If CSI fails to be upgraded from 2.x or 3.x to 4.13.0 and needs to be rolled back, uninstall CSI by referring to [4.1.3.1 Uninstallation Using Helm](#) and then download and install CSI of the source version.

NOTICE

- During the upgrade or rollback, the existing resources such as PVCs, snapshots, and Pods will run properly and will not affect your service access.
- During the upgrade or rollback, you cannot use Huawei CSI to create new resources or mount or unmount an existing PVC.
- During the upgrade or rollback, do not uninstall the snapshot-dependent component service.

4.1.5.1.1 Rolling Back Huawei CSI on Kubernetes, OpenShift, and Tanzu

Prerequisites

- CSI has been updated using Helm.

Procedure

Step 1 Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.

Step 2 Go to the **helm/esdk** working directory. For the directory path, see [Table 4-1](#).

```
cd helm/esdk
```

Step 3 Run the following command to query the historical versions of the CSI services deployed using Helm.

```
helm history helm-huawei-csi -n huawei-csi
```

The following is an example of the command output.

REVISION	UPDATED	STATUS	CHART	APP VERSION	DESCRIPTION
1	Mon Jan 8 04:15:40 2024	superseded	esdk-4.4.0	4.4.0	Install complete
2	Mon Jan 8 04:16:12 2024	deployed	esdk-4.13.0	4.13.0	Upgrade complete

Step 4 Run the following command to roll back the CSI services to the specified version.

In the preceding command, *revision-number* indicates a version number queried in [Step 3](#). For example, the version is **1**.

```
helm rollback helm-huawei-csi -n huawei-csi 1
```

The following is an example of the command output. If **Rollback was a success** is displayed in the command output, the CSI services are successfully rolled back to the specified version.

```
Rollback was a success! Happy Helming!
```

----End

4.1.5.1.2 Rolling Back Huawei CSI on CCE or CCE Agile

NOTICE

- During the upgrade or rollback, the existing resources such as PVCs, snapshots, and Pods will run properly and will not affect your service access.
 - During the upgrade or rollback, you cannot use Huawei CSI to create new resources or mount or unmount an existing PVC.
 - During the upgrade or rollback, do not uninstall the snapshot-dependent component service.
-

Prerequisites

You have downloaded the CSI software package of the source version.

Procedure

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
- Step 2** Uninstall CSI. For details, see [Procedure](#).
- Step 3** Reinstall CSI of the source version. For details, see [4.1.2.1.2 Installing Huawei CSI on the CCE or CCE Agile Platform](#).

----End

4.1.5.2 Manual Rollback

Uninstall CSI by referring to [4.1.3.2 Manual Uninstallation](#), and then download and install CSI of the source version.

NOTICE

- During the upgrade or rollback, the existing resources such as PVCs, snapshots, and Pods will run properly and will not affect your service access.
 - During the upgrade or rollback, you cannot use Huawei CSI to create new resources or mount or unmount an existing PVC.
 - During the upgrade or rollback, do not uninstall the snapshot-dependent component service.
-

Prerequisites

You have downloaded the CSI software package of the source version.

Procedure

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.

Step 2 Uninstall CSI. For details, see [4.1.3.2 Manual Uninstallation](#).

Step 3 Reinstall CSI of the source version. For details, see [4.1.2.2 Manual Installation](#).

----**End**

5 Basic Services

[5.1 Storage Backend Management](#)

[5.2 StorageClass Management](#)

[5.3 Persistent Volume Management](#)

5.1 Storage Backend Management

Backend is an abstract concept of Huawei storage resources. Each Huawei storage device can abstract multiple backend resources using features such as tenants, storage pools, and protocols. Each backend exists independently and defines Huawei storage information required for providing persistent volumes for Kubernetes clusters. The relationship between storage backends, storage devices, and storage pools is 1:1:n.

This section describes how to use the `oceanctl` tool to configure and manage storage backends.

Description of the `oceanctl` Tool

- You have obtained the `oceanctl` tool, copied the `oceanctl` tool to the environment directory, for example, `/usr/local/bin`, and obtained the execute permission. The `oceanctl` tool is stored in `/bin/oceanctl` of the software package.
- Before using the `oceanctl` tool, ensure that the `oceanctl` tool version is the same as the CSI version. For details about how to check the `oceanctl` version, see [Obtaining Help Information](#). The current version number should be V4.13.0.
- The `oceanctl` tool depends on `kubectl` (for the Kubernetes platform) or `oc` (for the OpenShift platform) commands. Therefore, you need to run the tool on a node where `kubectl` or `oc` commands can be executed.
- By default, the user who runs `oceanctl` commands must have the read and write permissions on the `/var/log` directory. If you do not have the permissions on the directory, run the `--log-dir=/path/to/custom` command to specify a directory on which you have the permissions as the log file directory.
- `huawei-csi` is the default namespace used by `oceanctl` to create a backend.

- For details about **oceanctl** commands, see [9.1 Description of oceanctl Commands](#).

5.1.1 Configuring the Storage Backend

NOTE

1. When oceanctl is used to create a storage backend, the entered account and key information is stored in the **Secret** object. It is recommended that the customer container platform encrypt the Secret object based on the suggestions of the supplier or K8s community. For details about how to encrypt the Secret object in the K8s community, see [Encrypting Confidential Data at Rest](#).
2. When a backend is created using a .json file, the backend name of an earlier version may contain uppercase letters or underscores (_). In this case, the old name is remapped to a new name. The mapping process automatically occurs and does not affect the original functions. For example, **ABC_123** is mapped to **abc-123-fd68e**. The mapping rules are as follows:
 - Uppercase letters are converted to lowercase letters.
 - An underscore (_) is converted to a hyphen (-).
 - A 5-digit hash code is added to the end.
3. If a storage backend is connected to a vStore, the vStore name cannot be changed after the storage backend is created.
4. When you use the CCE or CCE Agile platform to create a backend, you need to specify the **provisioner** parameter. For details, see commands of [Creating a Storage Backend](#) on the CCE Platform.

5.1.1.1 Flash Storage (OceanStor Dorado/OceanStor V5/OceanStor V6 and Later)

5.1.1.1.1 File System

NFS

This section describes how to create a storage backend of the NFS protocol type.

Configuration Item Description

Table 5-1 backend parameters

Parameter	Description	Mandator y	Default Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed to oceanstor-nas .

Parameter	Description	Mandatory	Default Value	Remarks
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.
vstoreName	vStore name on the storage side. This parameter needs to be specified when the connected backend is OceanStor V5 and resources need to be provisioned under a specified vStore.	No	-	This parameter needs to be specified only when the backend is OceanStor V5 and vStores need to be supported.
urls	Management URLs of the storage device. The value format is a list. The value can be a domain name or an IP address + port number.	Yes	-	• If resources need to be provisioned to a specified vStore, set this parameter to the logical management port URL of the specified vStore. • If the management URL is of the IPv6 type, the URL format is https://[IPv6 address]:Port number.

Parameter	Description	Mandatory	Default Value	Remarks
pools	Storage pools of storage devices. The value format is a list.	Yes	-	Enter the storage pool names.
parameters.protocol	Storage protocol. The value is a character string.	Yes	-	The value is fixed at nfs . • Ensure that an NFS client tool has been installed on the connected compute node.
parameters.portals	Service access port. Nodes will use this port to read and write storage resources. The value format is a list.	Yes	-	• If a vStore is used to connect to a backend, portals must be set to the logical port information of the vStore. • You can enter a domain name address. • IPv6 is supported. • Only one port can be configured.

Parameter	Description	Mandatory	Default Value	Remarks
parameters.nfsAutoAuthClient	Enables or disables the automatic management function of NFS share clients. • true: enabled • false: disabled	No	-	This parameter takes effect only when csiDriver.reportNodeIP and CSIDriverObject.attachRequired are set to true during CSI deployment. When this parameter is enabled, the CSI dynamically sets the permissions of the NFS share clients corresponding to the host IP addresses that meet the rules to read/write or none during PVC mounting or unmounting.

Parameter	Description	Mandatory	Default Value	Remarks
parameters.nfsAutoAuthClientCIDRs	List of IP CIDR blocks for NFS communication between hosts. Example: nfsAutoAuthClientCIDRs: - 10.10.0.0/16	No	-	• This parameter is valid only when parameters.nfsAutoAuthClient is set to true. • The CSI dynamically manages the host IP addresses within the configured CIDRs. • If no CIDRs are specified, the CSI dynamically manages all host IP addresses. • Ensure that the number of host IP addresses that take effect is 10 or less.
metroVStorePairID	HyperMetro vStore pair ID. This parameter is mandatory when a PV to be created on the storage side needs to support the NAS HyperMetro feature. In this case, you need to enter the ID of the HyperMetro vStore pair to which the PV to be created belongs.	No	-	You can query the HyperMetro vStore pair ID on DeviceManager.

Parameter	Description	Mandatory	Default Value	Remarks
metroBackend	Backend name of the HyperMetro peer. The value is a character string. This parameter is mandatory when a PV to be created on the storage side needs to support the NAS HyperMetro feature. In this case, you need to enter the name of the other backend to form a HyperMetro pair with the current backend.	No	-	The names of the two backends in the pair must be entered. After the two backends form a HyperMetro relationship, they cannot form a HyperMetro relationship with other backends.
supportedTopologies	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	30	The value ranges from 1 to 30. If this parameter is not set or the value is not in the specified range, the default value 30 is used.
authenticationMode	Authentication mode for logging in to a storage backend. The following modes are supported: • local: local authentication • ldap: LDAP authentication	No	local	When Huawei enterprise storage is OceanStor V5, the ID of the LDAP domain authentication server must be 0.

Creating Storage Backends of the Local Type

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "oceanstor-nas"
name: "backend-demo"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:8088"
pools:
  - "StoragePool001"
parameters:
  protocol: "nfs"
  portals:
    - "10.10.30.20"
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

NUMBER	CONFIGURED	NAME	STORAGE	URLS
1	false	backend-demo	oceanstor-nas	https://192.168.129.157:8088

Please enter the backend number to configure (Enter 'exit' to exit):

Step 3 Enter the serial number of the backend to be created and enter the account and password.

Please enter the backend number to configure (Enter 'exit' to exit):1

Please enter this backend user name: admin

Please enter this backend password:

Backend backend-demo is configured

NUMBER	CONFIGURED	NAME	STORAGE	URLS
1	true	backend-demo	oceanstor-nas	https://192.168.129.157:8088

Please enter the backend number to configure (Enter 'exit' to exit):

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

NAMESPACE	NAME	PROTOCOL	STORAGETYPE	SN	STATUS	ONLINE
huawei-csi	backend-demo	nfs	oceanstor-nas	xxxxxxxxxxxxxxxxxxxxxx	Bound	true

Url https://192.168.129.157:8088

----End

Creating Storage Backends of the HyperMetro Type

NOTE

- Before configuring NAS HyperMetro, you need to configure the HyperMetro relationship between two storage devices, including the remote device and HyperMetro domain. The HyperMetro domain of the file system can only work in HyperMetro active-active (AA) mode. For details about the configuration operation, see the product documentation of the corresponding storage model.
- The accounts for connecting to NAS HyperMetro backends must be the administrator accounts of the storage vStores, and the usernames and passwords used for login at both ends in the HyperMetro relationship must be the same.
- Except NAS HyperMetro backends, the management URLs of other backends cannot be the URL of a logical management port of a vStore that has established the HyperMetro relationship.
- When a HyperMetro storage backend is used, do not provision common file systems. Otherwise, services may be interrupted in logical port failover scenarios.

Step 1 Prepare a storage backend configuration file, for example, **backend.yaml**.

```
storage: "oceanstor-nas"
name: "backend-active"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.155:8088"
pools:
  - "StoragePool001"
metroStorePairID: "2100xxxxxxxxx000000000600000000"
metroBackend: "backend-standby"
parameters:
  protocol: "nfs"
  portals:
    - "192.168.129.156"
maxClientThreads: "30"
---
storage: "oceanstor-nas"
name: "backend-standby"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:8088"
pools:
  - "StoragePool001"
metroStorePairID: "2100xxxxxxxxx000000000600000000"
metroBackend: "backend-active"
parameters:
  protocol: "nfs"
  portals:
    - "192.168.129.158"
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

NUMBER	CONFIGURED	NAME	STORAGE	URLS
1	false	backend-active	oceanstor-nas	https://192.168.129.155:8088
2	false	backend-standby	oceanstor-nas	https://192.168.129.157:8088

Please enter the backend number to configure (Enter 'exit' to exit):

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
Please enter this backend user name: user1
Please enter this backend password:
```

```
Backend backend-standby is configured
NUMBER CONFIGURED NAME STORAGE URLS
1 true backend-active oceanstor-nas https://192.168.129.155:8088
2 true backend-standby oceanstor-nas https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

NAMESPACE	NAME	PROTOCOL	STORAGETYPE	SN	STATUS	ONLINE
Url						
huawei-csi	backend-active	nfs	oceanstor-nas	xxxxxxxxxxxxxxxxxxxxxx	Bound	true https://192.168.129.155:8088
huawei-csi	backend-standby	nfs	oceanstor-nas	xxxxxxxxxxxxxxxxxxxxxx	Bound	true https://192.168.129.157:8088

----End

NFS+

This section describes how to create a storage backend of the NFS+ protocol type.

Configuration Item Description

Table 5-2 backend parameters

Parameter	Description	Mandatory	Default Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed to oceanstor-nas .
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.

Parameter	Description	Mandatory	Default Value	Remarks
urls	Management URLs of the storage device. The value format is a list. The value can be a domain name or an IP address + port number.	Yes	-	- If resources need to be provisioned to a specified vStore, set this parameter to the logical management port URL of the specified vStore. - If the management URL is of the IPv6 type, the URL format is https://[IPv6 address]:Port number.
pools	Storage pools of storage devices. The value format is a list.	Yes	-	Enter the storage pool names.
parameters.protocol	Storage protocol. The value is a character string.	Yes	-	- The value is fixed to nfs+. - Ensure that an NFS+ client tool has been installed on the connected compute node.
parameters.ports	Service access port. Nodes will use this port to read and write storage resources. The value format is a list.	Yes	-	- If a vStore is used to connect to a backend, ports must be set to the logical port information of the vStore. - You can enter a domain name address. - IPv6 is supported. - Multiple ports can be configured.

Parameter	Description	Mandatory	Default Value	Remarks
metroVStorePairID	HyperMetro vStore pair ID. This parameter is mandatory when a PV to be created on the storage side needs to support the NAS HyperMetro feature. In this case, you need to enter the ID of the HyperMetro vStore pair to which the PV to be created belongs.	No	-	You can query the HyperMetro vStore pair ID on DeviceManager.
metroBackend	Backend name of the HyperMetro peer. The value is a character string. This parameter is mandatory when a PV to be created on the storage side needs to support the NAS HyperMetro feature. In this case, you need to enter the name of the other backend to form a HyperMetro pair with the current backend.	No	-	The names of the two backends in the pair must be entered. After the two backends form a HyperMetro relationship, they cannot form a HyperMetro relationship with other backends.
supportedTopologies	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	30	The value ranges from 1 to 30 . If this parameter is not set or the value is not in the specified range, the default value 30 is used.
authenticationMode	Authentication mode for logging in to a storage backend. The following modes are supported: ● local: local authentication ● ldap: LDAP authentication	No	local	-

Creating Storage Backends of the Local Type

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "oceanstor-nas"
name: "backend-demo"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:8088"
pools:
  - "StoragePool001"
parameters:
  protocol: "nfs+"
  portals:
    - "10.10.30.20"
    - "10.10.30.30"
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

NUMBER	CONFIGURED	NAME	STORAGE	URLS
1	false	backend-demo	oceanstor-nas	https://192.168.129.157:8088

Please enter the backend number to configure (Enter 'exit' to exit):

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
Please enter this backend user name: admin
Please enter this backend password:
```

```
Backend backend-demo is configured
NUMBER CONFIGURED NAME STORAGE URLS
1 true backend-demo oceanstor-nas https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

NAMESPACE	NAME	PROTOCOL	STORAGETYPE	SN	STATUS	ONLINE
huawei-csi	backend-demo	nfs+	oceanstor-nas	xxxxxxxxxxxxxxxxxxxxxx	Bound	true

192.168.129.157:8088 https://

----End

Creating Storage Backends of the HyperMetro Type

NOTE

- Before configuring NAS HyperMetro, you need to configure the HyperMetro relationship between two storage devices, including the remote device and HyperMetro domain. The HyperMetro domain of the file system can only work in HyperMetro active-active (AA) mode. For details about the configuration operation, see the product documentation of the corresponding storage model.
- The accounts for connecting to NAS HyperMetro backends must be the administrator accounts of the storage vStores, and the usernames and passwords used for login at both ends in the HyperMetro relationship must be the same.
- Except NAS HyperMetro backends, the management URLs of other backends cannot be the URL of a logical management port of a vStore that has established the HyperMetro relationship.
- When a HyperMetro storage backend is used, do not provision common file systems. Otherwise, services may be interrupted in logical port failover scenarios.

Procedure

Step 1 Prepare a storage backend configuration file, for example, **backend.yaml**.

```
storage: "oceanstor-nas"
name: "backend-active"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.155:8088"
pools:
  - "StoragePool001"
metroStorePairID: "2100xxxxxxxxx0000000000600000000"
metroBackend: "backend-standby"
parameters:
  protocol: "nfs+"
  portals:
    - "192.168.129.156"
    - "192.168.129.157"
  maxClientThreads: "30"
---
storage: "oceanstor-nas"
name: "backend-standby"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.158:8088"
pools:
  - "StoragePool001"
metroStorePairID: "2100xxxxxxxxx0000000000600000000"
metroBackend: "backend-active"
parameters:
  protocol: "nfs+"
  portals:
    - "192.168.129.159"
    - "192.168.129.160"
  maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

NUMBER	CONFIGURED	NAME	STORAGE	URLS
1	false	backend-active	oceanstor-nas	https://192.168.129.155:8088
2	false	backend-standby	oceanstor-nas	https://192.168.129.158:8088

Please enter the backend number to configure (Enter 'exit' to exit):

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
Please enter this backend user name: user1
Please enter this backend password:

Backend backend-standby is configured
NUMBER CONFIGURED NAME STORAGE URLS
1 true backend-active oceanstor-nas https://192.168.129.155:8088
2 true backend-standby oceanstor-nas https://192.168.129.158:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

NAMESPACE	NAME	PROTOCOL	STORAGETYPE	SN	STATUS	ONLINE	Url
huawei-csi	backend-active	nfs+	oceanstor-nas	xxxxxxxxxxxxxxxxxxxx	Bound	true	https://192.168.129.155:8088
huawei-csi	backend-standby	nfs+	oceanstor-nas	xxxxxxxxxxxxxxxxxxxx	Bound	true	https://192.168.129.158:8088

----End

5.1.1.1.2 Dtree

NFS

This section describes how to create a storage backend of the NFS protocol type.

Configuration Item Description

Table 5-3 backend parameters

Parameter	Description	Mandatory	Default Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed at oceanstor-dtree .
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.

Parameter	Description	Mandatory	Default Value	Remarks
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.
urls	Management URLs of the storage device. The value format is a list. The value can be a domain name or an IP address + port number.	Yes	-	• If resources need to be provisioned to a specified vStore, set this parameter to the logical management port URL of the specified vStore. • If the management URL is of the IPv6 type, the URL format is https://[IPv6 address]:Port number.
parameters.protocol	Storage protocol. The value is a character string.	Yes	-	• The value is fixed at nfs. • Ensure that an NFS client tool has been installed on the connected compute node.

Parameter	Description	Mandatory	Default Value	Remarks
parameters.ports	Service access port. Nodes will use this port to read and write storage resources. The value format is a list.	Yes	-	• If a vStore is used to connect to a backend, ports must be set to the logical port information of the vStore. • You can enter a domain name address. • IPv6 is supported. • Only one port can be configured.
parameters.nfsAutoAuthClient	Enables or disables the automatic management function of NFS share clients. • true: enabled • false: disabled	No	-	This parameter takes effect only when csiDriver.reportNodeIP and CSIDriverObject.attachRequired are set to true during CSI deployment. When this parameter is enabled, the CSI dynamically sets the permissions of the NFS share clients corresponding to the host IP addresses that meet the rules to read/write or none during PVC mounting or unmounting.

Parameter	Description	Mandatory	Default Value	Remarks
parameters.nfsAutoAuthClientCIDRs	List of IP CIDR blocks for NFS communication between hosts. Example: nfsAutoAuthClientCIDRs: - 10.10.0.0/16	No	-	• This parameter is valid only when parameters.nfsAutoAuthClient is set to true. • The CSI dynamically manages the host IP addresses within the configured CIDRs. • If no CIDRs are specified, the CSI dynamically manages all host IP addresses. • Ensure that the number of host IP addresses that take effect is 10 or less.
parameters.parentname	Name of a file system on the current storage device. A dtree is created in the file system.	No	-	• Query the name on the File Systems page of DeviceManager. • You can configure the parameter in StorageClass.

Parameter	Description	Mandatory	Default Value	Remarks
supportedTopologies	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	30	The value ranges from 1 to 30 . If this parameter is not set or the value is not in the specified range, the default value 30 is used.
authenticationMode	Authentication mode for logging in to a storage backend. The following modes are supported: • local: local authentication • ldap: LDAP authentication	No	local	-

Creating a Storage Backend

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "oceanstor-dtree"
name: "backend-demo"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:8088"
parameters:
  protocol: "nfs"
  parentname: "parent-filesystem"
portals:
  - "10.10.30.20"
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

```
NUMBER CONFIGURED NAME STORAGE URLS
1 false backend-demo oceanstor-dtree https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
Please enter this backend user name: admin
Please enter this backend password:

Backend backend-demo is configured
NUMBER CONFIGURED NAME STORAGE URLS
1 true backend-demo oceanstor-dtree https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

```
NAMESPACE NAME PROTOCOL STORAGETYPE SN STATUS ONLINE
Url
huawei-csi backend-demo nfs oceanstor-dtree xxxxxxxxxxxxxxxxxxxx Bound true https://
192.168.129.157:8088
```

----End

NFS+

This section describes how to create a storage backend of the NFS+ protocol type.

Configuration Item Description

Table 5-4 backend parameters

Parameter	Description	Mandatory	Default Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed to oceanstor-dtree .
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.

Parameter	Description	Mandatory	Default Value	Remarks
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.
urls	Management URLs of the storage device. The value format is a list. The value can be a domain name or an IP address + port number.	Yes	-	- If resources need to be provisioned to a specified vStore, set this parameter to the logical management port URL of the specified vStore. - If the management URL is of the IPv6 type, the URL format is https://[IPv6 address]:Port number.
parameters.protocol	Storage protocol. The value is a character string.	Yes	-	- The value is fixed to nfs+. - Ensure that an NFS+ client tool has been installed on the connected compute node.
parameters.ports	Service access port. Nodes will use this port to read and write storage resources. The value format is a list.	Yes	-	- If a vStore is used to connect to a backend, ports must be set to the logical port information of the vStore. - You can enter a domain name address. - IPv6 is supported. - Multiple ports can be configured.
parameters.parentname	Name of a file system on the current storage device. A dtree is created in the file system.	No	-	- Query the name on the File Systems page of DeviceManager. - You can configure the parameter in StorageClass.

Parameter	Description	Mandatory	Default Value	Remarks
supportedTopologies	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	30	The value ranges from 1 to 30 . If this parameter is not set or the value is not in the specified range, the default value 30 is used.
authenticationMode	Authentication mode for logging in to a storage backend. The following modes are supported: • local: local authentication • ldap: LDAP authentication	No	local	-

Creating a Storage Backend

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "oceanstor-dtree"
name: "backend-demo"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:8088"
parameters:
  protocol: "nfs+"
  parentname: "parent-filesystem"
portals:
  - "10.10.30.20"
  - "10.10.30.30"
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

```
NUMBER CONFIGURED NAME STORAGE URLS
1 false backend-demo oceanstor-dtree https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
Please enter this backend user name: admin
```

Please enter this backend password:

Backend backend-demo is configured

NUMBER	CONFIGURED	NAME	STORAGE	URLS
1	true	backend-demo	oceanstor-dtree	https://192.168.129.157:8088

Please enter the backend number to configure (Enter 'exit' to exit):

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

NAMESPACE	NAME	PROTOCOL	STORAGETYPE	SN	STATUS	ONLINE
Url						
huawei-csi	backend-demo	nfs+	oceanstor-dtree	xxxxxxxxxxxxxxxxxxxxxx	Bound	true https://192.168.129.157:8088

----End

5.1.1.1.3 Block Service

iSCSI

This section describes how to create a storage backend of the iSCSI protocol type.

Configuration Item Description

Table 5-5 backend parameters

Parameter	Description	Mandatory	Default Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed at oceanstor-san .
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.
urls	Management URLs of the storage device. The value format is a list. The value can be a domain name or an IP address + port number.	Yes	-	If the management URL is of the IPv6 type, the URL format is https://[IPv6 address]:Port number.

Parameter	Description	Mandatory	Default Value	Remarks
pools	Storage pools of storage devices. The value format is a list.	Yes	-	Enter the storage pool names.
parameters.protocol	Storage protocol. The value is a character string.	Yes	-	- The value is fixed at iscsi. - Ensure that an iSCSI client has been installed on the connected compute node.
parameters.ports	Service access port. Nodes will use this port to read and write storage resources. The value format is a list.	Yes	-	- IPv6 is supported. - Multiple ports can be configured.
parameters.ALUA	ALUA configuration of the storage backend. If the worker node uses the native multipathing software provided by the OS and ALUA is enabled, you need to configure this parameter.	Conditionally mandatory	-	If ALUA is enabled for the host multipathing software, ensure that the backend ALUA configuration is the same as that of the host ALUA configuration. For details, see 7.10 Configuring ALUA .
supportedTopologies	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	30	The value ranges from 1 to 30 . If this parameter is not set or the value is not in the specified range, the default value 30 is used.

Parameter	Description	Mandatory	Default Value	Remarks
authenticationMode	Authentication mode for logging in to a storage backend. The following modes are supported: • local: local authentication • ldap: LDAP authentication	No	local	When Huawei enterprise storage is OceanStor V5, the ID of the LDAP domain authentication server must be 0.
metroBackend	Backend name of the HyperMetro peer. The value is a character string. This parameter is mandatory when a PV to be created on the storage side needs to support the SAN HyperMetro feature. In this case, you need to enter the name of the other backend to form a HyperMetro pair with the current backend.	No	-	The names of the two backends in the pair must be entered. After the two backends form a HyperMetro relationship, they cannot form a HyperMetro relationship with other backends.
hyperMetroDomain	HyperMetro domain name. This parameter is mandatory when a PV to be created on the storage side needs to support the SAN HyperMetro feature. In this case, you need to enter the name of the HyperMetro domain to which the PV to be created belongs.	No	-	You can query the HyperMetro domain name on DeviceManager.

Creating a Storage Backend

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "oceanstor-san"
name: "backend-demo"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:8088"
pools:
  - "StoragePool001"
parameters:
  protocol: "iscsi"
portals:
  - "10.10.30.20"
```

```
- "10.10.30.21"  
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

```
NUMBER CONFIGURED NAME STORAGE URLS  
1 false backend-demo oceanstor-san https://192.168.129.157:8088  
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1  
Please enter this backend user name: admin  
Please enter this backend password:
```

```
Backend backend-demo is configured  
NUMBER CONFIGURED NAME STORAGE URLS  
1 true backend-demo oceanstor-san https://192.168.129.157:8088  
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

```
NAMESPACE NAME PROTOCOL STORAGETYPE SN STATUS ONLINE  
Url  
huawei-csi backend-demo iscsi oceanstor-san xxxxxxxxxxxxxxxxxxxx Bound true https://  
192.168.129.157:8088
```

----End

Creating Storage Backends of the HyperMetro Type

NOTE

- Before configuring SAN HyperMetro, you need to configure the HyperMetro relationship between two storage devices, including the remote device and HyperMetro domain. For details, see the product documentation of the corresponding storage model.

Step 1 Prepare a storage backend configuration file, for example, **backend.yaml**.

```
storage: "oceanstor-san"  
name: "backend-active"  
namespace: "huawei-csi"  
urls:  
- "https://192.168.129.155:8088"  
pools:  
- "StoragePool001"  
hyperMetroDomain: "BlockHyperMetroDomain"  
metroBackend: "backend-standby"  
parameters:  
protocol: "iscsi"  
portals:  
- "192.168.129.156"  
- "192.168.129.158"  
maxClientThreads: "30"  
---  
storage: "oceanstor-san"  
name: "backend-standby"  
namespace: "huawei-csi"  
urls:
```

```
- "https://192.168.129.157:8088"
pools:
- "StoragePool001"
hyperMetroDomain: "BlockHyperMetroDomain"
metroBackend: "backend-active"
parameters:
  protocol: "iscsi"
  portals:
    - "192.168.129.157"
    - "192.168.129.159"
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

```
NUMBER CONFIGURED NAME STORAGE URLS
1 false backend-active oceanstor-san https://192.168.129.155:8088
2 false backend-standby oceanstor-san https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
```

```
Please enter this backend user name: user1
```

```
Please enter this backend password:
```

```
Backend backend-standby is configured
```

```
NUMBER CONFIGURED NAME STORAGE URLS
1 true backend-active oceanstor-san https://192.168.129.155:8088
2 true backend-standby oceanstor-san https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

NAMESPACE	NAME	PROTOCOL	STORAGETYPE	SN	STATUS	ONLINE
Url						
huawei-csi	backend-active	iscsi	oceanstor-san	xxxxxxxxxxxxxxxxxxxxxx	Bound	true https://192.168.129.155:8088
huawei-csi	backend-standby	iscsi	oceanstor-san	xxxxxxxxxxxxxxxxxxxxxx	Bound	true https://192.168.129.157:8088

----End

FC

This section describes how to create a storage backend of the FC protocol type.

Configuration Item Description

Table 5-6 backend parameters

Parameter	Description	Mandator y	Default Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed at oceanstor-san .

Parameter	Description	Mandatory	Default Value	Remarks
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.
urls	Management URLs of the storage device. The value format is a list. The value can be a domain name or an IP address + port number.	Yes	-	If the management URL is of the IPv6 type, the URL format is https://[IPv6 address]:Port number.
pools	Storage pools of storage devices. The value format is a list.	Yes	-	Enter the storage pool names.
parameters.protocol	Storage protocol. The value is a character string.	Yes	-	The value is fixed at fc.
parameters.ALUA	ALUA configuration of the storage backend. If the worker node uses the native multipathing software provided by the OS and ALUA is enabled, you need to configure this parameter.	Conditionally mandatory	-	If ALUA is enabled for the host multipathing software, ensure that the backend ALUA configuration is the same as that of the host ALUA configuration. For details, see 7.10 Configuring ALUA .

Parameter	Description	Mandatory	Default Value	Remarks
supportedTopologies	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	30	The value ranges from 1 to 30 . If this parameter is not set or the value is not in the specified range, the default value 30 is used.
authenticationMode	Authentication mode for logging in to a storage backend. The following modes are supported: • local: local authentication • ldap: LDAP authentication	No	local	When Huawei enterprise storage is OceanStor V5, the ID of the LDAP domain authentication server must be 0.
metroBackend	Backend name of the HyperMetro peer. The value is a character string. This parameter is mandatory when a PV to be created on the storage side needs to support the SAN HyperMetro feature. In this case, you need to enter the name of the other backend to form a HyperMetro pair with the current backend.	No	-	The names of the two backends in the pair must be entered. After the two backends form a HyperMetro relationship, they cannot form a HyperMetro relationship with other backends.

Parameter	Description	Mandatory	Default Value	Remarks
hyperMetroDomain	HyperMetro domain name. This parameter is mandatory when a PV to be created on the storage side needs to support the SAN HyperMetro feature. In this case, you need to enter the name of the HyperMetro domain to which the PV to be created belongs.	No	-	You can query the HyperMetro domain name on DeviceManager.

Creating a Storage Backend

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "oceanstor-san"
name: "backend-demo"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:8088"
pools:
  - "StoragePool001"
parameters:
  protocol: "fc"
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

```
NUMBER CONFIGURED NAME STORAGE URLS
1 false backend-demo oceanstor-san https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
Please enter this backend user name: admin
Please enter this backend password:
```

```
Backend backend-demo is configured
NUMBER CONFIGURED NAME STORAGE URLS
1 true backend-demo oceanstor-san https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

NAMESPACE	NAME	PROTOCOL	STORAGETYPE	SN	STATUS	ONLINE
Url	huawei-csi backend-demo	fc	oceanstor-san	xxxxxxxxxxxxxxxxxxxxxx	Bound	true
	192.168.129.157:8088					https://

----End

Creating Storage Backends of the HyperMetro Type

NOTE

- Before configuring SAN HyperMetro, you need to configure the HyperMetro relationship between two storage devices, including the remote device and HyperMetro domain. For details, see the product documentation of the corresponding storage model.

Step 1 Prepare a storage backend configuration file, for example, **backend.yaml**.

```
storage: "oceanstor-san"
name: "backend-active"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.155:8088"
pools:
  - "StoragePool001"
hyperMetroDomain: "BlockHyperMetroDomain"
metroBackend: "backend-standby"
parameters:
  protocol: "fc"
  maxClientThreads: "30"
---
storage: "oceanstor-san"
name: "backend-standby"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:8088"
pools:
  - "StoragePool001"
hyperMetroDomain: "BlockHyperMetroDomain"
metroBackend: "backend-active"
parameters:
  protocol: "fc"
  maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

NUMBER	CONFIGURED	NAME	STORAGE	URLS
1	false	backend-active	oceanstor-san	https://192.168.129.155:8088
2	false	backend-standby	oceanstor-san	https://192.168.129.157:8088

Please enter the backend number to configure (Enter 'exit' to exit):

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
Please enter this backend user name: user1
Please enter this backend password:
```

```
Backend backend-standby is configured
NUMBER CONFIGURED NAME STORAGE URLS
1 true backend-active oceanstor-san https://192.168.129.155:8088
2 true backend-standby oceanstor-san https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

NAMESPACE	NAME	PROTOCOL	STORAGETYPE	SN	STATUS	ONLINE	Url
huawei-csi	backend-active	fc	oceanstor-san	xxxxxxxxxxxxxxxxxxxx	Bound	true	https://192.168.129.155:8088
huawei-csi	backend-standby	fc	oceanstor-san	xxxxxxxxxxxxxxxxxxxx	Bound	true	https://192.168.129.157:8088

----End

NVMe over RoCE

This section describes how to create a storage backend of the NVMe over RoCE protocol type.

Configuration Item Description

Table 5-7 backend parameters

Parameter	Description	Mandatory	Default Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed to oceanstor-san .
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.
urls	Management URLs of the storage device. The value format is a list. The value can be a domain name or an IP address + port number.	Yes	-	If the management URL is of the IPv6 type, the URL format is https://[IPv6 address]:Port number.
pools	Storage pools of storage devices. The value format is a list.	Yes	-	Enter the storage pool names.

Parameter	Description	Mandatory	Default Value	Remarks
parameters.protocol	Storage protocol. The value is a character string.	Yes	-	- The value is fixed to roce-nvme or roce (roce is deprecated). - Ensure that the nvme-cli tool has been installed on the compute node to be connected. The supported nvme-cli tool version is 1.9 and later.
parameters.ports	Service access port. Nodes will use this port to read and write storage resources. The value format is a list.	Yes	-	- IPv6 is supported. - Multiple ports can be configured.
supportedTopologies	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	30	The value ranges from 1 to 30 . If this parameter is not set or the value is not in the specified range, the default value 30 is used.
authenticationMode	Authentication mode for logging in to a storage backend. The following modes are supported: local: local authentication ldap: LDAP authentication	No	local	When Huawei enterprise storage is OceanStor V5, the ID of the LDAP domain authentication server must be 0.

Creating a Storage Backend

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "oceanstor-san"
name: "backend-demo"
namespace: "huawei-csi"
urls:
```

```
- "https://192.168.129.157:8088"
pools:
- "StoragePool001"
parameters:
  protocol: "roce-nvme"
portals:
- "10.10.30.20"
- "10.10.30.21"
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

```
NUMBER CONFIGURED NAME STORAGE URLS
1 false backend-demo oceanstor-san https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
Please enter this backend user name: admin
Please enter this backend password:
```

```
Backend backend-demo is configured
NUMBER CONFIGURED NAME STORAGE URLS
1 true backend-demo oceanstor-san https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

```
NAMESPACE NAME PROTOCOL STORAGETYPE SN STATUS ONLINE
Url
huawei-csi backend-demo roce-nvme oceanstor-san xxxxxxxxxxxxxxxxxxxx Bound true https://
192.168.129.157:8088
```

----End

NVMe over TCP

This section describes how to create a storage backend of the NVMe over TCP protocol type.

Configuration Item Description

Table 5-8 backend parameters

Parameter	Description	Mandatory	Default Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed at oceanstor-san .

Parameter	Description	Mandatory	Default Value	Remarks
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.
urls	Management URLs of the storage device. The value format is a list. The value can be a domain name or an IP address + port number.	Yes	-	If the management URL is of the IPv6 type, the URL format is https://[IPv6 address]:Port number.
pools	Storage pools of storage devices. The value format is a list.	Yes	-	Enter the storage pool names.
parameters.protocol	Storage protocol. The value is a character string.	Yes	-	- The value is fixed at tcp-nvme. - Ensure that the nvme-cli tool has been installed on the compute node to be connected. The supported nvme-cli tool version is 2.0 and later.
parameters.ports	Service access port. Nodes will use this port to read and write storage resources. The value format is a list.	Yes	-	- IPv6 is supported. - Multiple ports can be configured.
supportedTopology	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .

Parameter	Description	Mandatory	Default Value	Remarks
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	30	The value ranges from 1 to 30 . If this parameter is not set or the value is not in the specified range, the default value 30 is used.
authenticationMode	Authentication mode for logging in to a storage backend. The following modes are supported: • local: local authentication • ldap: LDAP authentication	No	local	-

Creating a Storage Backend

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "oceanstor-san"
name: "backend-demo"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:8088"
pools:
  - "StoragePool001"
parameters:
  protocol: "tcp-nvme"
  portals:
    - "10.10.30.20"
    - "10.10.30.21"
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

```
NUMBER CONFIGURED NAME STORAGE URLS
1 false backend-demo oceanstor-san https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
Please enter this backend user name: admin
Please enter this backend password:
```

```
Backend backend-demo is configured
NUMBER CONFIGURED NAME STORAGE URLS
1 true backend-demo oceanstor-san https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

NAMESPACE	NAME	PROTOCOL	STORAGETYPE	SN	STATUS	ONLINE	Url
huawei-csi	backend-demo	tcp-nvme	oceanstor-san	xxxxxxxxxxxxxxxxxxxxxx	Bound	true	https://192.168.129.157:8088

----End

FC-NVMe

This section describes how to create a storage backend of the FC-NVMe protocol type.

Configuration Item Description

Table 5-9 backend parameters

Parameter	Description	Mandatory	Default Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed to oceanstor-san .
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.
urls	Management URLs of the storage device. The value format is a list. The value can be a domain name or an IP address + port number.	Yes	-	If the management URL is of the IPv6 type, the URL format is https://[IPv6 address]:Port number.
pools	Storage pools of storage devices. The value format is a list.	Yes	-	Enter the storage pool names.

Parameter	Description	Mandatory	Default Value	Remarks
parameters.protocol	Storage protocol. The value is a character string.	Yes	-	- The value is fixed to fc-nvme. - Ensure that the nvme-cli tool has been installed on the compute node to be connected. The supported nvme-cli tool version is 1.9 and later.
supportedTopologies	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	30	The value ranges from 1 to 30 . If this parameter is not set or the value is not in the specified range, the default value 30 is used.
authenticationMode	Authentication mode for logging in to a storage backend. The following modes are supported: local: local authentication ldap: LDAP authentication	No	local	When Huawei enterprise storage is OceanStor V5, the ID of the LDAP domain authentication server must be 0.

Creating a Storage Backend

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "oceanstor-san"
name: "backend-demo"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:8088"
pools:
  - "StoragePool001"
parameters:
  protocol: "fc-nvme"
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

```
NUMBER CONFIGURED NAME STORAGE URLS
1 false backend-demo oceanstor-san https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
Please enter this backend user name: admin
Please enter this backend password:

Backend backend-demo is configured
NUMBER CONFIGURED NAME STORAGE URLS
1 true backend-demo oceanstor-san https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

```
NAMESPACE NAME PROTOCOL STORAGETYPE SN STATUS ONLINE
Url
huawei-csi backend-demo fc-nvme oceanstor-san xxxxxxxxxxxxxxxxxxxx Bound true https://
192.168.129.157:8088
```

----End

5.1.1.2 Flash Storage (OceanStor A600/A800)

5.1.1.2.1 Local File System

NFS

This section describes how to create a storage backend of the NFS protocol type.

Configuration Item Description

Table 5-10 backend parameters

Parameter	Description	Mandatory	Default Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed at oceanstor-a-series-nas .

Parameter	Description	Mandatory	Default Value	Remarks
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.
urls	Management URLs of the storage device. The value format is a list.	Yes	-	The value can be a domain name or an IP address + port number.
pools	Storage pools of storage devices. The value format is a list.	Yes	-	Enter the storage pool names.
parameters.protocol	Storage protocol. The value is a character string.	Yes	-	• The value is fixed at nfs. • Ensure that an NFS client tool has been installed on the connected compute node.
parameters.ports	Service access port. Nodes will use this port to read and write storage resources. The value format is a list.	Yes	-	Only one port can be configured.

Parameter	Description	Mandatory	Default Value	Remarks
supportedTopologies	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	30	The value ranges from 1 to 30 . If this parameter is not set or the value is not in the specified range, the default value 30 is used.
authenticationMode	Authentication mode for logging in to a storage backend. The following modes are supported: • local: local authentication • ldap: LDAP authentication	No	local	-

Creating a Storage Backend

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "oceanstor-a-series-nas"
name: "backend-demo"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:8088"
pools:
  - "StoragePool001"
parameters:
  protocol: "nfs"
portals:
  - "10.10.30.20"
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

```
NUMBER CONFIGURED NAME STORAGE URLS
1 false backend-demo oceanstor-a-series-nas https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
Please enter this backend user name: admin
Please enter this backend password:

Backend backend-demo is configured
NUMBER CONFIGURED NAME STORAGE URLS
1 true backend-demo oceanstor-a-series-nas https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

```
NAMESPACE NAME PROTOCOL STORAGETYPE SN STATUS ONLINE
Url
huawei-csi backend-demo nfs oceanstor-a-series-nas xxxxxxxxxxxxxxxxxxxx Bound true
https://192.168.129.157:8088
```

----End

DataTurbo

This section describes how to create a storage backend of the DataTurbo protocol type.

Configuration Item Description

Table 5-11 backend parameters

Parameter	Description	Mandator y	Defau lt Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed to oceanstor-a-series-nas .
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.

Parameter	Description	Mandatory	Default Value	Remarks
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.
urls	Management URLs of the storage device. The value format is a list.	Yes	-	The value can be a domain name or an IP address + port number.
pools	Storage pools of storage devices. The value format is a list.	Yes	-	Enter the storage pool names.
parameters.protocol	Storage protocol. The value is a character string.	Yes	-	• The value is fixed to dtfs. • Ensure that a DataTurbo client has been installed on the connected compute node and the connectivity of the storage logical port has been configured.
supportedTopology	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	30	The value ranges from 1 to 30 . If this parameter is not set or the value is not in the specified range, the default value 30 is used.

Parameter	Description	Mandatory	Default Value	Remarks
authenticationMode	Authentication mode for logging in to a storage backend. The following modes are supported: • local: local authentication • ldap: LDAP authentication	No	local	-

Creating a Storage Backend

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "oceanstor-a-series-nas"
name: "backend-demo"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:8088"
pools:
  - "StoragePool001"
parameters:
  protocol: "dtfs"
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

```
NUMBER CONFIGURED NAME STORAGE URLS
1 false backend-demo oceanstor-a-series-nas https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
Please enter this backend user name: admin
Please enter this backend password:
```

```
Backend backend-demo is configured
NUMBER CONFIGURED NAME STORAGE URLS
1 true backend-demo oceanstor-a-series-nas https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

```
NAMESPACE NAME PROTOCOL STORAGETYPE SN STATUS ONLINE
Url
```

```
huawei-csi backend-demo dtfs oceanstor-a-series-nas xxxxxxxxxxxxxxxxxxxx Bound true https://  
192.168.129.157:8088
```

----End

5.1.1.2.2 Dtree

NFS

This section describes how to create a storage backend of the NFS protocol type.

Configuration Item Description

Table 5-12 backend parameters

Parameter	Description	Mandato ry	Defaul t Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed to oceanstor-a-series-dtree .
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.
urls	Management URLs of the storage device. The value format is a list.	Yes	-	The value can be a domain name or an IP address + port number.
parameters. protocol	Storage protocol. The value is a character string.	Yes	-	- The value is fixed to nfs. - Ensure that an NFS client tool has been installed on the connected compute node.

Parameter	Description	Mandatory	Default Value	Remarks
parameters.portals	Service access port. Nodes will use this port to read and write storage resources. The value format is a list.	Yes	-	Only one port can be configured.
parameters.parentname	Name of a file system on the current storage device. Dtree is created in the file system.	No	-	- Query the name on the File Systems page of DeviceManager. - You can configure the parameter in StorageClass.
supportedTopologies	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	30	The value ranges from 1 to 30 . If this parameter is not set or the value is not in the specified range, the default value 30 is used.
authenticationMode	Authentication mode for logging in to a storage backend. The following modes are supported: local: local authentication ldap: LDAP authentication	No	local	-

Creating a Storage Backend

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "oceanstor-a-series-dtree"  
name: "backend-demo"
```

```
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:8088"
parameters:
  protocol: "nfs"
  parentname: "fs-demo"
  portals:
    - "10.10.30.20"
  maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output:

```
NUMBER CONFIGURED NAME STORAGE URLS
1 false backend-demo oceanstor-a-series-dtree https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
```

```
Please enter this backend user name: admin
```

```
Please enter this backend password:
```

```
Backend backend-demo is configured
```

```
NUMBER CONFIGURED NAME STORAGE URLS
1 true backend-demo oceanstor-a-series-dtree https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

```
NAMESPACE NAME PROTOCOL STORAGETYPE SN STATUS ONLINE
Url
huawei-csi backend-demo nfs oceanstor-a-series-dtree xxxxxxxxxxxxxxxxxxxx Bound true
https://192.168.129.157:8088
```

----End

DataTurbo

This section describes how to create a storage backend of the DataTurbo protocol type.

Configuration Item Description

Table 5-13 backend parameters

Parameter	Description	Mandator y	Default Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed to oceanstor-a-series-dtree .

Parameter	Description	Mandatory	Default Value	Remarks
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.
urls	Management URLs of the storage device. The value format is a list.	Yes	-	The value can be a domain name or an IP address + port number.
parameters.protocol	Storage protocol. The value is a character string.	Yes	-	• The value is fixed to dtfs. • Ensure that a DataTurbo client has been installed on the connected compute node and the connectivity of the storage logical port has been configured.
parameters.parentname	Name of a file system on the current storage device. Dtree is created in the file system.	No	-	• Query the name on the File Systems page of DeviceManager. • You can configure the parameter in StorageClass.

Parameter	Description	Mandatory	Default Value	Remarks
supportedTopologies	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	30	The value ranges from 1 to 30 . If this parameter is not set or the value is not in the specified range, the default value 30 is used.
authenticationMode	Authentication mode for logging in to a storage backend. The following modes are supported: • local: local authentication • ldap: LDAP authentication	No	local	-

Creating a Storage Backend

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "oceanstor-a-series-dtree"
name: "backend-demo"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:8088"
parameters:
  parentname: "fs-demo"
  protocol: "dtfs"
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output:

```
NUMBER CONFIGURED NAME STORAGE URLS
1 false backend-demo oceanstor-a-series-dtree https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
Please enter this backend user name: admin
Please enter this backend password:

Backend backend-demo is configured
NUMBER CONFIGURED NAME STORAGE URLS
1 true backend-demo oceanstor-a-series-dtree https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

NAMESPACE	NAME	PROTOCOL	STORAGETYPE	SN	STATUS	ONLINE
Url						
huawei-csi	backend-demo	dtfs	oceanstor-a-series-dtree	xxxxxxxxxxxxxxxxxxxxxx	Bound	true
https://192.168.129.157:8088						

----End

5.1.1.3 iMaster DME (Interconnected with an OceanStor A800 Cluster)

5.1.1.3.1 Global File System

NFS

This section describes how to create a storage backend of the NFS protocol type.

Configuration Item Description

Table 5-14 Backend parameters

Parameter	Description	Mandatory	Default Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed to oceanstor-a-series-nas .
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.

Parameter	Description	Mandatory	Default Value	Remarks
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.
urls	Management URLs of the storage device. The value format is a list.	Yes	-	The value can be a domain name or an IP address + port number.
storageDevicesSN	Serial number of the storage device.	Yes		On the iMaster DME, choose Infrastructure > Storage Devices > ESN to obtain the serial number of the storage device.
pools	Storage pools of storage devices. The value format is a list.	Yes	-	Enter the storage pool names.
parameters.protocol	Storage protocol. The value is a character string.	Yes	-	- The value is fixed at nfs. - Ensure that an NFS client tool has been installed on the connected compute node.
parameters.ports	Service access port. Nodes will use this port to read and write storage resources. The value format is a list.	Yes	-	- You can enter a domain name address. - Only one port can be configured.
supportedTopologies	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	5	The value ranges from 1 to 30 . If this parameter is not set or the value is not in the specified range, the default value 5 is used.

Creating a Storage Backend

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "oceanstor-a-series-nas"
name: "backend-demo"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:26335"
storageDeviceSN: 21000000000000000000
pools:
  - "StoragePool001"
parameters:
  protocol: "nfs"
  portals:
    - "10.10.30.20"
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output:

NUMBER	CONFIGURED	NAME	STORAGE	URLS
1	false	backend-demo	oceanstor-a-series-nas	https://192.168.129.157:26335

Please enter the backend number to configure (Enter 'exit' to exit):

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
Please enter this backend user name: admin
Please enter this backend password:
```

```
Backend backend-demo is configured
NUMBER CONFIGURED NAME STORAGE URLS
1 true backend-demo oceanstor-a-series-nas https://192.168.129.157:26335
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

NAMESPACE	NAME	PROTOCOL	STORAGETYPE	SN	STATUS	ONLINE
Url						
huawei-csi	backend-demo	nfs	oceanstor-a-series-nas	xxxxxxxxxxxxxxxxxxxxxx	Bound	true

https://192.168.129.157:26335

----End

DataTurbo

This section describes how to create a storage backend of the DataTurbo protocol type.

Configuration Item Description

Table 5-15 Backend parameters

Parameter	Description	Mandatory	Default Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed to oceanstor-a-series-nas .
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.
urls	Management URLs of the storage device. The value format is a list.	Yes	-	The value can be a domain name or an IP address + port number.
storageDeviceSN	Serial number of the storage device.	Yes		On the iMaster DME, choose Infrastructure > Storage Devices > ESN to obtain the serial number of the storage device.
pools	Storage pools of storage devices. The value format is a list.	Yes	-	Enter the storage pool names.
parameters.protocol	Storage protocol. The value is a character string.	Yes	-	- The value is fixed to dtfs. - Ensure that a DataTurbo client has been installed on the connected compute node and the connectivity of the storage logical port has been configured.

Parameter	Description	Mandatory	Default Value	Remarks
supportedTopologies	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	30	The value ranges from 1 to 30 . If this parameter is not set or the value is not in the specified range, the default value 30 is used.

Creating a Storage Backend

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "oceanstor-a-series-nas"
name: "backend-demo"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:26335"
storageDeviceSN: 21000000000000000000
pools:
  - "StoragePool001"
parameters:
  protocol: "dtfs"
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output:

```
NUMBER CONFIGURED NAME STORAGE URLS
1 false backend-demo oceanstor-a-series-nas https://192.168.129.157:26335
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
Please enter this backend user name: admin
Please enter this backend password:
```

```
Backend backend-demo is configured
NUMBER CONFIGURED NAME STORAGE URLS
1 true backend-demo oceanstor-a-series-nas https://192.168.129.157:26335
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

NAMESPACE	NAME	PROTOCOL	STORAGETYPE	SN	STATUS	ONLINE	Url
huawei-csi	backend-demo	dtfs	oceanstor-a-series-nas	xxxxxxxxxxxxxxxxxxxxxx	Bound	true	https://192.168.129.157:26335

----End

5.1.1.3.2 Local File System

NOTE

In the current version, only local file systems with KV cache resources can be provisioned. Local file systems cannot be directly provisioned.

NFS

This section describes how to create a storage backend of the NFS protocol type.

Configuration Item Description

Table 5-16 Backend parameters

Parameter	Description	Mandatory	Default Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed to oceanstor-a-series-nas .
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.
urls	Management URLs of the storage device. The value format is a list.	Yes	-	The value can be a domain name or an IP address + port number.
storageDeviceSN	Serial number of the storage device.	Yes		On the iMaster DME, choose Infrastructure > Storage Devices > ESN to obtain the serial number of the storage device.

Parameter	Description	Mandatory	Default Value	Remarks
zoneSN	Serial number of the storage zone.	Conditionally mandatory		This parameter is mandatory if a local file system is used to create KV cache. On iMaster DME, choose Infrastructure > Storage Devices > Hardware > Storage Zones . On the menu bar, select ESN to obtain the serial number of the storage zone.
pools	Storage pools of storage devices. The value format is a list.	Yes	-	Enter the storage pool names.
parameters.protocol	Storage protocol. The value is a character string.	Yes	-	- The value is fixed at nfs. - Ensure that an NFS client tool has been installed on the connected compute node.
parameters.ports	Service access port. Nodes will use this port to read and write storage resources. The value format is a list.	Yes	-	- You can enter a domain name address. - Only one port can be configured.
supportedTopologies	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	5	The value ranges from 1 to 30 . If this parameter is not set or the value is not in the specified range, the default value 5 is used.

Creating a Storage Backend

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "oceanstor-a-series-nas"
name: "backend-demo"
```



```
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:26335"
storageDeviceSN: 21000000000000000000
zoneSN: 22000000000000000000
pools:
  - "StoragePool001"
parameters:
  protocol: "nfs"
  portals:
    - "10.10.30.20"
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

NUMBER	CONFIGURED	NAME	STORAGE	URLS
1	false	backend-demo	oceanstor-a-series-nas	https://192.168.129.157:26335

Please enter the backend number to configure (Enter 'exit' to exit):

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
Please enter this backend user name: admin
Please enter this backend password:
```

```
Backend backend-demo is configured
NUMBER CONFIGURED NAME STORAGE URLS
1 true backend-demo oceanstor-a-series-nas https://192.168.129.157:26335
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

NAMESPACE	NAME	PROTOCOL	STORAGETYPE	SN	STATUS	ONLINE
Url	huawei-csi	backend-demo	nfs	oceanstor-a-series-nas	xxxxxxxxxxxxxxxxxxxx	Bound true
https://192.168.129.157:26335						

----End

5.1.1.3.3 Dtree

NFS

This section describes how to create a storage backend of the NFS protocol type.

Configuration Item Description

Table 5-17 Backend parameters

Parameter	Description	Mandatory	Default Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed to oceanstor-a-series-dtree .
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.
urls	Management URLs of the storage device. The value format is a list.	Yes	-	The value can be a domain name or an IP address + port number.
storageDeviceSN	Serial number of the storage device.	Yes	-	On the iMaster DME, choose Infrastructure > Storage Devices > ESN to obtain the serial number of the storage device.
parameters.protocol	Storage protocol. The value is a character string.	Yes	-	- The value is fixed to nfs. - Ensure that an NFS client tool has been installed on the connected compute node.
parameters.portals	Service access port. Nodes will use this port to read and write storage resources. The value format is a list.	Yes	-	Only one port can be configured.

Parameter	Description	Mandatory	Default Value	Remarks
parameters.parentname	Name of a file system on the current storage device. Dtree is created in the file system.	No	-	- Obtain the value on the iMaster DME. - You can configure the parameter in StorageClass.
supportedTopology	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	30	The value ranges from 1 to 30 . If this parameter is not set or the value is not in the specified range, the default value 30 is used.

Creating a Storage Backend

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "oceanstor-a-series-dtree"
name: "backend-demo"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:8088"
storageDeviceSN: 21000000000000000000
parameters:
  protocol: "nfs"
  parentname: "fs-demo"
  portals:
    - "10.10.30.20"
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

```
NUMBER CONFIGURED NAME STORAGE URLS
1 false backend-demo oceanstor-a-series-dtree https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
Please enter this backend user name: admin
Please enter this backend password:
Backend backend-demo is configured
```

NUMBER	CONFIGURED	NAME	STORAGE	URLS
1	true	backend-demo	oceanstor-a-series-dtree	https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):				

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

NAMESPACE	NAME	PROTOCOL	STORAGETYPE	SN	STATUS	ONLINE
Url	huawei-csi	backend-demo	nfs	oceanstor-a-series-dtree	xxxxxxxxxxxxxxxxxxxxxx	Bound true
https://192.168.129.157:8088						

----End

DataTurbo

This section describes how to create a storage backend of the DataTurbo protocol type.

Configuration Item Description

Table 5-18 Backend parameters

Parameter	Description	Mandatory	Default Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed to oceanstor-a-series-dtree .
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.
urls	Management URLs of the storage device. The value format is a list.	Yes	-	The value can be a domain name or an IP address + port number.
storageDeviceSN	Serial number of the storage device.	Yes	-	On the iMaster DME, choose Infrastructure > Storage Devices > ESN to obtain the serial number of the storage device.

Parameter	Description	Mandatory	Default Value	Remarks
parameters.protocol	Storage protocol. The value is a character string.	Yes	-	- The value is fixed to dtfs. - Ensure that a DataTurbo client has been installed on the connected compute node and the connectivity of the storage logical port has been configured.
parameters.parentname	Name of a file system on the current storage device. Dtree is created in the file system.	No	-	- Obtain the value on the iMaster DME. - You can configure the parameter in StorageClass.
supportedTopologies	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	30	The value ranges from 1 to 30 . If this parameter is not set or the value is not in the specified range, the default value 30 is used.

Creating a Storage Backend

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "oceanstor-a-series-dtree"
name: "backend-demo"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:8088"
storageDeviceSN: 21000000000000000000
parameters:
  parentname: "fs-demo"
  protocol: "dtfs"
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

```
NUMBER CONFIGURED NAME STORAGE URLS
1 false backend-demo oceanstor-a-series-dtree https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
Please enter this backend user name: admin
Please enter this backend password:

Backend backend-demo is configured
NUMBER CONFIGURED NAME STORAGE URLS
1 true backend-demo oceanstor-a-series-dtree https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

```
NAMESPACE NAME PROTOCOL STORAGETYPE SN STATUS ONLINE
Url
huawei-csi backend-demo dtfs oceanstor-a-series-dtree xxxxxxxxxxxxxxxxxxxx Bound true
https://192.168.129.157:8088
```

----End

5.1.1.4 Mass Storage (OceanStor Pacific/FusionStorage Block Series)

5.1.1.4.1 File System

NFS

This section describes how to create a storage backend of the NFS protocol type.

Configuration Item Description

Table 5-19 backend parameters

Parameter	Description	Mandatory	Default Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed at fusionstorage-nas .
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.

Parameter	Description	Mandatory	Default Value	Remarks
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.
accountName	Account name on the storage side.	No	-	This parameter is mandatory when NAS resources need to be provisioned under a specified account.
urls	Management URLs of the storage device. The value format is a list.	Yes	-	The value can be a domain name or an IP address + port number.
pools	Storage pools of storage devices. The value format is a list.	Yes	-	Enter the storage pool names.
parameters.protocol	Storage protocol. The value is a character string.	Yes	-	• The value is fixed at nfs. • Ensure that an NFS client tool has been installed on the connected compute node.
parameters.ports	Service access port. Nodes will use this port to read and write storage resources. The value format is a list.	Yes	-	• If an account is used to connect to a backend, ports must be set to the logical port information of the account. • You can enter a domain name address. • Only one port can be configured.
supportedTopologies	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .

Parameter	Description	Mandatory	Default Value	Remarks
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	30	The value ranges from 1 to 30 . If this parameter is not set or the value is not in the specified range, the default value 30 is used.

Creating a Storage Backend

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "fusionstorage-nas"
name: "backend-demo"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:8088"
pools:
  - "StoragePool001"
parameters:
  protocol: "nfs"
portals:
  - "10.10.30.20"
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

```
NUMBER CONFIGURED NAME STORAGE URLS
1 false backend-demo fusionstorage-nas https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
Please enter this backend user name: admin
Please enter this backend password:

Backend backend-demo is configured
NUMBER CONFIGURED NAME STORAGE URLS
1 true backend-demo fusionstorage-nas https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

```
NAMESPACE NAME PROTOCOL STORAGETYPE SN STATUS ONLINE
Url
huawei-csi backend-demo nfs fusionstorage-nas Bound true https://
192.168.129.157:8088
```

----End

DPC

This section describes how to create a storage backend of the DPC protocol type.

Configuration Item Description

Table 5-20 backend parameters

Parameter	Description	Mandatory	Default Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed to fusionstorage-nas .
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.
account Name	Account name on the storage side.	No	-	This parameter is mandatory when NAS resources need to be provisioned under a specified account.
urls	Management URLs of the storage device. The value format is a list.	Yes	-	The value can be a domain name or an IP address + port number.
pools	Storage pools of storage devices. The value format is a list.	Yes	-	Enter the storage pool names.
parameters.protocol	Storage protocol. The value is a character string.	Yes	-	- The value is fixed to dpc. - Ensure that DPC has been installed on the connected compute node and the node has been added as a DPC compute node on the storage device to be connected.

Parameter	Description	Mandatory	Default Value	Remarks
supportedTopologies	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	30	The value ranges from 1 to 30 . If this parameter is not set or the value is not in the specified range, the default value 30 is used.

Creating a Storage Backend

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "fusionstorage-nas"
name: "backend-demo"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:8088"
pools:
  - "StoragePool001"
parameters:
  protocol: "dpc"
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

```
NUMBER CONFIGURED NAME STORAGE URLS
1 false backend-demo fusionstorage-nas https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
Please enter this backend user name: admin
Please enter this backend password:
```

```
Backend backend-demo is configured
NUMBER CONFIGURED NAME STORAGE URLS
1 true backend-demo fusionstorage-nas https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

NAMESPACE	NAME	PROTOCOL	STORAGETYPE	SN	STATUS	ONLINE
Url	huawei-csi backend-demo	dpc	fusionstorage-nas		Bound true	https://192.168.129.157:8088

----End

5.1.1.4.2 Dtree

NFS

This section describes how to create a storage backend of the NFS protocol type.

Configuration Item Description

Table 5-21 backend parameters

Parameter	Description	Mandatory	Default Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed to fusionstorage-dtree .
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.
account Name	Account name on the storage side.	No	-	This parameter is mandatory when NAS resources need to be provisioned under a specified account.
urls	Management URLs of the storage device. The value format is a list.	Yes	-	The value can be a domain name or an IP address + port number.

Parameter	Description	Mandatory	Default Value	Remarks
parameters.protocol	Storage protocol. The value is a character string.	Yes	-	- The value is fixed to nfs. - Ensure that an NFS client tool has been installed on the connected compute node.
parameters.ports	Service access port. Nodes will use this port to read and write storage resources. The value format is a list.	Yes	-	- If an account is used to connect to a backend, ports must be set to the logical port information of the account. - You can enter a domain name address.
parameters.parentname	Name of a file system on the current storage device. A dtree is created in the file system.	No	-	- Query the name on the File Systems page of DeviceManager. - You can configure the parameter in StorageClass.
supportedTopologies	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	30	The value ranges from 1 to 30 . If this parameter is not set or the value is not in the specified range, the default value 30 is used.

Creating a Storage Backend

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "fusionstorage-dtree"
name: "backend-demo"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:8088"
parameters:
  protocol: "nfs"
```

```
parentname: "parent-filesystem"
portals:
  - "10.10.30.20"
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

```
NUMBER CONFIGURED NAME STORAGE URLS
1 false backend-demo fusionstorage-dtree https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
Please enter this backend user name: admin
Please enter this backend password:
```

```
Backend backend-demo is configured
NUMBER CONFIGURED NAME STORAGE URLS
1 true backend-demo fusionstorage-dtree https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

NAMESPACE	NAME	PROTOCOL	STORAGETYPE	SN	STATUS	ONLINE
Url						
huawei-csi	backend-demo	nfs	fusionstorage-dtree		Bound	true https://192.168.129.157:8088

----End

DPC

This section describes how to create a storage backend of the DPC protocol type.

Configuration Item Description

Table 5-22 backend parameters

Parameter	Description	Mandatory	Default Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed to fusionstorage-dtree .

Parameter	Description	Mandatory	Default Value	Remarks
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.
accountName	Account name on the storage side.	No	-	This parameter is mandatory when NAS resources need to be provisioned under a specified account.
urls	Management URLs of the storage device. The value format is a list.	Yes	-	The value can be a domain name or an IP address + port number.
parameters.protocol	Storage protocol. The value is a character string. dpc	Yes	-	- The value is fixed to dpc. - Ensure that DPC has been installed on the connected compute node and the node has been added as a DPC compute node on the storage device to be connected.
parameters.parentname	Name of a file system on the current storage device. A dtree is created in the file system.	No	-	- Query the name on the File Systems page of DeviceManager. - You can configure the parameter in StorageClass.
supportedTopologies	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .

Parameter	Description	Mandatory	Default Value	Remarks
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	30	The value ranges from 1 to 30 . If this parameter is not set or the value is not in the specified range, the default value 30 is used.

Creating a Storage Backend

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "fusionstorage-dtree"
name: "backend-demo"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:8088"
parameters:
  protocol: "dpc"
  parentname: "parent-filesystem"
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

```
NUMBER CONFIGURED NAME STORAGE URLS
1 false backend-demo fusionstorage-dtree https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
Please enter this backend user name: admin
Please enter this backend password:
```

```
Backend backend-demo is configured
NUMBER CONFIGURED NAME STORAGE URLS
1 true backend-demo fusionstorage-dtree https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

```
NAMESPACE NAME PROTOCOL STORAGETYPE SN STATUS ONLINE
Url
huawei-csi backend-demo dpc fusionstorage-dtree Bound true https://
192.168.129.157:8088
```

----End

5.1.1.4.3 Block Service

iSCSI

This section describes how to create a storage backend of the iSCSI protocol type.

Configuration Item Description

Table 5-23 backend parameters

Parameter	Description	Mandatory	Default Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed to fusionstorage-san .
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.
urls	Management URLs of the storage device. The value format is a list.	Yes	-	The value can be a domain name or an IP address + port number.
pools	Storage pools of storage devices. The value format is a list.	Yes	-	Enter the storage pool names.
parameters.protocol	Storage protocol. The value is a character string.	Yes	-	- The value is fixed to iscsi. - Ensure that an iSCSI client has been installed on the connected compute node.

Parameter	Description	Mandatory	Default Value	Remarks
parameters.ports	Service access port. Nodes will use this port to read and write storage resources. The value format is a list.	Conditionally mandatory	-	- Multiple ports can be configured. - Either iscsiLinks or ports must be set.
parameters.iscsiLinks	Number of dynamic access links. This parameter is supported only in OceanStor Pacific 8.1.5 and later versions.	Conditionally mandatory		Either iscsiLinks or ports must be set. NOTE - The iscsiLinks parameter takes effect only when the ports parameter is left empty. - The number of valid links is the maximum value among the following three values: - Value of iscsiLinks - Minimum number of links dynamically allocated by the storage - Number of established links on the host
supportedTopologies	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	30	The value ranges from 1 to 30 . If this parameter is not set or the value is not in the specified range, the default value 30 is used.

Creating a Storage Backend

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "fusionstorage-san"
name: "backend-demo"
```

```
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:8088"
pools:
  - "StoragePool001"
parameters:
  protocol: "iscsi"
  portals:
    - "10.10.30.20"
    - "10.10.30.21"
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

```
NUMBER CONFIGURED NAME STORAGE URLS
1 false backend-demo fusionstorage-san https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
Please enter this backend user name: admin
Please enter this backend password:
```

```
Backend backend-demo is configured
NUMBER CONFIGURED NAME STORAGE URLS
1 true backend-demo fusionstorage-san https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

```
NAMESPACE NAME PROTOCOL STORAGETYPE SN STATUS ONLINE Url
huawei-csi backend-demo iscsi fusionstorage-san Bound true https://
192.168.129.157:8088
```

----End

SCSI

This section describes how to create a storage backend of the SCSI protocol type.

Configuration Item Description

Table 5-24 backend parameters

Parameter	Description	Mandatory	Default Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed to fusionstorage-san .

Parameter	Description	Mandatory	Default Value	Remarks
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.
urls	Management URLs of the storage device. The value format is a list.	Yes	-	The value can be a domain name or an IP address + port number.
pools	Storage pools of storage devices. The value format is a list.	Yes	-	Enter the storage pool names.
parameters.protocol	Storage protocol. The value is a character string.	Yes	-	- The value is fixed to scsi. - Ensure that a distributed storage VBS client has been installed on the connected compute node.
parameters.ports	Service access port. Nodes will use this port to read and write storage resources. The value format is a list.	Yes	-	If the protocol is scsi , the port is in dictionary format where the key indicates the name of the host where the VBS client is deployed and the value indicates the IP address of the host where the VBS client is deployed.
supportedTopology	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .

Parameter	Description	Mandatory	Default Value	Remarks
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	30	The value ranges from 1 to 30 . If this parameter is not set or the value is not in the specified range, the default value 30 is used.

Creating a Storage Backend

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "fusionstorage-san"
name: "backend-demo"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:8088"
pools:
  - "StoragePool001"
parameters:
  protocol: "scsi"
  portals:
    - {"hostname01": "192.168.125.21", "hostname02": "192.168.125.22"}
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

```
NUMBER CONFIGURED NAME STORAGE URLS
1 false backend-demo fusionstorage-san https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
Please enter this backend user name: admin
Please enter this backend password:

Backend backend-demo is configured
NUMBER CONFIGURED NAME STORAGE URLS
1 true backend-demo fusionstorage-san https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

```
NAMESPACE NAME PROTOCOL STORAGETYPE SN STATUS ONLINE Url
huawei-csi backend-demo scsi fusionstorage-san Bound true https://192.168.129.157:8088
```

----End

5.1.1.5 Mass Storage (OceanDisk Series)

5.1.1.5.1 File System

NFS

This section describes how to create a storage backend of the NFS protocol type.

Configuration Item Description

Table 5-25 backend parameters

Parameter	Description	Mandatory	Default Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed to oceanstor-nas .
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.
urls	Management URLs of the storage device. The value format is a list. The value can be a domain name or an IP address + port number.	Yes	-	If resources need to be provisioned to a specified vStore, set this parameter to the logical management port URL of the specified vStore.
pools	Storage pools of storage devices. The value format is a list.	Yes	-	Enter the storage pool names.
parameters.protocol	Storage protocol. The value is a character string.	Yes	-	The value is fixed to nfs . Ensure that an NFS client tool has been installed on the connected compute node.

Parameter	Description	Mandatory	Default Value	Remarks
parameters.ports	Service access port. Nodes will use this port to read and write storage resources. The value format is a list.	Yes	-	- If a vStore is used to connect to a backend, ports must be set to the logical port information of the vStore. - You can enter a domain name address. - Only one port can be configured.
supportedTopologies	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	30	The value ranges from 1 to 30 . If this parameter is not set or the value is not in the specified range, the default value 30 is used.

Creating Storage Backends of the Local Type

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "oceanstor-nas"
name: "backend-demo"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:8088"
pools:
  - "StoragePool001"
parameters:
  protocol: "nfs"
  portals:
    - "10.10.30.20"
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

```
NUMBER CONFIGURED NAME STORAGE URLS
1 false backend-demo oceanstor-nas https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
Please enter this backend user name: admin
Please enter this backend password:

Backend backend-demo is configured
NUMBER CONFIGURED NAME STORAGE URLS
1 true backend-demo oceanstor-nas https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

NAMESPACE	NAME	PROTOCOL	STORAGETYPE	SN	STATUS	ONLINE	Url
huawei-csi	backend-demo	nfs	oceanstor-nas	xxxxxxxxxxxxxxxxxxxxxx	Bound	true	https://192.168.129.157:8088

----End

5.1.1.5.2 Block Service

iSCSI

This section describes how to create a storage backend of the iSCSI protocol type.

Configuration Item Description

Table 5-26 backend parameters

Parameter	Description	Mandatory	Default Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed to oceandisk-san .
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.

Parameter	Description	Mandatory	Default Value	Remarks
urls	Management URLs of the storage device. The value format is a list.	Yes	-	The value can be a domain name or an IP address + port number.
pools	Storage pools of storage devices. The value format is a list.	Yes	-	Enter the storage pool names.
parameters.protocol	Storage protocol. The value is a character string.	Yes	-	- The value is fixed to iscsi. - Ensure that an iSCSI client has been installed on the connected compute node.
parameters.portals	Service access port. Nodes will use this port to read and write storage resources. The value format is a list.	Yes	-	Multiple ports can be configured.
supportedTopologies	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	30	The value ranges from 1 to 30 . If this parameter is not set or the value is not in the specified range, the default value 30 is used.

Creating a Storage Backend

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "oceandisk-san"
name: "backend-demo"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:8088"
pools:
  - "StoragePool001"
parameters:
  protocol: "iscsi"
  portals:
    - "10.10.30.20"
```



```
- "10.10.30.21"  
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

```
NUMBER CONFIGURED NAME STORAGE URLS  
1 false backend-demo oceandisk-san https://192.168.129.157:8088  
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1  
Please enter this backend user name: admin  
Please enter this backend password:
```

```
Backend backend-demo is configured  
NUMBER CONFIGURED NAME STORAGE URLS  
1 true backend-demo oceandisk-san https://192.168.129.157:8088  
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

```
NAMESPACE NAME PROTOCOL STORAGETYPE SN STATUS ONLINE  
Url  
huawei-csi backend-demo iscsi oceandisk-san xxxxxxxxxxxxxxxxxxxx Bound true https://  
192.168.129.157:8088
```

----End

FC

This section describes how to create a storage backend of the FC protocol type.

Configuration Item Description

Table 5-27 backend parameters

Parameter	Description	Mandatory	Default Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed to oceandisk-san .
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.

Parameter	Description	Mandatory	Default Value	Remarks
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.
urls	Management URLs of the storage device. The value format is a list.	Yes	-	The value can be a domain name or an IP address + port number.
pools	Storage pools of storage devices. The value format is a list.	Yes	-	Enter the storage pool names.
parameters.protocol	Storage protocol. The value is a character string.	Yes	-	The value is fixed to fc.
supportedTopologies	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	30	The value ranges from 1 to 30 . If this parameter is not set or the value is not in the specified range, the default value 30 is used.

Creating a Storage Backend

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "oceandisk-san"
name: "backend-demo"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:8088"
pools:
  - "StoragePool001"
parameters:
  protocol: "fc"
maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

```
NUMBER CONFIGURED NAME STORAGE URLS
1 false backend-demo oceandisk-san https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
```

```
Please enter this backend user name: admin
```

```
Please enter this backend password:
```

```
Backend backend-demo is configured
```

```
NUMBER CONFIGURED NAME STORAGE URLS
1 true backend-demo oceandisk-san https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

```
NAMESPACE NAME PROTOCOL STORAGETYPE SN STATUS ONLINE
Url
huawei-csi backend-demo fc oceandisk-san xxxxxxxxxxxxxxxxxxxx Bound true https://
192.168.129.157:8088
```

----End

NVMe over RoCE

This section describes how to create a storage backend of the NVMe over RoCE protocol type.

Configuration Item Description

Table 5-28 backend parameters

Parameter	Description	Mandatory	Default Value	Remarks
storage	Storage service type.	Yes	-	The value is fixed to oceandisk-san .
name	Storage backend name. The value can contain a maximum of 63 characters, including lowercase letters, digits, and hyphens (-). It must start with a letter or digit.	Yes	-	Ensure that the storage backend name is unique.
namespace	Namespace.	No	huawei-csi	The storage backend must be in the same namespace as Huawei CSI.

Parameter	Description	Mandatory	Default Value	Remarks
urls	Management URLs of the storage device. The value format is a list.	Yes	-	The value can be a domain name or an IP address + port number.
pools	Storage pools of storage devices. The value format is a list.	Yes	-	Enter the storage pool names.
parameters.protocol	Storage protocol. The value is a character string.	Yes	-	- The value is fixed to roce-nvme or roce (roce is deprecated). - Ensure that the nvme-cli tool has been installed on the compute node to be connected. The supported nvme-cli tool version is 1.9 and later.
parameters.ports	Service access port. Nodes will use this port to read and write storage resources. The value format is a list.	Yes	-	Multiple ports can be configured.
supportedTopologies	Storage topology awareness configuration. The parameter format is JSON of the list type.	No	-	This parameter is mandatory if storage topology awareness is enabled. For details, see 7.9 Configuring Storage Topology Awareness .
maxClientThreads	Maximum number of concurrent connections to a storage backend.	No	30	The value ranges from 1 to 30 . If this parameter is not set or the value is not in the specified range, the default value 30 is used.

Creating a Storage Backend

Step 1 Prepare a backend configuration file, for example, **backend.yaml**.

```
storage: "oceandisk-san"
name: "backend-demo"
namespace: "huawei-csi"
urls:
  - "https://192.168.129.157:8088"
pools:
  - "StoragePool001"
```

```
parameters:
  protocol: "roce-nvme"
  portals:
    - "10.10.30.20"
    - "10.10.30.21"
  maxClientThreads: "30"
```

Step 2 Run the following command to create a storage backend.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml
```

The following is an example of the command output.

```
NUMBER CONFIGURED NAME STORAGE URLS
1 false backend-demo oceandisk-san https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 3 Enter the serial number of the backend to be created and enter the account and password.

```
Please enter the backend number to configure (Enter 'exit' to exit):1
Please enter this backend user name: admin
Please enter this backend password:
```

```
Backend backend-demo is configured
NUMBER CONFIGURED NAME STORAGE URLS
1 true backend-demo oceandisk-san https://192.168.129.157:8088
Please enter the backend number to configure (Enter 'exit' to exit):
```

Step 4 Check the storage backend creation result.

```
oceanctl get backend
```

The following is an example of the command output. If the backend status is **Bound**, the creation is successful.

NAMESPACE	NAME	PROTOCOL	STORAGETYPE	SN	STATUS	ONLINE	Url
huawei-csi	backend-demo	roce-nvme	oceandisk-san	xxxxxxxxxxxxxxxxxxxxxx	Bound	true	https://192.168.129.157:8088

----End

5.1.2 Managing Storage Backends

5.1.2.1 Querying a Storage Backend

- Run the following command to obtain the help information about querying a backend.

```
oceanctl get backend -h
```
- Run the following command to query a single storage backend in the default namespace.

```
oceanctl get backend <backend-name>
```
- Run the following command to query all storage backends in the specified namespace.

```
oceanctl get backend -n <namespace>
```
- Run the following command to format the output. Currently, **json**, **yaml**, and **wide** are supported.

```
oceanctl get backend <backend-name> -o json
```

5.1.2.2 Updating a Storage Backend

NOTICE

- When you use `oceanctl` to update the storage backend information, only the storage backend password and authentication mode for logging in to the storage backend can be updated.
- If the backend account password is updated on the storage device, the CSI plug-in will retry due to login failures. As a result, the account may be locked. If the account is locked, change the password by referring to [8.2.3 An Account Is Locked After the Password Is Updated on the Storage Device](#).

5.1.2.2.1 Updating the Password of a Storage Backend Using `oceanctl`

Obtaining the Help Information About Updating a Backend

- Step 1** Run the following command to obtain the help information about updating a storage backend.

```
oceanctl update backend -h
```

The following is an example of the command output.

Update a backend for Ocean Storage in Kubernetes

Usage:

```
oceanctl update backend <name> [flags]
```

Examples:

```
# Update backend account information in default(huawei-csi) namespace
oceanctl update backend <name> --password
```

```
# Update backend account information in specified namespace
oceanctl update backend <name> -n namespace --password
```

```
# Update backend account information with ldap authentication mode in default(huawei-csi) namespace
oceanctl update backend <name> --password --authenticationMode=ldap
```

```
# Update backend account information with local authentication mode in default(huawei-csi) namespace
oceanctl update backend <name> --password --authenticationMode=local
```

```
# Update backend account information with ldap authentication mode in specified namespace
oceanctl update backend <name> -n namespace --password --authenticationMode=ldap
```

Flags:

```
--authenticationMode string  Specify authentication mode
-h, --help                    help for backend
-n, --namespace string        namespace of resources
--password                    Update account password
```

Global Flags:

```
--log-dir string  Specify the directory for printing log files. (default "/var/log/huawei")
```

----End

Example of Updating the Password of a Backend

- Step 1** Run the following command to update a storage backend. In the command, *backend-name* indicates the name of the storage backend to be updated.

```
oceanctl update backend backend-name --password
```

Enter the user name and new password as prompted:

```
Please enter this backend user name:admin
Please enter this backend password:
backend/backend-name updated
```

----End

5.1.2.2.2 Updating the Login Authentication Mode of a Storage Backend

Example of Updating the Backend Login Authentication Type to LDAP

Step 1 Run the following command to update a storage backend. In the command, *backend-name* indicates the name of the storage backend to be updated.

```
oceanctl update backend backend-name --password --authenticationMode=ldap
```

Enter the user name and new password as prompted:

```
Please enter this backend user name:admin
Please enter this backend password:
backend/backend-name updated
```

----End

5.1.2.2.3 Updating a Storage Backend

NOTE

- PVC provisioning must be based on a configured storage backend. Therefore, if a PVC has been provisioned on a storage backend, do not change the storage backend.
- The name uniquely identifies a storage backend. The name of a storage backend with a PVC provisioned cannot be changed.
- After a storage backend is modified, the new configuration applies only to volumes to be provisioned.
- Do not perform volume management operations during the modification of a storage backend.

Procedure

Step 1 Delete the storage backend to be modified. For details, see [5.1.2.3 Deleting a Storage Backend](#).

Step 2 Create a storage backend with the same name. For details, see [5.1.1 Configuring the Storage Backend](#). The storage backend name cannot be changed.

----End

5.1.2.3 Deleting a Storage Backend

NOTICE

Do not delete a storage backend when a volume management operation is being performed on it.

Example of Deleting a Backend

Step 1 Run the following command to obtain information about a storage backend.

```
oceanctl get backend
```

The following is an example of the command output.

NAMESPACE	NAME	PROTOCOL	STORAGETYPE	SN	STATUS	ONLINE	Url
huawei-csi	backend-1	roce-nvme	oceanstor-san	xxxxxxxxxxxxxxxxxxxx	Bound	true	https://192.168.129.157:8088
huawei-csi	backend-2	roce-nvme	oceanstor-san	xxxxxxxxxxxxxxxxxxxx	Bound	true	https://192.168.129.158:8088

Step 2 Run the following command to delete the specified storage backend.

```
oceanctl delete backend backend-1
```

Step 3 Run the following command to check the deletion result.

```
oceanctl get backend backend-1
```

The following is an example of the command output. If **not found** is displayed, the deletion is successful.

```
Error from server (NotFound): backend "backend-1" not found
```

----End

5.1.2.4 Adding a Storage Backend Certificate

CSI allows you to add a storage certificate to use the TLS/SSL protocol to encrypt data transmission channels, improving the security of communication with storage devices.

Prerequisites

A certificate has been created. Take OceanStor Dorado as an example. For details about how to create a certificate, [click here](#).

Example of Creating a Certificate

Step 1 Prepare a certificate file in advance, for example, **cert.crt**.

Step 2 Run the following command to obtain information about a storage backend.

```
oceanctl get backend
```

The following is an example of the command output.

NAMESPACE	NAME	PROTOCOL	STORAGETYPE	SN	STATUS	ONLINE	Url
huawei-csi	backend-1	roce-nvme	oceanstor-san	xxxxxxxxxxxxxxxxxxxx	Bound	true	https://192.168.129.157:8088
huawei-csi	backend-2	roce-nvme	oceanstor-san	xxxxxxxxxxxxxxxxxxxx	Bound	true	https://192.168.129.158:8088

Step 3 Run the following command to create a certificate for the specified storage backend.

```
oceanctl create cert cert-1 -b backend-1 -f /path/to/cert.crt
```

Step 4 Check the certificate creation result.

```
oceanctl get cert -b backend-1
```


The following is an example of the command output.

```
NAMESPACE NAME BOUNDBACKEND
huawei-csi cert-1 backend-1
```

----End

5.1.2.5 Deleting a Storage Backend Certificate

Procedure

- Step 1** Run the following command to obtain information about a storage backend.

```
oceanctl get backend
```

The following is an example of the command output.

```
NAMESPACE NAME PROTOCOL STORAGETYPE SN STATUS ONLINE
Url
huawei-csi backend-1 roce-nvme oceanstor-san xxxxxxxxxxxxxxxxxxxx Bound true https://
192.168.129.157:8088
huawei-csi backend-2 roce-nvme oceanstor-san xxxxxxxxxxxxxxxxxxxx Bound true https://
192.168.129.158:8088
```

- Step 2** Run the following command to obtain information about the certificate of the specified storage backend.

```
oceanctl get cert -b backend-1
```

The following is an example of the command output.

```
NAMESPACE NAME BOUNDBACKEND
huawei-csi cert-1 backend-1
```

- Step 3** Run the following command to delete the certificate of the specified storage backend.

```
oceanctl delete cert -b backend-1
```

- Step 4** Check the deletion result.

```
oceanctl get cert -b backend-1
```

The following is an example of the command output. If **no cert found** is displayed, the deletion is successful.

```
Error from server (NotFound): no cert found on backend backend-1 in huawei-csi namespace
```

----End

5.2 StorageClass Management

A [StorageClass](#) provides administrators with methods to describe a storage "class". Different types may map to a different group of capability definitions. Kubernetes cluster users can dynamically provision volumes based on a StorageClass.

5.2.1 Configuring a StorageClass

Cluster administrators can define multiple StorageClass objects as required. When configuring a PVC, the cluster administrators need to specify a StorageClass that meets service requirements. When applying for resources from Huawei storage

devices, Huawei CSI creates storage resources that meet service requirements based on the preset StorageClass.

5.2.1.1 Flash Storage (OceanStor Dorado/OceanStor V5/OceanStor V6 and Later)

5.2.1.1.1 File System

Creating a StorageClass

Step 1 Prepare a StorageClass configuration file, for example, **mysc.yaml**. For details about the StorageClass configuration, see the following example.

Step 2 Run the following command to create a StorageClass using the configuration file.

```
kubectl apply -f mysc.yaml
```

Step 3 Run the following command to view the information about the created StorageClass.

```
kubectl get sc mysc
```

The following is an example of the command output.

NAME	PROVISIONER	RECLAIMPOLICY	VOLUMEBINDINGMODE	ALLOWVOLUMEEXPANSION	AGE
mysc	csi.huawei.com	Delete	Immediate	true	8s

----End

NFS Protocol Configuration Example

When a container uses the NFS protocol to connect to file system resources, refer to the following StorageClass configuration example. In this example, NFS version 4.1 is specified for mounting.

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
  name: mysc
provisioner: csi.huawei.com
parameters:
  backend: nfs-nas-181
  pool: StoragePool001
  volumeType: fs
  allocType: thin
  authClient: "*"
mountOptions:
  - nfsvers=4.1 # Specify the version 4.1 for NFS mounting.
```

NFS+ Protocol Configuration Example

When a container uses the NFS+ protocol to connect to file system resources, refer to the following StorageClass configuration example.

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
  name: mysc
provisioner: csi.huawei.com
parameters:
  backend: nfs-nas-181
  pool: StoragePool001
```

```
volumeType: fs
allocType: thin
authClient: ""
```

HyperMetro File System Configuration Example

When a container uses an NFS HyperMetro file system as a storage resource, refer to the following configuration example. In this example, the used backend supports HyperMetro, and **hyperMetro** is set to **true**.

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
  name: mysc
provisioner: csi.huawei.com
parameters:
  backend: nfs-hypermetro-dorado-181
  pool: pool001
  volumeType: fs
  hyperMetro: "true" # Provision HyperMetro volumes.
  allocType: thin
  authClient: ""
```

NOTICE

- Before provisioning a NAS HyperMetro volume, you need to configure the HyperMetro relationship between two storage devices, including the remote device and HyperMetro domain. The HyperMetro domain of the file system can only work in HyperMetro AA mode. For details about the configuration operation, see the product documentation of the corresponding storage model.
- If a storage device is faulty, the logical management port may fail over. Deleting a PVC in the failover state will delete resources only at one end. You need to manually clear the storage resources at the peer end.

StorageClass Parameters Supported by File Systems

Table 5-29 StorageClass configuration parameters

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
metadata.name	User-defined name of a StorageClass object.	Yes	-	Yes	Take Kubernetes v1.22.1 as an example. The value can contain digits, lowercase letters, hyphens (-), and periods (.), and must start and end with a letter or digit.
provisioner	Name of the provisioner.	Yes	csi.huawei.com	Yes	Set this parameter to the driver name set during Huawei CSI installation. The value is the same as that of driverName in the values.yaml file.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
reclaimPolicy	Reclamation policy. The following types are supported: • Delete: Resources are automatically reclaimed. • Retain: Resources are manually reclaimed.	No	Delete	Yes	• Delete: When a PV/PVC is deleted, resources on the storage device are also deleted. • Retain: When a PV/PVC is deleted, resources on the storage device are not deleted.
allowVolumeExpansion	Whether to allow volume expansion. If this parameter is set to true , the capacity of the PV that uses the StorageClass can be expanded.	No	false	Yes	This function can only be used to expand PV capacity but cannot be used to reduce PV capacity.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
mountOptions	List of mount parameters, which can be used to specify the parameters of the -o option when the mount command is executed on a host.	No	-	Yes	For details about common parameters in mountOptions , see Table 5-30 . You can also specify other mount parameters.
parameters.backend	Name of the backend where the resource to be created is located. This field must be set if parameters.pool is set.	Conditionally mandatory	-	No	If this parameter is not set, Huawei CSI will randomly select a backend that meets the capacity requirements to create resources. You are advised to specify a backend to ensure that the created resource is located on the expected backend.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameter s.pool	Name of the storage resource pool where the resource to be created is located.	No	-	No	If this parameter is not set, Huawei CSI will select a storage pool with the largest remaining capacity from the selected backend to create resources. You are advised to specify a storage pool to ensure that the created resource is located in the expected storage pool.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameter.s.volumeName	Name of the storage resource created by dynamic volume provisioning. You can configure a placeholder to customize the storage resource name. The following placeholders are supported: • PVC namespace: <code>{{ .PVCNamespace }}</code> • PVC name: <code>{{ .PVCName }}</code>	No	-	No	• The value can contain letters, digits, hyphens (-), underscores (_), and periods (.). This parameter cannot be left empty. The length of the generated storage resource name ranges from 1 to 255 characters. • Both the PVC namespace and PVC name must be configured. • To avoid duplicate resource names, the PVC UID is added to the end of the name as a unique identifier by default. Configuration example: PVC namespace: namespace. PVC name:

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
					pvc-1. PVC UID: c2fd3f46-bf17-4a7d-b88e-2e3232bae434. volumeName is set to prefix-{{ .PVCNamespaced }}_{{ .PVCName }}. The ultimate storage resource name is prefix-namespace_pvc-1-c2fd3f46bf174a7db88e2e3232bae434.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameter.s.volumeType	Type of the volume to be created. The following types are supported: • lun: A LUN is provisioned on the storage side. • fs: A file system is provisioned on the storage side. • dtree: A volume of the dtree type is provisioned on the storage side.	Yes	-	Yes	The value is fixed at fs .

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameter.s.allocType	Allocation type of the volume to be created. The following types are supported: • thin: Not all required space is allocated during creation. Instead, the space is dynamically allocated based on the usage. • thick: All required space is allocated during creation.	No	thin	No	If this parameter is set to thin, the required space is not allocated immediately when a volume is created. Instead, the space is dynamically allocated based on the usage. OceanStor Dorado does not support thick.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameter.s.authClient	IP address of the NFS client that can access the volume. You can enter the client host name (a full domain name is recommended), client IP address, or client IP address segment.	Yes	-	No	The asterisk (*) can be used to indicate any client. If you are not sure about the IP address of the access client, you are advised to use the asterisk (*) to prevent the client access from being rejected by the storage system. If the client host name is used, you are advised to use the full domain name. The IP addresses can be IPv4 addresses, IPv6 addresses, or a combination of IPv4 and IPv6 addresses. You can enter multiple host names, IP addresses, or IP address segments and separate them with semicolons (;). Example: 192.168.0.10;192.168.0.0/24;

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
					myserver1.test
parameter.s.waitForSplit	Whether to wait for the clone pair synchronization to complete during PVC creation.	No	"true"	No	In HyperMetro scenarios, this parameter can only be set to "true" . • "true": Wait until the clone pair synchronization is complete. • "false": Do not wait for the clone pair synchronization to complete.
parameter.s.cloneSpeed	Clone speed. The value ranges from 1 to 4. The value can be: • "1": low • "2": medium • "3": high • "4": highest	No	"3"	No	This parameter takes effect when a persistent volume is cloned or a persistent volume is created from a snapshot.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameter.s.applicationType	Application type name when a file system is created.	No	-	Yes	Log in to DeviceManager and choose Services > File Service > File Systems > Create to obtain the application type name.
parameter.s.qos	QoS settings of the file system on the storage side of the PV. The value of the parameter is JSON character strings in dictionary format. A character string is enclosed by single quotation marks and the dictionary key by double quotation marks. For example, '{"MAXBANDWIDTH": 999, "MAXIOPS": 999}' .	No	-	No	For details about the supported QoS configurations, see Table 5-31 .

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameter.s.hyperMetro	Whether a HyperMetro volume is to be created. This parameter needs to be configured when the backend is of the HyperMetro type. • "true": The created volume is a HyperMetro volume. If the storage backend is a HyperMetro backend, the value must be "true". • "false": The created volume is a common volume.	Conditionally mandatory	"false"	No	When the used backend is a HyperMetro backend and a HyperMetro volume needs to be provisioned, set this parameter to "true" . If this parameter is set to "false" , services may be interrupted if the logical management port connected to the backend fails over.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameter.s.metroPairSyncSpeed	Data synchronization speed of a HyperMetro pair. The value ranges from 1 to 4. The value can be: • "1": low • "2": medium • "3": high • "4": highest	No	-	No	The configuration takes effect when a HyperMetro volume is created. Note: • If this parameter is not configured, the storage speed of the HyperMetro pair is determined by the storage device. • The highest synchronization speed may increase the host latency.
parameter.fsPermission	Permission on the directory mounted to a container.	No	-	No	For details about the configuration format, refer to the Linux permission settings, for example, 777 and 755.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameter.s.rootSquash	Controls the root permission of the client. The value can be: • root_squash: The client cannot access the storage system as user root. If a client accesses the storage system as user root, the client will be mapped as an anonymous user. • no_root_squash: A client can access the storage system as user root and has the permission of	No	-	No	

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
	user root.				
parameter s.allSquash	Whether to retain the user ID (UID) and group ID (GID) of a shared directory. The value can be: • all_squash: The UID and GID of the shared directory are mapped to anonymous users. • no_all_squash: The UID and GID of the shared directory are retained.	No	-	No	

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameter.s.accesskrb5	Configures the krb5 security protocol. • read_only: read-only • read_write: read and write • none: no permission	No	-	No	During mounting, you can specify the sec parameter in mountOptions .
parameter.s.accesskrb5i	Configures the krb5i security protocol. • read_only: read-only • read_write: read and write • none: no permission	No	-	No	During mounting, you can specify the sec parameter in mountOptions .

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameter.s.accesskrb5p	Configures the krb5p security protocol. • read_only: read-only • read_write: read and write • none: no permission	No	-	No	During mounting, you can specify the sec parameter in mountOptions .
parameter.s.snapshotDirectoryVisibility	Whether the snapshot directory is visible. The value can be: • visible: The snapshot directory is visible. • invisible: The snapshot directory is invisible.	No	-	No	Only NAS storage is supported.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameter.s.reservedSnapshotSpaceRatio	Configures reserved snapshot space. Value type: character string Value range: 0 to 50	No	-	No	OceanStor Dorado 6.1.5+ and OceanStor 6.1.5+ NAS storage devices are supported.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameter.s.disableVerifyCapacity	Whether to disable volume capacity verification. After this function is disabled, the system will not verify whether the volume capacity is an integer multiple of the sector size. The value can be: • "true": disables volume capacity verification. • "false": enables volume capacity verification. NOTICE When Red Hat OpenShift Virtualization is used to connect to CSI, this parameter must be set to "true".	No	"true"	No	For OceanStor Dorado and OceanStor storage, the sector size is 512 bytes.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameter.s.description	Description of the file system to be created. Value type: character string The value contains 0 to 255 characters.	No	Created from Kubernetes CSI	No	
parameter.s.advanced Options	Advanced volume creation parameters. The value of the parameter is JSON character strings in dictionary format. A character string is enclosed by single quotation marks and the dictionary key by double quotation marks. Example: '{"CAPACITYTHRESHOLD": 90}'	No	-	No	For details about the supported advanced parameters, see Table 5-32 .

Table 5-30 Common parameters in mountOptions

Parameter	Description	Mandatory	Default Value	Remarks
mountOptions.nfsvers	NFS mount option on the host. The following mount option is supported: nfsvers : protocol version for NFS mounting. The value can be 3 , 4 , 4.0 , 4.1 , or 4.2 .	No	-	This parameter specified after the -o parameter is optional when the mount command is executed on the host. The value is in list format. If the NFS version is specified for mounting, NFS 3, 4.0, 4.1, and 4.2 protocols are supported (the protocol must be supported and enabled on storage devices). If nfsvers is set to 4 , the latest protocol version NFS 4 may be used for mounting due to different OS configurations, for example, 4.2. If the 4.0 protocol is required, you are advised to set nfsvers to 4.0 .

Parameter	Description	Mandatory	Default Value	Remarks
mountOptions.sec	Kerberos 5 protocol for mounting NFS file systems.	No	-	• If Kerberos 5 is used, set this parameter to krb5. • If Kerberos 5i is used, set this parameter to krb5i. • If Kerberos 5p is used, set this parameter to krb5p. • Kerberos supports only NFSv4.0 and later versions. • OceanStor Dorado and OceanStor 6.1.3 and later versions support Kerberos.
mountOptions.proto	Transmission protocol used for NFS mounting. The value can be rdma .	No	-	• The parameter is supported for NAS storage of OceanStor Dorado and OceanStor 6.1.7 and later.

Parameter	Description	Mandatory	Default Value	Remarks
mountOptions.port	Protocol port used for NFS mounting.	Conditionally mandatory	-	If the transmission protocol is rdma , set this parameter to 20049 .

Table 5-31 Supported QoS configurations

Storage Type	Parameter	Description	Remarks
OceanStor V5	IOTYPE	Read/write type.	This parameter is optional. If it is not specified, the default value of the storage backend is used. For details, see related storage documents. The value can be: • 0: read I/O • 1: write I/O • 2: read and write I/Os
	MAXBANDWIDTH	Maximum bandwidth. This is a restriction policy parameter.	The value is an integer greater than 0, expressed in MB/s.
	MINBANDWIDTH	Minimum bandwidth. This is a protection policy parameter.	The value is an integer greater than 0, expressed in MB/s.
	MAXIOPS	Maximum IOPS. This is a restriction policy parameter.	The value is an integer greater than 0.
	MINIOPS	Minimum IOPS. This is a protection policy parameter.	The value is an integer greater than 0.
	LATENCY	Maximum latency. This is a protection policy parameter.	The value is an integer greater than 0, expressed in ms.

Storage Type	Parameter	Description	Remarks
OceanStor or Dorado/OceanStor	IOTYPE	Read/write type.	The value can be: • 2: read and write I/Os
	MAXBANDWIDTH	Maximum bandwidth. This is a restriction policy parameter.	The value is an integer ranging from 1 to 999999999, expressed in MB/s.
	MINBANDWIDTH	Minimum bandwidth. This is a protection policy parameter.	The value is an integer ranging from 1 to 999999999, expressed in MB/s.
	MAXIOPS	Maximum IOPS. This is a restriction policy parameter.	The value is an integer ranging from 100 to 999999999.
	MINIOPS	Minimum IOPS. This is a protection policy parameter.	The value is an integer ranging from 100 to 999999999.
	LATENCY	Maximum latency. This is a protection policy parameter.	The value can be 0.5 or 1.5 , expressed in ms.

Table 5-32 Supported advanced volume creation parameters

Parameter	Description	Remarks
CAPACITYTHRESHOLD	Total capacity alarm threshold.	Parameter type: uint64. For details about the default value and value range, see the corresponding storage product manual.

5.2.1.1.2 Dtree

Creating a StorageClass

Step 1 Prepare a StorageClass configuration file, for example, **mysc.yaml**. For details about the StorageClass configuration, see the following example.

Step 2 Run the following command to create a StorageClass using the configuration file.

```
kubectl apply -f mysc.yaml
```

Step 3 Run the following command to view the information about the created StorageClass.

```
kubectl get sc mysc
```

The following is an example of the command output.

NAME	PROVISIONER	RECLAIMPOLICY	VOLUMEBINDINGMODE	ALLOWVOLUMEEXPANSION	AGE
mysc	csi.huawei.com	Delete	Immediate	true	8s

----End

Example of StorageClass Configuration Supported by the NFS Protocol

When a container uses the NFS protocol to connect to dtree resources, refer to the following StorageClass configuration example. In this example, NFS version 4.1 is specified for mounting.

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
  name: mysc
provisioner: csi.huawei.com
parameters:
  backend: nfs-dtree-181
  parentname: parent-filesystem-name
  volumeType: dtree
  authClient: "*"
mountOptions:
  - nfsvers=4.1 # Specify the version 4.1 for NFS mounting.
```

Example of StorageClass Configuration Supported by the NFS+ Protocol

When a container uses the NFS+ protocol to connect to dtree resources, refer to the following StorageClass configuration example.

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
  name: mysc
provisioner: csi.huawei.com
parameters:
  backend: nfs-dtree-181
  parentname: parent-filesystem-name
  volumeType: dtree
  authClient: "*"

```

Storage Class Parameters Supported by Dtrees

Table 5-33 StorageClass configuration parameters

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
metadata.name	User-defined name of a StorageClass object.	Yes	-	Yes	Take Kubernetes v1.22.1 as an example. The value can contain digits, lowercase letters, hyphens (-), and periods (.), and must start and end with a letter or digit.
provisioner	Name of the provisioner.	Yes	csi.huawei.com	Yes	Set this parameter to the driver name set during Huawei CSI installation. The value is the same as that of driverName in the values.yaml file.
reclaimPolicy	Reclamation policy. The following types are supported: ● Delete: Resources are automatically reclaimed. ● Retain: Resources are manually reclaimed.	No	Delete	Yes	● Delete: When a PV/PVC is deleted, resources on the storage device are also deleted. ● Retain: When a PV/PVC is deleted, resources on the storage device are not deleted.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
allowVolumeExpansion	Whether to allow volume expansion. If this parameter is set to true , the capacity of the PV that uses the StorageClass can be expanded.	No	false	Yes	This function can only be used to expand PV capacity but cannot be used to reduce PV capacity.
mountOptions	List of mount parameters, which can be used to specify the parameters of the -o option when the mount command is executed on a host.	No	-	Yes	For details about common parameters in mountOptions , see Table 5-34 . You can also specify other mount parameters.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.backend	Name of the backend where the resource to be created is located.	Conditionally mandatory	-	No	If this parameter is not set, Huawei CSI will randomly select a backend that meets the capacity requirements to create resources. You are advised to specify a backend to ensure that the created resource is located on the expected backend. This parameter is mandatory if parameters.parentname is configured.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.parentname	Name of a file system on the current storage device. Dtree is created in the file system.	Conditionally mandatory	-	Yes	This parameter is mandatory when parentname is not configured for the backend. If parentname is configured only in the StorageClass but not configured in the storage backend, set CSIDriverObject.attachRequired to true during CSI installation.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.volumeName	Name of the storage resource created by dynamic volume provisioning. You can configure a placeholder to customize the storage resource name. The following placeholders are supported: • PVC namespace: <code>{{ .PVCNamespace }}</code> • PVC name: <code>{{ .PVCName }}</code>	No	-	No	• The value can contain letters, digits, hyphens (-), underscores (_), and periods (.). This parameter cannot be left empty. The length of the generated storage resource name ranges from 1 to 255 characters. • Both the PVC namespace and PVC name must be configured. • To avoid duplicate resource names, the PVC UID is added to the end of the name as a unique identifier by default. Configuration example: PVC namespace: namespace. PVC name: pvc-1. PVC UID: c2fd3f46-bf17-4a7d-b88e-2e3232bae434. volumeName is set to prefix-{{ .PVCNamespace }}_{{ .PVCName }}. The ultimate storage resource name is prefix-namespace_pvc-1-c2fd3f46bf174a7db88e2e3232bae434.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.volumeType	Type of the volume to be created. The following types are supported: • lun: A LUN is provisioned on the storage side. • fs: A file system is provisioned on the storage side. • dtree: A volume of the Dtree type is provisioned on the storage side.	Yes	-	Yes	When dtree is used, the value must be dtree .

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.authClient	IP address of the NFS client that can access the volume. You can enter the client host name (a full domain name is recommended), client IP address, or client IP address segment.	Yes	-	No	The asterisk (*) can be used to indicate any client. If you are not sure about the IP address of the access client, you are advised to use the asterisk (*) to prevent the client access from being rejected by the storage system. If the client host name is used, you are advised to use the full domain name. The IP addresses can be IPv4 addresses, IPv6 addresses, or a combination of IPv4 and IPv6 addresses. You can enter multiple host names, IP addresses, or IP address segments and separate them with semicolons (;). Example: 192.168.0.10;192.168.0.0/24;myserver1.test
parameters.fsPermission	Permission on the directory mounted to a container.	No	-	No	For details about the configuration format, refer to the Linux permission settings, for example, 777 and 755.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.rootSquash	Controls the root permission of the client. The value can be: • root_squash: The client cannot access the storage system as user root. If a client accesses the storage system as user root, the client will be mapped as an anonymous user. • no_root_squash: A client can access the storage system as user root and has the permission of user root.	No	-	No	

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.allSquash	Whether to retain the user ID (UID) and group ID (GID) of a shared directory. The value can be: • all_squash: The UID and GID of the shared directory are mapped to anonymous users. • no_all_squash: The UID and GID of the shared directory are retained.	No	-	No	
parameters.accesskrb5	Configures the krb5 security protocol. • read_only: read-only • read_write: read and write • none: no permission	No	-	No	During mounting, you can specify the sec parameter in mountOptions .

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.accesskrb5i	Configures the krb5i security protocol. • read_only: read-only • read_write: read and write • none: no permission	No	-	No	During mounting, you can specify the sec parameter in mountOptions .
parameters.accesskrb5p	Configures the krb5p security protocol. • read_only: read-only • read_write: read and write • none: no permission	No	-	No	During mounting, you can specify the sec parameter in mountOptions .

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.disableVerifyCapacity	Whether to disable volume capacity verification. After this function is disabled, the system will not verify whether the volume capacity is an integer multiple of the sector size. The value can be: • "true": disables volume capacity verification. • "false": enables volume capacity verification. NOTICE When Red Hat OpenShift Virtualization is used to connect to CSI, this parameter must be set to true.	No	"true"	No	For OceanStor Dorado and OceanStor storage, the sector size is 512 bytes.

Parameter	Description	M a n d a t o r y	Def ault Valu e	Wh eth er the Vol um e Ma nag em ent Tak es Eff ect	Remarks
parameters.d escription	Description of the dtree share to be created. Value type: character string The value contains 0 to 255 characters.	N o	Crea ted from Kub erne tes CSI	No	

Table 5-34 Common parameters in mountOptions

Parameter	Description	Mandatory	Default Value	Remarks
mountOptions.nfsvers	NFS mount option on the host. The following mount option is supported: nfsvers: protocol version for NFS mounting. The value can be 3, 4, 4.0, 4.1, or 4.2 .	No	-	This parameter specified after the -o parameter is optional when the mount command is executed on the host. The value is in list format. If the NFS version is specified for mounting, NFS 3, 4.0, 4.1, and 4.2 protocols are supported (the protocol must be supported and enabled on storage devices). If nfsvers is set to 4 , the latest protocol version NFS 4 may be used for mounting due to different OS configurations, for example, 4.2. If the 4.0 protocol is required, you are advised to set nfsvers to 4.0 .

Parameter	Description	Mandatory	Default Value	Remarks
mountOptions.sec	Kerberos 5 protocol for mounting NFS file systems.	No	-	• If Kerberos 5 is used, set this parameter to krb5. • If Kerberos 5i is used, set this parameter to krb5i. • If Kerberos 5p is used, set this parameter to krb5p. • Kerberos supports only NFSv4.0 and later versions. • OceanStor Dorado and OceanStor 6.1.3 and later versions support Kerberos.
mountOptions.proto	Transmission protocol used for NFS mounting. The value can be rdma .	No	-	• The parameter is supported for NAS storage of OceanStor Dorado and OceanStor 6.1.7 and later.

Parameter	Description	Mandatory	Default Value	Remarks
mountOptions.port	Protocol port used for NFS mounting.	Conditionally mandatory	-	If the transmission protocol is rdma , set this parameter to 20049 .

5.2.1.1.3 Block Service

Creating a StorageClass

Step 1 Prepare a StorageClass configuration file, for example, **mysc.yaml**. For details about the StorageClass configuration, see the following example.

Step 2 Run the following command to create a StorageClass using the configuration file.

```
kubectl apply -f mysc.yaml
```

Step 3 Run the following command to view the information about the created StorageClass.

```
kubectl get sc mysc
```

The following is an example of the command output.

NAME	PROVISIONER	RECLAIMPOLICY	VOLUMEBINDINGMODE	ALLOWVOLUMEEXPANSION	AGE
mysc	csi.huawei.com	Delete	Immediate	true	8s

----End

Block Storage Class Configuration Example

If LUNs are used as storage resources and the file system needs to be formatted to a local file system, refer to the following example.

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
  name: mysc
provisioner: csi.huawei.com
parameters:
  backend: lun-181
  pool: StoragePool001
  volumeType: lun
  allocType: thin
  fsType: ext4
```

StorageClass Parameters Supported by Block Services

Table 5-35 StorageClass configuration parameters

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
metadata.name	User-defined name of a StorageClass object.	Yes	-	Yes	Take Kubernetes v1.22.1 as an example. The value can contain digits, lowercase letters, hyphens (-), and periods (.), and must start and end with a letter or digit.
provisioner	Name of the provisioner.	Yes	csi.huawei.com	Yes	Set this parameter to the driver name set during Huawei CSI installation. The value is the same as that of driverName in the values.yaml file.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
reclaimPolicy	Reclamation policy. The following types are supported: • Delete: Resources are automatically reclaimed. • Retain: Resources are manually reclaimed.	No	Delete	Yes	• Delete: When a PV/PVC is deleted, resources on the storage device are also deleted. • Retain: When a PV/PVC is deleted, resources on the storage device are not deleted.
allowVolumeExpansion	Whether to allow volume expansion. If this parameter is set to true , the capacity of the PV that uses the StorageClass can be expanded.	No	false	Yes	This function can only be used to expand PV capacity but cannot be used to reduce PV capacity.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
mountOptions	List of mount parameters, which can be used to specify the parameters of the -o option when the mount command is executed on a host.	No	-	Yes	For details about common parameters in mountOptions , see Table 5-36 . You can also specify other mount parameters.
parameters.backend	Name of the backend where the resource to be created is located. This field must be set if parameters.pool is set.	Conditionally mandatory	-	No	If this parameter is not set, Huawei CSI will randomly select a backend that meets the capacity requirements to create resources. You are advised to specify a backend to ensure that the created resource is located on the expected backend.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.pool	Name of the storage resource pool where the resource to be created is located.	No	-	No	If this parameter is not set, Huawei CSI will randomly select a storage pool with the largest remaining capacity from the selected backend to create resources. You are advised to specify a storage pool to ensure that the created resource is located in the expected storage pool.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.volume Name	Name of the storage resource created by dynamic volume provisioning. You can configure a placeholder to customize the storage resource name. The following placeholders are supported: • PVC namespace: {{ .PVC Namespace }} • PVC name: {{ .PVC Name }}	No	-	No	• The value can contain letters, digits, hyphens (-), underscores (_), and periods (.). This parameter cannot be left empty. The length of the generated storage resource name ranges from 1 to 255 characters. • Both the PVC namespace and PVC name must be configured. • To avoid duplicate resource names, the PVC UID is added to the end of the name as a unique identifier by default. Configuration example: PVC namespace: namespace. PVC name: pvc-1. PVC UID:

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
					c2fd3f46-bf17-4a7d-b88e-2e3232bae434. volumeName is set to prefix-{{ .PVCNamespace }}-{{ .PVCName }}. The ultimate storage resource name is prefix-namespace_pvc-1-c2fd3f46bf174a7db88e2e3232bae434.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.volumeType	Type of the volume to be created. The following types are supported: • lun: A LUN is provisioned on the storage side. • fs: A file system is provisioned on the storage side. • dtree: A volume of the dtree type is provisioned on the storage side.	Yes	-	Yes	The value is fixed at lun .

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.allocType	Allocation type of the volume to be created. The following types are supported: • thin: Not all required space is allocated during creation. Instead, the space is dynamically allocated based on the usage. • thick: All required space is allocated during creation.	No	thin	No	If this parameter is set to thin, the required space is not allocated immediately when a volume is created. Instead, the space is dynamically allocated based on the usage. OceanStor Dorado does not support thick.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.fsType	Type of a host file system. The supported types are: • ext2 • ext3 • ext4 • xfs	No	ext4	Yes	This parameter is available only when volumeMode of the PVC is set to Filesystem .

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.restore Mode	Restoration mode when a PVC is restored using a snapshot. The following modes are supported: • clone • snapshot	No	clone	No	• clone: Create a new resource on the storage device and establish a clone pair between the new resource and the snapshot resource for data restoration. • snapshot: Directly use the snapshot resource on the storage device and allocate it to the PVC.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
					NOTICE • The snapshot mode supports only non-HyperMetro volumes of OceanStor Dorado and OceanStor storage. • When the snapshot mode is used, snapshot resources are directly used as PVCs. To ensure data consistency, you are advised to set reclaimPolicy in the StorageClass to Retain and accessMode in the PVC to ROX. • To avoid data inconsistency , you are not advised to use the capacity expansion or snapshot capability for PVCs restored in snapshot mode.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.cloneSpeed	Clone speed. The value ranges from 1 to 4. The value can be: • "1": low • "2": medium • "3": high • "4": highest	No	"3"	No	This parameter takes effect when a PVC is cloned or a PVC is created from a snapshot.
parameters.applicationType	Application type name of the LUN to be created.	No	-	Yes	Log in to DeviceManager and choose Services > Block Service > LUN Groups (or Namespace Groups) > LUNs (or Namespaces) > Create to obtain the application type name.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.qos	LUN/NAS QoS settings of the PV on the storage side. The value of the parameter is JSON character strings in dictionary format. A character string is enclosed by single quotation marks and the dictionary key by double quotation marks. For example, '{"MAXBANDWIDTH": 999, "MAXIOPS": 999}' .	No	-	No	For details about the supported QoS configurations, see Table 5-37 .
parameters.fsPermission	Permission on the directory mounted to a container.	No	-	No	For details about the configuration format, refer to the Linux permission settings, for example, 777 and 755.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.disableVerifyCapacity	Whether to disable volume capacity verification. After this function is disabled, the system will not verify whether the volume capacity is an integer multiple of the sector size. The value can be: • "true": disables volume capacity verification. • "false": enables volume capacity verification.	No	"true"	Yes	For OceanStor Dorado and OceanStor storage, the sector size is 512 bytes.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
	NOTICE When Red Hat OpenShift Virtualization is used to connect to CSI, this parameter must be set to "true".				
parameters.description	Description of the LUN to be created. Value type: character string The value contains 0 to 255 characters.	No	Created from Kubernetes CSI	No	

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.hyperMetro	Whether a HyperMetro volume is to be created. This parameter needs to be configured when the backend is of the HyperMetro type. • "true": The created volume is a HyperMetro volume. If the storage backend is a HyperMetro backend, the value must be "true". • "false": The created volume is a common volume.	Conditionally mandatory	"false"	No	If the backend is of the HyperMetro type and HyperMetro volumes need to be provisioned, set this parameter to "true" . If the parameter is not configured, common volumes will be provisioned.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.metroPairSyncSpeed	Data synchronization speed of a HyperMetro pair. The value ranges from 1 to 4. The value can be: • "1": low • "2": medium • "3": high • "4": highest	No	-	No	The configuration takes effect when a HyperMetro volume is created. Note: • If this parameter is not configured, the storage speed of the HyperMetro pair is determined by the storage device. • The highest synchronization speed may increase the host latency.

Table 5-36 Common parameters in mountOptions

Parameter	Description	Mandatory	Default Value	Remarks
mountOptions.discard	Automatically triggers the Trim or Discard operation when a file system is mounted. This operation instructs a block device to release unused blocks.	No	-	The xfs and ext4 file systems are supported.

Table 5-37 Supported QoS configurations

Storage Type	Parameter	Description	Remarks
OceanStor V5	IOTYPE	Read/write type.	This parameter is optional. If it is not specified, the default value of the storage backend is used. For details, see related storage documents. The value can be: • 0: read I/O • 1: write I/O • 2: read and write I/Os
	MAXBANDWIDTH	Maximum bandwidth. This is a restriction policy parameter.	The value is an integer greater than 0, expressed in MB/s.
	MINBANDWIDTH	Minimum bandwidth. This is a protection policy parameter.	The value is an integer greater than 0, expressed in MB/s.
	MAXIOPS	Maximum IOPS. This is a restriction policy parameter.	The value is an integer greater than 0.

Storage Type	Parameter	Description	Remarks
	MINIOPS	Minimum IOPS. This is a protection policy parameter.	The value is an integer greater than 0.
	LATENCY	Maximum latency. This is a protection policy parameter.	The value is an integer greater than 0, expressed in ms.
OceanStor or Dorado/OceanStor	IOTYPE	Read/write type.	The value can be: • 2: read and write I/Os
	MAXBANDWIDTH	Maximum bandwidth. This is a restriction policy parameter.	The value is an integer ranging from 1 to 999999999, expressed in MB/s.
	MINBANDWIDTH	Minimum bandwidth. This is a protection policy parameter.	The value is an integer ranging from 1 to 999999999, expressed in MB/s.
	MAXIOPS	Maximum IOPS. This is a restriction policy parameter.	The value is an integer ranging from 100 to 999999999.
	MINIOPS	Minimum IOPS. This is a protection policy parameter.	The value is an integer ranging from 100 to 999999999.
	LATENCY	Maximum latency. This is a protection policy parameter.	The value can be 0.5 or 1.5 , expressed in ms.

5.2.1.2 Flash Storage (OceanStor A600/A800)

5.2.1.2.1 File System

Creating a StorageClass

Step 1 Prepare a StorageClass configuration file, for example, **mysc.yaml**. For details about the StorageClass configuration, see the following example.

Step 2 Run the following command to create a StorageClass using the configuration file.

```
kubectl apply -f mysc.yaml
```

Step 3 Run the following command to view the information about the created StorageClass.

```
kubectl get sc mysc
```

The following is an example of the command output.

NAME	PROVISIONER	RECLAIMPOLICY	VOLUMEBINDINGMODE	ALLOWVOLUMEEXPANSION	AGE
mysc	csi.huawei.com	Delete	Immediate	true	8s

----End

NFS Protocol Configuration Example

When a container uses the NFS protocol to connect to file system resources, refer to the following StorageClass configuration example. In this example, NFS version 4.1 is specified for mounting.

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
  name: mysc
provisioner: csi.huawei.com
parameters:
  backend: nfs-nas-181
  pool: StoragePool001
  volumeType: fs
  allocType: thin
  authClient: "*"
mountOptions:
  - nfsvers=4.1 # Specify the version 4.1 for NFS mounting.
```

DataTurbo Protocol Configuration Example

If a container uses OceanStor A series storage and the storage supports DataTurbo-based access, you can refer to the following configuration example. In this example, the DataTurbo share user name is **user01**.

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
  name: mysc
provisioner: csi.huawei.com
parameters:
  backend: dtfs-nas-181
  pool: pool001
  volumeType: fs
  allocType: thin
  authUser: user01
```

KV Cache Configuration Example

If a container uses OceanStor A series storage and the storage supports KV cache, you can refer to the following configuration example for the NFS protocol. KV cache also supports the DTFS protocol. For details about the configuration, see the parameter description.

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
  name: mysc
provisioner: csi.huawei.com
parameters:
  enableKVCache: "true"
  backend: nfs-nas-181
  pool: StoragePool001
  volumeType: fs
  allocType: thin
  authClient: "*"
mountOptions:
  - nfsvers=4.1 # Specify the version 4.1 for NFS mounting.
```

StorageClass Parameters Supported by File Services

Table 5-38 StorageClass configuration parameters

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
metadata.name	User-defined name of a StorageClass object.	Yes	-	Yes	Take Kubernetes v1.22.1 as an example. The value can contain digits, lowercase letters, hyphens (-), and periods (.), and must start and end with a letter or digit.
provisioner	Name of the provisioner.	Yes	csi.huawei.com	Yes	Set this parameter to the driver name set during Huawei CSI installation. The value is the same as that of driverName in the values.yaml file.
reclaimPolicy	Reclamation policy. The following types are supported: • Delete: Resources are automatically reclaimed. • Retain: Resources are manually reclaimed.	No	Delete	Yes	• Delete: When a PV/PVC is deleted, resources on the storage device are also deleted. • Retain: When a PV/PVC is deleted, resources on the storage device are not deleted.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
allowVolumeExpansion	Whether to allow volume expansion. If this parameter is set to true , the capacity of the PV that uses the StorageClass can be expanded.	No	false	Yes	This function can only be used to expand PV capacity but cannot be used to reduce PV capacity.
mountOptions	List of mount parameters, which can be used to specify the parameters of the -o option when the mount command is executed on a host.	No	-	Yes	For details about common parameters in mountOptions , see Table 5-39 . You can also specify other mount parameters.
parameters.backend	Name of the backend where the resource to be created is located. This field must be set if parameters.pool is set.	Conditionally mandatory	-	No	If this parameter is not set, Huawei CSI will randomly select a backend that meets the capacity requirements to create resources. You are advised to specify a backend to ensure that the created resource is located on the expected backend.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.pool	Name of the storage resource pool where the resource to be created is located.	No	-	No	If this parameter is not set, Huawei CSI will select a storage pool with the largest remaining capacity from the selected backend to create resources. You are advised to specify a storage pool to ensure that the created resource is located in the expected storage pool.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.volumeName	Name of the storage resource created by dynamic volume provisioning. You can configure a placeholder to customize the storage resource name. The following placeholders are supported: - PVC namespace: <code>{{ .PVCNamespace }}</code> - PVC name: <code>{{ .PVCName }}</code>	No	-	No	- The value can contain letters, digits, hyphens (-), underscores (_), and periods (.). This parameter cannot be left empty. The length of the generated storage resource name ranges from 1 to 255 characters. - Both the PVC namespace and PVC name must be configured. - To avoid duplicate resource names, the PVC UID is added to the end of the name as a unique identifier by default. Configuration example: PVC namespace: namespace. PVC name: pvc-1. PVC UID: c2fd3f46-bf17-4a7d-b88e-2e3232bae434. volumeName is set to prefix-{{ .PVCNamespace }}-{{ .PVCName }}. The ultimate storage resource name is prefix-namespace_pvc-1-

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
					c2fd3f46bf174a7db88e2e3232bae434.
parameters.volumeType	Type of the volume to be created. The following types are supported: • lun: A LUN is provisioned on the storage side. • fs: A file system is provisioned on the storage side. • dtree: A volume of the dtree type is provisioned on the storage side.	Yes	-	Yes	To use the file service, you must set this parameter to fs .

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.allocType	Allocation type of the volume to be created. The following types are supported: • thin: Not all required space is allocated during creation. Instead, the space is dynamically allocated based on the usage.	No	thin	No	If this parameter is set to thin , the required space is not allocated immediately when a volume is created. Instead, the space is dynamically allocated based on the usage.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.authClient	IP address of the NFS client that can access the volume. This parameter is mandatory when the nfs or nfs+ protocol is used. You can enter the client host name (a full domain name is recommended), client IP address, or client IP address segment.	Conditionally mandatory	-	No	The asterisk (*) can be used to indicate any client. If you are not sure about the IP address of the access client, you are advised to use the asterisk (*) to prevent the client access from being rejected by the storage system. If the client host name is used, you are advised to use the full domain name. The IP addresses can be IPv4 addresses, IPv6 addresses, or a combination of IPv4 and IPv6 addresses. You can enter multiple host names, IP addresses, or IP address segments and separate them with semicolons (;). Example: 192.168.0.10;192.168.0.0/24;myserver1.test
parameters.authUser	DataTurbo user who can access the DataTurbo share. This parameter is mandatory when the DataTurbo (DTFS) protocol is used.	Conditionally mandatory	-	No	You can enter multiple DataTurbo users at a time and separate them with semicolons (;). Example: auth_user1;auth_user2;auth_user3

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.applicationType	Application type name when a file system is created.	No	-	Yes	Log in to DeviceManager and choose Services > File Service > File Systems > Create to obtain the application type name.
parameters.fsPermission	Permission on the directory mounted to a container.	No	-	No	For details about the configuration format, refer to the Linux permission settings, for example, 777 and 755.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.rootSquash	Controls the root permission of the client. The value can be: • root_squash: The client cannot access the storage system as user root. If a client accesses the storage system as user root, the client will be mapped as an anonymous user. • no_root_squash: A client can access the storage system as user root and has the permission of user root.	No	-	No	

Parameter	Description	Mand at ory	Def ault Val ue	Wh eth er the Vol um e Ma na ge me nt Tak es Eff ect	Remarks
parameters.allSquash	Whether to retain the user ID (UID) and group ID (GID) of a shared directory. The value can be: • all_squash: The UID and GID of the shared directory are mapped to anonymous users. • no_all_squash: The UID and GID of the shared directory are retained.	No	-	No	

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.disableVerifyCapacity	Whether to disable volume capacity verification. After this function is disabled, the system will not verify whether the volume capacity is an integer multiple of the sector size. The value can be: • "true": disables volume capacity verification. • "false": enables volume capacity verification. NOTICE When Red Hat OpenShift Virtualization is used to connect to CSI, this parameter must be set to "true".	No	"true"	Yes	The sector size of OceanStor A series is 512 bytes.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.description	Description of the file system to be created. Value type: character string The value contains 0 to 255 characters.	No	Created from Kubernetes CSI	No	
parameters.advancedOptions	Advanced volume creation parameters. The value of the parameter is JSON character strings in dictionary format. A character string is enclosed by single quotation marks and the dictionary key by double quotation marks. Example: '{"CAPACITYTHRESHOLD": 90}'	No	-	No	For details about the supported advanced parameters, see Table 5-40 .

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.enableKVCache	Whether to create a KV cache store. "false": disable "true": enable	No	"false"	No	If this parameter is set to "true" , parameters.applicationType can only be set to NAS_AI_Inference or left empty.
parameters.enableTimeAwareGc	Whether to enable proactive clearing for a KV cache. "false": disable "true": enable	No	"false"	No	No matter whether this parameter is set to "true" or "false" , the background clears the KV cache when the memory store capacity is insufficient. When this parameter is set to "true" , the background clears the expired KV cache.
parameters.gcTimeThreshold	KV cache expiration time. The value ranges from 1 to 3650 days. Example: 1	Conditionally mandatory	-	No	This parameter is mandatory when parameters.enableTimeAwareGc is set to "true" .

Table 5-39 Common parameters in mountOptions

Parameter	Description	Mandatory	Default Value	Remarks
mountOptions.nfsvers	NFS mount option on the host. The following mount option is supported: nfsvers: protocol version for NFS mounting. The value can be 3, 4, 4.0, 4.1, or 4.2 .	No	-	This parameter specified after the -o parameter is optional when the mount command is executed on the host. The value is in list format. If the NFS version is specified for mounting, NFS 3, 4.0, 4.1, and 4.2 protocols are supported (the protocol must be supported and enabled on storage devices). If nfsvers is set to 4 , the latest protocol version NFS 4 may be used for mounting due to different OS configurations, for example, 4.2. If the 4.0 protocol is required, you are advised to set nfsvers to 4.0 .

Parameter	Description	Mandatory	Default Value	Remarks
mountOptions.dn	Domain name of the logical port used for mounting when the DataTurbo(d tfs) protocol is used.	No	WWN of a storage device.	To mount the HyperScale cluster file system, enter the domain name of the HyperScale cluster. The description of the dn parameter is for reference only. For details about other mounting parameters of the DataTurbo protocol, see AI Storage Kit 25.x.x DTFS User Guide .

Table 5-40 Supported advanced volume creation parameters

Parameter	Description	Remarks
CAPACITYTHRESHOLD	Total capacity alarm threshold.	Parameter type: uint64. For details about the default value and value range, see the corresponding storage product manual.

5.2.1.2.2 Dtree

Creating a StorageClass

- Step 1** Prepare a StorageClass configuration file, for example, **mysc.yaml**. For details about the StorageClass configuration, see the following example.
- Step 2** Run the following command to create a StorageClass using the configuration file.

```
kubectl apply -f mysc.yaml
```

Step 3 Run the following command to view the information about the created StorageClass.

```
kubectl get sc mysc
```

The following is an example of the command output.

NAME	PROVISIONER	RECLAIMPOLICY	VOLUMEBINDINGMODE	ALLOWVOLUMEEXPANSION	AGE
mysc	csi.huawei.com	Delete	Immediate	true	8s

----End

NFS Protocol Configuration Example

When a container uses the NFS protocol to connect to dtree resources, refer to the following StorageClass configuration example. In this example, NFS version 4.1 is specified for mounting.

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
  name: mysc
provisioner: csi.huawei.com
parameters:
  backend: nfs-dtree-181
  volumeType: dtree
  allocType: thin
  authClient: ""
mountOptions:
  - nfsvers=4.1 # Specify the version 4.1 for NFS mounting.
```

DataTurbo Protocol Configuration Example

If a container uses OceanStor A series storage and the storage supports DataTurbo-based access, you can refer to the following configuration example. In this example, the DataTurbo share user name is **user01**.

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
  name: mysc
provisioner: csi.huawei.com
parameters:
  backend: dtfs-nas-181
  volumeType: dtree
  allocType: thin
  authUser: user01
```

Storage Class Parameters Supported by Dtrees

Table 5-41 StorageClass configuration parameters

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
metadata.name	User-defined name of a StorageClass object.	Yes	-	Yes	Take Kubernetes v1.22.1 as an example. The value can contain digits, lowercase letters, hyphens (-), and periods (.), and must start and end with a letter or digit.
provisioner	Name of the provisioner.	Yes	csi.huawei.com	Yes	Set this parameter to the driver name set during Huawei CSI installation. The value is the same as that of driverName in the values.yaml file.
reclaimPolicy	Reclamation policy. The following types are supported: • Delete: Resources are automatically reclaimed. • Retain: Resources are manually reclaimed.	No	Delete	Yes	• Delete: When a PV/PVC is deleted, resources on the storage device are also deleted. • Retain: When a PV/PVC is deleted, resources on the storage device are not deleted.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
allowVolumeExpansion	Whether to allow volume expansion. If this parameter is set to true , the capacity of the PV that uses the StorageClass can be expanded.	No	false	Yes	This function can only be used to expand PV capacity but cannot be used to reduce PV capacity.
mountOptions	List of mount parameters, which can be used to specify the parameters of the -o option when the mount command is executed on a host.	No	-	Yes	For details about common parameters in mountOptions , see Table 5-42 . You can also specify other mount parameters.
parameters.backend	Name of the backend where the resource to be created is located. This field must be set if parameters.pool is set.	Conditionally mandatory	-	No	If this parameter is not set, Huawei CSI will randomly select a backend that meets the capacity requirements to create resources. You are advised to specify a backend to ensure that the created resource is located on the expected backend.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.parentname	Name of a file system on the current storage device. Dtree is created in the file system.	Conditionally mandatory	-	Yes	This parameter is mandatory when parentname is not configured for the backend. If parentname is configured only in the StorageClass but not configured in the storage backend, set CSIDriverObject.attachRequired to true during CSI installation.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.volumeName	Name of the storage resource created by dynamic volume provisioning. You can configure a placeholder to customize the storage resource name. The following placeholders are supported: - PVC namespace: <code>{{ .PVCNamespace }}</code> - PVC name: <code>{{ .PVCName }}</code>	No	-	No	- The value can contain letters, digits, hyphens (-), underscores (_), and periods (.). This parameter cannot be left empty. The length of the generated storage resource name ranges from 1 to 255 characters. - Both the PVC namespace and PVC name must be configured. - To avoid duplicate resource names, the PVC UID is added to the end of the name as a unique identifier by default. Configuration example: PVC namespace: namespace. PVC name: pvc-1. PVC UID: c2fd3f46-bf17-4a7d-b88e-2e3232bae434. volumeName is set to prefix-{{ .PVCNamespace }}_{{ .PVCName }}. The ultimate storage resource name is prefix-namespace_pvc-1-

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
					c2fd3f46bf174a7db88e2e3232bae434.
parameters.volumeType	Type of the volume to be created. The following types are supported: • lun: A LUN is provisioned on the storage side. • fs: A file system is provisioned on the storage side. • dtree: A volume of the dtree type is provisioned on the storage side.	Yes	-	Yes	When dtree is used, the value must be dtree .

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.alllocType	Allocation type of the volume to be created. The following types are supported: • thin: Not all required space is allocated during creation. Instead, the space is dynamically allocated based on the usage.	No	thin	No	If this parameter is set to thin , the required space is not allocated immediately when a volume is created. Instead, the space is dynamically allocated based on the usage.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.authClient	IP address of the NFS client that can access the volume. This parameter is mandatory when the nfs or nfs+ protocol is used. You can enter the client host name (a full domain name is recommended), client IP address, or client IP address segment.	Conditionally mandatory	-	No	The asterisk (*) can be used to indicate any client. If you are not sure about the IP address of the access client, you are advised to use the asterisk (*) to prevent the client access from being rejected by the storage system. If the client host name is used, you are advised to use the full domain name. The IP addresses can be IPv4 addresses, IPv6 addresses, or a combination of IPv4 and IPv6 addresses. You can enter multiple host names, IP addresses, or IP address segments and separate them with semicolons (;). Example: 192.168.0.10;192.168.0.0/24;myserver1.test
parameters.authUser	DataTurbo user who can access the DataTurbo share. This parameter is mandatory when the DataTurbo (DTFS) protocol is used.	Conditionally mandatory	-	No	You can enter multiple DataTurbo users at a time and separate them with semicolons (;). Example: auth_user1;auth_user2;auth_user3

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.fsPermission	Permission on the directory mounted to a container.	No	-	No	For details about the configuration format, refer to the Linux permission settings, for example, 777 and 755.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.rootSquash	Controls the root permission of the client. The value can be: • root_squash: The client cannot access the storage system as user root. If a client accesses the storage system as user root, the client will be mapped as an anonymous user. • no_root_squash: A client can access the storage system as user root and has the permission of user root.	No	no_root_squash	No	

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.allSquash	Whether to retain the user ID (UID) and group ID (GID) of a shared directory. The value can be: • all_squash: The UID and GID of the shared directory are mapped to anonymous users. • no_all_squash: The UID and GID of the shared directory are retained.	No	no_all_squash	No	

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.disableVerifyCapacity	Whether to disable volume capacity verification. After this function is disabled, the system will not verify whether the volume capacity is an integer multiple of the sector size. The value can be: • "true": disables volume capacity verification. • "false": enables volume capacity verification. NOTICE When Red Hat OpenShift Virtualization is used to connect to CSI, this parameter must be set to "true".	No	"true"	Yes	The sector size of OceanStor A series is 512 bytes.

Table 5-42 Common parameters in mountOptions

Parameter	Description	Mandatory	Default Value	Remarks
mountOptions.nfsvers	NFS mount option on the host. The following mount option is supported: nfsvers : protocol version for NFS mounting. The value can be 3 , 4 , 4.0 , 4.1 , or 4.2 .	No	-	This parameter specified after the -o parameter is optional when the mount command is executed on the host. The value is in list format. If the NFS version is specified for mounting, NFS 3, 4.0, 4.1, and 4.2 protocols are supported (the protocol must be supported and enabled on storage devices). If nfsvers is set to 4 , the latest protocol version NFS 4 may be used for mounting due to different OS configurations, for example, 4.2. If the 4.0 protocol is required, you are advised to set nfsvers to 4.0 .

Parameter	Description	Mandatory	Default Value	Remarks
mountOptions.proto	Transmission protocol used for NFS mounting. The value can be rdma .	No	-	
mountOptions.port	Protocol port used for NFS mounting.	Conditionally mandatory	-	If the transmission protocol is rdma , set this parameter to 20049 .
mountOptions.dn	Domain name of the logical port used for mounting when the DataTurbo(dtfs) protocol is used.	No	WWN of a storage device.	To mount the HyperScale cluster file system, enter the domain name of the HyperScale cluster. The description of the dn parameter is for reference only. For details about other mounting parameters of the DataTurbo protocol, see AI Storage Kit 25.x.x DTFS User Guide .

5.2.1.3 iMaster DME (Interconnected with an OceanStor A800 Cluster)

5.2.1.3.1 File System

Creating a StorageClass

Step 1 Prepare a StorageClass configuration file, for example, **myisc.yaml**. For details about the StorageClass configuration, see the following example.

Step 2 Run the following command to create a StorageClass using the configuration file.

```
kubectl apply -f mysc.yaml
```

Step 3 Run the following command to view the information about the created StorageClass.

```
kubectl get sc mysc
```

The following is an example of the command output.

NAME	PROVISIONER	RECLAIMPOLICY	VOLUMEBINDINGMODE	ALLOWVOLUMEEXPANSION	AGE
mysc	csi.huawei.com	Delete	Immediate	true	8s

----End

NFS Protocol Configuration Example

When a container uses the NFS protocol to connect to global file system resources, refer to the following StorageClass configuration example. In this example, NFS version 3 is specified for mounting.

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
  name: mysc
provisioner: csi.huawei.com
parameters:
  backend: nfs-nas-181
  pool: StoragePool001
  volumeType: fs
  allocType: thin
  authClient: ""
mountOptions:
  - nfsvers=3 # Specify the version 3 for NFS mounting.
```

DataTurbo Protocol Configuration Example

When a container uses the DataTurbo protocol to connect to global file system resources, refer to the following configuration example. In this example, the DataTurbo share user name is **user01**.

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
  name: mysc
provisioner: csi.huawei.com
parameters:
  backend: dtfs-nas-181
  pool: pool001
  volumeType: fs
  allocType: thin
  authUser: user01
mountOptions:
  - dn=xxx # Specify the domain name of the logical port as xxx.
```

StorageClass Parameters Supported by File Services

Table 5-43 StorageClass configuration parameters

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
metadata.name	User-defined name of a StorageClass object.	Yes	-	Yes	Take Kubernetes v1.22.1 as an example. The value can contain digits, lowercase letters, hyphens (-), and periods (.), and must start and end with a letter or digit.
provisioner	Name of the provisioner.	Yes	csi.huawei.com	Yes	Set this parameter to the driver name set during Huawei CSI installation. The value is the same as that of driverName in the values.yaml file.
reclaimPolicy	Reclamation policy. The following types are supported: • Delete: Resources are automatically reclaimed. • Retain: Resources are manually reclaimed.	No	Delete	Yes	• Delete: When a PV/PVC is deleted, resources on the storage device are also deleted. • Retain: When a PV/PVC is deleted, resources on the storage device are not deleted.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
allowVolumeExpansion	Whether to allow volume expansion. If this parameter is set to true , the capacity of the PV that uses the StorageClass can be expanded.	No	false	Yes	This function can only be used to expand PV capacity but cannot be used to reduce PV capacity.
mountOptions	List of mount parameters, which can be used to specify the parameters of the -o option when the mount command is executed on a host.	No	-	Yes	For details about common parameters in mountOptions , see Table 5-44 . You can also specify other mount parameters.
parameters.backend	Name of the backend where the resource to be created is located. This field must be set if parameters.pool is set.	Conditionally mandatory	-	No	If this parameter is not set, Huawei CSI will randomly select a backend that meets the capacity requirements to create resources. You are advised to specify a backend to ensure that the created resource is located on the expected backend.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.pool	Name of the storage resource pool where the resource to be created is located.	No	-	No	If this parameter is not set, Huawei CSI will randomly select a storage pool with the largest remaining capacity from the selected backend to create resources. You are advised to specify a storage pool to ensure that the created resource is located in the expected storage pool.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.volumeName	Name of the storage resource created by dynamic volume provisioning. You can configure a placeholder to customize the storage resource name. The following placeholders are supported: - PVC namespace: <code>{{ .PVCNamespace }}</code> - PVC name: <code>{{ .PVCName }}</code>	No	-	No	- The value can contain letters, digits, hyphens (-), underscores (_), and periods (.). This parameter cannot be left empty. The length of the generated storage resource name ranges from 1 to 255 characters. - Both the PVC namespace and PVC name must be configured. - To avoid duplicate resource names, the PVC UID is added to the end of the name as a unique identifier by default. Configuration example: PVC namespace: namespace. PVC name: pvc-1. PVC UID: c2fd3f46-bf17-4a7d-b88e-2e3232bae434. volumeName is set to prefix-{{ .PVCNamespace }}_{{ .PVCName }}. The ultimate storage resource name is prefix-namespace_pvc-1-

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
					c2fd3f46bf174a7db88e2e3232bae434.
parameters.volumeType	Type of the volume to be created. The following types are supported: • lun: A LUN is provisioned on the storage side. • fs: A file system is provisioned on the storage side. • dtree: A volume of the dtree type is provisioned on the storage side.	Yes	-	Yes	To use the file service, you must set this parameter to fs .

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.allocType	Allocation type of the volume to be created. The following types are supported: • thin: Not all required space is allocated during creation. Instead, the space is dynamically allocated based on the usage.	No	thin	No	If this parameter is set to thin , the required space is not allocated immediately when a volume is created. Instead, the space is dynamically allocated based on the usage.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.authClient	IP address of the NFS client that can access the volume. This parameter is mandatory when the nfs or nfs+ protocol is used. You can enter the client host name (a full domain name is recommended), client IP address, or client IP address segment.	Conditionally mandatory	-	No	The asterisk (*) can be used to indicate any client. If you are not sure about the IP address of the access client, you are advised to use the asterisk (*) to prevent the client access from being rejected by the storage system. If the client host name is used, you are advised to use the full domain name. The IP addresses can be IPv4 addresses, IPv6 addresses, or a combination of IPv4 and IPv6 addresses. You can enter multiple host names, IP addresses, or IP address segments and separate them with semicolons (;). Example: 192.168.0.10;192.168.0.0/24;myserver1.test
parameters.authUser	DataTurbo user who can access the DataTurbo share. This parameter is mandatory when the DataTurbo(dtfs) protocol is used.	Conditionally mandatory	-	No	You can enter multiple DataTurbo users at a time and separate them with semicolons (;). Example: auth_user1;auth_user2;auth_user3

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.rootSquash	Controls the root permission of the client. The value can be: • root_squash: The client cannot access the storage system as user root. If a client accesses the storage system as user root, the client will be mapped as an anonymous user. • no_root_squash: A client can access the storage system as user root and has the permission of user root.	No	-	No	

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.allSquash	Whether to retain the user ID (UID) and group ID (GID) of a shared directory. The value can be: • all_squash: The UID and GID of the shared directory are mapped to anonymous users. • no_all_squash: The UID and GID of the shared directory are retained.	No	-	No	

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.disableVerifyCapacity	Whether to disable volume capacity verification. After this function is disabled, the system will not verify whether the volume capacity is an integer multiple of the sector size. The value can be: • "true": disables volume capacity verification. • "false": enables volume capacity verification. NOTICE When Red Hat OpenShift Virtualization is used to connect to CSI, this parameter must be set to true.	No	"true"	Yes	The sector size of OceanStor A series is 512 bytes.

Parameter	Description	Mand at ory	Def ault Val ue	Wh eth er the Vol um e Ma na ge me nt Tak es Eff ect	Remarks
parameters.d escription	Description of the file system to be created. Value type: character string The value contains 0 to 255 characters.	No	Crea ted fro m Kub erne tes CSI	No	

Table 5-44 Common parameters in mountOptions

Parameter	Description	Mandatory	Default Value	Remarks
mountOptions.nfsvers	NFS mount option on the host. The following mount option is supported: nfsvers: protocol version for NFS mounting. The value can be 3.	No	-	This parameter specified after the -o parameter is optional when the mount command is executed on the host. The value is in list format. If the NFS version is specified for mounting, the NFS 3 protocol is supported (the protocol must be supported and enabled on storage devices).

Parameter	Description	Mandatory	Default Value	Remarks
mountOptions.dn	Domain name of the logical port used for mounting when the DataTurbo(dvfs) protocol is used.	Yes	-	To mount the HyperScale cluster file system, enter the domain name of the HyperScale cluster. The description of the dn parameter is for reference only. For details about other mounting parameters of the DataTurbo protocol, see AI Storage Kit 25.x.x DTFS User Guide .

5.2.1.3.2 Local File System

Creating a StorageClass

- Step 1** Prepare a StorageClass configuration file, for example, **mysc.yaml**. For details about the StorageClass configuration, see the following example.
- Step 2** Run the following command to create a StorageClass using the configuration file.
- ```
kubectl apply -f mysc.yaml
```
- Step 3** Run the following command to view the information about the created StorageClass.
- ```
kubectl get sc mysc
```

The following is an example of the command output.

NAME	PROVISIONER	RECLAIMPOLICY	VOLUMEBINDINGMODE	ALLOWVOLUMEEXPANSION	AGE
mysc	csi.huawei.com	Delete	Immediate	true	8s

----End

KV Cache Configuration Example

If the container uses KV cache to connect to the iMaster DME local file system and the storage system supports KV cache, refer to the following configuration example.

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
  name: kvcache-nas-sc
provisioner: csi.huawei.com
reclaimPolicy: Delete
parameters:
  backend: dme-a800-backend
  pool: StoragePool001
  zoneVstoreName: System_vstore
  volumeType: fs
  enableKVCache: "true"
  enableTimeAwareGc: "true"
  gcTimeThreshold: "30"
  disableVerifyCapacity: "true"
mountOptions:
  - nfsvers=3 # Specify the version 3 for NFS mounting.
```

Table 5-45 StorageClass configuration parameters

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
metadata.name	User-defined name of a StorageClass object.	Yes	-	Yes	Take Kubernetes v1.22.1 as an example. The value can contain digits, lowercase letters, hyphens (-), and periods (.), and must start and end with a letter or digit.
provisioner	Name of the provisioner.	Yes	csi.huawei.com	Yes	Set this parameter to the driver name set during Huawei CSI installation. The value is the same as that of driverName in the values.yaml file.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
reclaimPolicy	Reclamation policy. The following types are supported: Delete: Resources are automatically reclaimed. Retain: Resources are manually reclaimed.	No	Delete	Yes	Delete: When a PV/PVC is deleted, resources on the storage device are also deleted. Retain: When a PV/PVC is deleted, resources on the storage device are not deleted. Note: When a KV cache resource is deleted, the file system share is also deleted.
allowVolumeExpansion	Whether to allow volume expansion. If this parameter is set to true , the capacity of the PV that uses the StorageClass can be expanded.	No	false	Yes	This function can only be used to expand PV capacity but cannot be used to reduce PV capacity.
mountOptions	List of mount parameters, which can be used to specify the parameters of the -o option when the mount command is executed on a host.	No	-	Yes	For details about common parameters in mountOptions , see Table 5-46 . You can also specify other mount parameters.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.backend	Name of the backend where the resource to be created is located. This field must be set if parameters.pool is set.	Yes	-	No	When creating a KV cache in the local file system of iMaster DME, you need to specify the backend to be used.
parameters.pool	Name of the storage resource pool where the resource to be created is located.	No	-	No	If this parameter is not set, Huawei CSI will select a storage pool with the largest remaining capacity from the selected backend to create resources. You are advised to specify a storage pool to ensure that the created resource is located in the expected storage pool.
parameters.zoneVstoreName	vStore name of the storage device.	No	System_vStore	No	This parameter is valid only when a zone SN is specified. On the iMaster DME, choose Infrastructure > Storage Devices > System > vStores . Obtain the name of a non-global vStore in the corresponding zone.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.volumeName	Name of the storage resource created by dynamic volume provisioning. You can configure a placeholder to customize the storage resource name. The following placeholders are supported: PVC namespace: <code>{{ .PVCNamespace }}</code> ; PVC name: <code>{{ .PVCName }}</code>	No	-	No	The value can contain letters, digits, underscores (<code>_</code>), and hyphens (<code>-</code>). It cannot be left empty. The name length of the generated storage resource ranges from 1 to 255 characters. Both the PVC namespace and PVC name must be configured. To avoid duplicate resource names, the PVC UID is added to the end of the name as a unique identifier by default.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.volumeType	Type of the volume to be created. The following types are supported: • lun: A LUN is provisioned on the storage side. • fs: A file system is provisioned on the storage side. • dtree: A volume of the dtree type is provisioned on the storage side.	Yes	-	Yes	To use the file service, you must set this parameter to fs .

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.disableVerifyCapacity	Whether to disable volume capacity verification. After this function is disabled, the system will not verify whether the volume capacity is an integer multiple of the sector size. The value can be "true" (disables volume capacity verification) or "false" (enables volume capacity verification). When Red Hat OpenShift Virtualization is used to connect to CSI, this parameter must be set to "true" .	No	"true"	Yes	The sector size of OceanStor A series is 512 bytes.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.description	Description of the file system to be created. Value type: string; length limit: 1 to 255 characters.	No	Created from Kubernetes CSI	No	
parameters.enableKVCache	Whether to create a KV cache store. The options include "false" (disabled) and "true" (enabled).	Yes	"false"	No	This parameter must be set to true when a single-zone resource is created in the current scenario.
parameters.enableTimeAwareGc	Whether to enable proactive clearing for a KV cache. The options include "false" (disabled) and "true" (enabled).	No	"false"	No	No matter whether this parameter is set to "true" or "false" , the background clears the KV cache when the memory store capacity is insufficient. When this parameter is set to "true" , the background clears the expired KV cache.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.gcTimeThreshold	KV cache expiration time. The value ranges from 1 to 3650 days. Example: 1	Conditionally mandatory	-	No	This parameter is mandatory when parameters.enableTimeAwareGc is set to "true" .

Table 5-46 Common parameters in mountOptions

Parameter	Description	Mandatory	Default Value	Remarks
mountOptions.nfsvers	NFS mount option on the host. The following mount option is supported: nfsvers : protocol version for NFS mounting. The value can be 3, 4, 4.0, 4.1, or 4.2 .	No	-	This parameter is optional after the -o parameter when the mount command is executed on the host. The value is in list format. If the NFS version is specified for mounting, NFS 3, 4.0, 4.1, and 4.2 protocols are supported (the protocol must be supported and enabled on storage devices). If nfsvers is set to 4 , the latest protocol version NFS 4 may be used for mounting due to different OS configurations, for example, 4.2. If the 4.0 protocol is required, you are advised to set nfsvers to 4.0 .

5.2.1.3.3 Dtree

Creating a StorageClass

Step 1 Prepare a StorageClass configuration file, for example, **mysc.yaml**. For details about the StorageClass configuration, see the following example.

Step 2 Run the following command to create a StorageClass using the configuration file.

```
kubectl apply -f mysc.yaml
```

Step 3 Run the following command to view the information about the created StorageClass.

```
kubectl get sc mysc
```

The following is an example of the command output.

NAME	PROVISIONER	RECLAIMPOLICY	VOLUMEBINDINGMODE	ALLOWVOLUMEEXPANSION	AGE
mysc	csi.huawei.com	Delete	Immediate	true	8s

----End

NFS Protocol Configuration Example

When a container uses the NFS protocol to connect to dtree resources, refer to the following StorageClass configuration example. In this example, NFS version 3 is specified for mounting.

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
  name: mysc
provisioner: csi.huawei.com
parameters:
  backend: nfs-dtree-181
  volumeType: dtree
  allocType: thin
  authClient: ""
mountOptions:
  - nfsvers=3 # Specify the version 3 for NFS mounting.
```

DataTurbo Protocol Configuration Example

When a container uses the DataTurbo protocol to connect to dtree resources, refer to the following configuration example. In this example, the DataTurbo share user name is **user01**.

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
  name: mysc
provisioner: csi.huawei.com
parameters:
  backend: dtfs-nas-181
  volumeType: dtree
  allocType: thin
  authUser: user01
```

StorageClass Parameters Supported by Dtrees

Table 5-47 StorageClass configuration parameters

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
metadata.name	User-defined name of a StorageClass object.	Yes	-	Yes	Take Kubernetes v1.22.1 as an example. The value can contain digits, lowercase letters, hyphens (-), and periods (.), and must start and end with a letter or digit.
provisioner	Name of the provisioner.	Yes	csi.huawei.com	Yes	Set this parameter to the driver name set during Huawei CSI installation. The value is the same as that of driverName in the values.yaml file.
reclaimPolicy	Reclamation policy. The following types are supported: • Delete: Resources are automatically reclaimed. • Retain: Resources are manually reclaimed.	No	Delete	Yes	• Delete: When a PV/PVC is deleted, resources on the storage device are also deleted. • Retain: When a PV/PVC is deleted, resources on the storage device are not deleted.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
allowVolumeExpansion	Whether to allow volume expansion. If this parameter is set to true , the capacity of the PV that uses the StorageClass can be expanded.	No	false	Yes	This function can only be used to expand PV capacity but cannot be used to reduce PV capacity.
mountOptions	List of mount parameters, which can be used to specify the parameters of the -o option when the mount command is executed on a host.	No	-	Yes	For details about common parameters in mountOptions , see Table 5-48 . You can also specify other mount parameters.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.backend	Name of the backend where the resource to be created is located.	Conditionally mandatory	-	No	If this parameter is not set, Huawei CSI will randomly select a backend that meets the capacity requirements to create resources. You are advised to specify a backend to ensure that the created resource is located on the expected backend. This parameter is mandatory if parameters.parentname is set.
parameters.parentname	Name of a file system on the current storage device. Dtree is created in the file system.	Conditionally mandatory	-	Yes	This parameter is mandatory when parentname is not set for the backend. If parentname is configured only in the StorageClass but not configured in the storage backend, set CSIDriverObject.attachRequired to true during CSI installation.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.volumeName	Name of the storage resource created by dynamic volume provisioning. You can configure a placeholder to customize the storage resource name. The following placeholders are supported: - PVC namespace: <code>{{ .PVCNamespace }}</code> - PVC name: <code>{{ .PVCName }}</code>	No	-	No	- The value can contain letters, digits, hyphens (-), underscores (_), and periods (.). This parameter cannot be left empty. The length of the generated storage resource name ranges from 1 to 255 characters. - Both the PVC namespace and PVC name must be configured. - To avoid duplicate resource names, the PVC UID is added to the end of the name as a unique identifier by default. Configuration example: PVC namespace: namespace. PVC name: pvc-1. PVC UID: c2fd3f46-bf17-4a7d-b88e-2e3232bae434. volumeName is set to prefix-{{ .PVCNamespace }}_{{ .PVCName }}. The ultimate storage resource name is prefix-namespace_pvc-1-

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
					c2fd3f46bf174a7db88e2e3232bae434.
parameters.volumeType	Type of the volume to be created. The following types are supported: • lun: A LUN is provisioned on the storage side. • fs: A file system is provisioned on the storage side. • dtree: A volume of the dtree type is provisioned on the storage side.	Yes	-	Yes	dtree : A volume of the dtree type is provisioned on the storage side.

Parameter	Description	Mand at ory	Def ault Val ue	Wh eth er the Vol um e Ma na ge me nt Tak es Eff ect	Remarks
parameters.allocType	Allocation type of the volume to be created. The following types are supported: • thin: Not all required space is allocated during creation. Instead, the space is dynamically allocated based on the usage.	No	thin	No	If this parameter is set to thin , the required space is not allocated immediately when a volume is created. Instead, the space is dynamically allocated based on the usage.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.authClient	IP address of the NFS client that can access the volume. This parameter is mandatory when the nfs or nfs+ protocol is used. You can enter the client host name (a full domain name is recommended), client IP address, or client IP address segment.	Conditionally mandatory	-	No	The asterisk (*) can be used to indicate any client. If you are not sure about the IP address of the access client, you are advised to use the asterisk (*) to prevent the client access from being rejected by the storage system. If the client host name is used, you are advised to use the full domain name. The IP addresses can be IPv4 addresses, IPv6 addresses, or a combination of IPv4 and IPv6 addresses. You can enter multiple host names, IP addresses, or IP address segments and separate them with semicolons (;). Example: 192.168.0.10;192.168.0.0/24;myserver1.test
parameters.authUser	DataTurbo user who can access the DataTurbo share. This parameter is mandatory when the DataTurbo(dtfs) protocol is used.	Conditionally mandatory	-	No	You can enter multiple DataTurbo users at a time and separate them with semicolons (;). Example: auth_user1;auth_user2;auth_user3

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.fsPermission	Permission on the directory mounted to a container.	No	-	No	For details about the configuration format, refer to the Linux permission settings, for example, 777 and 755.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.rootSquash	Controls the root permission of the client. The value can be: • root_squash: The client cannot access the storage system as user root. If a client accesses the storage system as user root, the client will be mapped as an anonymous user. • no_root_squash: A client can access the storage system as user root and has the permission of user root.	No	no_root_squash	No	

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.allSquash	Whether to retain the user ID (UID) and group ID (GID) of a shared directory. The value can be: • all_squash: The UID and GID of the shared directory are mapped to anonymous users. • no_all_squash: The UID and GID of the shared directory are retained.	No	no_all_squash	No	

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.disableVerifyCapacity	Whether to disable volume capacity verification. After this function is disabled, the system will not verify whether the volume capacity is an integer multiple of the sector size. The value can be: • "true": disables volume capacity verification. • "false": enables volume capacity verification. NOTICE When Red Hat OpenShift Virtualization is used to connect to CSI, this parameter must be set to "true".	No	"true"	Yes	For iMaster DME dtrees, the sector size is 1 KB.

Table 5-48 Common parameters in mountOptions

Parameter	Description	Mandatory	Default Value	Remarks
mountOptions.nfsvers	NFS mount option on the host. The following mount option is supported: nfsvers: protocol version for NFS mounting. The value can be 3 .	No	-	This parameter is optional after the -o parameter when the mount command is executed on the host. The value is in list format. If the NFS version is specified for mounting, the NFS 3 protocol is supported (the protocol must be supported and enabled on storage devices).
mountOptions.proto	Transmission protocol used for NFS mounting. The value can be rdma .	No	-	
mountOptions.port	Protocol port used for NFS mounting.	Conditionally mandatory	-	If the transmission protocol is rdma , set this parameter to 20049 .

Parameter	Description	Mandatory	Default Value	Remarks
mountOptions.dn	Domain name of the logical port used for mounting when the DataTurbo(dtfs) protocol is used.	No	WWN of a storage device.	To mount the HyperScale cluster file system, enter the domain name of the HyperScale cluster. The description of the dn parameter is for reference only. For details about other mounting parameters of the DataTurbo protocol, see AI Storage Kit 25.x.x DTFS User Guide .

5.2.1.4 Mass Storage (OceanStor Pacific/FusionStorage Block Series)

5.2.1.4.1 File System

Creating a StorageClass

Step 1 Prepare a StorageClass configuration file, for example, **mysc.yaml**. For details about the StorageClass configuration, see the following example.

Step 2 Run the following command to create a StorageClass using the configuration file.

```
kubectl apply -f mysc.yaml
```

Step 3 Run the following command to view the information about the created StorageClass.

```
kubectl get sc mysc
```

The following is an example of the command output.

NAME	PROVISIONER	RECLAIMPOLICY	VOLUMEBINDINGMODE	ALLOWVOLUMEEXPANSION	AGE
mysc	csi.huawei.com	Delete	Immediate	true	8s

----End

NFS Protocol Configuration Example

When a container uses the NFS protocol to connect to file system resources, refer to the following StorageClass configuration example. In this example, NFS version 4.1 is specified for mounting.

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
  name: mysc
provisioner: csi.huawei.com
parameters:
  backend: nfs-nas-181
  pool: pool001
  volumeType: fs
  allocType: thin
  authClient: "*"
mountOptions:
  - nfsvers=4.1 # Specify the version 4.1 for NFS mounting.
```

DPC Protocol Configuration Example

When the storage supports access using the DPC protocol, you can configure the mount parameters for DPC access in the StorageClass. In this example, **acl** is used as the authentication parameter for mounting, and **cnflush** is used to set the asynchronous disk flushing mode.

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
  name: mysc
provisioner: csi.huawei.com
parameters:
  backend: nfs-dpc-101
  pool: pool001
  volumeType: fs
  allocType: thin
mountOptions:
  - acl # Set the authentication parameter.
  - cnflush # Set the asynchronous disk flushing mode.
```

StorageClass Parameters Supported by File Systems

Table 5-49 StorageClass configuration parameters

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
metadata.name	User-defined name of a StorageClass object.	Yes	-	Yes	Take Kubernetes v1.22.1 as an example. The value can contain digits, lowercase letters, hyphens (-), and periods (.), and must start and end with a letter or digit.
provisioner	Name of the provisioner.	Yes	csi.huawei.com	Yes	Set this parameter to the driver name set during Huawei CSI installation. The value is the same as that of driverName in the values.yaml file.
reclaimPolicy	Reclamation policy. The following types are supported: • Delete: Resources are automatically reclaimed. • Retain: Resources are manually reclaimed.	No	Delete	Yes	• Delete: When a PV/PVC is deleted, resources on the storage device are also deleted. • Retain: When a PV/PVC is deleted, resources on the storage device are not deleted.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
allowVolumeExpansion	Whether to allow volume expansion. If this parameter is set to true , the capacity of the PV that uses the StorageClass can be expanded.	No	false	Yes	This function can only be used to expand PV capacity but cannot be used to reduce PV capacity.
mountOptions	List of mount parameters, which can be used to specify the parameters of the -o option when the mount command is executed on a host.	No	-	Yes	For details about common parameters in mountOptions , see Table 5-50 . You can also specify other mount parameters.
parameters.backend	Name of the backend where the resource to be created is located. This field must be set if parameters.pool is set.	Conditionally mandatory	-	No	If this parameter is not set, Huawei CSI will randomly select a backend that meets the capacity requirements to create resources. You are advised to specify a backend to ensure that the created resource is located on the expected backend.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.pool	Name of the storage resource pool where the resource to be created is located.	No	-	No	If this parameter is not set, Huawei CSI will select a storage pool with the largest remaining capacity from the selected backend to create resources. You are advised to specify a storage pool to ensure that the created resource is located in the expected storage pool.
parameters.volumeType	Type of the volume to be created. The following types are supported: • lun: A LUN is provisioned on the storage side. • fs: A file system is provisioned on the storage side. • dtree: A volume of the dtree type is provisioned on the storage side.	Yes	-	Yes	The value is fixed at fs .

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.alllocType	Allocation type of the volume to be created. The following types are supported: • thin: Not all required space is allocated during creation. Instead, the space is dynamically allocated based on the usage.	No	thin	No	If this parameter is set to thin , the required space is not allocated immediately when a volume is created. Instead, the space is dynamically allocated based on the usage.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.authClient	IP address of the NFS client that can access the volume. This parameter is mandatory when the NFS protocol is used. You can enter the client host name (a full domain name is recommended), client IP address, or client IP address segment.	Conditionally mandatory	-	No	The asterisk (*) can be used to indicate any client. If you are not sure about the IP address of the access client, you are advised to use the asterisk (*) to prevent the client access from being rejected by the storage system. If the client host name is used, you are advised to use the full domain name. The IP addresses can be IPv4 addresses, IPv6 addresses, or a combination of IPv4 and IPv6 addresses. You can enter multiple host names, IP addresses, or IP address segments and separate them with semicolons (;). Example: 192.168.0.10;192.168.0.0/24;myserver1.test

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.storageQuota	File system quota settings. The value of the parameter is JSON character strings in dictionary format. A character string is enclosed by single quotation marks and the dictionary key by double quotation marks. Example: <code>'{"spaceQuota": "softQuota", "gracePeriod": 100}'</code>	No	-	No	For details about the supported quota configurations, see Table 5-52 .

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.qos	QoS settings of the file system on the storage side of the PV. The value of the parameter is JSON character strings in dictionary format. A character string is enclosed by single quotation marks and the dictionary key by double quotation marks. Example: '{"maxMBPS": 999, "maxIOPS": 999}'	No	-	No	For details about the supported QoS configurations, see Table 5-51 .
parameters.fsPermission	Permission on the directory mounted to a container.	No	-	No	For details about the configuration format, refer to the Linux permission settings, for example, 777 and 755.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.rootSquash	Controls the root permission of the client. The value can be: • root_squash: The client cannot access the storage system as user root. If a client accesses the storage system as user root, the client will be mapped as an anonymous user. • no_root_squash: A client can access the storage system as user root and has the permission of user root.	No	-	No	

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.allSquash	Whether to retain the user ID (UID) and group ID (GID) of a shared directory. The value can be: • all_squash: The UID and GID of the shared directory are mapped to anonymous users. • no_all_squash: The UID and GID of the shared directory are retained.	No	-	No	

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.snapshotDirectoryVisibility	Whether the snapshot directory is visible. The value can be: • visible: The snapshot directory is visible. • invisible: The snapshot directory is invisible.	No	-	No	

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.disableVerifyCapacity	Whether to disable volume capacity verification. After this function is disabled, the system will not verify whether the volume capacity is an integer multiple of the sector size. The value can be: • "true": disables volume capacity verification. • "false": enables volume capacity verification. NOTICE When Red Hat OpenShift Virtualization is used to connect to CSI, this parameter must be set to "true".	No	"true"	No	For OceanStor Pacific NAS, the sector size is 1 KB.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.advancedOptions	Advanced volume creation parameters. The value of the parameter is JSON character strings in dictionary format. A character string is enclosed by single quotation marks and the dictionary key by double quotation marks. Example: <code>'{"is_trash_enable": true}'</code>	No	-	No	For details about the supported advanced parameters, see Table 5-53 .

Table 5-50 Common parameters in mountOptions

Parameter	Description	Mandatory	Default Value	Remarks
mountOptions.nfsvers	NFS mount option on the host. The following mount option is supported: nfsvers: protocol version for NFS mounting. The value can be 3 or 4.1 .	No	-	This parameter is optional after the -o parameter when the mount command is executed on the host. The value is in list format. If the NFS version is specified for mounting, NFS 3 and 4.1 protocols are supported (the protocol must be supported and enabled on storage devices).
mountOptions.proto	Transmission protocol used for NFS mounting. The value can be rdma .	No	-	This example applies to 8.2.0 or later.
mountOptions.port	Protocol port used for NFS mounting.	Conditionally mandatory	-	If the transmission protocol is rdma , set this parameter to 20049 .

Parameter	Description	Mandatory	Default Value	Remarks
mountOptions.acl	The DPC namespace supports the ACL function. The DPC client supports POSIX ACL, NFSv4 ACL, and NT ACL authentication.	No	-	The descriptions of acl , aclonlyposix , cnflush , and cflush are for reference only. For details about the parameters, see OceanStor Pacific Series Product Documentation and choose Configuration > Basic Service Configuration Guide for File > Configuring Basic Services (DPC Scenario) > Accessing a DPC Share on a Client > Step 2 .

Parameter	Description	Mandatory	Default Value	Remarks
mountOptions.aclonlyposix	The DPC namespace supports POSIX ACL, and the DPC client supports POSIX ACL authentication. The following protocols support POSIX ACL: DPC, NFSv3, and HDFS. If NFSv4 ACL or NT ACL is used, the DPC client cannot identify the ACL of this type. As a result, the ACL of this type does not take effect.	No	-	If aclonlyposix and acl are used together, only acl takes effect. That is, the namespace supports the ACL function.

Parameter	Description	Mandatory	Default Value	Remarks
mountOptions.cnflush	Asynchronous disk flushing mode. That is, data is not flushed to disks immediately when files in the namespace are closed.	No	-	Asynchronous flushing mode: When a file is closed, data in the cache is not flushed to storage media in synchronous mode. Instead, data is written from the cache to the storage media in asynchronous flushing mode. After the write service is complete, data is flushed from the cache to disks periodically based on the flushing period. In a multi-client scenario, if concurrent operations are performed on the same file, the file size update is affected by the disk flushing period. That is, the file size is updated only after the disk flushing is complete. Generally, the

Parameter	Description	Mandatory	Default Value	Remarks
				update is completed within several seconds. Synchronous I/Os are not affected by the disk flushing period.
mountOption s.cflush	Synchronous disk flushing mode. That is, data is flushed to disks immediately when files in the namespace are closed.	No	-	By default, the synchronous disk flushing mode is used.

Table 5-51 Supported QoS configurations

Parameter	Description	Remarks
maxMBPS	Maximum bandwidth. This is a restriction policy parameter.	This parameter is mandatory. The value is an integer greater than 0, expressed in MB/s. For details about the maximum value, see the actual limit of the storage device. For example, the maximum value of OceanStor Pacific NAS is 1073741824.
maxIOPS	Maximum IOPS. This is a restriction policy parameter.	This parameter is mandatory. The value is an integer greater than 0. For details about the maximum value, see the actual limit of the storage device. For example, the maximum value of OceanStor Pacific NAS is 1073741824000.

Table 5-52 Supported quota configurations

Parameter	Description	Remarks
spaceQuota	File quota type.	This parameter is mandatory. Only softQuota or hardQuota can be configured.
gracePeriod	Grace period allowed when the soft quota is configured.	This parameter is conditionally optional only when spaceQuota is set to softQuota . The value is an integer ranging from 0 to 4294967294.

Table 5-53 Supported advanced volume creation parameters

Parameter	Type	Description	Remarks
is_trash_enable	bool	Whether to enable recycle bin.	true : The recycle bin function is enabled. false : The recycle bin function is disabled (default).
checkpoint_trash	int	Retention period of the recycle bin (minutes)	Recycle bin protection duration. The value 0 indicates that the recycle bin is permanently retained and is not automatically deleted. The maximum value is 4294967295 . This parameter takes effect for both the namespace recycle bin and HDFS recycle bin.

Parameter	Type	Description	Remarks
interval_trash	int	Interval for clearing the recycle bin (minutes)	Interval for scanning the recycle bin. The value 0 indicates that the recycle bin is not scanned. The maximum value is 4294967295 . This parameter takes effect for both the namespace recycle bin and HDFS recycle bin.
is_trash_visible	bool	Whether the recycle bin is visible.	true : The recycle bin directory is visible. false : The recycle bin directory is invisible (default).
audit_log_switch	int	Audit log switch.	Whether the audit log function is enabled. This function is disabled by default. 0 : disabled. 1 : enabled.

Parameter	Type	Description	Remarks
audit_log_rule	int	Audit log rule. The default value is 0. 0x00000001:open 0x00000002:create 0x00000004:read 0x00000008:write 0x00000010:close 0x00000020:delete 0x00000040:rename 0x00000080:get_attr 0x00000100:set_attr 0x00000200:get_security 0x00000400:set_security 0x00000800:get_xattr 0x00001000:set_xattr 0x00002000:listdir 0x00004000:contact 0x00008000:Login Logoff 0x00010000:mount unmount	Multiple rules can be configured. For example: create+read +delete=0x2+0x4+0x20 =0x26 (38 in decimal notation) Set it to 38.

5.2.1.4.2 Dtree

Creating a StorageClass

Step 1 Prepare a StorageClass configuration file, for example, **mysc.yaml**. For details about the StorageClass configuration, see the following example.

Step 2 Run the following command to create a StorageClass using the configuration file.

```
kubectl apply -f mysc.yaml
```

Step 3 Run the following command to view the information about the created StorageClass.

```
kubectl get sc mysc
```

The following is an example of the command output.

NAME	PROVISIONER	RECLAIMPOLICY	VOLUMEBINDINGMODE	ALLOWVOLUMEEXPANSION	AGE
mysc	csi.huawei.com	Delete	Immediate	true	8s

----End

Example of StorageClass Configuration Supported by the NFS Protocol

When a container uses the NFS protocol to connect to dtree resources, refer to the following StorageClass configuration example. In this example, NFS version 4.1 is specified for mounting.

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
  name: mysc
provisioner: csi.huawei.com
parameters:
  backend: nfs-dtree-181
  parentname: parent-filesystem-name
  volumeType: dtree
  authClient: "*"
mountOptions:
  - nfsvers=4.1 # Specify the version 4.1 for NFS mounting.
```

Example of StorageClass Configuration Supported by the DPC Protocol

When a container uses the DPC protocol to connect to dtree resources, refer to the following StorageClass configuration example. In this example, **acl** is used as the authentication parameter for mounting, and **cnflush** is used to set the asynchronous disk flushing mode.

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
  name: mysc
provisioner: csi.huawei.com
parameters:
  backend: nfs-dtree-181
  parentname: parent-filesystem-name
  volumeType: dtree
  authClient: "*"
mountOptions:
  - acl # Set the authentication parameter.
  - cnflush # Set the asynchronous disk flushing mode.
```

StorageClass Parameters Supported by Dtrees

Table 5-54 StorageClass configuration parameters

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
metadata.name	User-defined name of a StorageClass object.	Yes	-	Yes	Take Kubernetes v1.22.1 as an example. The value can contain digits, lowercase letters, hyphens (-), and periods (.), and must start and end with a letter or digit.
provisioner	Name of the provisioner.	Yes	csi.huawei.com	Yes	Set this parameter to the driver name set during Huawei CSI installation. The value is the same as that of driverName in the values.yaml file.
reclaimPolicy	Reclamation policy. The following types are supported: • Delete: Resources are automatically reclaimed. • Retain: Resources are manually reclaimed.	No	Delete	Yes	• Delete: When a PV/PVC is deleted, resources on the storage device are also deleted. • Retain: When a PV/PVC is deleted, resources on the storage device are not deleted.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
allowVolumeExpansion	Whether to allow volume expansion. If this parameter is set to true , the capacity of the PV that uses the StorageClass can be expanded.	No	false	Yes	This function can only be used to expand PV capacity but cannot be used to reduce PV capacity.
mountOptions	List of mount parameters, which can be used to specify the parameters of the -o option when the mount command is executed on a host.	No	-	Yes	For details about common parameters in mountOptions , see Table 5-55 . You can also specify other mount parameters.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.backend	Name of the backend where the resource to be created is located.	Conditionally mandatory	-	No	If this parameter is not set, Huawei CSI will randomly select a backend that meets the capacity requirements to create resources. You are advised to specify a backend to ensure that the created resource is located on the expected backend. This parameter is mandatory if parameters.parentname is set.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.parentname	Name of a file system on the current storage device. Dtree is created in the file system.	Conditionally mandatory	-	No	This parameter is mandatory when parentname is not set for the backend. If parentname is configured only in the StorageClass but not configured in the storage backend, set CSIDriverObject.attachRequired to true during CSI installation.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.volumeType	Type of the volume to be created. The following types are supported: • lun: A LUN is provisioned on the storage side. • fs: A file system is provisioned on the storage side. • dtree: A volume of the dtree type is provisioned on the storage side.	Yes	-	Yes	When dtree is used, the value must be dtree .

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.authClient	IP address of the NFS client that can access the volume. You can enter the client host name (a full domain name is recommended), client IP address, or client IP address segment.	Yes	-	No	The asterisk (*) can be used to indicate any client. If you are not sure about the IP address of the access client, you are advised to use the asterisk (*) to prevent the client access from being rejected by the storage system. If the client host name is used, you are advised to use the full domain name. The IP addresses can be IPv4 addresses, IPv6 addresses, or a combination of IPv4 and IPv6 addresses. You can enter multiple host names, IP addresses, or IP address segments and separate them with semicolons (;). Example: 192.168.0.10;192.168.0.0/24;myserver1.test
parameters.fsPermission	Permission on the directory mounted to a container.	No	-	No	For details about the configuration format, refer to the Linux permission settings, for example, 777 and 755.

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.rootSquash	Controls the root permission of the client. The value can be: • root_squash: The client cannot access the storage system as user root. If a client accesses the storage system as user root, the client will be mapped as an anonymous user. • no_root_squash: A client can access the storage system as user root and has the permission of user root.	No	-	No	

Parameter	Description	M a n d a t o r y	Def ault Valu e	Wh eth er the Vol um e Ma nag em ent Tak es Eff ect	Remarks
parameters.allSquash	Whether to retain the user ID (UID) and group ID (GID) of a shared directory. The value can be: • all_squash: The UID and GID of the shared directory are mapped to anonymous users. • no_all_squash: The UID and GID of the shared directory are retained.	No	-	No	

Parameter	Description	Mandatory	Default Value	Whether the Volume Management Takes Effect	Remarks
parameters.disableVerifyCapacity	Whether to disable volume capacity verification. After this function is disabled, the system will not verify whether the volume capacity is an integer multiple of the sector size. The value can be: • "true": disables volume capacity verification. • "false": enables volume capacity verification. NOTICE When Red Hat OpenShift Virtualization is used to connect to CSI, this parameter must be set to "true".	No	"true"	No	For OceanStor Pacific dtrees, the sector size is 1 byte.

Table 5-55 Common parameters in mountOptions

Parameter	Description	Mandatory	Default Value	Remarks
mountOptions.nfsvers	NFS mount option on the host. The following mount option is supported: nfsvers: protocol version for NFS mounting. The value can be 3 or 4.1 .	No	-	This parameter is optional after the -o parameter when the mount command is executed on the host. The value is in list format. If the NFS version is specified for mounting, NFS 3 and 4.1 protocols are supported (the protocol must be supported and enabled on storage devices).
mountOptions.proto	Transmission protocol used for NFS mounting. The value can be rdma .	No	-	This parameter is supported by 8.2.0 or later.
mountOptions.port	Protocol port used for NFS mounting.	Conditionally mandatory	-	If the transmission protocol is rdma , set this parameter to 20049 .

Parameter	Description	Mandatory	Default Value	Remarks
mountOptions.acl	The DPC namespace supports the ACL function. The DPC client supports POSIX ACL, NFSv4 ACL, and NT ACL authentication.	No	-	The descriptions of acl , aclonlyposix , cnflush , and cflush are for reference only. For details about the parameters, see OceanStor Pacific Series Product Documentation and choose Configuration > Basic Service Configuration Guide for File > Configuring Basic Services (DPC Scenario) > Accessing a DPC Share on a Client > Step 2 .

Parameter	Description	Mandatory	Default Value	Remarks
mountOptions.aclonlyposix	The DPC namespace supports POSIX ACL, and the DPC client supports POSIX ACL authentication. The following protocols support POSIX ACL: DPC, NFSv3, and HDFS. If NFSv4 ACL or NT ACL is used, the DPC client cannot identify the ACL of this type. As a result, the ACL of this type does not take effect.	No	-	If aclonlyposix and acl are used together, only acl takes effect. That is, the namespace supports the ACL function.

Parameter	Description	Mandatory	Default Value	Remarks
mountOptions.cnflush	Asynchronous disk flushing mode. That is, data is not flushed to disks immediately when files in the namespace are closed.	No	-	Asynchronous flushing mode: When a file is closed, data in the cache is not flushed to storage media in synchronous mode. Instead, data is written from the cache to the storage media in asynchronous flushing mode. After the write service is complete, data is flushed from the cache to disks periodically based on the flushing period. In a multi-client scenario, if concurrent operations are performed on the same file, the file size update is affected by the disk flushing period. That is, the file size is updated only after the disk flushing is complete. Generally, the

Parameter	Description	Mandatory	Default Value	Remarks
				update is completed within several seconds. Synchronous I/Os are not affected by the disk flushing period.
mountOption s.cflush	Synchronous disk flushing mode. That is, data is flushed to disks immediately when files in the namespace are closed.	No	-	By default, the synchronous disk flushing mode is used.

5.2.1.4.3 Block Service

Creating a StorageClass

- Step 1** Prepare a StorageClass configuration file, for example, **mysc.yaml**. For details about the StorageClass configuration, see the following example.
- Step 2** Run the following command to create a StorageClass using the configuration file.
- ```
kubectl apply -f mysc.yaml
```
- Step 3** Run the following command to view the information about the created StorageClass.

```
kubectl get sc mysc
```

The following is an example of the command output.

| NAME | PROVISIONER    | RECLAIMPOLICY | VOLUMEBINDINGMODE | ALLOWVOLUMEEXPANSION | AGE |
|------|----------------|---------------|-------------------|----------------------|-----|
| mysc | csi.huawei.com | Delete        | Immediate         | true                 | 8s  |

----End

#### Block StorageClass Configuration Example

If LUNs are used as storage resources and the file system needs to be formatted to a local file system, refer to the following example.

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
```

```
metadata:
 name: mysc
provisioner: csi.huawei.com
parameters:
 backend: lun-181
 pool: StoragePool001
 volumeType: lun
 allocType: thin
 fsType: ext4
```

StorageClass Parameters Supported by Block Services

Table 5-56 StorageClass configuration parameters

| Parameter     | Description                                 | Mandatory | Default Value  | Whether the Volume Management Takes Effect | Remarks                                                                                                                                                              |
|---------------|---------------------------------------------|-----------|----------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| metadata.name | User-defined name of a StorageClass object. | Yes       | -              | Yes                                        | Take Kubernetes v1.22.1 as an example. The value can contain digits, lowercase letters, hyphens (-), and periods (.), and must start and end with a letter or digit. |
| provisioner   | Name of the provisioner.                    | Yes       | csi.huawei.com | Yes                                        | Set this parameter to the driver name set during Huawei CSI installation. The value is the same as that of <b>driverName</b> in the <b>values.yaml</b> file.         |

| Parameter            | Description                                                                                                                                                                                                               | Mandatory | Default Value | Whether the Volume Management Takes Effect | Remarks                                                                                                                                                                                                                                          |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| reclaimPolicy        | Reclamation policy. The following types are supported: <ul style="list-style-type: none"><li>• <b>Delete:</b> Resources are automatically reclaimed.</li><li>• <b>Retain:</b> Resources are manually reclaimed.</li></ul> | No        | Delete        | Yes                                        | <ul style="list-style-type: none"><li>• <b>Delete:</b> When a PV/PVC is deleted, resources on the storage device are also deleted.</li><li>• <b>Retain:</b> When a PV/PVC is deleted, resources on the storage device are not deleted.</li></ul> |
| allowVolumeExpansion | Whether to allow volume expansion. If this parameter is set to <b>true</b> , the capacity of the PV that uses the StorageClass can be expanded.                                                                           | No        | false         | Yes                                        | This function can only be used to expand PV capacity but cannot be used to reduce PV capacity.                                                                                                                                                   |
| mountOptions         | List of mount parameters, which can be used to specify the parameters of the <b>-o</b> option when the <b>mount</b> command is executed on a host.                                                                        | No        | -             | Yes                                        | For details about common parameters in <b>mountOptions</b> , see <a href="#">Table 5-57</a> .<br>You can also specify other mount parameters.                                                                                                    |

| Parameter          | Description                                                                                                               | Mandatory               | Default Value | Whether the Volume Management Takes Effect | Remarks                                                                                                                                                                                                                                                                                |
|--------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------|---------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parameters.backend | Name of the backend where the resource to be created is located. This field must be set if <b>parameters.pool</b> is set. | Conditionally mandatory | -             | No                                         | If this parameter is not set, Huawei CSI will randomly select a backend that meets the capacity requirements to create resources.<br><br>You are advised to specify a backend to ensure that the created resource is located on the expected backend.                                  |
| parameters.pool    | Name of the storage resource pool where the resource to be created is located.                                            | No                      | -             | No                                         | If this parameter is not set, Huawei CSI will randomly select a storage pool with the largest remaining capacity from the selected backend to create resources. You are advised to specify a storage pool to ensure that the created resource is located in the expected storage pool. |



| Parameter             | Description                                                                                                                                                                                                                                                                                                                                | Mandatory | Default Value | Whether the Volume Management Takes Effect | Remarks                            |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|--------------------------------------------|------------------------------------|
| parameters.volumeType | Type of the volume to be created. The following types are supported: <ul style="list-style-type: none"><li>• <b>lun</b>: A LUN is provisioned on the storage side.</li><li>• <b>fs</b>: A file system is provisioned on the storage side.</li><li>• <b>dtree</b>: A volume of the dtree type is provisioned on the storage side.</li></ul> | Yes       | -             | Yes                                        | The value is fixed at <b>lun</b> . |

| Parameter            | Description                                                                                                                                                                                                                                                       | Mandatory | Default Value | Whether the Volume Management Takes Effect | Remarks                                                                                                                                                                                                                                                                             |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parameters.allocType | Allocation type of the volume to be created. The following types are supported: <ul style="list-style-type: none"><li><b>thin</b>: Not all required space is allocated during creation. Instead, the space is dynamically allocated based on the usage.</li></ul> | No        | thin          | No                                         | If this parameter is set to <b>thin</b> , the required space is not allocated immediately when a volume is created. Instead, the space is dynamically allocated based on the usage.                                                                                                 |
| parameters.fsType    | Type of a host file system. The supported types are: <ul style="list-style-type: none"><li>ext2</li><li>ext3</li><li>ext4</li><li>xfs</li></ul>                                                                                                                   | No        | ext4          | No                                         | This parameter is available only when <b>volumeMode</b> of the PVC is set to <b>Filesystem</b> .<br><b>NOTICE</b><br>When Ubuntu 22.04 LTS with a kernel version earlier than 5.15.194 is used to interconnect with CSI, the XFS file system cannot be mounted due to a kernel bug. |

| Parameter               | Description                                                                                                                                                                                                                                                                                         | Mandatory | Default Value | Whether the Volume Management Takes Effect | Remarks                                                                                                       |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| parameters.qos          | LUN/NAS QoS settings of the PV on the storage side.<br><br>The value of the parameter is JSON character strings in dictionary format. A character string is enclosed by single quotation marks and the dictionary key by double quotation marks. Example: <b>'{"maxMBPS": 999, "maxIOPS": 999}'</b> | No        | -             | No                                         | For details about the supported QoS configurations, see <a href="#">Table 5-58</a> .                          |
| parameters.fsPermission | Permission on the directory mounted to a container.                                                                                                                                                                                                                                                 | No        | -             | No                                         | For details about the configuration format, refer to the Linux permission settings, for example, 777 and 755. |

| Parameter                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Mandatory | Default Value | Whether the Volume Management Takes Effect | Remarks                                              |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|--------------------------------------------|------------------------------------------------------|
| parameters.disableVerifyCapacity | <p>Whether to disable volume capacity verification. After this function is disabled, the system will not verify whether the volume capacity is an integer multiple of the sector size.</p> <p>The value can be:</p> <ul style="list-style-type: none"><li>• "true": disables volume capacity verification.</li><li>• "false": enables volume capacity verification.</li></ul> <p><b>NOTICE</b><br/>When Red Hat OpenShift Virtualization is used to connect to CSI, this parameter must be set to <b>"true"</b>.</p> | No        | "true"        | Yes                                        | For OceanStor Pacific SAN, the sector size is 1 MiB. |

**Table 5-57** Common parameters in mountOptions

| Parameter            | Description                                                                                                                                           | Mandatory | Default Value | Remarks                                      |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|----------------------------------------------|
| mountOptions.discard | Automatically triggers the Trim or Discard operation when a file system is mounted. This operation instructs a block device to release unused blocks. | No        | -             | The xfs and ext4 file systems are supported. |

**Table 5-58** Supported QoS configurations

| Parameter | Description                                                | Remarks                                                                                                                                                                                                                                         |
|-----------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| maxMBPS   | Maximum bandwidth. This is a restriction policy parameter. | This parameter is mandatory. The value is an integer greater than 0, expressed in MB/s. For details about the maximum value, see the actual limit of the storage device. For example, the maximum value of OceanStor Pacific NAS is 1073741824. |
| maxIOPS   | Maximum IOPS. This is a restriction policy parameter.      | This parameter is mandatory. The value is an integer greater than 0. For details about the maximum value, see the actual limit of the storage device. For example, the maximum value of OceanStor Pacific NAS is 1073741824000.                 |

## 5.2.1.5 Mass Storage (OceanDisk Series)

### 5.2.1.5.1 File System

#### Creating a StorageClass

**Step 1** Prepare a StorageClass configuration file, for example, **mysc.yaml**. For details about the StorageClass configuration, see the following example.

**Step 2** Run the following command to create a StorageClass using the configuration file.

```
kubectl apply -f mysc.yaml
```

**Step 3** Run the following command to view the information about the created StorageClass.

```
kubectl get sc mysc
```

The following is an example of the command output.

| NAME | PROVISIONER    | RECLAIMPOLICY | VOLUMEBINDINGMODE | ALLOWVOLUMEEXPANSION | AGE |
|------|----------------|---------------|-------------------|----------------------|-----|
| mysc | csi.huawei.com | Delete        | Immediate         | true                 | 8s  |

----End

## NFS Protocol Configuration Example

When a container uses the NFS protocol to connect to file system resources, refer to the following StorageClass configuration example. In this example, NFS version 4.0 is specified for mounting.

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
 name: mysc
provisioner: csi.huawei.com
parameters:
 backend: nfs-nas-181
 pool: StoragePool001
 volumeType: fs
 allocType: thin
 authClient: "*"
mountOptions:
 - nfsvers=4.0 # Specify the version 4.0 for NFS mounting.
```

## StorageClass Parameters Supported by File Systems

Table 5-59 StorageClass configuration parameters

| Parameter     | Description                                                                                                                                                                                                               | Mandatory | Default Value  | Whether the Volume Management Takes Effect | Remarks                                                                                                                                                                                                                                          |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| metadata.name | User-defined name of a StorageClass object.                                                                                                                                                                               | Yes       | -              | Yes                                        | Take Kubernetes v1.22.1 as an example. The value can contain digits, lowercase letters, hyphens (-), and periods (.), and must start and end with a letter or digit.                                                                             |
| provisioner   | Name of the provisioner.                                                                                                                                                                                                  | Yes       | csi.huawei.com | Yes                                        | Set this parameter to the driver name set during Huawei CSI installation. The value is the same as that of <b>driverName</b> in the <b>values.yaml</b> file.                                                                                     |
| reclaimPolicy | Reclamation policy. The following types are supported: <ul style="list-style-type: none"><li>• <b>Delete:</b> Resources are automatically reclaimed.</li><li>• <b>Retain:</b> Resources are manually reclaimed.</li></ul> | No        | Delete         | Yes                                        | <ul style="list-style-type: none"><li>• <b>Delete:</b> When a PV/PVC is deleted, resources on the storage device are also deleted.</li><li>• <b>Retain:</b> When a PV/PVC is deleted, resources on the storage device are not deleted.</li></ul> |

| Parameter            | Description                                                                                                                                        | Mandatory               | Default Value | Whether the Volume Management Takes Effect | Remarks                                                                                                                                                                                                                                           |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|---------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| allowVolumeExpansion | Whether to allow volume expansion. If this parameter is set to <b>true</b> , the capacity of the PV that uses the StorageClass can be expanded.    | No                      | false         | Yes                                        | This function can only be used to expand PV capacity but cannot be used to reduce PV capacity.                                                                                                                                                    |
| mountOptions         | List of mount parameters, which can be used to specify the parameters of the <b>-o</b> option when the <b>mount</b> command is executed on a host. | No                      | -             | Yes                                        | For details about common parameters in <b>mountOptions</b> , see <a href="#">Table 5-60</a> .<br>You can also specify other mount parameters.                                                                                                     |
| parameters.backend   | Name of the backend where the resource to be created is located. This field must be set if <b>parameters.pool</b> is set.                          | Conditionally mandatory | -             | No                                         | If this parameter is not set, Huawei CSI will randomly select a backend that meets the capacity requirements to create resources.<br>You are advised to specify a backend to ensure that the created resource is located on the expected backend. |



| Parameter       | Description                                                                    | Mandatory | Default Value | Whether the Volume Management Takes Effect | Remarks                                                                                                                                                                                                                                                                                |
|-----------------|--------------------------------------------------------------------------------|-----------|---------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parameters.pool | Name of the storage resource pool where the resource to be created is located. | No        | -             | No                                         | If this parameter is not set, Huawei CSI will randomly select a storage pool with the largest remaining capacity from the selected backend to create resources. You are advised to specify a storage pool to ensure that the created resource is located in the expected storage pool. |

| Parameter             | Description                                                                                                                                                                                                                                                                                                                                        | Mandatory | Default Value | Whether the Volume Management Takes Effect | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parameters.volumeName | <p>Name of the storage resource created by dynamic volume provisioning.</p> <p>You can configure a placeholder to customize the storage resource name. The following placeholders are supported:</p> <ul style="list-style-type: none"><li>PVC namespace: <code>{{ .PVCNamespace }}</code></li><li>PVC name: <code>{{ .PVCName }}</code></li></ul> | No        | -             | No                                         | <ul style="list-style-type: none"><li>The value can contain letters, digits, hyphens (-), underscores (_), and periods (.). This parameter cannot be left empty. The length of the generated storage resource name ranges from 1 to 255 characters.</li><li>Both the PVC namespace and PVC name must be configured.</li><li>To avoid duplicate resource names, the PVC UID is added to the end of the name as a unique identifier by default.</li></ul> <p>Configuration example:</p> <p>PVC namespace: <b>namespace</b>. PVC name: <b>pvc-1</b>. PVC UID: <b>c2fd3f46-bf17-4a7d-b88e-2e3232bae434</b>.</p> <p><b>volumeName</b> is set to <b>prefix-{{ .PVCNamespace }}_{{ .PVCName }}</b>.</p> <p>The ultimate storage resource name is <b>prefix-namespace_pvc-1-</b></p> |

| Parameter             | Description                                                                                                                                                                                                                                                                                                                                | Mandatory | Default Value | Whether the Volume Management Takes Effect | Remarks                           |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|--------------------------------------------|-----------------------------------|
|                       |                                                                                                                                                                                                                                                                                                                                            |           |               |                                            | c2fd3f46bf174a7db88e2e3232bae434. |
| parameters.volumeType | Type of the volume to be created. The following types are supported: <ul style="list-style-type: none"><li>• <b>lun</b>: A LUN is provisioned on the storage side.</li><li>• <b>fs</b>: A file system is provisioned on the storage side.</li><li>• <b>dtree</b>: A volume of the dtree type is provisioned on the storage side.</li></ul> | Yes       | -             | Yes                                        | The value is fixed at <b>fs</b> . |

| Parameter            | Description                                                                                                                                                                                                                                                                | Mand<br>at<br>ory | Def<br>ault<br>Val<br>ue | Wh<br>eth<br>er<br>the<br>Vol<br>um<br>e<br>Ma<br>na<br>ge<br>me<br>nt<br>Tak<br>es<br>Eff<br>ect | Remarks                                                                                                                                                                             |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parameters.allocType | <p>Allocation type of the volume to be created. The following types are supported:</p> <ul style="list-style-type: none"><li>• <b>thin</b>: Not all required space is allocated during creation. Instead, the space is dynamically allocated based on the usage.</li></ul> | No                | thin                     | No                                                                                                | If this parameter is set to <b>thin</b> , the required space is not allocated immediately when a volume is created. Instead, the space is dynamically allocated based on the usage. |

| Parameter                  | Description                                                                                                                                                                              | Mandatory | Default Value | Whether the Volume Management Takes Effect | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parameters.authClient      | IP address of the NFS client that can access the volume.<br><br>You can enter the client host name (a full domain name is recommended), client IP address, or client IP address segment. | Yes       | -             | No                                         | The asterisk (*) can be used to indicate any client. If you are not sure about the IP address of the access client, you are advised to use the asterisk (*) to prevent the client access from being rejected by the storage system.<br><br>If the client host name is used, you are advised to use the full domain name.<br><br>The IP addresses can be IPv4 addresses, IPv6 addresses, or a combination of IPv4 and IPv6 addresses.<br><br>You can enter multiple host names, IP addresses, or IP address segments and separate them with semicolons (;). Example:<br><b>192.168.0.10;192.168.0.0/24;myserver1.test</b> |
| parameters.applicationType | Application type name when a file system is created.                                                                                                                                     | No        | -             | Yes                                        | Log in to DeviceManager and choose <b>Services &gt; File Service &gt; File Systems &gt; Create</b> to obtain the application type name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| Parameter               | Description                                                                                                                                                                                                                                                                                                                 | Mandatory | Default Value | Whether the Volume Management Takes Effect | Remarks                                                                                                       |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| parameters.qos          | QoS settings of the file system on the storage side of the PV.<br>The value of the parameter is JSON character strings in dictionary format. A character string is enclosed by single quotation marks and the dictionary key by double quotation marks. For example, <code>'{"MAXBANDWIDTH": 999, "MAXIOPS": 999}'</code> . | No        | -             | No                                         | For details about the supported QoS configurations, see <a href="#">Table 5-61</a> .                          |
| parameters.fsPermission | Permission on the directory mounted to a container.                                                                                                                                                                                                                                                                         | No        | -             | No                                         | For details about the configuration format, refer to the Linux permission settings, for example, 777 and 755. |

| Parameter             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Mandatory | Default Value | Whether the Volume Management Takes Effect | Remarks |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|--------------------------------------------|---------|
| parameters.rootSquash | <p>Controls the <b>root</b> permission of the client.</p> <p>The value can be:</p> <ul style="list-style-type: none"><li>• <b>root_squash</b>: The client cannot access the storage system as user <b>root</b>. If a client accesses the storage system as user <b>root</b>, the client will be mapped as an anonymous user.</li><li>• <b>no_root_squash</b>: A client can access the storage system as user <b>root</b> and has the permission of user <b>root</b>.</li></ul> | No        | -             | No                                         |         |

| Parameter             | Description                                                                                                                                                                                                                                                                                                                          | Mandatory | Default Value | Whether the Volume Management Takes Effect | Remarks                                                                            |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|--------------------------------------------|------------------------------------------------------------------------------------|
| parameters.allSquash  | Whether to retain the user ID (UID) and group ID (GID) of a shared directory.<br>The value can be: <ul style="list-style-type: none"><li>• <b>all_squash</b>: The UID and GID of the shared directory are mapped to anonymous users.</li><li>• <b>no_all_squash</b>: The UID and GID of the shared directory are retained.</li></ul> | No        | -             | No                                         |                                                                                    |
| parameters.accesskrb5 | Configures the krb5 security protocol. <ul style="list-style-type: none"><li>• <b>read_only</b>: read-only</li><li>• <b>read_write</b>: read and write</li><li>• <b>none</b>: no permission</li></ul>                                                                                                                                | No        | -             | No                                         | During mounting, you can specify the <b>sec</b> parameter in <b>mountOptions</b> . |



| Parameter              | Description                                                                                                                                                                                            | Mandatory | Default Value | Whether the Volume Management Takes Effect | Remarks                                                                            |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|--------------------------------------------|------------------------------------------------------------------------------------|
| parameters.accesskrb5i | Configures the krb5i security protocol. <ul style="list-style-type: none"><li>• <b>read_only</b>: read-only</li><li>• <b>read_write</b>: read and write</li><li>• <b>none</b>: no permission</li></ul> | No        | -             | No                                         | During mounting, you can specify the <b>sec</b> parameter in <b>mountOptions</b> . |
| parameters.accesskrb5p | Configures the krb5p security protocol. <ul style="list-style-type: none"><li>• <b>read_only</b>: read-only</li><li>• <b>read_write</b>: read and write</li><li>• <b>none</b>: no permission</li></ul> | No        | -             | No                                         | During mounting, you can specify the <b>sec</b> parameter in <b>mountOptions</b> . |

| Parameter                              | Description                                                                                                                                                                                                                           | Mandatory | Default Value | Whether the Volume Management Takes Effect | Remarks                        |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|--------------------------------------------|--------------------------------|
| parameters.snapshotDirectoryVisibility | Whether the snapshot directory is visible.<br>The value can be: <ul style="list-style-type: none"><li>• <b>visible</b>: The snapshot directory is visible.</li><li>• <b>invisible</b>: The snapshot directory is invisible.</li></ul> | No        | -             | No                                         | Only NAS storage is supported. |
| parameters.reservedSnapshotSpaceRatio  | Configures reserved snapshot space.<br>Value type: character string<br>Value range: 0 to 50                                                                                                                                           | No        | -             | No                                         | -                              |

| Parameter                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Mand<br>at<br>ory | Def<br>ault<br>Val<br>ue | Wh<br>eth<br>er<br>the<br>Vol<br>um<br>e<br>Ma<br>na<br>ge<br>me<br>nt<br>Tak<br>es<br>Eff<br>ect | Remarks                                      |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------|
| parameters.disableVerifyCapacity | <p>Whether to disable volume capacity verification. After this function is disabled, the system will not verify whether the volume capacity is an integer multiple of the sector size.</p> <p>The value can be:</p> <ul style="list-style-type: none"><li>• "true": disables volume capacity verification.</li><li>• "false": enables volume capacity verification.</li></ul> <p><b>NOTICE</b><br/>When Red Hat OpenShift Virtualization is used to connect to CSI, this parameter must be set to <b>"true"</b>.</p> | No                | "true"                   | Yes                                                                                               | For OceanDisk, the sector size is 512 bytes. |

| Parameter                  | Description                                                                                                                                                                                                                                                               | Mandatory | Default Value               | Whether the Volume Management Takes Effect | Remarks                                                                               |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------|--------------------------------------------|---------------------------------------------------------------------------------------|
| parameters.description     | Description of the file system to be created.<br>Value type: character string<br>The value contains 0 to 255 characters.                                                                                                                                                  | No        | Created from Kubernetes CSI | No                                         |                                                                                       |
| parameters.advancedOptions | Advanced volume creation parameters.<br>The value of the parameter is JSON character strings in dictionary format. A character string is enclosed by single quotation marks and the dictionary key by double quotation marks. Example: <b>'{"CAPACITYTHRESHOLD": 90}'</b> | No        | -                           | No                                         | For details about the supported advanced parameters, see <a href="#">Table 5-62</a> . |

**Table 5-60** Common parameters in mountOptions

| Parameter            | Description                                                                                                                                                            | Mandatory | Default Value | Remarks                                                                                                                                                                                                                                                                                              |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| mountOptions.nfsvers | NFS mount option on the host. The following mount option is supported:<br><b>nfsvers:</b> protocol version for NFS mounting. The value can be <b>3</b> or <b>4.0</b> . | No        | -             | This parameter is optional after the <b>-o</b> parameter when the <b>mount</b> command is executed on the host. The value is in list format.<br>If the NFS version is specified for mounting, NFS 3 and 4.0 protocols are supported (the protocol must be supported and enabled on storage devices). |
| mountOptions.sec     | Kerberos 5 protocol for mounting NFS file systems.                                                                                                                     | No        | -             | <ul style="list-style-type: none"><li>• If Kerberos 5 is used, set this parameter to <b>krb5</b>.</li><li>• If Kerberos 5i is used, set this parameter to <b>krb5i</b>.</li><li>• If Kerberos 5p is used, set this parameter to <b>krb5p</b>.</li><li>• Only NFS 4.0 supports Kerberos.</li></ul>    |

| Parameter          | Description                                                                    | Mandatory               | Default Value | Remarks                                                                            |
|--------------------|--------------------------------------------------------------------------------|-------------------------|---------------|------------------------------------------------------------------------------------|
| mountOptions.proto | Transmission protocol used for NFS mounting.<br>The value can be <b>rdma</b> . | No                      | -             | -                                                                                  |
| mountOptions.port  | Protocol port used for NFS mounting.                                           | Conditionally mandatory | -             | If the transmission protocol is <b>rdma</b> , set this parameter to <b>20049</b> . |

**Table 5-61** Supported QoS configurations

| Storage Type | Parameter    | Description                                                | Remarks                                                                                                                                                                                                                                                                                                               |
|--------------|--------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OceanStor V5 | IOTYPE       | Read/write type.                                           | This parameter is optional. If it is not specified, the default value of the storage backend is used. For details, see related storage documents.<br><br>The value can be: <ul style="list-style-type: none"><li>• <b>0</b>: read I/O</li><li>• <b>1</b>: write I/O</li><li>• <b>2</b>: read and write I/Os</li></ul> |
|              | MAXBANDWIDTH | Maximum bandwidth. This is a restriction policy parameter. | The value is an integer greater than 0, expressed in MB/s.                                                                                                                                                                                                                                                            |
|              | MINBANDWIDTH | Minimum bandwidth. This is a protection policy parameter.  | The value is an integer greater than 0, expressed in MB/s.                                                                                                                                                                                                                                                            |
|              | MAXIOPS      | Maximum IOPS. This is a restriction policy parameter.      | The value is an integer greater than 0.                                                                                                                                                                                                                                                                               |
|              | MINIOPS      | Minimum IOPS. This is a protection policy parameter.       | The value is an integer greater than 0.                                                                                                                                                                                                                                                                               |

| Storage Type                  | Parameter    | Description                                                | Remarks                                                                                    |
|-------------------------------|--------------|------------------------------------------------------------|--------------------------------------------------------------------------------------------|
|                               | LATENCY      | Maximum latency. This is a protection policy parameter.    | The value is an integer greater than 0, expressed in ms.                                   |
| OceanStor or Dorado/OceanStor | IOTYPE       | Read/write type.                                           | The value can be: <ul style="list-style-type: none"><li>• 2: read and write I/Os</li></ul> |
|                               | MAXBANDWIDTH | Maximum bandwidth. This is a restriction policy parameter. | The value is an integer ranging from 1 to 999999999, expressed in MB/s.                    |
|                               | MINBANDWIDTH | Minimum bandwidth. This is a protection policy parameter.  | The value is an integer ranging from 1 to 999999999, expressed in MB/s.                    |
|                               | MAXIOPS      | Maximum IOPS. This is a restriction policy parameter.      | The value is an integer ranging from 100 to 999999999.                                     |
|                               | MINIOPS      | Minimum IOPS. This is a protection policy parameter.       | The value is an integer ranging from 100 to 999999999.                                     |
|                               | LATENCY      | Maximum latency. This is a protection policy parameter.    | The value can be <b>0.5</b> or <b>1.5</b> , expressed in ms.                               |

**Table 5-62** Supported advanced volume creation parameters

| Parameter         | Description                     | Remarks                                                                                                                       |
|-------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| CAPACITYTHRESHOLD | Total capacity alarm threshold. | Parameter type: uint64.<br>For details about the default value and value range, see the corresponding storage product manual. |

### 5.2.1.5.2 Block Service

#### Creating a StorageClass

**Step 1** Prepare a StorageClass configuration file, for example, **mysc.yaml**. For details about the StorageClass configuration, see the following example.

**Step 2** Run the following command to create a StorageClass using the configuration file.

```
kubectl apply -f mysc.yaml
```

**Step 3** Run the following command to view the information about the created StorageClass.

```
kubectl get sc mysc
```

The following is an example of the command output.

| NAME | PROVISIONER    | RECLAIMPOLICY | VOLUMEBINDINGMODE | ALLOWVOLUMEEXPANSION | AGE |
|------|----------------|---------------|-------------------|----------------------|-----|
| mysc | csi.huawei.com | Delete        | Immediate         | true                 | 8s  |

----End

## Block StorageClass Configuration Example

If LUNs are used as storage resources and the file system needs to be formatted to a local file system, refer to the following example.

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
 name: mysc
provisioner: csi.huawei.com
parameters:
 backend: lun-181
 pool: StoragePool001
 volumeType: lun
 allocType: thin
 fsType: ext4
```

## StorageClass Parameters Supported by Block Services

**Table 5-63** StorageClass configuration parameters

| Parameter     | Description                                 | Mandatory | Default Value | Whether the Volume Management Takes Effect | Remarks                                                                                                                                                              |
|---------------|---------------------------------------------|-----------|---------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| metadata.name | User-defined name of a StorageClass object. | Yes       | -             | Yes                                        | Take Kubernetes v1.22.1 as an example. The value can contain digits, lowercase letters, hyphens (-), and periods (.), and must start and end with a letter or digit. |



| Parameter            | Description                                                                                                                                                                                                               | Mandatory | Default Value  | Whether the Volume Management Takes Effect | Remarks                                                                                                                                                                                                                                          |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| provisioner          | Name of the provisioner.                                                                                                                                                                                                  | Yes       | csi.huawei.com | Yes                                        | Set this parameter to the driver name set during Huawei CSI installation.<br>The value is the same as that of <b>driverName</b> in the <b>values.yaml</b> file.                                                                                  |
| reclaimPolicy        | Reclamation policy. The following types are supported: <ul style="list-style-type: none"><li>• <b>Delete:</b> Resources are automatically reclaimed.</li><li>• <b>Retain:</b> Resources are manually reclaimed.</li></ul> | No        | Delete         | Yes                                        | <ul style="list-style-type: none"><li>• <b>Delete:</b> When a PV/PVC is deleted, resources on the storage device are also deleted.</li><li>• <b>Retain:</b> When a PV/PVC is deleted, resources on the storage device are not deleted.</li></ul> |
| allowVolumeExpansion | Whether to allow volume expansion. If this parameter is set to <b>true</b> , the capacity of the PV that uses the StorageClass can be expanded.                                                                           | No        | false          | Yes                                        | This function can only be used to expand PV capacity but cannot be used to reduce PV capacity.                                                                                                                                                   |

| Parameter          | Description                                                                                                                                        | Mandatory               | Default Value | Whether the Volume Management Takes Effect | Remarks                                                                                                                                                                                                                                                                                |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|---------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| mountOptions       | List of mount parameters, which can be used to specify the parameters of the <b>-o</b> option when the <b>mount</b> command is executed on a host. | No                      | -             | Yes                                        | For details about common parameters in <b>mountOptions</b> , see <a href="#">Table 5-64</a> .<br>You can also specify other mount parameters.                                                                                                                                          |
| parameters.backend | Name of the backend where the resource to be created is located. This field must be set if <b>parameters.pool</b> is set.                          | Conditionally mandatory | -             | No                                         | If this parameter is not set, Huawei CSI will randomly select a backend that meets the capacity requirements to create resources.<br>You are advised to specify a backend to ensure that the created resource is located on the expected backend.                                      |
| parameters.pool    | Name of the storage resource pool where the resource to be created is located.                                                                     | No                      | -             | No                                         | If this parameter is not set, Huawei CSI will randomly select a storage pool with the largest remaining capacity from the selected backend to create resources. You are advised to specify a storage pool to ensure that the created resource is located in the expected storage pool. |

| Parameter             | Description                                                                                                                                                                                                                                                                                                                                | Mandatory | Default Value | Whether the Volume Management Takes Effect | Remarks                            |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|--------------------------------------------|------------------------------------|
| parameters.volumeType | Type of the volume to be created. The following types are supported: <ul style="list-style-type: none"><li>• <b>lun</b>: A LUN is provisioned on the storage side.</li><li>• <b>fs</b>: A file system is provisioned on the storage side.</li><li>• <b>dtree</b>: A volume of the dtree type is provisioned on the storage side.</li></ul> | Yes       | -             | Yes                                        | The value is fixed at <b>lun</b> . |

| Parameter            | Description                                                                                                                                                                                                                                                         | Mandatory | Default Value | Whether the Volume Management Takes Effect | Remarks                                                                                                                                                                             |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parameters.allocType | Allocation type of the volume to be created. The following types are supported: <ul style="list-style-type: none"><li>• <b>thin</b>: Not all required space is allocated during creation. Instead, the space is dynamically allocated based on the usage.</li></ul> | No        | thin          | No                                         | If this parameter is set to <b>thin</b> , the required space is not allocated immediately when a volume is created. Instead, the space is dynamically allocated based on the usage. |
| parameters.fsType    | Type of a host file system. The supported types are: <ul style="list-style-type: none"><li>• ext2</li><li>• ext3</li><li>• ext4</li><li>• xfs</li></ul>                                                                                                             | No        | ext4          | No                                         | This parameter is available only when <b>volumeMode</b> of the PVC is set to <b>Filesystem</b> .                                                                                    |

| Parameter                  | Description                                                                                                                                                                                                                                                                                                      | Mandatory | Default Value | Whether the Volume Management Takes Effect | Remarks                                                                                                                                                                                                        |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parameters.applicationType | Application type name of the LUN to be created.                                                                                                                                                                                                                                                                  | No        | -             | Yes                                        | Log in to DeviceManager and choose <b>Services &gt; Block Service &gt; LUN Groups</b> (or <b>Namespace Groups</b> ) > <b>LUNs</b> (or <b>Namespaces</b> ) > <b>Create</b> to obtain the application type name. |
| parameters.qos             | LUN/NAS QoS settings of the PV on the storage side.<br>The value of the parameter is JSON character strings in dictionary format. A character string is enclosed by single quotation marks and the dictionary key by double quotation marks. For example, <code>'{"MAXBANDWIDTH": 999, "MAXIOPS": 999}'</code> . | No        | -             | No                                         | For details about the supported QoS configurations, see <a href="#">Table 5-65</a> .                                                                                                                           |

| Parameter               | Description                                         | Mandatory | Default Value | Whether the Volume Management Takes Effect | Remarks                                                                                                       |
|-------------------------|-----------------------------------------------------|-----------|---------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| parameters.fsPermission | Permission on the directory mounted to a container. | No        | -             | No                                         | For details about the configuration format, refer to the Linux permission settings, for example, 777 and 755. |

| Parameter                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Mand<br>at<br>ory | Def<br>ault<br>Val<br>ue | Wh<br>eth<br>er<br>the<br>Vol<br>um<br>e<br>Ma<br>na<br>ge<br>me<br>nt<br>Tak<br>es<br>Eff<br>ect | Remarks                                      |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------|
| parameters.disableVerifyCapacity | <p>Whether to disable volume capacity verification. After this function is disabled, the system will not verify whether the volume capacity is an integer multiple of the sector size.</p> <p>The value can be:</p> <ul style="list-style-type: none"><li>• "true": disables volume capacity verification.</li><li>• "false": enables volume capacity verification.</li></ul> <p><b>NOTICE</b><br/>When Red Hat OpenShift Virtualization is used to connect to CSI, this parameter must be set to <b>true</b>.</p> | No                | "true"                   | Yes                                                                                               | For OceanDisk, the sector size is 512 bytes. |

| Parameter              | Description                                                                                                      | Mandatory | Default Value               | Whether the Volume Management Takes Effect | Remarks |
|------------------------|------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------|--------------------------------------------|---------|
| parameters.description | Description of the LUN to be created.<br>Value type: character string<br>The value contains 0 to 255 characters. | No        | Created from Kubernetes CSI | No                                         |         |

Table 5-64 Common parameters in mountOptions

| Parameter            | Description                                                                                                                                           | Mandatory | Default Value | Remarks                                      |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|----------------------------------------------|
| mountOptions.discard | Automatically triggers the Trim or Discard operation when a file system is mounted. This operation instructs a block device to release unused blocks. | No        | -             | The xfs and ext4 file systems are supported. |



**Table 5-65** Supported QoS configurations

| Parameter    | Description                                                | Remarks                                                                                    |
|--------------|------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| IOTYPE       | Read/write type.                                           | The value can be: <ul style="list-style-type: none"><li>• 2: read and write I/Os</li></ul> |
| MAXBANDWIDTH | Maximum bandwidth. This is a restriction policy parameter. | The value is an integer ranging from 1 to 999999999, expressed in MB/s.                    |
| MINBANDWIDTH | Minimum bandwidth. This is a protection policy parameter.  | The value is an integer ranging from 1 to 999999999, expressed in MB/s.                    |
| MAXIOPS      | Maximum IOPS. This is a restriction policy parameter.      | The value is an integer ranging from 100 to 999999999.                                     |
| MINIOPS      | Minimum IOPS. This is a protection policy parameter.       | The value is an integer ranging from 100 to 999999999.                                     |
| LATENCY      | Maximum latency. This is a protection policy parameter.    | The value can be <b>0.5</b> or <b>1.5</b> , expressed in ms.                               |

## 5.2.2 Managing a StorageClass

### 5.2.2.1 Querying a StorageClass

**Step 1** Run the following command to view the information about the created StorageClass.

```
kubectl get sc mysc
```

The following is an example of the command output.

|      |                |               |                   |                      |     |
|------|----------------|---------------|-------------------|----------------------|-----|
| NAME | PROVISIONER    | RECLAIMPOLICY | VOLUMEBINDINGMODE | ALLOWVOLUMEEXPANSION | AGE |
| mysc | csi.huawei.com | Delete        | Immediate         | true                 | 8s  |

----End

### 5.2.2.2 Modifying a StorageClass

Kubernetes allows only the **allowVolumeExpansion** field of a StorageClass to be modified. The modification method is as follows:

**Step 1** Run the following command to check whether the StorageClass can be expanded:

```
kubectl get sc mysc
```

If the command output is displayed as the following example, the StorageClass **mysc** cannot be expanded:

|      |                |               |                   |                      |     |
|------|----------------|---------------|-------------------|----------------------|-----|
| NAME | PROVISIONER    | RECLAIMPOLICY | VOLUMEBINDINGMODE | ALLOWVOLUMEEXPANSION | AGE |
| mysc | csi.huawei.com | Delete        | Immediate         | false                | 8s  |

**Step 2** Run the following command to enable capacity expansion for the StorageClass:

```
kubectcl patch sc mysc --patch '{"allowVolumeExpansion":true}'
```

**Step 3** Check the StorageClass again and ensure that the modification is successful.

| NAME | PROVISIONER    | RECLAIMPOLICY | VOLUMEBINDINGMODE | ALLOWVOLUMEEXPANSION | AGE |
|------|----------------|---------------|-------------------|----------------------|-----|
| mysc | csi.huawei.com | Delete        | Immediate         | true                 | 8s  |

----End

### 5.2.2.3 Deleting a StorageClass

**Step 1** Run the following command to view the StorageClass:

```
kubectcl get sc mysc
```

The following is an example of the command output.

| NAME | PROVISIONER    | RECLAIMPOLICY | VOLUMEBINDINGMODE | ALLOWVOLUMEEXPANSION | AGE |
|------|----------------|---------------|-------------------|----------------------|-----|
| mysc | csi.huawei.com | Delete        | Immediate         | false                | 8s  |

**Step 2** Run the following command to delete the StorageClass:

```
kubectcl delete sc mysc
```

If the following information is displayed, the deletion is successful:

```
storageclass.storage.k8s.io "mysc" deleted
```

----End

## 5.3 Persistent Volume Management

### NOTICE

- Do not delete a storage backend when using Huawei CSI to manage volumes.
- When block volumes are mapped, Huawei CSI automatically creates associated objects, such as hosts, host groups, and LUN groups, as well as mapping views. If these objects are manually created on the storage, the mapping logic of Huawei CSI will be affected. Therefore, ensure that these objects are deleted before mapping volumes using Huawei CSI.
- The host name of a node in the cluster must contain no more than 27 characters. The extra part will be truncated when the host is created on the storage device. As a result, multiple nodes in the cluster may be mapped to the same host on the storage device.

Based on service requirements, files in containers need to be persistently stored on disks. When the containers are re-built or re-allocated to new nodes, the persistent data can still be used.

To persistently store data on storage devices, you need to use the [PersistentVolume \(PV\)](#) and [PersistentVolumeClaim \(PVC\)](#) when provisioning containers.

- PV: a piece of storage in the Kubernetes cluster that has been provisioned by an administrator or dynamically provisioned using a [StorageClass](#).

- PVC: a request for storage by a user. A PVC consumes PV resources. A PVC can request specific size and access modes. For example, a PV can be mounted in ReadWriteOnce, ReadOnlyMany, or ReadWriteMany mode. For details, see [Access Modes](#).

This section describes how to use Huawei CSI to create, clone, and expand the capacity of a PV and PVC.

## 5.3.1 Configuring PVs

Huawei CSI allows storage resources (LUNs or file systems) to be created on Huawei storage and provided for containers based on user settings. For details about the supported features, see [3 Compatibility and Features](#) of the storage device.

PV configurations can be classified into configuring dynamic PVs or static PVs, and managing PVs.

- Configuring dynamic PV does not require a PV to be created in advance. Huawei CSI automatically creates resources required by a PV on storage devices based on a StorageClass. In addition, you can create a PV when creating a PVC.
- Configuring a static PV requires the administrator to create required resources on a storage device in advance and use existing resources by creating a PV. In addition, you can specify the associated PV when creating a PVC.
- Managing PVs does not require a PV to be created in advance. You can specify the StorageClass and resource information on the storage device in a PVC. Create the PVC and PV at the same time, and manage all existing storage resources in the cluster.

### 5.3.1.1 Configuring Dynamic PVs

Dynamic volume provisioning allows storage volumes to be created on demand. Dynamic volume provisioning depends on the StorageClass objects. The cluster administrator can define multiple StorageClass objects as required and specify a StorageClass that meets service requirements when declaring a PV or PVC. When applying for resources from Huawei storage devices, Huawei CSI creates storage resources that meet service requirements based on the preset StorageClass.

## Configuration Description

Perform the following steps to configure and use dynamic PVs:

- [Preparation](#)
- [Configuring a PVC](#)
- [Using a PVC](#)

## Preparation

Before configuring dynamic PVs, configure StorageClass by referring to [5.2.1 Configuring a StorageClass](#).

## Configuring a PVC

- Step 1** Prepare the PVC configuration file **mypvc.yaml**. The following is an example. For details about other parameters, see [Table 5-66](#).

```
kind: PersistentVolumeClaim
apiVersion: v1
metadata:
 name: mypvc
spec:
 accessModes:
 - ReadWriteOnce
 volumeMode: Filesystem
 storageClassName: mysc
 resources:
 requests:
 storage: 100Gi
```

- Step 2** Run the following command to create a PVC using the configuration file.

```
kubectl create -f mypvc.yaml
```

- Step 3** After a period of time, run the following command to view the information about the created PVC.

```
kubectl get pvc mypvc
```

The following is an example of the command output. If the PVC status is **Bound**, the PVC has been created and can be used by a Pod.

| NAME  | STATUS | VOLUME                                   | CAPACITY | ACCESS MODES | STORAGECLASS | AGE |
|-------|--------|------------------------------------------|----------|--------------|--------------|-----|
| mypvc | Bound  | pvc-840054d3-1d5b-4153-b73f-826f980abf9e | 100Gi    | RWO          | mysc         | 12s |

### NOTICE

- After the PVC is created, if the PVC is in the **Pending** state after a long time (for example, one minute), refer to [8.3.1 When a PVC Is Created, the PVC Is in the Pending State](#).
- You are advised to create or delete a maximum of 100 PVCs in a batch.
- When a local file system volume with KV cache resources is provisioned, the volume capacity must be greater than or equal to 10 Gi and less than or equal to 32 Pi.

----End

**Table 5-66** PVC parameters for configuring a dynamic PV

| Parameter     | Description                        | Mandatory | Default Value | Remarks                                                                                                                                                              |
|---------------|------------------------------------|-----------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| metadata.name | User-defined name of a PVC object. | Yes       | -             | Take Kubernetes v1.22.1 as an example. The value can contain digits, lowercase letters, hyphens (-), and periods (.), and must start and end with a letter or digit. |

| Parameter                       | Description                                                                                                                                                                                                                        | Mandatory | Default Value | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| spec.volumeMode                 | Volume mode. This parameter is optional. When LUN volumes are used, the following types are supported: <ul style="list-style-type: none"><li>• <b>Filesystem</b>: local file system.</li><li>• <b>Block</b>: raw device.</li></ul> | No        | Filesystem    | This parameter takes effect when a PV is mounted. The default value is <b>Filesystem</b> . <ul style="list-style-type: none"><li>• <b>Filesystem</b> indicates that a container accesses a PV using a local file system. The local file system type is specified by the <b>fsType</b> field in the specified StorageClass. Storage of the Dtree type also uses this parameter.</li><li>• <b>Block</b> indicates that a PV is accessed in raw volume mode.</li></ul>                                                                                                                                                             |
| spec.storageClassName           | Name of the StorageClass object.                                                                                                                                                                                                   | Yes       | -             | Name of the StorageClass object required by services.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| spec.resources.requests.storage | Size of the volume to be created. The format is ***Gi and the unit is GiB.                                                                                                                                                         | Yes       | 10Gi          | The PVC capacity depends on storage specifications and host specifications. For example, OceanStor Dorado 6.1.2 or OceanStor Pacific series 8.1.0 is connected to CentOS 7. If ext4 file systems are used, see <a href="#">Table 5-67</a> . If XFS file systems are used, see <a href="#">Table 5-68</a> . If NFS or raw devices are used, the capacity must meet the specifications of the used Huawei storage device model and version.<br><br>If the PVC capacity does not meet the specifications, a PVC or Pod may fail to be created due to the limitations of storage specifications or host file system specifications. |

| Parameter        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Mandatory | Default Value | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| spec.accessModes | <p>Access mode of the volume.</p> <ul style="list-style-type: none"><li>• <b>RWO</b> (ReadWriteOnce): A volume can be mounted to a node in read/write mode. This mode also allows multiple Pods running on the same node to access the volume.</li><li>• <b>ROX</b> (ReadOnlyMany): A volume can be mounted to multiple nodes in read-only mode.</li><li>• <b>RWX</b> (ReadWriteMany): A volume can be mounted to multiple nodes in read/write mode.</li><li>• <b>RWOP</b> (ReadWriteOncePod): A volume can only be mounted to a single Pod in read/write mode. Kubernetes 1.22 and later versions support this feature.</li></ul> | Yes       | ReadWriteOnce | <ul style="list-style-type: none"><li>• RWO/ROX/RWOP: supported by all types of volumes. RWOP is supported only by Kubernetes 1.22 and later versions. For versions earlier than Kubernetes 1.29, you need to enable this feature by following the instructions in <a href="#">7.6 Enabling the ReadWriteOncePod Feature Gate</a>.</li><li>• The support for RWX is as follows:<ul style="list-style-type: none"><li>– NAS storage: supported by all volumes</li><li>– SAN storage: supported only by volumes whose <b>volumeMode</b> is set to <b>Block</b></li></ul></li></ul> |

**Table 5-67** ext4 capacity specifications

| Storage Type             | Storage Specifications | ext4 Specifications | CSI Specifications |
|--------------------------|------------------------|---------------------|--------------------|
| OceanStor Dorado         | 512 Ki to 256 Ti       | 50 Ti               | 512 Ki to 50 Ti    |
| OceanStor Pacific series | 64 Mi to 512 Ti        | 50 Ti               | 64 Mi to 50 Ti     |
| OceanDisk                | 512 Ki to 256 Ti       | 50 Ti               | 512 Ki to 50 Ti    |

**Table 5-68** XFS capacity specifications

| Storage Type             | Storage Specifications | XFS Specifications | CSI Specifications |
|--------------------------|------------------------|--------------------|--------------------|
| OceanStor Dorado         | 512 Ki to 256 Ti       | 500 Ti             | 512 Ki to 256 Ti   |
| OceanStor Pacific series | 64 Mi to 512 Ti        | 500 Ti             | 64 Mi to 500 Ti    |
| OceanDisk                | 512 Ki to 256 Ti       | 500 Ti             | 512 Ki to 256 Ti   |

## Using a PVC

After a PVC is created, you can use the PVC to create a Pod. The following is a simple example of using a PVC. In this example, the created Pod uses the newly created *mypvc*.

```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: nginx-deployment
spec:
 selector:
 matchLabels:
 app: nginx
 replicas: 2
 template:
 metadata:
 labels:
 app: nginx
 spec:
 containers:
 - image: nginx:alpine
 name: container-0
 volumeMounts:
 - mountPath: /tmp
```

```
name: pvc-mypvc
restartPolicy: Always
volumes:
- name: pvc-mypvc
 persistentVolumeClaim:
 claimName: mypvc # name of PVC
```

#### NOTE

If Pods are batch created using PVCs, the Pods are in the **ContainerCreating** status for a long time, and the huawei-csi-node service is in the **OOMKilled** status, the memory of the huawei-csi-node service is insufficient. Increase the memory limit of huawei-csi-node by referring to [Table 10-3](#).

### 5.3.1.2 Configuring a Static PV

Static volume provisioning allows administrators to use a resource created on the storage side as a PV for containers in the cluster.

---

#### NOTICE

When static PVs are used to manage HyperMetro volumes, ensure that storage volume names on both ends in the HyperMetro relationship are the same.

---

## Configuration Description

Perform the following steps to configure a static PV:

- [Preparation](#)
- [Configuring a PV](#)
- [Configuring a PVC](#)
- [Using a PVC](#)

## Preparation

A storage resource, such as a LUN or file system, required by the PV to be created exists on the storage device. If the storage resource is a file system, you also need to create the share and client information of the file system.

## Configuring a PV

**Step 1** Prepare the PV configuration file **mypv.yaml**. The following is an example. For details about other parameters, see [Table 5-69](#).

```
kind: PersistentVolume
apiVersion: v1
metadata:
 name: mypv
spec:
 volumeMode: Filesystem
 storageClassName: "" # The value must be "".
 accessModes:
 - ReadWriteOnce
 csi:
 driver: csi.huawei.com # Enter the CSI driver name.
 volumeHandle: iscsi-dorado-181.lun0001 # Enter the volume name.
 fsType: xfs # Set the file system type.
```



```
capacity:
storage: 100Gi
```

**NOTE**

In the configuration file for static volume provisioning, **storageClassName** must be set to "". Otherwise, Kubernetes will use the default StorageClass.

**Step 2** Run the following command to create a PV based on the prepared .yaml file.

```
kubectl create -f mypv.yaml
```

**Step 3** After a period of time, run the following command to view the information about the created PV.

```
kubectl get pv
```

The following is an example of the command output. If the PV status is **Available**, the PV is successfully created.

| NAME   | CAPACITY | ACCESS MODES | RECLAIM POLICY | STATUS    | CLAIM | STORAGECLASS |
|--------|----------|--------------|----------------|-----------|-------|--------------|
| REASON | AGE      |              |                |           |       |              |
| mypv   | 100Gi    | RWO          | Retain         | Available |       | 4s           |

----End

**Table 5-69** Parameters for configuring a static PV

| Parameter       | Description                                                                                                                                                                                                                    | Mandatory | Default Value | Remarks                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| metadata.name   | User-defined name of a PV object.                                                                                                                                                                                              | Yes       | -             | Take Kubernetes v1.22.1 as an example. The value can contain digits, lowercase letters, hyphens (-), and periods (.), and must start and end with a letter or digit.                                                                                                                                                                                                                                        |
| spec.volumeMode | Volume mode. This parameter is optional. When LUN volumes are used, the following types are supported: <ul style="list-style-type: none"><li><b>Filesystem</b>: local file system.</li><li><b>Block</b>: raw device.</li></ul> | No        | Filesystem    | This parameter takes effect when a PV is mounted. The default value is <b>Filesystem</b> . <ul style="list-style-type: none"><li><b>Filesystem</b> indicates that a container accesses a PV using a local file system. The local file system type is specified by the <b>fsType</b> field in the specified StorageClass.</li><li><b>Block</b> indicates that a PV is accessed in raw volume mode.</li></ul> |

| Parameter             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Mandatory | Default Value  | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| spec.storageClassName | Name of the StorageClass object. This parameter is mandatory.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Yes       | -              | Set the parameter to an empty string, that is, enter "".                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| spec.accessModes      | <p>Access mode of the volume.</p> <ul style="list-style-type: none"><li>• <b>RWO</b> (ReadWriteOnce): A volume can be mounted to a node in read/write mode. This mode also allows multiple Pods running on the same node to access the volume.</li><li>• <b>ROX</b> (ReadOnlyMany): A volume can be mounted to multiple nodes in read-only mode.</li><li>• <b>RWX</b> (ReadWriteMany): A volume can be mounted to multiple nodes in read/write mode.</li><li>• <b>RWOP</b> (ReadWriteOncePod): A volume can only be mounted to a single Pod in read/write mode. Kubernetes 1.22 and later versions support this feature.</li></ul> | Yes       | ReadWriteOnce  | <ul style="list-style-type: none"><li>• RWO/ROX/RWOP: supported by all types of volumes. RWOP is supported only by Kubernetes 1.22 and later versions. Check whether this feature is enabled for your Kubernetes cluster by referring to <a href="#">7.6 Enabling the ReadWriteOncePod Feature Gate</a>.</li><li>• The support for RWX is as follows:<ul style="list-style-type: none"><li>– NAS storage: supported by all volumes</li><li>– SAN storage: supported only by volumes whose <b>volumeMode</b> is set to <b>Block</b></li></ul></li></ul> |
| spec.csi.driver       | CSI driver name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Yes       | csi.huawei.com | Set this parameter to the driver name set during Huawei CSI installation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

| Parameter                                 | Description                                                                                                                                                                         | Mandatory | Default Value | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| spec.csi.volumeHandle                     | Unique identifier of a storage resource. This parameter is mandatory.<br>Format:<br><backendName>.<volume-name>                                                                     | Yes       | -             | The value of this parameter consists of the following parts: <ul style="list-style-type: none"><li>• &lt;backendName&gt;: indicates the name of the backend where the volume resides. You can run the following command to obtain the configured backend information.<br/><b>oceanctl get backend</b></li><li>• &lt;volume-name&gt;: indicates the name of a resource (LUN/file system/dtree) on the storage. You can obtain the value from DeviceManager.</li></ul> |
| spec.csi.fsType                           | Type of a host file system. This parameter is optional. The supported types are: <ul style="list-style-type: none"><li>• ext2</li><li>• ext3</li><li>• ext4</li><li>• xfs</li></ul> | No        | -             | If this parameter is not set, the default value <b>ext4</b> is used. This parameter is available only when <b>volumeMode</b> is set to <b>Filesystem</b> .                                                                                                                                                                                                                                                                                                           |
| spec.csi.volumeAttributes.dTreeParentName | Name of the parent file system when the volume resource type is dtree.                                                                                                              | Yes       | -             | This parameter is mandatory when the managed object is a dtree resource.                                                                                                                                                                                                                                                                                                                                                                                             |

| Parameter                 | Description                                                                                                                                                                                                    | Mandatory | Default Value | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| spec.capacity.storage     | Volume size.                                                                                                                                                                                                   | Yes       | 100Gi         | Ensure that the size is the same as that of the corresponding resource on the storage. Kubernetes will not invoke CSI to check whether the value of this parameter is correct. Therefore, the PV can be successfully created even if its capacity is inconsistent with that of the corresponding resource on the storage.                                                                                                                                                                                                                                          |
| spec.mountOptions.nfsvers | NFS mount option on the host. The following mount option is supported:<br><b>nfsvers</b> : protocol version for NFS mounting. The value can be <b>3</b> , <b>4</b> , <b>4.0</b> , <b>4.1</b> , or <b>4.2</b> . | No        | -             | This parameter is optional after the <b>-o</b> parameter when the <b>mount</b> command is executed on the host. The value is in list format.<br><br>If the NFS version is specified for mounting, NFS 3, 4.0, 4.1, and 4.2 protocols are supported (the protocol must be supported and enabled on storage devices). If <b>nfsvers</b> is set to <b>4</b> , the latest protocol version NFS 4 may be used for mounting due to different OS configurations, for example, 4.2. If the 4.0 protocol is required, you are advised to set <b>nfsvers</b> to <b>4.0</b> . |

| Parameter                      | Description                                                                                                                                                                                                                                                                                                             | Mandatory | Default Value | Remarks                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| spec.mountOptions.acl          | The DPC namespace supports the ACL function. The DPC client supports POSIX ACL, NFSv4 ACL, and NT ACL authentication.                                                                                                                                                                                                   | No        | -             | The descriptions of <b>acl</b> , <b>aclonlyposix</b> , <b>cnflush</b> , and <b>cflush</b> are for reference only. For details about the parameters, see <a href="#">OceanStor Pacific Series Product Documentation</a> and choose <b>Configuration &gt; Basic Service Configuration Guide for File &gt; Configuring Basic Services (DPC Scenario) &gt; Accessing a DPC Share on a Client &gt; Step 2</b> . |
| spec.mountOptions.aclonlyposix | <p>The DPC namespace supports POSIX ACL, and the DPC client supports POSIX ACL authentication.</p> <p>The following protocols support POSIX ACL: DPC, NFSv3, and HDFS. If NFSv4 ACL or NT ACL is used, the DPC client cannot identify the ACL of this type. As a result, the ACL of this type does not take effect.</p> | No        | -             | If <b>aclonlyposix</b> and <b>acl</b> are used together, only <b>acl</b> takes effect. That is, the namespace supports the ACL function.                                                                                                                                                                                                                                                                   |

| Parameter                 | Description                                                                                                                | Mandatory | Default Value | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------|-----------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| spec.mountOptions.cnflush | Asynchronous disk flushing mode. That is, data is not flushed to disks immediately when files in the namespace are closed. | No        | -             | Asynchronous flushing mode: When a file is closed, data in the cache is not flushed to storage media in synchronous mode. Instead, data is written from the cache to the storage media in asynchronous flushing mode. After the write service is complete, data is flushed from the cache to disks periodically based on the flushing period. In a multi-client scenario, if concurrent operations are performed on the same file, the file size update is affected by the disk flushing period. That is, the file size is updated only after the disk flushing is complete. Generally, the update is completed within several seconds. Synchronous I/Os are not affected by the disk flushing period. |
| spec.mountOptions.cflush  | Synchronous disk flushing mode. That is, data is flushed to disks immediately when files in the namespace are closed.      | No        | -             | By default, the synchronous disk flushing mode is used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

## Configuring a PVC

After a PV is created in static volume provisioning mode, you can create a PVC based on the PV for containers.

**Step 1** Prepare the PVC configuration file **mypvc.yaml**. The following is an example. For details about other parameters, see [Table 5-70](#).

```
kind: PersistentVolumeClaim
apiVersion: v1
```

```
metadata:
 name: mypvc
spec:
 storageClassName: ""
 accessModes:
 - ReadWriteOnce
 volumeMode: Filesystem
 resources:
 requests:
 storage: 100Gi
 volumeName: mypv # Enter the name of the corresponding PV.
```

**Step 2** Run the following command to create a PVC based on the configured .yaml file.

```
kubectl create -f mypvc.yaml
```

**Step 3** After a period of time, run the following command to view the information about the created PVC.

```
kubectl get pvc
```

The following is an example of the command output. If the PVC status is **Bound**, the PVC is successfully created.

| NAME  | STATUS | VOLUME                                   | CAPACITY | ACCESS MODES | STORAGECLASS | AGE |
|-------|--------|------------------------------------------|----------|--------------|--------------|-----|
| mypvc | Bound  | pvc-840054d3-1d5b-4153-b73f-826f980abf9e | 100Gi    | RWO          |              | 12s |

#### NOTE

- After the PVC is created, if the PVC is in the **Pending** state after a long time, see [8.3.1 When a PVC Is Created, the PVC Is in the Pending State](#).
- You are advised to create or delete a maximum of 100 PVCs in a batch.

----End

**Table 5-70** PVC parameters for configuring a static PV

| Parameter     | Description                        | Mandatory | Default Value | Remarks                                                                                                                                                              |
|---------------|------------------------------------|-----------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| metadata.name | User-defined name of a PVC object. | Yes       | -             | Take Kubernetes v1.22.1 as an example. The value can contain digits, lowercase letters, hyphens (-), and periods (.), and must start and end with a letter or digit. |

| Parameter        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Mandatory | Default Value | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| spec.accessModes | <p>Access mode of the volume.</p> <ul style="list-style-type: none"><li>• <b>RWO</b> (ReadWriteOnce): A volume can be mounted to a node in read/write mode. This mode also allows multiple Pods running on the same node to access the volume.</li><li>• <b>ROX</b> (ReadOnlyMany): A volume can be mounted to multiple nodes in read-only mode.</li><li>• <b>RWX</b> (ReadWriteMany): A volume can be mounted to multiple nodes in read/write mode.</li><li>• <b>RWOP</b> (ReadWriteOncePod): A volume can only be mounted to a single Pod in read/write mode. Kubernetes 1.22 and later versions support this feature.</li></ul> | Yes       | ReadWriteOnce | <ul style="list-style-type: none"><li>• RWO/ROX/RWOP: supported by all types of volumes. RWOP is supported only by Kubernetes 1.22 and later versions. For versions earlier than Kubernetes 1.29, you need to enable this feature by following the instructions in <a href="#">7.6 Enabling the ReadWriteOncePod Feature Gate</a>.</li><li>• The support for RWX is as follows:<ul style="list-style-type: none"><li>– NAS storage: supported by all volumes</li><li>– SAN storage: supported only by volumes whose <b>volumeMode</b> is set to <b>Block</b></li></ul></li></ul> |



| Parameter                       | Description                       | Mandatory | Default Value | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------|-----------------------------------|-----------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| spec.volumeMode                 | Volume mode.                      | No        | Filesystem    | This parameter is optional. The value can be <b>Filesystem</b> or <b>Block</b> . The default value is <b>Filesystem</b> . This parameter takes effect when a Pod is created. <b>Filesystem</b> indicates that a file system is created on a PVC to access the storage. <b>Block</b> indicates that a raw volume is used to access the storage.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| spec.resources.requests.storage | Size of the volume to be created. | Yes       | -             | <p>Size of the volume to be created. The format is ***Gi and the unit is GiB.</p> <p>The PVC capacity depends on storage specifications and host specifications. For example, OceanStor Dorado 6.1.2 or OceanStor Pacific series 8.1.0 is connected to CentOS 7. If ext4 file systems are used, see <a href="#">Table 5-67</a>. If XFS file systems are used, see <a href="#">Table 5-68</a>. If NFS or raw devices are used, the capacity must meet the specifications of the used Huawei storage device model and version.</p> <p>If the PVC capacity does not meet the specifications, a PVC or Pod may fail to be created due to the limitations of storage specifications or host file system specifications.</p> <p>When a PVC is created using a static PV and the PVC capacity is smaller than the capacity of the bound PV, the PVC capacity is set to the capacity of the bound PV. If the PVC capacity is greater than the capacity of the bound PV, the PVC cannot be created.</p> |

| Parameter             | Description                      | Mandatory | Default Value | Remarks                                                                                                                                           |
|-----------------------|----------------------------------|-----------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| spec.volumeName       | Name of the PV object.           | Yes       | -             | This parameter is mandatory when a PVC is created statically.                                                                                     |
| spec.storageClassName | Name of the StorageClass object. | Yes       | -             | When a PVC is created, an empty character string is transferred. If this parameter is not set, the default StorageClass object name will be used. |

## Using a PVC

The use method is the same as that for dynamic volume provisioning in [Using a PVC](#).

### 5.3.1.3 Managing PVs

Manage Volume Provisioning allows administrators to use resources created on storage as PVs and supports features of dynamic volumes, such as capacity expansion, snapshot, and clone. This is a custom capability of Huawei CSI. This feature applies to the following scenarios:

- In the reconstruction containerized applications, existing storage volumes need to be used.
- The Kubernetes cluster is rebuilt.
- Storage data is migrated in disaster recovery (DR) scenarios.

#### NOTE

- Manage Volume Provisioning allows existing storage resources to be managed by Kubernetes. You are not allowed to manage a storage resource for multiple times and concurrently delete or create a storage resource.  
When a storage resource is managed by multiple clusters, operations on the managed volume in a single cluster take effect only in the cluster and will not be synchronized to other clusters. Instead, you need to perform these operations on the managed volume in other clusters.  
For example, when you expand the capacity of a PVC in a cluster, the capacity of the corresponding PVC in other clusters will not be automatically expanded. In this case, you need to manually expand the capacity in other clusters by running the expansion commands in [5.3.2.1 Expanding the Capacity of a PV](#).
- When managing PVs, ensure that the PVs comply with the storage class declaration. Otherwise, Huawei CSI may fail to manage the PVs. For example, when you manage a common PV, the storage backend of the HyperMetro type is referenced in the storage class, and the HyperMetro parameter is enabled. After the PV is managed, if you clone the PV or create a PV based on a snapshot, Huawei CSI will fail to execute the operation.
- When managed PVs are used to manage HyperMetro volumes, ensure that storage volume names on both ends in the HyperMetro relationship are the same.

## Configuration Description

Perform the following steps to manage and use PVs:

- [Preparation](#)
- [Configuring a PVC](#)
- [Using a PVC](#)

## Preparation

- You have registered the storage where the volume to be managed resides with CSI.
- You have logged in to the storage device to obtain the name and capacity of the volume to be managed.
- The StorageClass has been configured. For details, see [5.2.1 Configuring a StorageClass](#) (pay attention to the **Whether the Volume Management Takes Effect** field in the table).

## Configuring a PVC

**Step 1** Prepare the PVC configuration file **mypvc.yaml**. The following is an example. For details about other parameters, see [Table 5-71](#).

```
kind: PersistentVolumeClaim
apiVersion: v1
metadata:
 name: mypvc
 annotations:
 csi.huawei.com/manageVolumeName: "*" # Enter the storage resource name.
 csi.huawei.com/manageBackendName: "*" # Enter the storage backend name.
 labels:
 provisioner: csi.huawei.com
spec:
 accessModes:
 - ReadWriteOnce
 volumeMode: Filesystem
 storageClassName: mysc
 resources:
 requests:
 storage: 100Gi
```

**Step 2** Run the following command to create a PVC using the configuration file.

```
kubectl create -f mypvc.yaml
```

**Step 3** After a period of time, run the following command to view the information about the created PVC.

```
kubectl get pvc mypvc
```

The following is an example of the command output. If the PVC status is **Bound**, the PVC has been created and can be used by a Pod.

| NAME  | STATUS | VOLUME                                   | CAPACITY | ACCESS MODES | STORAGECLASS | AGE |
|-------|--------|------------------------------------------|----------|--------------|--------------|-----|
| mypvc | Bound  | pvc-840054d3-1d5b-4153-b73f-826f980abf9e | 100Gi    | RWO          | mysc         | 12s |

**NOTICE**

- After the PVC is created, if the PVC is in the **Pending** state after a long time (for example, one minute), refer to [8.3.1 When a PVC Is Created, the PVC Is in the Pending State](#).
- You are advised to create or delete a maximum of 100 PVCs in a batch.

----End

**Table 5-71** PVC parameters for managing PVs

| Parameter            | Description                                                                                                                                                                                                                                                                                    | Mandatory | Default Value                                                                                  | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| metadata.annotations | <p>PVC object annotations. Set the following parameters:</p> <ul style="list-style-type: none"><li>• <i>Driver name/</i><b>manageVolumeName</b>: volume name on the storage.</li><li>• <i>Driver name/</i><b>manageBackendName</b>: name of the backend to which the volume belongs.</li></ul> | Yes       | <p>csi.huawei.com/<br/>manageVolumeName: *</p> <p>csi.huawei.com/<br/>manageBackendName: *</p> | <ul style="list-style-type: none"><li>• For details about how to obtain <i>Driver name</i>, see <a href="#">Table 4-7</a>.</li><li>• <i>Driver name/</i><b>manageVolumeName</b>: name of an existing volume on the storage. Only English characters are supported.</li><li>• <i>Driver name/</i><b>manageBackendName</b>: name of the storage backend in CSI.</li></ul> <p>You can run the <b>oceanctl get backend -n huawei-csi</b> command to obtain the backend name.</p> |

| Parameter       | Description                        | Mandatory | Default Value | Remarks                                                                                                                                                                                                                                                                                            |
|-----------------|------------------------------------|-----------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| metadata.labels | PVC object labels.                 | No        | -             | Format: <b>provisioner:</b><br><i>Driver name specified during installation</i><br>Example:<br><b>provisioner:</b><br><b>csi.huawei.com</b><br>This parameter takes effect when a PVC is created. It is used to listen to PVC resources and obtain information about <b>metadata.annotations</b> . |
| metadata.name   | User-defined name of a PVC object. | Yes       | -             | Take Kubernetes v1.22.1 as an example. The value can contain digits, lowercase letters, hyphens (-), and periods (.), and must start and end with a letter or digit.                                                                                                                               |

| Parameter       | Description                                                                                                                                                                                                                               | Mandatory | Default Value | Remarks                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| spec.volumeMode | <p>Volume mode. This parameter is optional. When LUN volumes are used, the following types are supported:</p> <ul style="list-style-type: none"><li>• <b>Filesystem</b>: local file system.</li><li>• <b>Block</b>: raw device.</li></ul> | No        | Filesystem    | <p>This parameter takes effect when a PV is mounted.</p> <ul style="list-style-type: none"><li>• <b>Filesystem</b> indicates that a container accesses a PV using a local file system. The local file system type is specified by the <b>fsType</b> field in the specified StorageClass.</li><li>• <b>Block</b> indicates that a PV is accessed in raw volume mode.</li></ul> |

| Parameter             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Mandatory | Default Value | Remarks                                                                               |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|---------------------------------------------------------------------------------------|
|                       | <p><b>NOTE</b></p> <p>This parameter takes effect when a PV is mounted. The use method of this parameter must be the same as that of the managed volume.</p> <ul style="list-style-type: none"><li>• If a volume is used as a raw volume before being managed, <b>volumeMode</b> must be set to <b>Block</b>.</li><li>• If a volume is used in ext2, ext3, or ext4 mode before being managed, <b>volumeMode</b> must be set to <b>Filesystem</b> and <b>fsType</b> in the StorageClass must be set to <b>ext2</b>, <b>ext3</b>, or <b>ext4</b>.</li><li>• If a volume is used in XFS mode before being managed, <b>volumeMode</b> must be set to <b>Filesystem</b> and <b>fsType</b> in the StorageClass must be set to <b>xfs</b>.</li></ul> |           |               |                                                                                       |
| spec.storageClassName | Name of the StorageClass object.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Yes       | -             | The configuration of the StorageClass must be the same as that of the managed volume. |

| Parameter                       | Description                                                                                                                                                                                                                                                              | Mandatory | Default Value | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| spec.resources.requests.storage | <p>Size of the volume to be created. The format is ***Gi and the unit is GiB.</p> <p><b>NOTE</b><br/>For OceanStor Dorado/OceanStor Pacific series, in the dtree management scenario, the capacity corresponds to the hard quota of the dtree on the storage system.</p> | Yes       | -             | <p>The PVC capacity depends on storage specifications and host specifications. For example, OceanStor Dorado 6.1.2 or OceanStor Pacific series 8.1.0 is connected to CentOS 7. If ext4 file systems are used, see <a href="#">Table 5-67</a>. If XFS file systems are used, see <a href="#">Table 5-68</a>. If NFS or raw devices are used, the capacity must meet the specifications of the used Huawei storage device model and version.</p> <p>If the PVC capacity does not meet the specifications, a PVC or Pod may fail to be created due to the limitations of storage specifications or host file system specifications.</p> |



| Parameter        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Mandatory | Default Value | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| spec.accessModes | <p>Access mode of the volume.</p> <ul style="list-style-type: none"><li>• <b>RWO</b> (ReadWriteOnce): A volume can be mounted to a node in read/write mode. This mode also allows multiple Pods running on the same node to access the volume.</li><li>• <b>ROX</b> (ReadOnlyMany): A volume can be mounted to multiple nodes in read-only mode.</li><li>• <b>RWX</b> (ReadWriteMany): A volume can be mounted to multiple nodes in read/write mode.</li><li>• <b>RWOP</b> (ReadWriteOnce Pod): A volume can only be mounted to a single Pod in read/write mode. Kubernetes 1.22 and later versions support this feature.</li></ul> | Yes       | ReadWriteOnce | <ul style="list-style-type: none"><li>• RWO/ROX/RWOP: supported by all types of volumes. RWOP is supported only by Kubernetes 1.22 and later versions. For versions earlier than Kubernetes 1.29, you need to enable this feature by following the instructions in <a href="#">7.6 Enabling the ReadWriteOncePod Feature Gate</a>.</li><li>• The support for RWX is as follows:<ul style="list-style-type: none"><li>– NAS storage: supported by all volumes</li><li>– SAN storage: supported only by volumes whose <b>volumeMode</b> is set to <b>Block</b></li></ul></li></ul> |

## Using a PVC

The procedure is the same as that for [Using a PVC](#) for dynamic volume provisioning.

### 5.3.2 Managing PVs

#### 5.3.2.1 Expanding the Capacity of a PV

##### NOTE

For OceanStor V700R001C10 as well as OceanStor Dorado V700R001C10 and later versions, the minimum capacity of a file system after capacity expansion is limited. For details, see the product documentation of the corresponding storage device.

When the capacity of a PVC used by a container is insufficient, you need to expand the capacity of the PVC.

#### Prerequisites

- A PVC has been created, the backend to which it resides exists and supports capacity expansion.
- [3 Compatibility and Features](#) lists the storage that supports capacity expansion, and [10.4 Kubernetes Feature Matrix](#) lists the Kubernetes versions that support capacity expansion.
- The csi-resizer service is enabled for huawei-csi-controller.

```
kubectl describe deploy huawei-csi-controller -n huawei-csi | grep csi-resizer
```

If the following information is displayed, the csi-resizer service is enabled.

```
csi-resizer:
Image: k8s.gcr.io/sig-storage/csi-resizer:v1.9.0
```

#### Procedure

- Step 1** Run the following command to check whether the StorageClass supports capacity expansion. In the preceding command, *myisc* indicates the name of the StorageClass to be queried.

```
kubectl get sc myisc
```

The following is an example of the command output.

| NAME                        | PROVISIONER    | RECLAIMPOLICY | VOLUMEBINDINGMODE          |
|-----------------------------|----------------|---------------|----------------------------|
| <b>ALLOWVOLUMEEXPANSION</b> | <b>AGE</b>     |               |                            |
| myisc                       | csi.huawei.com | Delete        | Immediate <b>true</b> 172m |

If the value of **ALLOWVOLUMEEXPANSION** is **true**, the current StorageClass supports capacity expansion. In this case, go to [Step 3](#).

- Step 2** Run the following command to change the value of **allowVolumeExpansion** to **true**. In the preceding command, *myisc* indicates the name of the StorageClass to be modified.

```
kubectl patch sc myisc --patch '{"allowVolumeExpansion":true}'
```

- Step 3** Run the following command to query the StorageClass name of the PVC. In the preceding command, *mypvc* indicates the name of the PVC to be expanded.

```
kubectl get pvc mypvc
```

The following is an example of the command output.

| NAME         | STATUS | VOLUME                                   | CAPACITY | ACCESS MODES |
|--------------|--------|------------------------------------------|----------|--------------|
| STORAGECLASS | AGE    |                                          |          |              |
| mypvc        | Bound  | pvc-3383be36-537c-4cb1-8f32-a415fa6ba384 | 2Gi      | RW0          |
| <i>mysc</i>  | 145m   |                                          |          |              |

**Step 4** Run the following command to expand the capacity.

```
kubectl patch pvc mypvc -p '{"spec":{"resources":{"requests":{"storage":"120Gi"}}}}'
```

In the preceding command, *mypvc* indicates the name of the PVC to be expanded, and *120Gi* indicates the capacity after expansion. Change the values based on the site requirements.

 **NOTE**

- The PVC capacity depends on storage specifications and host specifications. For example, OceanStor Dorado 6.1.2 or OceanStor Pacific series 8.1.0 is connected to CentOS 7. If ext4 file systems are used, see [Table 5-67](#). If XFS file systems are used, see [Table 5-68](#). If NFS or raw devices are used, the capacity must meet the specifications of the used Huawei storage device model and version.
- If the PVC capacity does not meet the specifications, a PVC or Pod may fail to be created due to the limitations of storage specifications or host file system specifications.
- If the capacity expansion fails because the target capacity exceeds the storage pool capacity, see [8.3.4 Failed to Expand the PVC Capacity Because the Target Capacity Exceeds the Storage Pool Capacity](#).

**Step 5** Run the following command to check whether the capacity modification takes effect.

```
kubectl get pvc
```

The following is an example of the command output. If the value of **CAPACITY** is changed to the specified capacity, the capacity expansion is successful.

| NAME  | STATUS | VOLUME                                   | CAPACITY     | ACCESS MODES | STORAGECLASS | AGE |
|-------|--------|------------------------------------------|--------------|--------------|--------------|-----|
| mypvc | Bound  | pvc-3383be36-537c-4cb1-8f32-a415fa6ba384 | <b>120Gi</b> | RWO          | mysc         | 24s |

----End

### 5.3.2.2 Cloning a PV

This section describes how to clone a PVC.

When cloning a PVC, you need to specify the data source. The following is an example of cloning a PVC. In this example, **mypvc** is used as the data source and a PVC named **myclone** is created.

```
kind: PersistentVolumeClaim
apiVersion: v1
metadata:
 name: myclone
spec:
 storageClassName: mysc
 dataSource:
 name: mypvc
 kind: PersistentVolumeClaim
 volumeMode: Filesystem
 accessModes:
 - ReadWriteOnce
 resources:
 requests:
 storage: 2Gi
```

**NOTICE**

- The specified **storageClassName** must be the same as the StorageClass of the source volume in **dataSource**.
- The capacity of the clone volume must be greater than or equal to that of the source volume. Equal capacity is recommended.

## Prerequisites

The source PVC already exists in the system, and the backend where the source PVC resides supports cloning. [3 Compatibility and Features](#) lists the storage that supports cloning, and [10.4 Kubernetes Feature Matrix](#) lists the Kubernetes versions that support cloning.

## Procedure

- Step 1** Run the following command to create a PVC based on the configuration file of the clone volume.

```
kubectl create -f myclone.yaml
```

----End

### 5.3.2.3 Changing a PV

#### 5.3.2.3.1 Enabling the PVC Change Feature

The PVC change feature is disabled by default during Huawei CSI installation. To use this feature, perform the following steps.

## Enabling the PVC Change Feature Using Helm

## Prerequisites

You have installed Huawei CSI using Helm.

## Procedure

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
- Step 2** Run the following command to check whether the PVC change feature is enabled.

**helm-huawei-csi** indicates the Helm chart name specified during installation, and **huawei-csi** indicates the Helm chart namespace specified during installation. For details about the component package path, see [Table 4-1](#).

```
helm get values helm-huawei-csi -n huawei-csi -a | grep volumeModify -A 1
```

The following is an example of the command output.

- If **enabled: true** is displayed in the command output, the feature is enabled. In this case, skip the following steps.

- If **enabled: false** is displayed in the command output, perform the following steps to enable the PVC change feature.

```
volumeModify:
 enabled: false
```

**Step 3** Go to the **/helm/esdk** directory and run the following command to configure the volume change CRD.

```
kubectl apply -f ./crds/volume-modify/
customresourcedefinition.apiextensions.k8s.io/volumemodifyclaims.xuanwu.huawei.io configured
customresourcedefinition.apiextensions.k8s.io/volumemodifycontents.xuanwu.huawei.io configured
```

#### NOTE

If the command output contains **Warning: resource customresourcedefinitions/volumemodifycontents.xuanwu.huawei.io is missing the kubectl.kubernetes.io/last-applied-configuration...**, you can ignore it. This message is displayed because the **kubectl create** command instead of the **kubectl apply** command is used for installation by Helm.

**Step 4** Run the following command to obtain the original service configuration file.

```
helm get values helm-huawei-csi -n huawei-csi -a > ./update-values.yaml
```

**Step 5** Run the **vi update-values.yaml** command to open the file obtained in [Step 4](#) and modify the following configuration. After the modification is complete, press **Esc** and enter **:wq!** to save the modification.

```
csiExtender:
 volumeModify:
 enabled: true
```

**Step 6** Run the following command to update Huawei CSI services.

```
helm upgrade helm-huawei-csi ./ -n huawei-csi -f ./update-values.yaml
```

**Step 7** Run the following command to check whether the services are started.

```
kubectl get pod -n huawei-csi
```

The following is an example of the command output. In the preceding command, **huawei-csi** indicates the namespace for deploying Huawei CSI.

| NAME                                   | READY | STATUS  | RESTARTS | AGE |
|----------------------------------------|-------|---------|----------|-----|
| huawei-csi-controller-6dfcc4b79f-9vjtq | 10/10 | Running | 0        | 24m |
| huawei-csi-node-tqs87                  | 3/3   | Running | 0        | 20m |

----End

## Enabling the PVC Change Feature Manually

### Prerequisites

Huawei CSI has been manually installed.

### Procedure

**Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.

**Step 2** Go to the **manual/esdk** working directory and run the following command to configure the volume change CRD.

```
kubectl apply -f ./crds/volume-modify/
```

**Step 3** Run the following commands to enable the CSI service with the extender container. For details about the component package path, see [Table 4-1](#).

```
kubectl apply -f ./deploy/huawei-csi-controller-extender.yaml
kubectl apply -f ./deploy/huawei-csi-node-extender.yaml
```

**Step 4** Run the following command to check whether the services are started.

```
kubectl get pod -n huawei-csi
```

The following is an example of the command output. In the preceding command, **huawei-csi** indicates the namespace for deploying Huawei CSI.

| NAME                                   | READY | STATUS  | RESTARTS | AGE |
|----------------------------------------|-------|---------|----------|-----|
| huawei-csi-controller-6dfcc4b79f-9vjtq | 10/10 | Running | 0        | 24m |
| huawei-csi-node-tqs87                  | 3/3   | Running | 0        | 24m |

----End

### 5.3.2.3.2 Configuring PVC Changes

The PVC change feature is implemented using CRD. Related resources are described as follows.

**Table 5-72** Resource description

| NAME                 | APIVERSION          | NAMESPACED | KIND                |
|----------------------|---------------------|------------|---------------------|
| volumemodifyclaims   | xuanwu.huawei.io/v1 | false      | VolumeModifyClaim   |
| volumemodifycontents | xuanwu.huawei.io/v1 | false      | VolumeModifyContent |

#### NOTE

- VolumeModifyClaim resources can be created, deleted, and queried, but cannot be updated.
- VolumeModifyContent resources can only be queried and are used to display the change details of a single PVC. Do not manually create, delete, or modify the resources.
- VolumeModifyContent resources are managed by VolumeModifyClaim. Do not manually manage VolumeModifyContent resources.

## Creating a PVC Change

### Prerequisites

The storage backends associated with the PVC to be changed are HyperMetro storage backends. If they are not HyperMetro storage backends, configure them by following the instructions in [5.1.2.2.3 Updating a Storage Backend](#).

### PVC Change File Description

The sample template of the PVC change file is **/examples/volumemodifyclaim.yaml**. The following table lists the configuration items.

**Table 5-73** Parameter description

| Parameter        | Description                                                     | Mandatory | Default Value       | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------|-----------------------------------------------------------------|-----------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| apiVersion       | API group, which is of the string type.                         | Yes       | xuanwu.huawei.io/v1 | The value is fixed at <b>xuanwu.huawei.io/v1</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| kind             | Resource type, which is of the string type.                     | Yes       | VolumeModifyClaim   | The value is fixed at <b>VolumeModifyClaim</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| metadata.name    | Name of a cluster resource object, which is of the string type. | Yes       | -                   | <p>The name must comply with the naming rules of a <b>DNS subdomain name</b>. The value can contain a maximum of 63 characters, including digits, lowercase letters, hyphens (-), and periods (.). It must start and end with a lowercase letter or digit.</p> <p>Note: During a PVC change, the original StorageClass is backed up. The name of the backup StorageClass is <i>&lt;Original StorageClass name&gt;&lt;VolumeModifyClaim name&gt;</i>, and must comply with the StorageClass naming rules.</p> |
| spec.source.kind | Data source type, which is of the string type.                  | Yes       | StorageClass        | This parameter can only be set to <b>StorageClass</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| spec.source.name | Data source name, which is of the string type.                  | Yes       | -                   | Only a StorageClass name can be configured.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| Parameter                                       | Description                                                                                                                                                                                                                                              | Mandatory | Default Value | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>spec.parameters.hyperMetro</code>         | Whether to change a common volume to a HyperMetro volume. Currently, the value can only be <b>"true"</b> .                                                                                                                                               | Yes       | -             | Only common storage volumes at the primary site can be changed to HyperMetro storage volumes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <code>spec.parameters.metroPairSyncSpeed</code> | Data synchronization speed of a HyperMetro pair. The value ranges from 1 to 4. The value can be: <ul style="list-style-type: none"><li>• <b>"1"</b>: low</li><li>• <b>"2"</b>: medium</li><li>• <b>"3"</b>: high</li><li>• <b>"4"</b>: highest</li></ul> | No        | -             | This parameter is available only when <b><code>spec.parameters.hyperMetro</code></b> is set to <b>"true"</b> .<br><b>NOTICE</b> <ul style="list-style-type: none"><li>• If this parameter is not configured, the storage speed of the HyperMetro pair is determined by the storage device.</li><li>• The highest synchronization speed may increase the host latency.</li><li>• In SAN scenarios, you need to supplement links after data on the source LUN is synchronized to the remote end. The timeout period for data synchronization is 12 hours. Set a proper HyperMetro pair synchronization speed based on the data volume.</li></ul> |

 **NOTE**

- The **`spec.source.kind`** and **`spec.source.name`** parameters are used to specify the volume change scope. For example, if they are set to a StorageClass and the corresponding name respectively, all PVCs in the **Bound** state provisioned using the target StorageClass will be changed.
- After all associated PVCs are changed, Huawei CSI will replace the original StorageClass and add the **`spec.parameters`** parameter of the VolumeModifyClaim so that the PVCs meet the StorageClass definition.



For details about the configuration in typical scenarios, see the following example:

## Changing a Common Volume to a HyperMetro Volume

The following is an example of changing a common volume to a HyperMetro volume:

```
apiVersion: xuanwu.huawei.io/v1
kind: VolumeModifyClaim
metadata:
 name: myvmc
spec:
 source:
 kind: StorageClass
 name: mysc
 parameters:
 hyperMetro: "true"
```

## Creating a PVC Change Resource

### NOTE

- In NAS scenarios, the changed HyperMetro volumes must be in HyperMetro AA mode.
- In NAS scenarios, when a common volume is changed to a HyperMetro volume, only the storage volume at the primary site can be changed.
- If there are storage volumes with the HyperMetro relationship within the change scope, ensure that storage volume names on both ends in the HyperMetro relationship are the same.
- Do not use Huawei CSI to manage a PVC during PVC change resource creation.
- Multiple VolumeModifyClaim resources cannot be created for the same PVC. If the target PVC needs to be changed for multiple times, perform the changes one by one.
- The configuration parameters of the remote volume are subject to the specified StorageClass. Before performing the change, ensure that the configuration parameters in the StorageClass are the same as those of the local volume.

To create a PVC change resource using a PVC change file, perform the following steps:

**Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.

**Step 2** Run the following command to create a PVC change.

```
kubectl create -f volumemodifyclaim.yaml
```

**Step 3** Query the creation result by following the instructions in [Querying a PVC Change](#).

----End

## Querying a PVC Change

This section describes how to use Kubectl to query the PVC change status. Currently, Huawei CSI provides the following APIs through CRD.

## Querying a VolumeModifyClaim

To query a VolumeModifyClaim using kubectl, perform the following steps.

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
- Step 2** Run the following command to query a PVC change. In the command, *vmc-name* indicates the name of the VolumeModifyClaim resource.

```
kubectl get volumemodifyclaims <vmc-name> -owide
```

The following is an example of the command output.

| NAME  | STATUS    | READY | SOURCEKIND   | SOURCENAME | STARTEDAT            | COMPLETEDAT          |
|-------|-----------|-------|--------------|------------|----------------------|----------------------|
| AGE   |           |       |              |            |                      |                      |
| myvmc | Completed | 1/1   | StorageClass | mysc       | 2024-06-06T03:19:13Z | 2024-06-06T03:19:16Z |
| 2m2s  |           |       |              |            |                      |                      |

**Table 5-74** Command output description

| Parameter   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NAME        | VolumeModifyClaim resource name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| STATUS      | VolumeModifyClaim resource status. The value can be: <ul style="list-style-type: none"><li>● <b>Pending:</b> initial status.</li><li>● <b>Creating:</b> The VolumeModifyClaim has completed basic verification and the server has received the change task, but the task has not been completed.</li><li>● <b>Completed:</b> All associated PVCs are changed.</li><li>● <b>Rollback:</b> When associated PVCs are partially changed, a user deletes PVCs.</li><li>● <b>Deleting:</b> When all associated PVCs are changed, a user deletes PVCs.</li></ul> |
| READY       | Ratio of the number of changed PVCs to the total number of PVCs that need to be changed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| SOURCEKIND  | Data source type, for example, StorageClass.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| SOURCENAME  | Data source name, for example, StorageClass name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| STARTEDAT   | Change start time, that is, the timestamp when the server receives the task and starts to process the task.                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| COMPLETEDAT | Change completion time, that is, the timestamp when the changes of all associated PVCs are complete. This parameter exists only when <b>STATUS</b> is <b>Completed</b> .                                                                                                                                                                                                                                                                                                                                                                                  |
| AGE         | Lifetime of a VolumeModifyClaim from the time when it is created to the current time.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

----End

 NOTE

You can use `kubectl` to view the **Events** information of a `VolumeModifyClaim`. If a `VolumeModifyClaim` cannot meet the creation requirements or an error occurs during the creation, the server will record the **Events** information. The following command is used as an example:

```
kubectl describe volumemodifyclaims local-to-hypermetro
```

## Querying a VolumeModifyContent

A `VolumeModifyContent` is created using a `VolumeModifyClaim` and records the change details of a single PVC. To query a `VolumeModifyContent` using `kubectl`, perform the following steps:

**Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.

**Step 2** Run the following command to query a PVC change. In the command, *myvmc-uid* indicates the `VolumeModifyContent` resource name.

```
kubectl get volumemodifycontents myvmc-uid -owide
```

The following is an example of the command output.

| NAME        | STATUS    | MODIFYCLAIMNAME | SOURCEVOLUME  | STARTEDAT            |
|-------------|-----------|-----------------|---------------|----------------------|
| COMPLETEDAT | AGE       |                 |               |                      |
| myvmc-uid   | Completed | myvmc           | default/mypvc | 2024-06-06T03:19:07Z |
| 36m         |           |                 |               | 2024-06-06T03:19:09Z |

**Table 5-75** Command output description

| Parameter       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NAME            | VolumeModifyContent resource name. The format is <i>VolumeModifyClaim name-UID of the associated PVC</i> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| STATUS          | VolumeModifyContent resource status. The value can be: <ul style="list-style-type: none"><li>● <b>Pending</b>: initial status.</li><li>● <b>Creating</b>: The VolumeModifyContent has completed basic verification and the server has received the change task, but the task has not been completed.</li><li>● <b>Staging</b>: This status appears only in SAN scenarios, indicating that the server has completed the change task and is waiting for the host to scan for disks and supplement links.</li><li>● <b>Completed</b>: The associated PVC is changed.</li><li>● <b>Rollback</b>: The PVC change is being rolled back.</li></ul> |
| MODIFYCLAIMNAME | Name of the associated VolumeModifyClaim.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| SOURCEVOLUME    | Information about the associated PVC. The format is <i>Namespace name/PVC name</i> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| STARTEDAT       | PVC change start time, that is, the timestamp when the server receives the task and starts to process the task.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| Parameter   | Description                                                                                                                                                                  |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| COMPLETEDAT | PVC change completion time, that is, the timestamp when the changes of all associated PVCs are complete. This parameter exists only when <b>STATUS</b> is <b>Completed</b> . |
| AGE         | Lifetime of a VolumeModifyContent from the time when it is created to the current time.                                                                                      |

----End

#### NOTE

You can use `kubect` to view the **Events** information of a VolumeModifyContent. If a VolumeModifyContent cannot meet the creation requirements or an error occurs during the PVC change, the server will record the **Events** information. The following command is used as an example:

```
kubect describe volumemodifycontents myvmc-uid
```

## Deleting a PVC Change

### NOTICE

- In NAS scenarios, if **STATUS** of a VolumeModifyClaim is **Creating**, deleting the VolumeModifyClaim resource will delete the resource created for the change on the storage side and then remove the cluster resource. After the deletion, if you continue to use the original StorageClass for PVC management, you need to restore the associated storage backend to a non-HyperMetro storage backend.
- In NAS scenarios, if some PVCs among the PVCs to be changed meet the change requirements and the batch change fails, all PVC changes will be removed. As a result, the PVCs that meet the change requirements will not meet the change requirements.
- In the SAN scenario, if **STATUS** of a VolumeModifyClaim is **Creating**, deleting the VolumeModifyClaim resource will remove the cluster resource after the single-to-dual task is complete.
- If **STATUS** of a VolumeModifyClaim is **Pending** or **Completed**, deleting the VolumeModifyClaim resource will only remove the cluster resource and will not delete the resource created for the change on the storage side (that is, there is no interaction with the storage side).
- VolumeModifyContent resources are managed by VolumeModifyClaim. Do not manually manage VolumeModifyContent resources.
- If a PVC to be changed has been manually managed on the storage side, the change may fail. Do not manually manage storage volumes when using the change feature.

This section describes how to use `kubect` to delete a PVC change. The procedure is as follows.

## Procedure

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
- Step 2** Run the following command to delete a PVC change. In the command, *vmc-name* indicates the name of the VolumeModifyClaim resource.
- ```
kubectrl delete volumemodifyclaims <vmc-name>
```
- Step 3** Query the deletion result by following the instructions in [Querying a PVC Change](#).
- End

6 Advanced Services

6.1 Snapshot Management

6.1 Snapshot Management

In Kubernetes, a [VolumeSnapshot](#) is a snapshot of a volume on a storage system. The VolumeSnapshot capability provides Kubernetes users with a standard way to replicate the content of a volume at a specified point in time without creating a volume. For example, this function enables database administrators to back up the database before making changes such as editing or deleting.

This section describes how to create a VolumeSnapshot using Huawei CSI. To create a VolumeSnapshot, perform the following steps:

- Configuring a VolumeSnapshotClass
- Configuring a VolumeSnapshot

Prerequisites

- [3 Compatibility and Features](#) lists storage that supports VolumeSnapshot creation. You need to search by the storage type and service type.
- For details about the Kubernetes versions that support VolumeSnapshot creation, see [Table 10-4](#).
- If you need to use volume snapshots and features associated with volume snapshots in the container environment, perform the operations in [4.1.1.4 Checking Volume Snapshot-Dependent Components](#) to check whether volume snapshot-dependent components have been deployed in your environment and check the api-versions information about volume snapshots.

6.1.1 Configuring a VolumeSnapshotClass

Creating a VolumeSnapshotClass

[VolumeSnapshotClass](#) provides a way to describe the "classes" of storage when provisioning a VolumeSnapshot. Each VolumeSnapshotClass contains the **driver**, **deletionPolicy**, and **parameters** fields, which are used when a VolumeSnapshot belonging to the class needs to be dynamically provisioned.

The name of a VolumeSnapshotClass object is significant, and is how users can request a particular class. Administrators set the name and other parameters of a class when first creating VolumeSnapshotClass objects, and the objects cannot be updated once they are created.

The following is an example of a VolumeSnapshotClass used by Huawei CSI:

```
apiVersion: snapshot.storage.k8s.io/v1
kind: VolumeSnapshotClass
metadata:
  name: mysnapclass
driver: csi.huawei.com
deletionPolicy: Delete
```

You can modify the parameters according to [Table 6-1](#).

Table 6-1 VolumeSnapshotClass parameters

Parameter	Description	Remarks
metadata.name	User-defined name of a VolumeSnapshotClass object.	Take Kubernetes v1.22.1 as an example. The value can contain digits, lowercase letters, hyphens (-), and periods (.), and must start and end with a letter or digit.
driver	driver identifier. This parameter is mandatory.	Set this parameter to the driver name set during Huawei CSI installation. The default driver name is csi.huawei.com .
deletionPolicy	Snapshot deletion policy. This parameter is mandatory. The value can be: - Delete - Retain	- If the deletion policy is Delete, the snapshot on the storage device will be deleted together with the VolumeSnapshotContent object. - If the deletion policy is Retain, the snapshot and VolumeSnapshotContent object on the storage device will be retained.
parameters.enableHyperMetroSnap	Whether to create SAN HyperMetro snapshots on both ends. "true": Create snapshots on the two storage systems that form the HyperMetro pair. "false": Create a snapshot on the storage system associated with the current storage class.	The default value is "false" . When the backend type is oceanstor-san and the storage version is V700R001C10 or later, this parameter can be set to "true" .

Procedure

- Step 1** Run the following command to create a VolumeSnapshotClass using the created VolumeSnapshotClass configuration file.

```
kubectl create -f mysnapclass.yaml
```

- Step 2** Run the following command to view the information about the created VolumeSnapshotClass.

```
kubectl get volumesnapshotclass
```

The following is an example of the command output.

NAME	DRIVER	DELETIONPOLICY	AGE
mysnapclass	csi.huawei.com	Delete	25s

----End

6.1.2 Configuring a VolumeSnapshot

VolumeSnapshot configurations can be classified into dynamically provisioned VolumeSnapshots and pre-provisioned VolumeSnapshots.

- A dynamically provisioned VolumeSnapshot is created by using VolumeSnapshot resources to dynamically obtain and create a snapshot from PVCs, without using existing snapshots.
- A pre-provisioned VolumeSnapshot requires the administrator to create the required snapshot on the storage device in advance and use the existing snapshot by creating a VolumeSnapshotContent. In addition, you can specify the associated VolumeSnapshotContent when creating a VolumeSnapshot.

6.1.2.1 Dynamically Provisioning a VolumeSnapshot

This section describes how to dynamically provision a VolumeSnapshot using Huawei CSI.

Prerequisites

- The source PVC exists, and the backend where the PVC resides supports VolumeSnapshot creation.

Creating a Snapshot for a Volume

The following is an example of the VolumeSnapshot configuration file:

```
apiVersion: snapshot.storage.k8s.io/v1
kind: VolumeSnapshot
metadata:
  name: mysnapshot
spec:
  volumeSnapshotClassName: mysnapclass
  source:
    persistentVolumeClaimName: mypvc
```

You can modify the parameters according to [Table 6-2](#).

Table 6-2 VolumeSnapshot parameters

Parameter	Description	Remarks
metadata.name	User-defined name of a VolumeSnapshot object.	Take Kubernetes v1.22.1 as an example. The value can contain digits, lowercase letters, hyphens (-), and periods (.), and must start and end with a letter or digit.
spec.volumeSnapshotClassName	Name of the VolumeSnapshotClass object.	--
spec.source.persistentVolumeClaimName	Name of the source PVC object.	Name of the source PVC of the snapshot

Step 1 Run the following command to create a VolumeSnapshot using the created VolumeSnapshot configuration file.

```
kubectl create -f mysnapshot.yaml
```

Step 2 Run the following command to view the information about the created VolumeSnapshot.

```
kubectl get volumesnapshot
```

The following is an example of the command output.

```
NAME      READYTOUSE SOURCEPVC SOURCESNAPSHOTCONTENT RESTORESIZE
SNAPSHOTCLASS SNAPSHOTCONTENT                                CREATIONTIME AGE
mysnapshot true      mypvc          100Gi          mysnapclass
snapshotcontent-1009af0a-24c2-4435-861c-516224503f2d <invalid> 78s
```

----End

6.1.2.2 Pre-provisioning a VolumeSnapshot

This section describes how to pre-provision a VolumeSnapshot using Huawei CSI.

Prerequisites

- A source VolumeSnapshot has been created on the Huawei storage device, and the created snapshot name can be obtained.

Creating a VolumeSnapshot Entity

The following is an example of the VolumeSnapshotContent file:

```
apiVersion: snapshot.storage.k8s.io/v1
kind: VolumeSnapshotContent
metadata:
  name: mysnapshotcontent
spec:
  deletionPolicy: Retain
  driver: csi.huawei.com
  volumeSnapshotRef:
    apiVersion: snapshot.storage.k8s.io/v1
    kind: VolumeSnapshot
```

```
name: mysnapshot
namespace: default
source:
  snapshotHandle: mybackend.1.snapshot_001
volumeSnapshotClassName: "mysnapclass"
```

You can modify the parameters according to [Table 6-3](#).

Table 6-3 VolumeSnapshotContent parameters

Parameter	Description	Remarks
metadata.name	User-defined name of a VolumeSnapshotContent object.	Take Kubernetes v1.22.1 as an example. The value can contain digits, lowercase letters, hyphens (-), and periods (.), and must start and end with a letter or digit.
spec.deletionPolicy	Deletion policy. The following types are supported: • Delete: Resources are automatically reclaimed. • Retain: Resources are manually reclaimed.	• Delete: When a VolumeSnapshotContent is deleted, the snapshot resources on the storage are also deleted. • Retain: When a VolumeSnapshotContent is deleted, the snapshot resources on the storage are not deleted.
spec.driver	Driver name.	Set this parameter to the driver name set during Huawei CSI installation. The value is the same as that of driverName in the values.yaml file.
spec.volumeSnapshotRef	Information about the target VolumeSnapshot to be bound, including the VolumeSnapshot name and namespace.	--

Parameter	Description	Remarks
spec.source.snapshotHandle	Unique identifier of a storage snapshot resource. This parameter is mandatory. The format is <i><backend-name>.<parent-id>.<snapshot-name></i> .	The value of this parameter consists of the following parts: • <backend-name>: indicates the backend name corresponding to the snapshot resource. You can run the oceanctl get backend command to obtain the configured backend information. • <parent-id>: indicates the ID of the parent resource object of the snapshot resource on the storage device. You can view the ID on DeviceManager. • <snapshot-name>: indicates the name of the snapshot resource on the storage device. You can view the name on DeviceManager.
spec.volumeSnapshotClassName	Name of the VolumeSnapshotClass object.	--

Step 1 Run the following command to create a VolumeSnapshotContent using the created VolumeSnapshotContent configuration file.

```
kubectl create -f mysnapshotcontent.yaml
```

Step 2 Run the following command to view the information about the created VolumeSnapshot.

```
kubectl get volumesnapshotcontent
```

The following is an example of the command output.

```
NAME                READYTOUSE  RESTORESIZE  DELETIONPOLICY  DRIVER
VOLUMESNAPSHOTCLASS VOLUMESNAPSHOT VOLUMESNAPSHOTNAMESPACE AGE
mysnapshotcontent  true        0            Retain          csi.huawei.com  mysnapclass
mysnapshot         default     4s
```

----End

Creating a Snapshot for a Volume

After a VolumeSnapshotContent is created in provisioned mode, you can create a VolumeSnapshot based on the VolumeSnapshotContent. The following is an example of the VolumeSnapshot configuration file:

```
apiVersion: snapshot.storage.k8s.io/v1
kind: VolumeSnapshot
metadata:
  name: mysnapshot
  namespace: default
spec:
  volumeSnapshotClassName: mysnapclass
  source:
    volumeSnapshotContentName: mysnapshotcontent
```

You can modify the parameters according to [Table 6-4](#).

Table 6-4 VolumeSnapshot parameters

Parameter	Description	Remarks
metadata.name	User-defined name of a VolumeSnapshot object.	Take Kubernetes v1.22.1 as an example. The value can contain digits, lowercase letters, hyphens (-), and periods (.), and must start and end with a letter or digit. The value must be the same as the name of the VolumeSnapshot specified by VolumeSnapshotContent.
metadata.namespace	Namespace to which the VolumeSnapshot belongs.	The value must be the same as the namespace of the VolumeSnapshot specified by VolumeSnapshotContent.
spec.volumeSnapshotClassName	Name of the VolumeSnapshotClass object.	--
spec.source.volumeSnapshotContentName	Name of the source VolumeSnapshotContent object.	Name of the source VolumeSnapshotContent of the snapshot.

Step 1 Run the following command to create a VolumeSnapshot using the created VolumeSnapshot configuration file.

```
kubectl create -f mysnapshot.yaml
```

Step 2 Run the following command to view the information about the created VolumeSnapshot.

```
kubectl get volumesnapshot
```

The following is an example of the command output.

```
NAME          READYTOUSE SOURCEPVC SOURCESNAPSHOTCONTENT RESTORESIZE SNAPSHOTCLASS
SNAPSHOTCONTENT CREATIONTIME AGE
mysnapshot    true          mysnapshotcontent 0          mysnapclass mysnapshotcontent
2m39s        8s
```

----End

6.1.3 Managing Volume Snapshots

6.1.3.1 Querying a VolumeSnapshot

Viewing a VolumeSnapshotClass

Step 1 Run the following command to view the information about the created VolumeSnapshotClass.

```
kubectl get volumesnapshotclass
```

The following is an example of the command output.

NAME	DRIVER	DELETIONPOLICY	AGE
mynapclass	csi.huawei.com	Delete	25s

----End

Querying a Volume Snapshot

Step 1 Run the following command to view the information about the created VolumeSnapshot.

```
kubectl get volumesnapshot
```

The following is an example of the command output.

NAME	READYTOUSE	SOURCEPVC	SOURCESNAPSHOTCONTENT	RESTORESIZE	SNAPSHOTCLASS	SNAPSHOTCONTENT	CREATIONTIME	AGE
mynapshot	true	mypvc	100Gi	mynapclass	snapshotcontent-1009af0a-24c2-4435-861c-516224503f2d	<invalid>	78s	

----End

6.1.3.2 Creating a PV Using a VolumeSnapshot

Prerequisites

- [3 Compatibility and Features](#) lists storage that support PVC creation using a VolumeSnapshot. You need to search by the storage type and service type.
- [10.4 Kubernetes Feature Matrix](#) lists the Kubernetes versions that support PVC creation using a VolumeSnapshot.
- A VolumeSnapshot exists, and the backend where the VolumeSnapshot resides supports cloning.

When creating a PVC, you need to specify a VolumeSnapshot as the data source. The following is an example of creating a PVC using a VolumeSnapshot. In this example, **mynapshot** is used as the data source and a PVC named **myrestore** is created.

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: myrestore
spec:
  storageClassName: mysc
  dataSource:
    name: mynapshot
    kind: VolumeSnapshot
    apiGroup: snapshot.storage.k8s.io
  volumeMode: Filesystem
  accessModes:
    - ReadWriteOnce
```

```
resources:
  requests:
    storage: 100Gi
```

NOTICE

- The specified **storageClassName** must be the same as the StorageClass of the snapshot source volume in **dataSource**.
- The capacity of the clone volume must be greater than or equal to that of the snapshot. Equal capacity is recommended.

Procedure

- Step 1** Run the following command to create a PVC based on the configuration file for creating a volume using a snapshot.

```
kubectl create -f myrestore.yaml
```

----End

6.1.3.3 Deleting a VolumeSnapshot

Deleting a VolumeSnapshot

Perform this operation when the volume snapshot is no longer required.

Procedure

- Step 1** Run the following command to view the information about the created VolumeSnapshot.

```
kubectl get volumesnapshot
```

The following is an example of the command output.

NAME	READYTOUSE	SOURCEPVC	SOURCESNAPSHOTCONTENT	RESTORESIZE
				CREATIONTIME AGE
mynsnapshot	true	mypvc	100Gi	mynsnapshotclass
snapshotcontent-1009af0a-24c2-4435-861c-516224503f2d	<invalid>			78s

- Step 2** Run the following command to delete the VolumeSnapshot:

```
kubectl delete volumesnapshot mynsnapshot
```

If the following information is displayed, the deletion is successful:

```
volumesnapshot.snapshot.storage.k8s.io "mynsnapshot" deleted
```

----End

Deleting a VolumeSnapshotClass

Perform this operation when the VolumeSnapshotClass is not bound to any VolumeSnapshot and the VolumeSnapshotClass is no longer required.

Procedure

Step 1 Run the following command to view the VolumeSnapshotClass:

```
kubectl get vsclass
```

The following is an example of the command output.

NAME	DRIVER	DELETIONPOLICY	AGE
mynapclass	csi.huawei.com	Delete	25s

Step 2 Run the following command to delete the StorageClass:

```
kubectl delete vsclass mynapclass
```

If the following information is displayed, the deletion is successful:

```
volumesnapshotclass.snapshot.storage.k8s.io "mynapclass" deleted
```

----End

7 Common O&M Operations

- [7.1 Installing Helm](#)
- [7.2 Collecting Information](#)
- [7.3 Downloading a Container Image](#)
- [7.4 Updating the huawei-csi-controller or huawei-csi-node Service](#)
- [7.5 Modifying the Log Output Mode](#)
- [7.6 Enabling the ReadWriteOncePod Feature Gate](#)
- [7.7 Configuring Access to the Kubernetes Cluster as a Non-root User](#)
- [7.8 Configuring IPv6](#)
- [7.9 Configuring Storage Topology Awareness](#)
- [7.10 Configuring ALUA](#)
- [7.11 Managing Storage Backends Using kubectl](#)

7.1 Installing Helm

This section describes how to install Helm. Helm 3 is used as an example.

For details, see <https://helm.sh/docs/intro/install/>.

Prerequisites

Ensure that the master node in the Kubernetes cluster can access the Internet.

Procedure

- Step 1** Run the following command to download the Helm installation script.

```
curl -fsSL -o get_helm.sh https://raw.githubusercontent.com/helm/helm/master/scripts/get-helm-3
```
- Step 2** Run the following command to modify the permission on the Helm installation script.

```
chmod 700 get_helm.sh
```


- Step 3** Determine the Helm version to be installed based on the version mapping between Helm and Kubernetes. For details about the version mapping, see [Helm Version Support Policy](#). Then run the following command to change the **DESIRED_VERSION** environment variable to the Helm version to be installed and run the installation command.

```
DESIRED_VERSION=v3.9.0 ./get_helm.sh
```

- Step 4** Run the following command to check whether Helm of the specified version is successfully installed.

```
helm version
```

If the following information is displayed, the installation is successful.

```
version.BuildInfo{Version:"v3.9.0", GitCommit:"7ceeda6c585217a19a1131663d8cd1f7d641b2a7",  
GitTreeState:"clean", GoVersion:"go1.17.5"}
```

----End

7.2 Collecting Information

7.2.1 Obtaining the CSI Version

This section describes how to view the CSI version.

Procedure

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
- Step 2** Run the following command to query information about the node where huawei-csi-node resides.

```
kubectl get pod -A -owide | grep huawei-csi-node
```

The following is an example of the command output.

NAMESPACE	NAME	READY	STATUS	RESTARTS	AGE	IP
huawei-csi	huawei-csi-node-87mss	3/3	Running	0	6m41s	
192.168.129.155	node-1	<none>	<none>			
huawei-csi	huawei-csi-node-xp8cc	3/3	Running	0	6m41s	192.168.129.156
node-2	<none>	<none>				

- Step 3** Use a remote access tool, such as PuTTY, to log in to any node where huawei-csi-node resides through the node IP address.
- Step 4** Run the following command to view the CSI version.

```
cat /var/lib/kubelet/plugins/csi.huawei.com/version
```

The version information is displayed as follows:

```
4.13.0
```

----End

7.2.2 Viewing Huawei CSI Logs

Viewing Logs of the huawei-csi-controller Service

- Step 1** Run the following command to obtain the node where huawei-csi-controller is located.

```
kubectl get pod -A -o wide | grep huawei
```

The following is an example of the command output, where **IP** indicates the node IP address and **NODE** indicates the node name.

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED
NODE READINESS GATES							
huawei-csi-controller-695b84b4d8-tg64l	9/9	Running	0	14s	<host1-ip>	<host1-name>	
<none>							<none>

- Step 2** Use a remote access tool, such as PuTTY, to log in to the node where the huawei-csi-controller service resides in the Kubernetes cluster through the management IP address.

- Step 3** Go to the log directory.

```
cd /var/log/huawei
```

- Step 4** Run the following command to view the customized output logs of the container.

```
vi huawei-csi-controller
```

- Step 5** Go to the container directory.

```
cd /var/log/containers
```

- Step 6** Run the following command to view the standard output logs of the container.

```
vi huawei-csi-controller-<name>_huawei-csi_huawei-csi-driver-<container-id>.log
```

----End

Viewing Logs of the huawei-csi-node Service

- Step 1** Run the following command to obtain the node where huawei-csi-node is located.

```
kubectl get pod -A -o wide | grep huawei
```

The following is an example of the command output, where **IP** indicates the node IP address and **NODE** indicates the node name.

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE
READINESS GATES							
huawei-csi-node-g6f7z	3/3	Running	0	14s	<host2-ip>	<host2-name>	<none>
<none>							

- Step 2** Use a remote access tool, such as PuTTY, to log in to the node where the huawei-csi-node service resides in the Kubernetes cluster through the management IP address.

- Step 3** Go to the log directory.

```
cd /var/log/huawei
```

- Step 4** Run the following command to view the customized output logs of the container.

```
vi huawei-csi-node
```

- Step 5** Go to the container directory.

```
cd /var/log/containers
```

Step 6 Run the following command to view the standard output logs of the container.

```
vi huawei-csi-node-<name>_huawei-csi_huawei-csi-driver-<container-id>.log
```

----End

7.2.3 Collecting Logs

Performing Check Before Collection

Step 1 Use a remote access tool, such as PuTTY, to log in to the node where the oceanctl tool is installed in the Kubernetes cluster through the management IP address.

Step 2 Run the following command. The displayed version is **4.13.0**.

```
oceanctl version
```

The following is an example of the command output.

```
Oceanctl Version: 4.13.0
```

Step 3 Run the **oceanctl collect logs --help** command. The following information is displayed.

```
$ oceanctl collect logs --help
```

Collect logs of one or more nodes in specified namespace in Kubernetes

Usage:

```
oceanctl collect logs [flags]
```

Examples:

```
# Collect logs of all nodes in specified namespace
oceanctl collect logs -n <namespace>
```

```
# Collect logs of specified node in specified namespace
oceanctl collect logs -n <namespace> -N <node>
```

```
# Collect logs of all nodes in specified namespace
oceanctl collect logs -n <namespace> -a
```

```
# Collect logs of all nodes in specified namespace with a maximum of 50 nodes collected at the same time
oceanctl collect logs -n <namespace> -a --threads-max=50
```

```
# Collect logs of specified node in specified namespace
oceanctl collect logs -n <namespace> -N <node> -a
```

Flags:

```
-a, --all          Collect all nodes messages
```

```
-h, --help        help for logs
```

```
-n, --namespace string namespace of resources
```

```
-N, --nodename string Specify the node for which information is to be collected.
```

```
--threads-max int set maximum number[1~1000] of threads for nodes to be collected. (default 50)
```

Global Flags:

```
--log-dir string Specify the directory for printing log files. (default "/var/log/huawei")
```

Step 4 Run the following command to check whether a Pod is started properly. In the command, *huawei-csi* indicates the namespace for installing CSI.

```
kubectl get deployment -n huawei-csi
```

The following is an example of the command output.

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
huawei-csi-controller	1/1	1	1	21h

----End

Collecting All Logs in the CSI Namespace Using oceanctl

- Step 1** Use a remote access tool, such as PuTTY, to log in to the node checked in [Performing Check Before Collection](#) through the management IP address.
- Step 2** Run the **oceanctl collect logs -n <namespace> -a --threads-max=<max_node_processing_num>** command to collect CSI logs of all nodes where CSI containers reside in the cluster. In the command, **threads-max** indicates the maximum number of nodes for which logs can be collected at the same time. The default value is **50**. You can set the value based on the host performance and load.

```
oceanctl collect logs -n huawei-csi -a --threads-max=10
```

- Step 3** Check the log package generated in the **/tmp** directory. You can run the **unzip <zip_name> -d collect_logs** command to decompress the log package. In the preceding command, **<zip_name>** indicates the package name.

```
# date
Wed Sep 20 02:49:24 EDT 2023

# ls
huawei-csi-2023-09-20-02:48:22-all.zip
```

----End

Collecting the Log of a Single CSI Node Using oceanctl

- Step 1** Use a remote access tool, such as PuTTY, to log in to the node checked in [Performing Check Before Collection](#) through the management IP address.
- Step 2** Run the **oceanctl collect logs -n <namespace> -N <nodeName>** command to collect CSI logs of all nodes where CSI containers reside in the cluster.

```
oceanctl collect logs -n huawei-csi -N node-1
```

- Step 3** Check the log package generated in the **/tmp** directory. You can run the **unzip <zip_name> -d collect_logs** command to decompress the log package. In the preceding command, **<zip_name>** indicates the package name.

```
# date
Thu Sep 21 04:08:47 EDT 2023

# ls
huawei-csi-2023-09-21-04:05:15-node-1.zip
```

----End

7.3 Downloading a Container Image

Downloading a Container Image Using containerd

- Step 1** Run the following command to download an image to a local path. In the command, **image:tag** indicates the image to be pulled and its tag.

```
ctr image pull <image>:<tag>
```

- Step 2** Run the following command to export the image to a file. In the command, **image:tag** indicates the image to be exported, and **file** indicates the name of the exported image file.

```
ctr image export <file>.tar <image>:<tag>
```

----End

Downloading a Container Image Using Docker

- Step 1** Run the following command to download an image to a local path. In the command, *image:tag* indicates the image to be pulled.

```
docker pull <image>:<tag>
```

- Step 2** Run the following command to export the image to a file. In the command, *image:tag* indicates the image to be exported, and *file* indicates the name of the exported image file.

```
docker save <image>:<tag> -o <file>.tar
```

----End

Downloading a Container Image Using Podman

- Step 1** Run the following command to download an image to a local path. In the command, *image:tag* indicates the image to be pulled.

```
podman pull <image>:<tag>
```

- Step 2** Run the following command to export the image to a file. In the command, *image:tag* indicates the image to be exported, and *file* indicates the name of the exported image file.

```
podman save <image>:<tag> -o <file>.tar
```

----End

7.4 Updating the huawei-csi-controller or huawei-csi-node Service

Perform this operation when you need to update the huawei-csi-controller or huawei-csi-node service, for example, changing the number of copies for the huawei-csi-controller service.

Procedure

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
- Step 2** Go to the `/helm/esdk` directory and run the following command to obtain the original service configuration file. **helm-huawei-csi** indicates the Helm chart name specified during the installation of the earlier version, and **huawei-csi** indicates the Helm chart namespace specified during the installation of the earlier version. For details about the component package path, see [Table 4-1](#).
- ```
helm get values helm-huawei-csi -n huawei-csi -a > ./update-values.yaml
```
- Step 3** Run the **vi update-values.yaml** command to open the file obtained in [Step 2](#) and modify the configuration items by referring to [4.1.2.1.3 Parameters in the values.yaml File of Helm](#). After the modification, press **Esc** and enter **:wq!** to save the modification.

**Step 4** Run the following command to update Huawei CSI services.

```
helm upgrade helm-huawei-csi ./ -n huawei-csi -f ./update-values.yaml
```

----End

## 7.5 Modifying the Log Output Mode

huawei-csi supports two log output modes: **file** and **console**. **file** indicates that logs are output to the fixed directory (**/var/log/huawei**), and **console** indicates that logs are output to the standard directory of the container. You can set the log output mode as required. The default mode is **file**.

### Procedure

**Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.

**Step 2** Go to the **/helm/esdk** directory and run the following command to obtain the original service configuration file. **helm-huawei-csi** indicates the Helm chart name specified during the installation of the earlier version, and **huawei-csi** indicates the Helm chart namespace specified during the installation of the earlier version. For details about the component package path, see [Table 4-1](#).

```
helm get values helm-huawei-csi -n huawei-csi -a > ./update-values.yaml
```

**Step 3** Run the **vi update-values.yaml** command to open the file obtained in [Step 2](#) and modify the configuration items. After the modification, press **Esc** and enter **:wq!** to save the modification.

```
The CSI driver parameter configuration
csiDriver:
 # Driver name, it is strongly recommended not to modify this parameter
 # The CCE platform needs to modify this parameter, e.g. csi.oceanstor.com
 driverName: csi.huawei.com
 # Endpoint, it is strongly recommended not to modify this parameter
 endpoint: /csi/csi.sock
 # DR Endpoint, it is strongly recommended not to modify this parameter
 drEndpoint: /csi/dr-csi.sock
 # Maximum number of concurrent disk scans or detaches, support 1~10
 connectorThreads: 4
 # Flag to enable or disable volume multipath access, support [true, false]
 volumeUseMultipath: true
 # Multipath software used by fc/iscsi. support [DM-multipath, HW-UltraPath, HW-UltraPath-NVMe]
 scsiMultipathType: DM-multipath
 # Multipath software used by roce-nvme/fc-nvme. only support [HW-UltraPath-NVMe]
 nvmeMultipathType: HW-UltraPath-NVMe
 # Timeout interval for waiting for multipath aggregation when DM-multipath is used on the host. support
 1~600
 scanVolumeTimeout: 3
 # Timeout interval for running command on the host. support 1~600
 execCommandTimeout: 30
 # check the number of paths for multipath aggregation
 # Allowed values:
 # true: the number of paths aggregated by DM-multipath is equal to the number of online paths
 # false: the number of paths aggregated by DM-multipath is not checked.
 # Default value: false
 allPathOnline: false
 # Interval for updating backend capabilities. support 60~600
 backendUpdateInterval: 60
Huawei-csi-controller log configuration
controllerLogging:
 # Log record type, support [file, console]
 module: file
```

```
Log Level, support [debug, info, warning, error, fatal]
level: info
Directory for storing logs
fileDir: /var/log/huawei
Size of a single log file
fileSize: 20M
Maximum number of log files that can be backed up.
maxBackups: 9
Huawei-csi-node log configuration
nodeLogging:
Log record type, support [file, console]
module: file
Log Level, support [debug, info, warning, error, fatal]
level: info
Directory for storing logs
fileDir: /var/log/huawei
Size of a single log file
fileSize: 20M
Maximum number of log files that can be backed up.
maxBackups: 9
```

**Step 4** Run the following command to update the log configuration.

```
helm upgrade helm-huawei-csi ./ -n huawei-csi -f ./update-values.yaml
```

----End

## 7.6 Enabling the ReadWriteOncePod Feature Gate

The ReadWriteOnce access mode is the fourth access mode introduced by Kubernetes v1.22 for PVs and PVCs. If you create a Pod using a PVC in ReadWriteOncePod access mode, Kubernetes ensures that the Pod is the only Pod in the cluster that can read or write the PVC.

If the Kubernetes version is earlier than v1.29, you need to enable the ReadWriteOncePod feature in feature-gates of kube-apiserver, kube-scheduler, and kubelet.

### NOTE

Currently, the ReadWriteOncePod feature gate cannot be enabled on the CCE or CCE Agile platform.

## Procedure

**Step 1** Enable the ReadWriteOncePod feature gate for kube-apiserver.

1. Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
2. Run the **vi /etc/kubernetes/manifests/kube-apiserver.yaml** command, press **I** or **Insert** to enter the insert mode, and add **--feature-gates=ReadWriteOncePod=true** to the kube-apiserver container. After the modification is complete, press **Esc** and enter **:wq!** to save the modification.

```
...
spec:
 containers:
 - command:
 - kube-apiserver
 - --feature-gates=ReadWriteOncePod=true
 ...
```

 **NOTE**

After the editing is complete, Kubernetes will automatically apply the updates.

**Step 2** Enable the ReadWriteOncePod feature gate for kube-scheduler.

1. Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
2. Run the **vi /etc/kubernetes/manifests/kube-scheduler.yaml** command, press **I** or **Insert** to enter the insert mode, and add **--feature-gates=ReadWriteOncePod=true** to the kube-scheduler container. After the modification is complete, press **Esc** and enter **:wq!** to save the modification.

```
...
spec:
 containers:
 - command:
 - kube-scheduler
 - --feature-gates=ReadWriteOncePod=true
 ...
```

 **NOTE**

After the editing is complete, Kubernetes will automatically apply the updates.

**Step 3** Enable the ReadWriteOncePod feature gate for kubelet.**NOTICE**

The dynamic Kubelet configuration function is not used since v1.22 and deleted in v1.24. Therefore, you need to perform the following operations on kubelet on each worker node in the cluster.

1. Use a remote access tool, such as PuTTY, to log in to any worker node in the Kubernetes cluster through the management IP address.
2. Run the **vi /var/lib/kubelet/config.yaml** command, press **I** or **Insert** to enter the editing state, and add **ReadWriteOncePod: true** to the **featureGates** field of the KubeletConfiguration object. If the **featureGates** field does not exist, add it at the same time. After the modification is complete, press **Esc** and enter **:wq!** to save the modification.

```
apiVersion: kubelet.config.k8s.io/v1beta1
featureGates:
 ReadWriteOncePod: true
...
```

 **NOTE**

The default path of the kubelet configuration file is **/var/lib/kubelet/config.yaml**. Enter the path based on site requirements.

3. After the configuration is complete, run the **systemctl restart kubelet** command to restart kubelet.

----End



## 7.7 Configuring Access to the Kubernetes Cluster as a Non-root User

### Procedure

- Step 1** Copy the authentication file of the Kubernetes cluster and modify `/etc/kubernetes/admin.conf` to be the actual authentication file.

```
mkdir -p $HOME/.kube
sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/config
```

- Step 2** Change the user and user group of the authentication file.

```
sudo chown $(id -u):$(id -g) $HOME/.kube/config
```

- Step 3** Configure the **KUBECONFIG** environment variable of the current user. The following uses Ubuntu 20.04 as an example.

```
echo "export KUBECONFIG=$HOME/.kube/config" >> ~/.bashrc
source ~/.bashrc
```

----End

## 7.8 Configuring IPv6

This section describes how to use Huawei CSI to configure IPv6.

### 7.8.1 Configuring IPv6 During the Initial Installation of Huawei CSI

This section describes how to specify IPv6 during the initial installation of Huawei CSI.

### Prerequisites

- IPv6 can be configured only when the storage backend type is **oceanstor-nas**, **oceanstor-san**, or **oceanstor-dtree**.
- The host environment and Kubernetes cluster environment support IPv6. For details, see [IPv4/IPv6 dual-stack of Kubernetes](#).

### Procedure

- Step 1** Install the CSI-dependent components by following the instructions in [Step 1](#) to [Step 7](#) in [Installation Procedure](#).

- Step 2** Run the **vi values.yaml** command to open the configuration file and modify the **service.ipFamilyPolicy** and **service.ipFamilies** configuration items. For details about the parameters, see [Table 4-8](#).

The following is an example:

```
service:
 ipFamilyPolicy: SingleStack
 ipFamilies:
 - IPv6
```

**Step 3** Install and deploy Huawei CSI by following the instructions in [Step 8](#) to [Step 9](#) in [Installation Procedure](#).

**Step 4** Run the following command to check the status of the service in the **huawei-csi** namespace.

```
kubectl get service -n huawei-csi
```

The following is an example of the command output.

| NAME                  | TYPE      | CLUSTER-IP       | EXTERNAL-IP | PORT(S)  | AGE |
|-----------------------|-----------|------------------|-------------|----------|-----|
| huawei-csi-controller | ClusterIP | fd00:10:96::8136 | <none>      | 4433/TCP | 19m |

**Step 5** Run the following command to check the value of **ipFamilies** for the huawei-csi-controller service.

```
kubectl get svc -n huawei-csi huawei-csi-controller -o=jsonpath='{.spec.ipFamilies}'
```

The following is an example of the command output.

```
["IPv6"]
```

**Step 6** Run the following command to check the value of **ipFamilyPolicy** for the huawei-csi-controller service.

```
kubectl get svc -n huawei-csi huawei-csi-controller -o=jsonpath='{.spec.ipFamilyPolicy}'
```

The following is an example of the command output.

```
SingleStack
```

----End

## 7.8.2 Switching from IPv4 to IPv6 for Huawei CSI

This section describes how to switch from IPv4 to IPv6 when updating Huawei CSI.

### Prerequisites

- The host environment and Kubernetes cluster environment support IPv6. For details, see [IPv4/IPv6 dual-stack of Kubernetes](#).
- Huawei CSI has been installed and services are running properly.

### Procedure

**Step 1** Run the following command to obtain the original service configuration file. **helm-huawei-csi** indicates the Helm chart name specified during the installation of the earlier version, and **huawei-csi** indicates the Helm chart namespace specified during the installation of the earlier version.

```
helm get values helm-huawei-csi -n huawei-csi -a > ./update-values.yaml
```

**Step 2** Run the **vi update-values.yaml** command to open the file obtained in [Step 1](#) and modify the **service.ipFamilyPolicy** and **service.ipFamilies** configuration items. For details about the parameters, see [Table 4-8](#).

The following is an example:

```
service:
 ipFamilyPolicy: SingleStack
 ipFamilies:
 - IPv6
```

**Step 3** Before switching to IPv6, run the following command to delete the service in the **huawei-csi** namespace.

```
kubectrl delete service -n huawei-csi --all
```

- Step 4** Run the following command to upgrade Huawei CSI. In the following command, **helm-huawei-csi** indicates the specified Helm chart name, **huawei-csi** indicates the specified Helm chart namespace, and **update-values.yaml** indicates the file obtained in [Step 1](#).

```
helm upgrade helm-huawei-csi ./ -n huawei-csi -f ./values.yaml -f ./update-values.yaml
```

- Step 5** Run the following command to query the **ipFamilies** parameter of the huawei-csi-controller service after the update.

```
kubectrl get svc -n huawei-csi huawei-csi-controller -o=jsonpath='{.spec.ipFamilies}'
```

The expected output is as follows:

```
["IPv6"]
```

----End

## 7.9 Configuring Storage Topology Awareness

In the Kubernetes cluster, resources can be scheduled and provisioned based on the topology labels of nodes and the topology capabilities supported by storage backends.

### Prerequisites

To use the storage topology awareness capability, ensure the topology feature is enabled during CSI installation. Run the following command to check whether the topology feature is enabled:

```
helm get values helm-huawei-csi -n huawei-csi -a | grep topology
```

If **topology: true** is displayed in the command output, the feature is enabled. If **topology: false** is displayed or no command output is displayed in the command output, perform operations in [4.1.4.1 Upgrade Using Helm](#) to update CSI configuration parameters and enable the topology feature.

### Configuring Topology Labels for Nodes

You need to configure topology labels on worker nodes in the cluster. The method is as follows:

1. Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
2. Run the following command to view information about worker nodes in the current cluster.

```
kubectrl get node
```

The following is an example of the command output.

| NAME   | STATUS | ROLES                    | AGE | VERSION |
|--------|--------|--------------------------|-----|---------|
| node01 | Ready  | controlplane,etcd,worker | 42d | v1.22.3 |
| node02 | Ready  | worker                   | 42d | v1.22.3 |
| node03 | Ready  | worker                   | 42d | v1.22.3 |

3. Run the following command to configure a topology label for a worker node. In the preceding command, *nodename* indicates the name of a worker node. For details about the **key** and **value** parameters, see [Table 7-1](#).

```
kubectrl label node <nodename> <key>=<value>
```

**Table 7-1** Description of topology label parameters

| Parameter | Description                            | Remarks                                                                                                                                                                                                                  |
|-----------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <key>     | Unique identifier of a topology label. | The value can be <b>zone</b> , <b>region</b> , or <b>protocol</b> . <i>&lt;protocol&gt;</i> .<br><i>&lt;protocol&gt;</i> can be set to <b>iscsi</b> , <b>nfs</b> , <b>fc</b> , or <b>roce-nvme</b> .                     |
| <value>   | Value of a topology label.             | If <b>key</b> is set to <b>zone</b> or <b>region</b> , <b>value</b> is a user-defined parameter.<br>If <b>key</b> is set to <b>protocol</b> . <i>&lt;protocol&gt;</i> , <b>value</b> is fixed at <b>csi.huawei.com</b> . |

 **NOTE**

- A topology label must start with **topology.kubernetes.io**. Topology label examples:
    - Example 1: **topology.kubernetes.io/region=China-west**
    - Example 2: **topology.kubernetes.io/zone=ChengDu**
    - Example 3: **topology.kubernetes.io/protocol.iscsi=csi.huawei.com**
    - Example 4: **topology.kubernetes.io/protocol.fc=csi.huawei.com**
  - A key in a topology label on a node can have only one value.
  - If multiple protocols are configured in a topology label on a node, when you configure a StorageClass, the StorageClass needs to meet only one of the protocols.
  - If both the region and the zone are configured in a topology label on a node, when you configure a StorageClass, the StorageClass must meet all filter criteria.
4. Run the following command to view the label information about all worker nodes in the current cluster.
- ```
kubectl get nodes -o=jsonpath='{range .items[*]}[{.metadata.name}, {.metadata.labels}]{",\n"}{end}' | grep --color "topology.kubernetes.io"
```
- The following is an example of the command output.
- ```
[node01,"beta.kubernetes.io/arch":"amd64","beta.kubernetes.io/os":"linux","kubernetes.io/arch":"amd64","kubernetes.io/hostname":"node01","kubernetes.io/os":"linux","node-role.kubernetes.io/controlplane":"true","node-role.kubernetes.io/etcd":"true","node-role.kubernetes.io/worker":"true","topology.kubernetes.io/zone":"ChengDu"]
```
5. Run the following command to restart the huawei-csi-node service.
- ```
kubectl delete po -n huawei-csi -l app=huawei-csi-node
```
6. After all huawei-csi-node services are running, run the following command to check whether **topologyKeys** field is added to the CSINode resource. In the command, **nodename** indicates the name of the worker node.
- ```
kubectl get CSINode <nodename> -o yaml
```

## Topology Awareness Configuration Process

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
- Step 2** Go to the directory where the Helm project is located. If the previous Helm project cannot be found, copy the **helm** directory in the component package to any

directory on the master node. For details about the component package path, see [Table 4-1](#).

- Step 3** Go to the backend service configuration directory `/examples/backend/` and back up the **backend.yaml** file.

```
cp backend.yaml backend.yaml.bak
```

- Step 4** Run the **vi backend.yaml** command to open the file and configure topology awareness as required. The following is an example. After the modification is complete, press **Esc** and enter **:wq!** to save the modification.

```
storage: "oceanstor-san"
name: "dorado-iscsi-155"
namespace: "huawei-csi"
urls:
 - "https://192.168.129.155:8088"
pools:
 - "StoragePool001"
parameters:
 protocol: "iscsi"
 portals:
 - "10.10.30.20"
 - "10.10.30.21"
supportedTopologies:
 - { "topology.kubernetes.io/region": "China-west", "topology.kubernetes.io/zone": "ChengDu" }
 - { "topology.kubernetes.io/region": "China-south", "topology.kubernetes.io/zone": "ShenZhen" }
maxClientThreads: "30"
```

- Step 5** Run the following command to delete the storage backend to be modified. In the command, **dorado-iscsi-155** indicates the storage backend name.

```
oceanctl delete backend dorado-iscsi-155 -n huawei-csi
```

- Step 6** Run the following command to create a storage backend.

```
oceanctl create backend -f ../examples/backend/backend.yaml -i yaml
```

Enter the storage user name and password as prompted.

```
Please enter this backend user name:admin
Please enter this backend password:
```

- Step 7** Run the **vi StorageClass.yaml** command to modify the .yaml file. Press **I** or **Insert** to enter the insert mode and add related parameters in the .yaml file. For details about the parameters, see [Table 7-2](#). After the modification is complete, press **Esc** and enter **:wq!** to save the modification.

Add the following configuration items to the *StorageClass.yaml* file.

- Example 1: Configure zone and region information in the StorageClass.

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
 name: example-storageclass
provisioner: csi.huawei.com
parameters:
 volumeType: lun
 allocType: thin
volumeBindingMode: WaitForFirstConsumer
allowedTopologies:
 - matchLabelExpressions:
 - key: topology.kubernetes.io/zone
 values:
 - ChengDu
 - key: topology.kubernetes.io/region
 values:
 - China-west
```

- Example 2: Configure protocol information in the StorageClass.

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
 name: protocol-example-storageclass
provisioner: csi.huawei.com
parameters:
 volumeType: lun
 allocType: thin
volumeBindingMode: WaitForFirstConsumer
allowedTopologies:
- matchLabelExpressions:
 - key: topology.kubernetes.io/protocol.iscsi
 values:
 - csi.huawei.com
```

**Table 7-2** StorageClass parameter description

| Parameter                                      | Description                                                                                                                                                                                                              | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>volumeBindingMode</b>                       | PersistentVolume binding mode, used to control the time when PersistentVolume resources are dynamically allocated and bound.                                                                                             | You can set this parameter to <b>WaitForFirstConsumer</b> or <b>Immediate</b> .<br><b>WaitForFirstConsumer</b> : indicates that the binding and allocation of the PersistentVolume are delayed until a Pod that uses the PVC is created.<br><b>Immediate</b> : The PersistentVolume is bound and allocated immediately after a PVC is created.                                                                                                                                                                                                                                                                                                                       |
| <b>allowedTopologies.matchLabelExpressions</b> | Topology information label, which is used to filter CSI backends and Kubernetes nodes. If the matching fails, PVCs or Pods cannot be created.<br>Both <b>key</b> and <b>values</b> must be configured in a fixed format. | <b>key</b> : This parameter can be set to <b>topology.kubernetes.io/zone</b> or <b>topology.kubernetes.io/region</b> .<br><b>topology.kubernetes.io/protocol</b> . <i>&lt;protocol&gt;</i> : <i>&lt;protocol&gt;</i> indicates the protocol type and can be <b>iscsi</b> , <b>fc</b> , or <b>nfs</b> .<br><b>values</b> :<br>If <b>key</b> is <b>topology.kubernetes.io/zone</b> or <b>topology.kubernetes.io/region</b> , <b>values</b> must be the same as that in <a href="#">Configuring Topology Labels for Nodes</a> .<br>If <b>key</b> is <b>topology.kubernetes.io/protocol</b> . <i>&lt;protocol&gt;</i> , <b>value</b> is fixed at <b>csi.huawei.com</b> . |

**Step 8** Run the following command to create a StorageClass based on the .yaml file.

```
kubectl create -f StorageClass.yaml
```

**Step 9** Use the StorageClass to create a PVC with the topology capability. For details, see [Table 5-66](#).

----End

## 7.10 Configuring ALUA

Asymmetric Logical Unit Access (ALUA) is a model that supports access to multiple target ports. In the multipathing state, ALUA presents active/passive volumes to the host and provides a port access status switchover interface to switch over the working controllers for volumes. For example, when a volume of a controller fails, you can set the status of ports on the controller to **Unavailable**. After the host multipathing software that supports ALUA detects the status, it switches subsequent I/Os from the failed controller to the peer controller.

### 7.10.1 Configuring ALUA Parameters for a Huawei Enterprise Storage Backend

For details about how to configure ALUA for Huawei enterprise storage, see the host connectivity guide of the corresponding product.

The ALUA configuration may vary according to the OS. Visit [Huawei Technical Support](#), enter **Host Connectivity Guide** in the search box, and click the search button. In the search result, select the host connectivity guide for the desired OS. Configure ALUA according to the actual situation and the description in the guide. Huawei CSI will apply the configuration items you set to the initiator of the host on Huawei storage.

#### NOTE

A node with a Pod provisioned does not proactively change ALUA information. The host ALUA configuration changes only after a Pod is provisioned again to the node.

### ALUA Parameters for OceanStor V5 Series

**Table 7-3** lists the ALUA parameters supported by Huawei CSI for OceanStor V5 series.

**Table 7-3** ALUA parameters supported by Huawei CSI for OceanStor V5 series

| Parameter     | Description                                                                                                                                                                                                                                                                      | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HostName      | Host name rule. This parameter is mandatory. You can use a regular expression.                                                                                                                                                                                                   | <p>The host name can be obtained by running the <b>cat /etc/hostname</b> command. It can be matched by using regular expressions. When <b>HostName</b> is set to *, the configuration takes effect on hosts with any name. For details, see <a href="#">Regular expression</a>.</p> <p>If the host name of a compute node matches multiple ALUA configuration options, they will be sorted based on the matching accuracy and the first ALUA configuration option will be used. For details about the sorting rules, see <a href="#">Rules for Matching ALUA Configuration Items with Host Names</a>.</p> |
| MULTIPATHTYPE | <p>Multipathing type. This parameter is mandatory. The value can be:</p> <ul style="list-style-type: none"><li>• <b>0</b>: Third-party multipathing is not used.</li><li>• <b>1</b>: Third-party multipathing is used.</li></ul>                                                 | --                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| FAILOVERMODE  | <p>Initiator switchover mode. This parameter is conditionally mandatory. The value can be:</p> <ul style="list-style-type: none"><li>• <b>0</b>: early-version ALUA</li><li>• <b>1</b>: common ALUA</li><li>• <b>2</b>: ALUA not used</li><li>• <b>3</b>: special ALUA</li></ul> | This parameter needs to be specified only when third-party multipathing is used. Configure the initiator switchover mode by referring to the connectivity guide.                                                                                                                                                                                                                                                                                                                                                                                                                                          |



| Parameter           | Description                                                                                                                                                                                                                                              | Remarks                                                                                                                                                                                |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SPECIALMODE<br>TYPE | Special mode type of the initiator. This parameter is conditionally mandatory. The value can be: <ul style="list-style-type: none"><li>• 0: special mode 0</li><li>• 1: special mode 1</li><li>• 2: special mode 2</li><li>• 3: special mode 3</li></ul> | This parameter needs to be specified only when the initiator switchover mode is special ALUA. Configure the special mode type of the initiator by referring to the connectivity guide. |
| PATHTYPE            | Initiator path type. This parameter is conditionally mandatory. The value can be: <ul style="list-style-type: none"><li>• 0: preferred path</li><li>• 1: non-preferred path</li></ul>                                                                    | This parameter needs to be specified only when third-party multipathing is used. Configure the initiator path type by referring to the connectivity guide.                             |

The following uses OceanStor 18500 V5 as an example to describe how to connect to Red Hat. For details about the host connectivity guide, see *Huawei SAN Storage Host Connectivity Guide for Red Hat*.

The following ALUA configuration example is recommended in the OceanStor 18500 V5 host connectivity guide for Red Hat in non-HyperMetro storage scenarios. In this example, the OS on compute node **myhost01** in the Kubernetes cluster is RHEL 5.x, and that on other compute nodes is RHEL 7.x. According to the recommendation, the switchover mode of RHEL 5.x should be "ALUA not used", and that of RHEL 7.x should be "common ALUA".

```
storage: oceanstor-san
name: oceanstor-iscsi-155
urls:
 - https://192.168.129.155:8088
 - https://192.168.129.156:8088
pools:
 - StoragePool001
parameters:
 protocol: iscsi
 portals:
 - 192.168.128.120
 - 192.168.128.121
 ALUA:
 ^myhost01$:
 MULTIPATHTYPE: 1
 FAILOVERMODE: 2
 PATHTYPE: 0
 "**":
 MULTIPATHTYPE: 1
 FAILOVERMODE: 1
 PATHTYPE: 0
```

## ALUA Parameters for OceanStor and OceanStor Dorado Series

**Table 7-4** lists the ALUA parameters supported by Huawei CSI for OceanStor and OceanStor Dorado series.

### NOTE

By default, the initiator host access mode of OceanStor and OceanStor Dorado series storage is "balanced mode". Therefore, you are advised not to configure ALUA parameters for OceanStor and OceanStor Dorado series storage.

**Table 7-4** ALUA parameters for OceanStor and OceanStor Dorado series

| Parameter               | Description                                                                                                                                                                                              | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HostName                | Host name rule. This parameter is mandatory. You can use a regular expression.                                                                                                                           | The host name can be obtained by running the <b>cat /etc/hostname</b> command. It can be matched by using regular expressions. When <b>HostName</b> is set to *, the configuration takes effect on hosts with any name. For details, see <a href="#">Regular expression</a> .<br><br>If the host name of a compute node matches multiple ALUA configuration options, they will be sorted based on the matching accuracy and the first ALUA configuration option will be used. For details about the sorting rules, see <a href="#">Rules for Matching ALUA Configuration Items with Host Names</a> . |
| accessMode              | Host access mode. This parameter is mandatory. The value can be: <ul style="list-style-type: none"><li>• <b>0</b>: balanced mode</li><li>• <b>1</b>: asymmetric mode</li></ul>                           | The balanced mode is recommended in non-HyperMetro scenarios.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| hyperMetroPathOptimized | Whether the path of the host on the current storage array is preferred in HyperMetro scenarios. The value can be: <ul style="list-style-type: none"><li>• <b>1</b>: yes</li><li>• <b>0</b>: no</li></ul> | This parameter needs to be specified only when the host access mode is set to asymmetric.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

The following uses OceanStor Dorado 18000 as an example to describe how to connect to Red Hat. For details about the host connectivity guide, see *OceanStor Dorado and OceanStor Host Connectivity Guide for Red Hat*.

The following ALUA configuration example is recommended in the OceanStor Dorado 18000 host connectivity guide for Red Hat in non-HyperMetro storage scenarios.

```
storage: "oceanstor-san"
name: "dorado-iscsi-155"
urls:
 - "https://192.168.129.155:8088"
 - "https://192.168.129.156:8088"
pools:
 - "StoragePool001"
parameters:
 protocol: "iscsi"
 portals:
 - "192.168.128.120"
 - "192.168.128.121"
 ALUA:
 "":
 accessMode: 0
```

## Rules for Matching ALUA Configuration Items with Host Names

- If the configured host name rule exactly matches the host name of the service node, the ALUA configuration item corresponding to the host name rule is used.

For example, the host name rule in configuration item 1 is `*` and that in configuration item 2 is `^myhost01$`. If the host name of a compute node is **myhost01**, it exactly matches configuration item 2. In this case, Huawei CSI will apply the configuration information in configuration item 2 to the storage side.

- If the configured host name rule does not exactly match the host name of the service node, the first ALUA configuration item matched by regular expressions is used.

For example, the host name rule in configuration item 1 is `myhost0[0-9]` and that in configuration item 2 is `myhost0[5-9]`. In this case, configuration item 1 has a higher priority than configuration item 2. If the host name of a compute node is **myhost06**, both configuration items can be matched. In this case, Huawei CSI will apply the configuration information in configuration item 1 to the storage side.

## 7.10.2 Configuring ALUA Parameters for a Distributed Storage Backend

For details about how to configure ALUA for Huawei distributed storage, see the host connectivity guide of the corresponding product.

The ALUA configuration may vary according to the OS. Visit [Huawei Technical Support](#), enter **Host Connectivity Guide** in the search box, and click the search button. In the search result, select the host connectivity guide for the desired OS. Configure ALUA according to the actual situation and the description in the guide. Huawei CSI will apply the configuration items you set to the initiator of the host on Huawei storage.

 NOTE

A node with a Pod provisioned does not proactively change ALUA information. The host ALUA configuration changes only after a Pod is provisioned again to the node.

In non-HyperMetro scenarios of distributed storage, you are advised to set the switchover mode to "disable ALUA" (default value). This is because the storage system is in active/active mode and "enables ALUA" is meaningless. Therefore, you are advised not to configure ALUA parameters for distributed storage.

**Table 7-5** lists the ALUA parameters supported by Huawei CSI for distributed storage.

**Table 7-5** ALUA parameters for distributed storage

| Parameter      | Description                                                                                                                                                                                       | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HostName       | The value of <b>HostName</b> is the host name of a worker node, for example, <b>HostName1</b> and <b>HostName2</b> .                                                                              | The host name can be obtained by running the <b>cat /etc/hostname</b> command. It can be matched by using regular expressions. When <b>HostName</b> is set to *, the configuration takes effect on hosts with any name. For details, see <a href="#">Regular expression</a> .<br><br>If the host name of a compute node matches multiple ALUA configuration options, they will be sorted based on the matching accuracy and the first ALUA configuration option will be used. For details about the sorting rules, see <a href="#">Rules for Matching ALUA Configuration Items with Host Names</a> . |
| switchoverMode | Switchover mode. This parameter is mandatory. The value can be: <ul style="list-style-type: none"><li>• <b>Disable_alua</b>: disables ALUA.</li><li>• <b>Enable_alua</b>: enables ALUA.</li></ul> | In non-HyperMetro scenario, you are advised to set the switchover mode to "disable ALUA". This is because the storage system is in active/active mode and "enables ALUA" is meaningless. Currently, Huawei CSI does not support SAN HyperMetro scenarios. Exercise caution when enabling ALUA.                                                                                                                                                                                                                                                                                                       |

| Parameter | Description                                                                                                                                                                                                                | Remarks                                                                        |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| pathType  | <p>Path type. This parameter is conditionally mandatory. The value can be:</p> <ul style="list-style-type: none"><li>• <b>optimal_path</b>: preferred path</li><li>• <b>non_optimal_path</b>: non-preferred path</li></ul> | This parameter is mandatory when the switchover mode is set to "enables ALUA". |

## Rules for Matching ALUA Configuration Items with Host Names

- If the configured host name rule exactly matches the host name of the service node, the ALUA configuration item corresponding to the host name rule is used.

For example, the host name rule in configuration item 1 is `*` and that in configuration item 2 is `^myhost01$`. If the host name of a compute node is **myhost01**, it exactly matches configuration item 2. In this case, Huawei CSI will apply the configuration information in configuration item 2 to the storage side.

- If the configured host name rule does not exactly match the host name of the service node, the first ALUA configuration item matched by regular expressions is used.

For example, the host name rule in configuration item 1 is **myhost0[0-9]** and that in configuration item 2 is **myhost0[5-9]**. In this case, configuration item 1 has a higher priority than configuration item 2. If the host name of a compute node is **myhost06**, both configuration items can be matched. In this case, Huawei CSI will apply the configuration information in configuration item 1 to the storage side.

### 7.10.3 Example ALUA Configuration Policy of OceanStor V5 Series

**Example 1:** The configuration file content is as follows:

```
parameters:
 ALUA:
 "*/",
 MULTIPATHTYPE: 1
 FAILOVERMODE: 3
 SPECIALMODETYPE: 0
 PATHTYPE: 0
 node1:
 MULTIPATHTYPE: 1
 FAILOVERMODE: 3
 SPECIALMODETYPE: 0
 PATHTYPE: 1
```

If the host name is **node1**, both of the preceding ALUA configuration sections can be used to configure initiators. According to the configuration policy rules in [7.10.1 Configuring ALUA Parameters for a Huawei Enterprise Storage Backend](#), the priority of the second configuration section (where **HostName** is

**node1**) is higher than that of the first configuration section (where **HostName** is \*).

**Example 2:** The configuration file content is as follows:

```
parameters:
ALUA:
 node[0-9]:
 MULTIPATHTYPE: 1
 FAILOVERMODE: 3
 SPECIALMODETYPE: 0
 PATHTYPE: 0
 node[5-7]:
 MULTIPATHTYPE: 1
 FAILOVERMODE: 3
 SPECIALMODETYPE: 0
 PATHTYPE: 1
```

If the host name is **node6**, both of the preceding ALUA configuration sections can be used to configure initiators. According to the configuration policy rules in [7.10.1 Configuring ALUA Parameters for a Huawei Enterprise Storage Backend](#), select the first ALUA configuration section to configure initiators.

**Example 3:** The configuration file content is as follows:

```
parameters:
ALUA:
 node$:
 MULTIPATHTYPE: 1
 FAILOVERMODE: 3
 SPECIALMODETYPE: 0
 PATHTYPE: 0
 node10$:
 MULTIPATHTYPE: 1
 FAILOVERMODE: 3
 SPECIALMODETYPE: 0
 PATHTYPE: 1
```

According to the configuration policy rules in [7.10.1 Configuring ALUA Parameters for a Huawei Enterprise Storage Backend](#): For host **node1**, select the first ALUA configuration section to configure initiators. For host **node10**, select the second ALUA configuration section to configure initiators. ^ matches the beginning of a character string, and \$ matches the end of a character string.

## 7.10.4 Example ALUA Configuration Policy of OceanStor Dorado

**Example 1:** The configuration file content is as follows:

```
parameters:
ALUA:
 "1":
 accessMode: 1
 hyperMetroPathOptimized: 1
 node1:
 accessMode: 1
 hyperMetroPathOptimized: 0
```

If the host name is **node1**, both of the preceding ALUA configuration sections can be used to configure initiators. According to the configuration policy rules in [7.10.1 Configuring ALUA Parameters for a Huawei Enterprise Storage Backend](#), the priority of the second configuration section (where **HostName** is

**node1**) is higher than that of the first configuration section (where **HostName** is \*).

**Example 2:** The configuration file content is as follows:

```
parameters:
 ALUA:
 node[0-9]:
 accessMode: 1
 hyperMetroPathOptimized: 1
 node[5-7]:
 accessMode: 1
 hyperMetroPathOptimized: 0
```

If the host name is **node6**, both of the preceding ALUA configuration sections can be used to configure initiators. According to the configuration policy rules in [7.10.1 Configuring ALUA Parameters for a Huawei Enterprise Storage Backend](#), select the first ALUA configuration section to configure initiators.

**Example 3:** The configuration file content is as follows:

```
parameters:
 node1$:
 node[0-9]:
 accessMode: 1
 hyperMetroPathOptimized: 1
 node10$:
 accessMode: 1
 hyperMetroPathOptimized: 0
```

According to the configuration policy rules in [7.10.1 Configuring ALUA Parameters for a Huawei Enterprise Storage Backend](#): For host **node1**, select the first ALUA configuration section to configure initiators. For host **node10**, select the second ALUA configuration section to configure initiators. ^ matches the beginning of a character string, and \$ matches the end of a character string.

## 7.10.5 Example ALUA Configuration Policy of Distributed Storage

**Example 1:** The configuration file content is as follows:

```
parameters:
 ALUA:
 "*" :
 switchoverMode: Enable_alua
 pathType: optimal_path
 node1:
 switchoverMode: Enable_alua
 pathType: non_optimal_path
```

If the host name is **node1**, both of the preceding ALUA configuration sections can be used to configure initiators. According to the configuration policy rules in [7.10.2 Configuring ALUA Parameters for a Distributed Storage Backend](#), the priority of the second configuration section (where **HostName** is **node1**) is higher than that of the first configuration section (where **HostName** is \*).

**Example 2:** The configuration file content is as follows:

```
parameters:
 ALUA:
 node[0-9]:
 switchoverMode: Enable_alua
```

```
pathType: optimal_path
node[5-7]:
 switchoverMode: Enable_alua
 pathType: non_optimal_path
```

If the host name is **node6**, both of the preceding ALUA configuration sections can be used to configure initiators. According to the configuration policy rules in [7.10.2 Configuring ALUA Parameters for a Distributed Storage Backend](#), select the first ALUA configuration section to configure initiators.

**Example 3:** The configuration file content is as follows:

```
parameters:
 ALUA:
 node1$:
 switchoverMode: Enable_alua
 pathType: optimal_path
 node10$:
 switchoverMode: Enable_alua
 pathType: non_optimal_path
```

According to the configuration policy rules in [7.10.2 Configuring ALUA Parameters for a Distributed Storage Backend](#): For host **node1**, select the first ALUA configuration section to configure initiators. For host **node10**, select the second ALUA configuration section to configure initiators. **^** matches the beginning of a character string, and **\$** matches the end of a character string.

## 7.11 Managing Storage Backends Using kubectl

CSI storage backends are implemented based on Kubernetes custom resources (CRDs). In addition to using the **oceanctl** tool, you can also run **kubectl** commands to manage storage backend resources. Storage backends involve the following two types of CRD resources:

- **StorageBackendClaim** (sbc for short): namespace-level resource, which indicates the user's claim to a storage backend. You can create this resource to trigger the CSI controller to automatically connect the storage backend.
- **StorageBackendContent** (sbct for short): cluster-level resource, which is automatically created by the CSI controller after a **StorageBackendClaim** is successfully bound. It indicates the actual storage backend instance in the cluster. You should not directly create or modify this resource.

### NOTE

- Before using **kubectl** to manage storage backends, ensure that the CSI plug-in has been installed and the CRD has been correctly registered with the cluster.
- **StorageBackendContent** is automatically managed by the controller. Do not directly create or manually modify it. Otherwise, the storage backend may be abnormal.
- Do not delete a storage backend when a volume management operation is being performed on it.

### 7.11.1 Creating a Storage Backend

#### Prerequisites

- You have obtained the management IP address, username, and password of the storage device.



- You have obtained configuration information such as the storage pool name and storage protocol. For details about the configuration items, see section [5.1.1 Configuring the Storage Backend](#).

## Procedure

- Step 1** Prepare the secret configuration file, for example, **backend-secret.yaml**. For details about the fields in the secret configuration file, see [Table 7-6](#).

```
apiVersion: v1
kind: Secret
metadata:
 name: backend-demo
 namespace: huawei-csi
type: Opaque
stringData:
 user: "admin"
 password: "*****"
 authenticationMode: "0"
```

Create a secret.

```
kubectl apply -f backend-secret.yaml
```

**Table 7-6** Secret field description

| Parameter          | Description                                                                                                                                                                 | Mandatory | Default Value | Remarks                                                                                                            |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|--------------------------------------------------------------------------------------------------------------------|
| user               | Username for logging in to the storage.                                                                                                                                     | Yes       | -             |                                                                                                                    |
| password           | Password for logging in to the storage.                                                                                                                                     | Yes       | -             |                                                                                                                    |
| authenticationMode | Authentication mode for logging in to the storage. The value can be: <ul style="list-style-type: none"><li>0: local authentication</li><li>1: LDAP authentication</li></ul> | No        | 0             | Only OceanStor Dorado, OceanStor V5, OceanStor V6, OceanStor A600, and OceanStor A800 support LDAP authentication. |
| maxClientThreads   | Maximum number of concurrent connections to a storage backend.                                                                                                              | No        | 30            | The value ranges from 1 to 30.                                                                                     |

- Step 2** Create a ConfigMap resource for the storage backend to store the management configuration of the storage device.

Prepare the ConfigMap configuration file. The following describes how to create an OceanStor SAN storage backend of the iSCSI protocol type. The value of the

**csi.json** key is the storage backend configuration in JSON format. For details about the configuration items, see [5.1.1 Configuring the Storage Backend](#).

```
apiVersion: v1
kind: ConfigMap
metadata:
 name: backend-demo
 namespace: huawei-csi
data:
 csi.json: |-
 {
 "backends": {
 "storage": "oceanstor-san",
 "name": "backend-demo",
 "namespace": "huawei-csi",
 "urls": ["https://192.168.129.157:8088"],
 "pools": ["StoragePool001"],
 "provisioner": "csi.huawei.com",
 "parameters": {
 "protocol": "iscsi",
 "portals": ["10.10.30.20", "10.10.30.21"]
 },
 "maxClientThreads": "30"
 }
 }
```

Create a ConfigMap.

```
kubectl apply -f backend-configmap.yaml
```

**Step 3** Create a StorageBackendClaim resource to trigger the CSI controller to access the storage backend. For details about the fields in StorageBackendClaim, see [Table 7-7](#).

Prepare the StorageBackendClaim configuration file, for example, **backend-claim.yaml**.

```
apiVersion: xuanwu.huawei.io/v1
kind: StorageBackendClaim
metadata:
 name: backend-demo
 namespace: huawei-csi
spec:
 provider: csi.huawei.com
 configmapMeta: huawei-csi/backend-demo
 secretMeta: huawei-csi/backend-demo
 maxClientThreads: "30"
```

Create a StorageBackendClaim.

```
kubectl apply -f backend-claim.yaml
```

**Table 7-7** StorageBackendClaim spec field description

| Parameter | Description                                            | Man<br>dator<br>y | Defa<br>ult<br>Value | Remarks                                       |
|-----------|--------------------------------------------------------|-------------------|----------------------|-----------------------------------------------|
| provider  | Provider name, which is used to match the CSI plug-in. | Yes               | -                    | The value is fixed to <b>csi.huawei.com</b> . |

| Parameter        | Description                                                                                                                | Mandatory | Default Value | Remarks                                                                    |
|------------------|----------------------------------------------------------------------------------------------------------------------------|-----------|---------------|----------------------------------------------------------------------------|
| configmapMeta    | Reference to the ConfigMap of the storage configuration information. The format is <i>&lt;namespace&gt;/&lt;name&gt;</i> . | Yes       | -             | The value must be the same as the name of the ConfigMap created in step 2. |
| secretMeta       | Reference to the secret of the storage authentication information. The format is <i>&lt;namespace&gt;/&lt;name&gt;</i> .   | Yes       | -             | The value must be the same as the name of the secret created in step 1.    |
| maxClientThreads | Maximum number of concurrent connections to a storage backend.                                                             | No        | 30            | The value ranges from 1 to 30.                                             |

**Step 4** Check the storage backend creation result.

```
kubectl get sbct
```

The following is an example of the command output. If the online status of StorageBackendContent is **true**, the creation is successful.

| NAME            | CLAIM                   | SN             | VENDORNAME | PROVIDERVERSION | ONLINE | AGE |
|-----------------|-------------------------|----------------|------------|-----------------|--------|-----|
| content-xxxxxxx | huawei-csi/backend-demo | xxxxxxxxxxxxxx | Huawei     | 4.13.0          | true   | 53d |

----End

## 7.11.2 Querying a Storage Backend

### Querying StorageBackendClaim

**Step 1** Run the following command to query all storage backend claims in the default namespace:

```
kubectl get sbc -n huawei-csi
```

The following is an example of the command output:

| NAME         | STORAGEBACKENDCONTENTNAME | STATUS | AGE |
|--------------|---------------------------|--------|-----|
| backend-demo | content-xxxxxxxxxxx       | Bound  | 53d |

**Step 2** Run the following command to view storage backend claims in YAML format:

```
kubectl get sbc backend-demo -n huawei-csi -o yaml
```

----End

### Querying StorageBackendContent

**Step 1** Run the following command to query all storage backend instances:

```
kubectl get sbct
```

The following is an example of the command output:

| NAME            | CLAIM                   | SN             | VENDORNAME | PROVIDERVERSION | ONLINE | AGE |
|-----------------|-------------------------|----------------|------------|-----------------|--------|-----|
| content-xxxxxxx | huawei-csi/backend-demo | xxxxxxxxxxxxxx | Huawei     | 4.13.0          | true   | 53d |

**Step 2** Run the following command to view storage backend instances in YAML format:

```
kubectl get sbct content-xxxxxxx -o yaml
```

----End

### 7.11.3 Updating Storage Backend Authentication Information

To prevent the storage backend from becoming temporarily unavailable when the existing secret is modified directly, you are advised to update the authentication information by creating a new secret, replacing the reference to the existing secret, and then deleting the old secret.

#### Procedure

**Step 1** Prepare a new secret configuration file, for example, **backend-secret-new.yaml**.

```
apiVersion: v1
kind: Secret
metadata:
 name: backend-demo-new
 namespace: huawei-csi
type: Opaque
stringData:
 user: "admin"
 password: "*****"
 authenticationMode: "0"
```

**Step 2** Create a secret.

```
kubectl apply -f backend-secret-new.yaml
```

**Step 3** Update StorageBackendClaim and update secretMeta to reference the new secret. CSI automatically synchronizes the new secret to StorageBackendContent and uses the new authentication information to log in to the storage device again.

```
kubectl patch sbc backend-demo -n huawei-csi --type merge -p '{"spec":{"secretMeta":"huawei-csi/backend-demo-new"}}'
```

**Step 4** Check the storage backend creation result.

```
kubectl get StorageBackendContent
```

The following is an example of the command output. If the online status of StorageBackendContent is **true**, the update is successful.

| NAME            | CLAIM                   | SN             | VENDORNAME | PROVIDERVERSION | ONLINE | AGE |
|-----------------|-------------------------|----------------|------------|-----------------|--------|-----|
| content-xxxxxxx | huawei-csi/backend-demo | xxxxxxxxxxxxxx | Huawei     | 4.13.0          | true   | 53d |

**Step 5** Delete the old secret.

```
kubectl delete secret backend-demo -n huawei-csi
```

----End

### 7.11.4 Managing Storage Backend Certificates

CSI allows you to add a storage certificate to use the TLS/SSL protocol to encrypt data transmission channels, improving the security of communication with storage devices.

## Adding a Storage Certificate

**Step 1** Create a certificate. Take OceanStor Dorado as an example. For details about how to create a certificate, [click here](#).

**Step 2** Run the following command to create a secret from the certificate file:

```
kubectl create secret generic cert-1 --from-file=tls.crt=/path/to/cert.crt -n huawei-csi
```

**Step 3** Run the following command to update the StorageBackendClaim, enable the certificate, and specify the certificate secret. The format of the certSecret field is `<namespace>/<secret-name>`.

```
kubectl patch sbc backend-demo -n huawei-csi --type merge -p '{"spec": {"useCert": true, "certSecret": "huawei-csi/cert-1"}}'
```

**Step 4** Check the storage backend creation result.

```
kubectl get StorageBackendContent
```

The following is an example of the command output. If the online status of StorageBackendContent is **true**, the update is successful.

| NAME            | CLAIM                   | SN             | VENDORNAME | PROVIDERVERSION | ONLINE | AGE |
|-----------------|-------------------------|----------------|------------|-----------------|--------|-----|
| content-xxxxxxx | huawei-csi/backend-demo | xxxxxxxxxxxxxx | Huawei     | 4.13.0          | true   | 53d |

----End

## Deleting a Storage Certificate

**Step 1** Delete the certificate configuration of StorageBackendClaim.

```
kubectl patch sbc backend-demo -n huawei-csi --type merge -p '{"spec":{"useCert":false,"certSecret":""}}'
```

**Step 2** Delete the certificate secret.

```
kubectl delete secret cert-1 -n huawei-csi
```

----End

## 7.11.5 Deleting a Storage Backend

### NOTICE

Do not delete a storage backend when a volume management operation is being performed on it.

**Step 1** Delete the StorageBackendClaim.

```
kubectl delete sbc backend-demo -n huawei-csi
```

After the StorageBackendClaim is deleted, the controller automatically deletes the associated StorageBackendContent.

**Step 2** Check whether StorageBackendContent is deleted.

```
kubectl get sbct
```

If the associated StorageBackendContent is no longer listed, the deletion is successful.

**Step 3** Run the following commands to manually delete the associated ConfigMap and secret:

```
kubectl delete configmap backend-demo -n huawei-csi
kubectl delete secret backend-demo -n huawei-csi
```

**----End**

# 8 Troubleshooting

---

[8.1 Huawei CSI Service Issues](#)

[8.2 Storage Backend Issues](#)

[8.3 PVC Issues](#)

[8.4 Pod Issues](#)

[8.5 Common Problems and Solutions for Interconnecting with the Tanzu Kubernetes Cluster](#)

## 8.1 Huawei CSI Service Issues

### 8.1.1 Failed to Start the huawei-csi-node Service with Error Message `"/var/lib/iscsi is not a directory"` Reported

#### Symptom

The huawei-csi-node service cannot be started. When you run the **kubectl describe daemonset huawei-csi-node -n huawei-csi** command, error message `"/var/lib/iscsi is not a directory"` is reported.

#### Root Cause Analysis

The `/var/lib/iscsi` directory does not exist in the huawei-csi-node container.

#### Solution or Workaround

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
- Step 2** Go to the directory where the Helm project is located. If the previous Helm project cannot be found, copy the **helm** directory in the component package to any directory on the master node. For details about the component package path, see [Table 4-1](#).

**Step 3** Go to the **templates** directory and find the **huawei-csi-node.yaml** file.

```
cd /templates
```

**Step 4** Run the following command to set **path** in **huawei-csi-node.yaml** > **volumes** > **iscsi-dir** > **hostPath** to **/var/lib/iscsi**, save the file, and exit.

```
vi huawei-csi-node.yaml
```

**Step 5** Run the following command to upgrade the Helm chart. The upgrade command will update the Deployment, DaemonSet, and RBAC resources. In the preceding command, *helm-huawei-csi* indicates the custom chart name and *huawei-csi* indicates the custom namespace.

```
helm upgrade helm-huawei-csi ./ -n huawei-csi -f values.yaml
```

The following is an example of the command output.

```
Release "helm-huawei-csi" has been upgraded. Happy Helming!
NAME: helm-huawei-csi
LAST DEPLOYED: Thu Jun 9 07:58:15 2022
NAMESPACE: huawei-csi
STATUS: deployed
REVISION: 2
TEST SUITE: None
```

----End

## 8.1.2 Huawei CSI Services Fail to Be Started and Error Message **"/etc/localtime is not a file"** Is Displayed

### Symptom

During the installation and deployment of CSI, a Pod fails to run and is in the **ContainerCreating** state. Alarm **/etc/localtime is not a file** is generated for the Pod.

### Root Cause Analysis

When the container mounts the **/etc/localtime** file on the host, the type is incorrectly identified. As a result, the container fails to mount the **/etc/localtime** file on the host and the Pod cannot run.

### Procedure

**Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.

**Step 2** Run the following command to check the running status of the Pod of the CSI services.

```
kubectl get pod -n huawei-csi
```

The following is an example of the command output. *huawei-csi* indicates the namespace where the CSI services are deployed.

| NAME                                   | READY | STATUS            | RESTARTS | AGE |
|----------------------------------------|-------|-------------------|----------|-----|
| huawei-csi-controller-6dfcc4b79f-9vjfq | 9/9   | ContainerCreating | 0        | 24m |
| huawei-csi-controller-6dfcc4b79f-cspfc | 9/9   | ContainerCreating | 0        | 24m |
| huawei-csi-node-g6f4k                  | 3/3   | ContainerCreating | 0        | 20m |
| huawei-csi-node-tqs87                  | 3/3   | ContainerCreating | 0        | 20m |



**Step 3** Run the following command to check the **Events** parameter of the container.

```
kubectl describe pod huawei-csi-controller-6dfcc4b79f-9vjtg -n huawei-csi
```

The following is an example of the command output. In the command, *huawei-csi-controller-6dfcc4b79f-9vjtg* indicates the name of the Pod in the **ContainerCreating** state found in [Step 2](#), and *huawei-csi* indicates the namespace to which the Pod belongs.

```
...
Events:
 Type Reason Age From Message
 ---- -
 Normal Scheduled 96s default-scheduler Successfully assigned huawei-csi/huawei-csi-controller-6dfcc4b79f-9vjtg to node1
 Warning FailedMount 33s (x8 over 96s) kubelet MountVolume.SetUp failed for volume "host-time" : hostPath type check failed: /etc/localtime is not a file
```

**Step 4** Run the **cd /helm/esdk/templates** command to go to the CSI installation package path. For the path, see [Table 4-1](#).

**Step 5** Take the **huawei-csi-controller.yaml** file as an example. Run the following command to view the file content.

```
vi huawei-csi-controller.yaml
```

Find the **host-time** configuration item under **volumes**, and delete the **type: File** line. Perform the same operations on the **huawei-csi-node.yaml** deployment file that involves the configuration item in the **templates** directory.

```
...
...
volumes:
- hostPath:
 path: /var/log/
 type: Directory
 name: log
- hostPath:
 path: /etc/localtime
 type: File
 name: host-time
...
...
```

**Step 6** Uninstall and reinstall the service by referring to [4.1.3.1 Uninstallation Using Helm](#).

**Step 7** Run the following command to check whether the Pod running status of Huawei CSI services is **Running**.

```
kubectl get pod -n huawei-csi
```

The following is an example of the command output.

| NAME                                   | READY | STATUS  | RESTARTS | AGE |
|----------------------------------------|-------|---------|----------|-----|
| huawei-csi-controller-6dfcc4b79f-9vjts | 9/9   | Running | 0        | 24m |
| huawei-csi-controller-6dfcc4b79f-csphb | 9/9   | Running | 0        | 24m |
| huawei-csi-node-g6f41                  | 3/3   | Running | 0        | 20m |
| huawei-csi-node-tqs85                  | 3/3   | Running | 0        | 20m |

----End

## 8.1.3 Failed to Start huawei-csi Services with the Status Displayed as InvalidImageName

### Symptom

The huawei-csi services (huawei-csi-controller or huawei-csi-node) cannot be started. After the **kubectrl get pod -A | grep huawei** command is executed, the command output shows that the service status is **InvalidImageName**.

```
kubectrl get pod -A | grep huawei
```

The following is an example of the command output.

```
huawei-csi huawei-csi-controller-fd5f97768-qlldc 6/9 InvalidImageName 0 16s
huawei-csi huawei-csi-node-25txd 2/3 InvalidImageName 0 15s
```

### Root Cause Analysis

In the .yaml configuration files of the controller and node, the Huawei CSI image version number is incorrect. For example:

```
...
- name: huawei-csi-driver
 image: huawei-csi:4.13.0
...
```

### Solution or Workaround

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
- Step 2** Run the following command to modify the configuration file of the huawei-csi-node service. Press **I** or **Insert** to enter the insert mode and modify related parameters. After the modification is complete, press **Esc** and enter **:wq!** to save the modification.

```
kubectrl edit daemonset huawei-csi-node -o yaml -n=huawei-csi
```

#### NOTE

- In **huawei-csi-driver** in the sample .yaml file, modify **image** to Huawei CSI image **huawei-csi:4.13.0**.

containers:

```
...
- name: huawei-csi-driver
 image: huawei-csi:4.13.0
```

- Step 3** Run the following command to modify the configuration file of the huawei-csi-controller service: Press **I** or **Insert** to enter the insert mode and modify related parameters. After the modification is complete, press **Esc** and enter **:wq!** to save the modification.

```
kubectrl edit deployment huawei-csi-controller -o yaml -n=huawei-csi
```

#### NOTE

- In **huawei-csi-driver** in the sample .yaml file, modify **image** to Huawei CSI image **huawei-csi:4.13.0**.

containers:

```
...
- name: huawei-csi-driver
 image: huawei-csi:4.13.0
```

**Step 4** Wait until the huawei-csi-node and huawei-csi-controller services are started.

**Step 5** Run the following command to check whether the huawei-csi services are started.

```
kubectl get pod -A | grep huawei
```

The following is an example of the command output. If the Pod status is **Running**, the services are started successfully.

|            |                                        |     |         |   |       |
|------------|----------------------------------------|-----|---------|---|-------|
| huawei-csi | huawei-csi-controller-58799449cf-zvhmv | 9/9 | Running | 0 | 2m29s |
| huawei-csi | huawei-csi-node-7fxh6                  | 3/3 | Running | 0 | 12m   |

----End

## 8.2 Storage Backend Issues

### 8.2.1 A Backend Fails to Be Created Using the oceanctl Tool and Error Message "context deadline exceeded" Is Displayed

#### Symptom

A user fails to create a storage backend using the oceanctl tool, and "failed to call webhook: xxx :context deadline exceeded; error: exist status 1" is displayed on the console.

#### Root Cause Analysis

When a storage backend is created, the webhook service provided by CSI is invoked to verify the connectivity with the storage management network and the storage account and password. The possible causes are as follows:

- Huawei CSI fails to verify the connectivity of the storage management network.
- The communication between kube-apiserver and CSI webhook is abnormal.

#### Huawei CSI Fails to Verify the Connectivity of the Storage Management Network

Perform the following steps to check whether Huawei CSI fails to verify the connectivity of the storage management network.

**Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.

**Step 2** Run the following command to obtain CSI service information. *huawei-csi* indicates the namespace where the CSI services are deployed.

```
kubectl get pod -n huawei-csi -owide
```

The following is an example of the command output.

| NAME                      | READY | STATUS  | RESTARTS | AGE | IP       | NODE   | NOMINATED NODE |
|---------------------------|-------|---------|----------|-----|----------|--------|----------------|
| huawei-csi-controller-xxx | 9/9   | Running | 0        | 19h | host-ip1 | host-1 | <none>         |
| huawei-csi-node-mnqbz     | 3/3   | Running | 0        | 19h | host-ip1 | host-1 | <none>         |

**Step 3** Log in to the node where huawei-csi-controller resides, for example, **host-1** in [Step 2](#).

**Step 4** Go to the **/var/log/huawei** directory.

```
cd /var/log/huawei
```

**Step 5** View the **storage-backend-controller** log. The following uses the storage connection timeout as an example.

```
tail -n 1000 storage-backend-controller
```

The following is a log example.

```
2024-01-01 06:30:44.280661 1 [INFO]: Try to login https://192.168.129.155:8088/deviceManager/rest
2024-01-01 06:31:44.281626 1 [ERROR]: Send request method: POST, Url: https://192.168.129.155:8088/
deviceManager/rest/xx/sessions, error: Post "https://192.168.129.155:8088/deviceManager/rest/xx/sessions":
context deadline exceeded (Client.Timeout exceeded while awaiting headers)
2024-01-01 06:31:44.281793 1 [WARNING]: Login https://192.168.129.155:8088/deviceManager/rest error
due to connection failure, gonna try another Url
2024-01-01 06:31:44.291668 1 [INFO]: Finished validateCreate huawei-csi/backend-test.
2024-01-01 06:31:44.291799 1 [ERROR]: Failed to validate StorageBackendClaim, error: unconnected
```

**Step 6** If the log contains information about login timeout, login failure, or long request duration, check the connectivity between the host machine and the storage or the network status.

**Step 7** If no request is recorded in the log, the communication between kube-apiserver and CSI webhook is abnormal.

----End

## Abnormal Communication Between kube-apiserver and CSI Webhook

Contact the Kubernetes platform administrator to check the network between kube-apiserver and CSI webhook. For example, if kube-apiserver has an HTTPS proxy, the CSI webhook service may fail to be accessed.

### NOTE

In the temporary workaround, the webhook resource will be deleted. This resource is used to check whether the entered account information is correct and whether the connection to the storage can be set up when a storage backend is created. Therefore, deleting this resource affects only the verification during backend creation and does not affect other functions. Pay attention to the following:

- Ensure that the host machine where the huawei-csi-controller service is located can properly communicate with the storage.
- Ensure that the entered account and password are correct.

**Step 1** Run the following command to view CSI webhook information.

```
kubectl get validatingwebhookconfiguration storage-backend-controller.xuanwu.huawei.io
```

The following is an example of the command output.

| NAME                                        | WEBHOOKS | AGE   |
|---------------------------------------------|----------|-------|
| storage-backend-controller.xuanwu.huawei.io | 1        | 4d22h |

**Step 2** Contact the Kubernetes platform administrator to check whether the communication between kube-apiserver and CSI webhook is abnormal.

**Step 3** Perform the following temporary workaround: Run the following command to delete the webhook.

```
kubectl delete validatingwebhookconfiguration storage-backend-controller.xuanwu.huawei.io
```

**Step 4** Create a storage backend. For details, see [5.1 Storage Backend Management](#).

**Step 5** If the communication between kube-apiserver and CSI webhook is restored, you need to reconstruct the webhook. In this case, run the following command to restart CSI Controller and restore the number of CSI Controller copies by specifying `--replicas=*`. In the following example, the number is restored to **1**. Change it based on actual requirements.

Change the number of copies to 0 first.

```
kubectl scale deployment huawei-csi-controller -n huawei-csi --replicas=0
```

Then restore the number of copies to the original number.

```
kubectl scale deployment huawei-csi-controller -n huawei-csi --replicas=1
```

----End

## 8.2.2 The Value of the ONLINE Field Is "false" When the oceanctl Tool Is Used to Obtain Storage Backend Information

### Symptom

The following command is executed to check storage backend status:

```
oceanctl get backend
```

The value of the **ONLINE** field of the storage backend is **false**:

| NAMESPACE  | NAME                | PROTOCOL | STORAGETYPE   | SN                 | STATUS | ONLINE | Url                          |
|------------|---------------------|----------|---------------|--------------------|--------|--------|------------------------------|
| huawei-csi | backend-201-nas-nfs | nfs      | oceanstor-nas | XXXXXXXXXXXX000006 | Bound  | false  | https://192.168.129.157:8088 |

### Root Cause Analysis

When CSI uses the account and password entered during storage backend creation to log in to the storage backend, if the login fails due to either of the following reasons, the **ONLINE** field will be set to **false**.

1. Incorrect account password: The possible cause is that the password is changed on the storage backend but not updated in the Kubernetes cluster. For details about how to solve this problem, see [Solution or Workaround](#).
2. Locked account. For details about how to solve this problem, see [8.2.3 An Account Is Locked After the Password Is Updated on the Storage Device](#).

### Solution or Workaround

**Step 1** Obtain the latest account password.

**Step 2** Update the storage backend password by following the instructions in [5.1.2.2.1 Updating the Password of a Storage Backend Using oceanctl](#).

----End

## 8.2.3 An Account Is Locked After the Password Is Updated on the Storage Device

### Symptom

After a user changes the password on the storage device, the account is locked.

### Root Cause Analysis

CSI uses the account and password configured on the storage device to log in to the storage device. After the account password is changed on the storage device, CSI attempts to log in to the storage device again after the login fails. Take OceanStor Dorado as an example. The default login policy is that an account will be locked after three consecutive password verification failures. Therefore, when CSI retries for more than three times, the account will be locked.

### Solution or Workaround

- Step 1** If the backend account is **admin**, run the following command to set the number of huawei-csi-controller service copies to 0. If an account other than **admin** is used, skip this step.

```
kubectl scale deployment huawei-csi-controller -n huawei-csi --replicas=0
```

- Step 2** Log in to the storage device as user **admin** and modify the login policy. Take OceanStor Dorado as an example. On DeviceManager, choose **Settings > User and Security > Security Policies > Login Policy**, click **Modify**, and disable **Account Lockout**.

- Step 3** If the backend account is **admin**, run the following command to restore the number of CSI Controller copies using **--replicas=\***. In the following example, the number of copies is restored to **1**. Change it based on site requirements. If an account other than **admin** is used, skip this step.

```
kubectl scale deployment huawei-csi-controller -n huawei-csi --replicas=1
```

- Step 4** Use the oceanctl tool to change the storage backend password. For details about how to change the backend password, see [5.1.2.2 Updating a Storage Backend](#).

- Step 5** Log in to the storage device as user **admin** and modify the login policy. Take OceanStor Dorado as an example. On DeviceManager, choose **Settings > User and Security > Security Policies > Login Policy**, click **Modify**, and enable **Account Lockout**.

----End

## 8.3 PVC Issues

### 8.3.1 When a PVC Is Created, the PVC Is in the Pending State

#### Symptom

A PVC is created. After a period of time, the PVC is still in the **Pending** state.

## Root Cause Analysis

Cause 1: A StorageClass with the specified name is not created in advance. As a result, Kubernetes cannot find the specified StorageClass name when a PVC is created.

Cause 2: The storage pool capability does not match the StorageClass capability. As a result, huawei-csi fails to select a storage pool.

Cause 3: An error code (for example, 50331651) is returned by a RESTful interface of the storage. As a result, huawei-csi fails to create a PVC.

Cause 4: The storage does not return a response within the timeout period set by huawei-csi. As a result, huawei-csi returns a timeout error to Kubernetes.

Cause 5: Other causes.

## Solution or Workaround

When a PVC is created, if the PVC is in the **Pending** state, you need to take different measures according to the following causes.

**Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.

**Step 2** Run the following command to view details about the PVC.

```
kubectl describe pvc mypvc
```

**Step 3** Perform the corresponding operation according to the **Events** information in the detailed PVC information.

- If the PVC is in the **Pending** state due to cause 1, perform the following steps.

```
Events:
 Type Reason Age From Message
 ---- -
Warning ProvisioningFailed 0s (x15 over 3m24s) persistentvolume-controller
storageclass.storage.k8s.io "mysc" not found
```

- Delete the PVC.
- Create a StorageClass. For details, see [5.2.1 Configuring a StorageClass](#).
- Create a PVC. For details, see [5.3.1 Configuring PVs](#).

- If the PVC is in the **Pending** state due to cause 2, perform the following steps.

```
Events:
 Type Reason Age From Message
 ---- -
Normal Provisioning 63s (x3 over 64s) csi.huawei.com_huawei-csi-controller-b59577886-qqzm8_58533e4a-884c-4c7f-92c3-6e8a7b327515 External provisioner is provisioning volume for claim "default/mypvc"
Warning ProvisioningFailed 63s (x3 over 64s) csi.huawei.com_huawei-csi-controller-b59577886-qqzm8_58533e4a-884c-4c7f-92c3-6e8a7b327515 failed to provision volume with StorageClass "mysc": rpc error: code = Internal desc = failed to select pool, the capability filter failed, error: failed to select pool, the final filter field: replication, parameters map[allocType:thin replication:True size:1099511627776 volumeType:lun]. please check your storage class
```

- Delete the PVC.
- Delete the StorageClass.
- Modify the **StorageClass.yaml** file based on the **Events** information.

- d. Create a StorageClass. For details, see [5.2.1 Configuring a StorageClass](#).
  - e. Create a PVC. For details, see [5.3.1 Configuring PVs](#).
  - If the PVC is in the **Pending** state due to cause 3, contact Huawei engineers.  
Events:

| Type    | Reason             | Age               | Message                                                                                                                                                                                                                                                                                                                             |
|---------|--------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Normal  | Provisioning       | 63s (x4 over 68s) | csi.huawei.com_huawei-csi-controller-b59577886-qqzm8_58533e4a-884c-4c7f-92c3-6e8a7b327515 External provisioner is provisioning volume for claim "default/mypvc"                                                                                                                                                                     |
| Warning | ProvisioningFailed | 62s (x4 over 68s) | csi.huawei.com_huawei-csi-controller-b59577886-qqzm8_58533e4a-884c-4c7f-92c3-6e8a7b327515 failed to provision volume with StorageClass "mysc": rpc error: code = Internal desc = Create volume map[ALLOCTYPE:1 CAPACITY:20 DESCRIPTION:Created from Kubernetes CSI NAME:pvc-63ebfda5-4cf0-458e-83bd-ecc PARENTID:0] error: 50331651 |
  - If the PVC is in the **Pending** state due to cause 4, perform the following steps.  
Events:

| Type    | Reason             | Age               | Message                                                                                                                                                                                                                                                     |
|---------|--------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Normal  | Provisioning       | 63s (x3 over 52s) | csi.huawei.com_huawei-csi-controller-b59577886-qqzm8_58533e4a-884c-4c7f-92c3-6e8a7b327515 External provisioner is provisioning volume for claim "default/mypvc"                                                                                             |
| Warning | ProvisioningFailed | 63s (x3 over 52s) | csi.huawei.com_huawei-csi-controller-b59577886-qqzm8_58533e4a-884c-4c7f-92c3-6e8a7b327515 failed to provision volume with StorageClass "mysc": rpc error: code = Internal desc = context deadline exceeded (Client.Timeout exceeded while awaiting headers) |
  - a. Wait for 10 minutes and check the PVC details again by referring to this section.
  - b. If it is still in the **Pending** state, contact Huawei engineers.
  - If the PVC is in the **Pending** state due to cause 5, contact Huawei engineers.
- End

## 8.3.2 Before a PVC Is Deleted, the PVC Is in the Pending State

### Symptom

Before a PVC is deleted, the PVC is in the **Pending** state.

### Root Cause Analysis

Cause 1: A StorageClass with the specified name is not created in advance. As a result, Kubernetes cannot find the specified StorageClass name when a PVC is created.

Cause 2: The storage pool capability does not match the StorageClass capability. As a result, huawei-csi fails to select a storage pool.

Cause 3: An error code (for example, 50331651) is returned by a RESTful interface of the storage. As a result, huawei-csi fails to create a PVC.

Cause 4: The storage does not return a response within the timeout period set by huawei-csi. As a result, huawei-csi returns a timeout error to Kubernetes.

Cause 5: Other causes.



## Solution or Workaround

To delete a PVC in the **Pending** state, you need to take different measures according to the following causes.

**Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.

**Step 2** Run the following command to view details about the PVC.

```
kubectl describe pvc mypvc
```

**Step 3** Perform the corresponding operation according to the **Events** information in the detailed PVC information.

- If the PVC is in the **Pending** state due to cause 1, run the **kubectl delete pvc *mypvc*** command to delete the PVC.

Events:

| Type    | Reason             | Age                 | From | Message                                                                  |
|---------|--------------------|---------------------|------|--------------------------------------------------------------------------|
| Warning | ProvisioningFailed | 0s (x15 over 3m24s) |      | persistentvolume-controller storageclass.storage.k8s.io "mysc" not found |

- If the PVC is in the **Pending** state due to cause 2, run the **kubectl delete pvc *mypvc*** command to delete the PVC.

Events:

| Type    | Reason             | Age               | From                                                                                      | Message                                                                                                                                                                                                                                                                                                                                           |
|---------|--------------------|-------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Normal  | Provisioning       | 63s (x3 over 64s) | csi.huawei.com_huawei-csi-controller-b59577886-qqzm8_58533e4a-884c-4c7f-92c3-6e8a7b327515 | External provisioner is provisioning volume for claim "default/mypvc"                                                                                                                                                                                                                                                                             |
| Warning | ProvisioningFailed | 63s (x3 over 64s) | csi.huawei.com_huawei-csi-controller-b59577886-qqzm8_58533e4a-884c-4c7f-92c3-6e8a7b327515 | failed to provision volume with StorageClass "mysc": rpc error: code = Internal desc = <b>failed to select pool</b> , the capability filter failed, error: failed to select pool, the final filter field: <b>replication</b> , parameters map[allocType:thin replication:True size:1099511627776 volumeType:lun]. please check your storage class |

- If the PVC is in the **Pending** state due to cause 3, run the **kubectl delete pvc *mypvc*** command to delete the PVC.

Events:

| Type    | Reason             | Age               | From                                                                                      | Message                                                                                                                                                                                                                                          |
|---------|--------------------|-------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Normal  | Provisioning       | 63s (x4 over 68s) | csi.huawei.com_huawei-csi-controller-b59577886-qqzm8_58533e4a-884c-4c7f-92c3-6e8a7b327515 | External provisioner is provisioning volume for claim "default/mypvc"                                                                                                                                                                            |
| Warning | ProvisioningFailed | 62s (x4 over 68s) | csi.huawei.com_huawei-csi-controller-b59577886-qqzm8_58533e4a-884c-4c7f-92c3-6e8a7b327515 | failed to provision volume with StorageClass "mysc": rpc error: code = Internal desc = Create volume map[ALLOCTYPE:1 CAPACITY:20 DESCRIPTION:Created from Kubernetes CSI NAME:pvc-63ebfda5-4cf0-458e-83bd-ecc PARENTID:0] error: <b>50331651</b> |

- If the PVC is in the **Pending** state due to cause 4, contact Huawei engineers.

Events:

| Type    | Reason             | Age               | From                                                                                      | Message                                                                                                                                                                  |
|---------|--------------------|-------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Normal  | Provisioning       | 63s (x3 over 52s) | csi.huawei.com_huawei-csi-controller-b59577886-qqzm8_58533e4a-884c-4c7f-92c3-6e8a7b327515 | External provisioner is provisioning volume for claim "default/mypvc"                                                                                                    |
| Warning | ProvisioningFailed | 63s (x3 over 52s) | csi.huawei.com_huawei-csi-controller-b59577886-qqzm8_58533e4a-884c-4c7f-92c3-6e8a7b327515 | failed to provision volume with StorageClass "mysc": rpc error: code = Internal desc = <b>context deadline exceeded (Client.Timeout exceeded while awaiting headers)</b> |

- If the PVC is in the **Pending** state due to cause 5, contact Huawei engineers.

----End

### 8.3.3 Failed to Expand the Capacity of a Generic Ephemeral Volume

#### Symptom

In an environment where the Kubernetes version is earlier than 1.25, the capacity of a **generic ephemeral volume** of the LUN type fails to be expanded. The system displays a message indicating that the PV capacity has been expanded, but the PVC capacity fails to be updated.

#### Root Cause Analysis

This problem is caused by a Kubernetes [bug](#), which has been resolved in Kubernetes 1.25.

### 8.3.4 Failed to Expand the PVC Capacity Because the Target Capacity Exceeds the Storage Pool Capacity

#### Symptom

In a Kubernetes environment earlier than 1.23, PVC capacity expansion fails when the target capacity exceeds the storage pool capacity.

#### Root Cause Analysis

This is a known issue in the Kubernetes community. For details, see [Recovering from Failure when Expanding Volumes](#).

#### Solution or Workaround

For details, see [Recovering from Failure when Expanding Volumes](#).

## 8.4 Pod Issues

### 8.4.1 After a Worker Node in the Cluster Breaks Down and Recovers, Pod Failover Is Complete but the Source Host Where the Pod Resides Has Residual Drive Letters

#### Symptom

A Pod is running on worker node A, and an external block device is mounted to the Pod through CSI. After worker node A is powered off abnormally, the Kubernetes platform detects that the node is faulty and switches the Pod to worker node B. After worker node A recovers, the drive letters on worker node A change from normal to faulty.

## Environment Configuration

Kubernetes version: 1.18 or later

Storage type: block storage

## Root Cause Analysis

After worker node A recovers, Kubernetes initiates an unmapping operation on the storage, but does not initiate a drive letter removal operation on the host. After Kubernetes completes the unmapping, residual drive letters exist on worker node A.

## Solution or Workaround

Currently, you can only manually clear the residual drive letters on the host. Alternatively, restart the host again and use the disk scanning mechanism during the host restart to clear the residual drive letters. The specific method is as follows:

### Step 1 Check the residual drive letters on the host.

1. Run the following command to check whether a DM multipathing device with abnormal multipathing status exists.

```
multipath -ll
```

The following is an example of the command output. The path status is **failed faulty running**, the corresponding DM multipathing device is **dm-12**, and the associated SCSI disks are **sdi** and **sdj**. If multiple paths are configured, multiple SCSI disks exist. Record these SCSI disks.

```
mpathb (3618cf24100f8f457014a764c000001f6) dm-12 HUAWEI ,XSG1
size=100G features='0' hwhandler='0' wp=rw
`-+- policy='service-time 0' prio=-1 status=active
 |- 39:0:0:1 sdi 8:48 failed faulty running
 `-- 38:0:0:1 sdj 8:64 failed faulty running
```

- If yes, go to [step 1.2](#).
- If no, no further action is required.

2. Run the following command to check whether the residual DM multipathing device is readable.

```
dd if=/dev/dm-12 of=/dev/null count=1 bs=1M iflag=direct
```

The following is an example of the command output. If the returned result is **Input/output error** and the read data is **0 bytes (0 B) copied**, the device is unreadable. *dm-xx* indicates the device ID obtained in [step 1.1](#).

```
dd: error reading '/dev/dm-12': Input/output error
0+0 records in
0+0 records out
0 bytes (0 B) copied, 0.0236862 s, 0.0 kB/s
```

- If yes, record the residual *dm-xx* device and associated disk IDs (for details, see [step 1.1](#)) and perform the clearing operation.
- If the command execution is suspended, go to [step 1.3](#).
- If other cases, contact technical support engineers.

3. Log in to the node again in another window.

- a. Run the following command to view the suspended process.

```
ps -ef | grep dm-12 | grep -w dd
```

The following is an example of the command output.

```
root 21725 9748 0 10:33 pts/10 00:00:00 dd if=/dev/dm-12 of=/dev/null count=1 bs=10M
iflag=direct
```

- b. Kill the pid.  
`kill -9 pid`
- c. Record the residual *dm-xx* device and associated disk IDs (for details, see [step 1.1](#)) and perform the clearing operation.

## Step 2 Clear the residual drive letters on the host.

1. Run the following command to delete residual multipathing aggregation device information according to the DM multipathing device obtained in [step 1](#).

```
multipath -f /dev/dm-12
```

If an error is reported, contact technical support engineers.

2. Run the following command to clear the residual SCSI disks according to the drive letters of the residual disks obtained in [step 1](#).

```
echo 1 > /sys/block/xxxx/device/delete
```

When multiple paths are configured, clear the residual disks based on the drive letters. The residual paths are **sdi** and **sdj**.

```
echo 1 > /sys/block/sdi/device/delete
```

```
echo 1 > /sys/block/sdj/device/delete
```

If an error is reported, contact technical support engineers.

3. Check whether the DM multipathing device and SCSI disk information has been cleared.

Run the following commands in sequence to query the multipathing and disk information. If the residual **dm-12** device and SCSI disks **sdi** and **sdj** are cleared, the clearing is complete.

- a. View multipathing information.

```
multipath -ll
```

The following is an example of the command output. The residual **dm-12** device is cleared.

```
mpathb (3618cf24100f8f457014a764c000001f6) dm-3 HUAWEI ,XSG1
size=100G features='0' hwhandler='0' wp=rw
`-+ policy='service-time 0' prio=-1 status=active
 |- 39:0:0:1 sdd 8:48 active ready running
 `-- 38:0:0:1 sde 8:64 active ready running
mpathn (3618cf24100f8f457315a764c000001f6) dm-5 HUAWEI ,XSG1
size=100G features='0' hwhandler='0' wp=rw
`-+ policy='service-time 0' prio=-1 status=active
 |- 39:0:0:2 sdc 8:32 active ready running
 `-- 38:0:0:2 sdb 8:16 active ready running
```

- b. View device information.

```
ls -l /sys/block/
```

The following is an example of the command output. SCSI disks **sdi** and **sdj** are cleared.

```
total 0
lrwxrwxrwx 1 root root 0 Aug 11 19:56 dm-0 -> ../devices/virtual/block/dm-0
lrwxrwxrwx 1 root root 0 Aug 11 19:56 dm-1 -> ../devices/virtual/block/dm-1
lrwxrwxrwx 1 root root 0 Aug 11 19:56 dm-2 -> ../devices/virtual/block/dm-2
lrwxrwxrwx 1 root root 0 Aug 11 19:56 dm-3 -> ../devices/virtual/block/dm-3
lrwxrwxrwx 1 root root 0 Aug 11 19:56 sdb -> ../devices/platform/host35/session2/
target35:0:0/35:0:0:1/block/sdb
lrwxrwxrwx 1 root root 0 Aug 11 19:56 sdc -> ../devices/platform/host34/
target34:65535:5692/34:65535:5692:0/block/sdc
lrwxrwxrwx 1 root root 0 Aug 11 19:56 sdd -> ../devices/platform/host39/session6/
```

```
target39:0:0/39:0:0:1/block/sdd
lrwxrwxrwx 1 root root 0 Aug 11 19:56 sde -> ../devices/platform/host38/session5/
target38:0:0/38:0:0:1/block/sde
lrwxrwxrwx 1 root root 0 Aug 11 19:56 sdh -> ../devices/platform/host39/session6/
target39:0:0/39:0:0:3/block/sdh
lrwxrwxrwx 1 root root 0 Aug 11 19:56 sdi -> ../devices/platform/host38/session5/
target38:0:0/38:0:0:3/block/sdi
```

c. View disk information.

```
ls -l /dev/disk/by-id/
```

The following is an example of the command output. SCSI disks **sdi** and **sdj** are cleared.

```
total 0
lrwxrwxrwx 1 root root 10 Aug 11 19:57 dm-name-mpathb -> ../dm-3
lrwxrwxrwx 1 root root 10 Aug 11 19:58 dm-name-mpathn -> ../dm-5
lrwxrwxrwx 1 root root 10 Aug 11 19:57 dm-uuid-mpath-3618cf24100f8f457014a764c000001f6
-> ../dm-3
lrwxrwxrwx 1 root root 10 Aug 11 19:58 dm-uuid-mpath-3618cf24100f8f457315a764c000001f6
-> ../dm-5
lrwxrwxrwx 1 root root 9 Aug 11 19:57 scsi-3618cf24100f8f457014a764c000001f6 -> ../sdd
lrwxrwxrwx 1 root root 9 Aug 11 19:57 scsi-3618cf24100f8f45712345678000103e8 -> ../sdi
lrwxrwxrwx 1 root root 9 Aug 3 15:17 scsi-3648435a10058805278654321ffffffff -> ../sdb
lrwxrwxrwx 1 root root 9 Aug 2 14:49 scsi-368886030000020aff44cc0d060c987f1 -> ../sdc
lrwxrwxrwx 1 root root 9 Aug 11 19:57 wwn-0x618cf24100f8f457014a764c000001f6 -> ../sdd
lrwxrwxrwx 1 root root 9 Aug 11 19:57 wwn-0x618cf24100f8f45712345678000103e8 -> ../sdi
lrwxrwxrwx 1 root root 9 Aug 3 15:17 wwn-0x648435a10058805278654321ffffffff -> ../sdb
lrwxrwxrwx 1 root root 9 Aug 2 14:49 wwn-0x68886030000020aff44cc0d060c987f1 -> ../sdc
```

----End

## 8.4.2 When a Pod Is Created, the Pod Is in the ContainerCreating State

### Symptom

A Pod is created. After a period of time, the Pod is still in the **ContainerCreating** state. Check the log information (for details, see [7.2.2 Viewing Huawei CSI Logs](#)). The error message "Fibre Channel volume device not found" is displayed.

### Root Cause Analysis

This problem occurs because residual disks exist on the host node. As a result, disks fail to be found when a Pod is created next time.

### Solution or Workaround

**Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.

**Step 2** Run the following command to query information about the node where the Pod resides.

```
kubectl get pod -o wide
```

The following is an example of the command output.

| NAME            | READY | STATUS            | RESTARTS | AGE | IP           | NODE  | NOMINATED NODE |
|-----------------|-------|-------------------|----------|-----|--------------|-------|----------------|
| READINESS GATES |       |                   |          |     |              |       |                |
| mypod           | 0/1   | ContainerCreating | 0        | 51s | 10.244.1.224 | node1 | <none> <none>  |

**Step 3** Delete the Pod.

**Step 4** Use a remote access tool, such as PuTTY, to log in to the *node1* node in the Kubernetes cluster through the management IP address. *node1* indicates the node queried in [Step 2](#).

**Step 5** Clear the residual drive letters. For details, see [Solution or Workaround](#).

----End

### 8.4.3 A Pod Is in the ContainerCreating State for a Long Time When It Is Being Created

#### Symptom

When a Pod is being created, the Pod is in the **ContainerCreating** state for a long time. Check the huawei-csi-node log (for details, see [7.2.2 Viewing Huawei CSI Logs](#)). No Pod creation information is recorded in the huawei-csi-node log. After the **kubectrl get volumeattachment** command is executed, the name of the PV used by the Pod is not displayed in the **PV** column. After a long period of time (more than ten minutes), the Pod is normally created and the Pod status changes to **Running**.

#### Root Cause Analysis

The kube-controller-manager component of Kubernetes is abnormal.

#### Solution or Workaround

Contact container platform engineers to rectify the fault.

### 8.4.4 A Pod Fails to Be Created and the Log Shows That the Execution of the mount Command Times Out

#### Symptom

When a Pod is being created, the Pod keeps in the **ContainerCreating** status. In this case, check the log information of huawei-csi-node (for details, see [7.2.2 Viewing Huawei CSI Logs](#)). The log shows that the execution of the mount command times out.

#### Root Cause Analysis

Cause 1: The configured service IP address is disconnected. As a result, the **mount** command execution times out and fails.

Cause 2: For some operating systems, such as Kylin V10 SP1 and SP2, it takes a long time to run the **mount** command in a container using NFSv3. As a result, the **mount** command may time out and error message "error: exit status 255" is displayed. The possible cause is that the value of **LimitNOFILE** of container runtime containerd is too large (over 1 billion).

Cause 3: The mounting may fail due to network problems. The default mounting timeout period of CSI is 30 seconds. If the mounting still fails after 30 seconds, logs show that the execution of the **mount** command times out.

## Solution or Workaround

- Step 1** Run the **ping** command to check whether the service IP network is connected. If the ping fails, the fault is caused by cause 1. In this case, configure an available service IP address. If the ping succeeds, go to [Step 2](#).
- Step 2** Go to any container where the **mount** command can be executed and use NFSv3 to run the **mount** command. If the command times out, the fault may be caused by cause 2. Run the **systemctl status containerd.service** command to check the configuration file path, and then run the **cat /xxx/containerd.service** command to check the configuration file. If the file contains **LimitNOFILE=infinity** or the value of **LimitNOFILE** is 1 billion, go to [Step 3](#). Otherwise, contact Huawei technical support engineers.
- Step 3** For cause 2, perform the following operations:
- Try using NFSv4.0.
  - Change the value of **LimitNOFILE** to a proper one by referring to [change solution provided by the community](#). This solution will restart the container runtime. Evaluate the impact on services.
- Step 4** Manually mount the file system on the host machine where the mounting fails. If the required time exceeds 30 seconds, check whether the network between the host machine and the storage node is normal. An example of the **mount** command is as follows.
- Run the following command to create a test directory.  

```
mkdir /tmp/test_mount
```
  - Run the **mount** command to mount the file system and observe the time consumed. The value of *ip:nfs\_share\_path* can be obtained from the huawei-csi-node log. For details, see [7.2.2 Viewing Huawei CSI Logs](#).  

```
time mount ip:nfs_share_path /tmp/test_mount
```
  - After the test is complete, run the following command to unmount the file system.  

```
umount /tmp/test_mount
```

----End

### 8.4.5 A Pod Fails to Be Created and the Log Shows That the mount Command Fails to Be Executed

#### Symptom

In NAS scenarios, when a Pod is being created, the Pod keeps in the **ContainerCreating** status. In this case, check the log information of huawei-csi-node (for details, see [7.2.2 Viewing Huawei CSI Logs](#)). The log shows that the mount command fails to be executed.

#### Root Cause Analysis

The possible cause is that the NFS 4.0/4.1/4.2 protocol is not enabled on the storage side. After the NFS v4 protocol fails to be used for mounting, the host does not negotiate to use the NFS v3 protocol for mounting.

## Solution or Workaround

- Enable the NFS 3/4.0/4.1/4.2 protocol on the storage side and retry the default mounting.
- Specify an available NFS protocol for mounting. For details, see [5.2.1 Configuring a StorageClass](#).

## 8.4.6 A Pod Fails to Be Created and Message "publishInfo doesn't exist" Is Displayed in the Events Log

### Symptom

When a Pod is being created, the Pod keeps in the **ContainerCreating** state. It is found that the following alarm event is printed for the Pod: **rpc error: code = Internal desc = publishInfo doesn't exist**

### Root Cause Analysis

As required by CSI, when a workload needs to use a PV, the Container Orchestration system (CO system, communicating with the CSI plug-in using RPC requests) invokes the ControllerPublishVolume interface (provided by huawei-csi-controller) in the [CSI protocol](#) provided by the CSI plug-in to map the PV, and then invokes the NodeStageVolume interface (provided by huawei-csi-node) provided by the CSI plug-in to mount the PV. During a complete mounting operation, only the huawei-csi-node service receives the NodeStageVolume request. Before that, the huawei-csi-controller service does not receive the ControllerPublishVolume request. As a result, the huawei-csi-controller service does not map the PV volume and does not send the mapping information to the huawei-csi-node service. Therefore, error message **publishInfo doesn't exist** is reported.

### Solution

To solve this problem, Kubernetes needs to invoke the ControllerPublishVolume interface.

If this operation is triggered by all workloads created by earlier versions in the cluster, this problem will not occur.

### Procedure

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
- Step 2** Run the following command to obtain the information about the node where a workload is located.

```
kubectl get pod error-pod -n error-pod-in-namespace -owide
```

The following is an example of the command output.

| NAME    | READY | STATUS            | RESTARTS | AGE | IP     | NODE   | NOMINATED NODE | READINESS |
|---------|-------|-------------------|----------|-----|--------|--------|----------------|-----------|
| GATES   |       |                   |          |     |        |        |                |           |
| pod-nfs | 0/1   | ContainerCreating | 0        | 3s  | <none> | node-1 | <none>         | <none>    |



- Step 3** Fail over the workload to another node.
- Step 4** If the failover cannot be completed in the cluster, you can delete the workload and create a new one on the original node.
- Step 5** Check whether the workload is successfully started. If it fails to be started, contact Huawei technical support engineers.
- End

## Checking Cluster Workloads

When Kubernetes invokes the CSI plug-in to complete volume mapping, the VolumeAttachment resource is used to save the mapping information, indicating that a specified volume is attached to or detached from a specified node. This problem occurs because publishInfo does not exist. You can view the VolumeAttachment resource information to check whether this problem is also involved in other workloads in the cluster. The procedure is as follows:

- Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.
- Step 2** Run the following command to obtain the VolumeAttachment information and retain resources whose **ATTACHER** field is **csi.huawei.com**. **csi.huawei.com** indicates the Huawei CSI driver name and can be configured in the **values.yaml** file. The corresponding configuration item is **csiDriver.driverName**. For details about the configuration item, see [Table 4-7](#).

```
kubectl get volumeattachments.storage.k8s.io
```

The following is an example of the command output.

| NAME       | ATTACHER       | PV      | NODE   | ATTACHED | AGE |
|------------|----------------|---------|--------|----------|-----|
| csi-47abxx | csi.huawei.com | pvc-1xx | node-1 | true     | 12h |

- Step 3** Run the following command to view the VolumeAttachment resource details. In the following information, **csi-47abxx** is the resource name obtained in [Step 2](#).

```
kubectl get volumeattachments.storage.k8s.io csi-47abxx -o yaml
```

The following is an example of the command output.

```
kind: VolumeAttachment
metadata:
 annotations:
 csi.alpha.kubernetes.io/node-id: '{"HostName":"node-1"}'
 finalizers:
 - external-attacher/csi-huawei-com
 name: csi-47abxx
 uid: 0c87fa8a-c3d6-4623-acb8-71d6206d030d
spec:
 attacher: csi.huawei.com
 nodeName: debian-node
 source:
 persistentVolumeName: pvc-1xx
status:
 attached: true
 attachmentMetadata:
 publishInfo: '{<PUBLISH-INFO>}'
```

- Step 4** If **status.attachmentMetadata.publishInfo** exists in the resource obtained in [Step 3](#), the problem described in this FAQ is not involved in the workloads created using **pvc-1xx** on the **node-1** node. **node-1** and **pvc-1xx** are the query results in

**Step 2.** If **status.attachmentMetadata.publishInfo** does not exist, rectify the fault by referring to [Solution](#).

**Step 5** If multiple VolumeAttachment resources exist, repeat [Step 3](#) to [Step 4](#).

----End

## 8.4.7 After a Pod Fails to Be Created or kubelet Is Restarted, Logs Show That the Mount Point Already Exists

### Symptom

When a Pod is being created, the Pod is always in the **ContainerCreating** state. Alternatively, after kubelet is restarted, logs show that the mount point already exists. Check the log information of huawei-csi-node (for details, see [7.2.2 Viewing Huawei CSI Logs](#)). The error information is: **The mount /var/lib/kubelet/pods/xxx/mount is already exist, but the source path is not /var/lib/kubelet/plugins/kubernetes.io/xxx/globalmount**

### Root Cause Analysis

The root cause of this problem is that Kubernetes performs repeated mounting operations.

### Solution or Workaround

Run the following command to unmount the existing path. In the command, **/var/lib/kubelet/pods/xxx/mount** indicates the existing mount path displayed in the logs.

```
umount /var/lib/kubelet/pods/xxx/mount
```

## 8.4.8 "I/O error" Is Displayed When a Volume Directory Is Mounted to a Pod

### Symptom

When a Pod reads or writes a mounted volume, message "I/O error" is displayed.

### Root Cause Analysis

When a protocol such as SCSI is used, if the Pod continuously writes data to the mount directory, the storage device will restart. As a result, the link between the device on the host and the storage device is interrupted, triggering an I/O error. When the storage device is restored, the mount directory is still read-only.

### Solution

Remount the volume. That is, reconstruct the Pod to trigger re-mounting.

## 8.4.9 Failed to Create a Pod Because the iscsi\_tcp Service Is Not Started Properly When the Kubernetes Platform Is Set Up for the First Time

### Symptom

When you create a Pod, error **Cannot connect ISCSI portal \*.\*.\*: libkmod: kmod\_module\_insert\_module: could not find module by name='iscsi\_tcp'** is reported in the `/var/log/huawei-csi-node` log.

### Root Cause Analysis

The `iscsi_tcp` service may be stopped after the Kubernetes platform is set up and the iSCSI service is installed. You can run the following command to check whether the service is stopped.

```
lsmod | grep iscsi | grep iscsi_tcp
```

The following is an example of the command output.

```
iscsi_tcp 18333 6
libiscsi_tcp 25146 1 iscsi_tcp
libiscsi 57233 2 libiscsi_tcp,iscsi_tcp
scsi_transport_iscsi 99909 3 iscsi_tcp,libiscsi
```

### Solution or Workaround

Run the following command to manually load the `iscsi_tcp` service.

```
modprobe iscsi_tcp
lsmod | grep iscsi | grep iscsi_tcp
```

## 8.4.10 A Pod Fails to Be Created and Logs Show That an Initiator Has Been Associated with Another Host

### Symptom

When a Pod is created using SAN storage, the Pod is always in the **ContainerCreating** status. The Pod logs report alarm event "rpc error: code = Internal desc = initiator xxx is already associated to another host".

### Root Cause Analysis

Cause 1: CSI automatically creates hosts, host groups, and initiators based on certain rules. If the same resources exist on the storage side before CSI is used, conflicts will occur. The possible cause is that the same initiator has been added before CSI is used.

Cause 2: In the container cluster, the initiator names of different worker nodes are the same. In this case, perform the following steps to rectify the fault:

- Step 1** Log in to different worker nodes in the container cluster and run the following command to check whether different worker nodes use the same initiator name:
- To view the iSCSI initiator name, run the following command:

```
cat /etc/iscsi/initiatorname.iscsi
```

- To view the FC initiator name, run the following command:  

```
cat /sys/class/fc_host/host*/port_name
```
- To view the RoCE initiator name, run the following command:  

```
cat /etc/nvme/hostnqn
```

**Step 2** If different worker nodes use the same initiator name, rectify the fault according to [Solution or Workaround](#).

----End

## Solution or Workaround

**Step 1** Check whether the host associated with the initiator has volumes in use. If the host has volumes in use, migrate the Pod in use to another node.

**Step 2** After confirming that the host has no volume in use, change the initiator name to ensure that the initiator is unique.

**Step 3** Run the following command to restart the iscsid service.

```
systemctl restart iscsid
```

### NOTICE

Restarting the iscsid service may cause I/O interruption. Before restarting the service, ensure that the host associated with the initiator has no volume in use.

**Step 4** Restart the huawei-csi-node service.

----End

## 8.4.11 A Pod Fails to Be Created and Logs Show "Get DMDevice by alias: dm-x failed"

### Symptom

When a Pod is created, the Pod is in the **ContainerCreating** status for a long time. In addition, the following error message is reported in the logs of huawei-csi-node (for details, see [7.2.2 Viewing Huawei CSI Logs](#)):

```
check device: dm-1 is a partition device failed. error: Get DMDevice by alias:dm-1 failed. error: Can not get DMDevice by alias: dm-1
```

### Root Cause Analysis

In the DM-Multipath configuration file, the **user\_friendly\_names** parameter is not set to **yes**.

## Solution or Workaround

**Step 1** Check whether the worker node where the Pod is running has volumes in use. If the node has volumes in use, migrate the Pod in use to another node.

**Step 2** Configure the `/etc/multipath.conf` file by following the instructions in [4.1.1.5 Checking the Host Multipathing Configuration](#).

**Step 3** Run the following command to restart the multipathing software.

```
systemctl reload multipathd.service
systemctl restart multipathd
```

#### NOTICE

Restarting the multipathing software may cause I/O interruption. Ensure that the worker node where the Pod is running has no volume in use before restarting the multipathing software.

----End

## 8.4.12 After Pods on the Same Node Are Deleted in a Batch Using the NVMe Protocol, Residual NVMe Links Exist on the Node

### Symptom

In the NVMe protocol scenario, when pods on the same node are deleted in a batch, the pods are successfully deleted, but the NVMe links on the node are not cleared.

```
nvme list-subsys
nvme-subsys0 - NQN=nqn.xxx.nvme:nvm-subsystem-sn-xxxxxxx
\
+- nvme0 tcp traddr=xxx.xxx.xxx.xxx,trsvcid=4420,src_addr=xxx.xxx.xxx.xxx live
+- nvme1 tcp traddr=xxx.xxx.xxx.xxx,trsvcid=4420,src_addr=xxx.xxx.xxx.xxx live
```

### Root Cause Analysis

If the NVMe protocol is used, the device paths on the host are cleared only after the host is unmapped from the storage resources. When multiple pods are mounted to the same volume and the pods are deleted in a batch, the CSI in the NodeUnstageVolume phase (unmount phase) cannot detect the device path cleanup in the subsequent ControllerUnpublishVolume phase (unmap phase). As a result, the NVMe links cannot be cleared in a timely manner.

### Solution or Workaround

1. The residual NVMe links do not affect the CSI functions. However, the storage resources that are mounted subsequently scan for the corresponding device paths based on the residual links. If the number of links does not need to be reduced, you do not need to clear the residual links.
2. If the number of links to be mounted later needs to be reduced, you can manually delete the residual links on the node. For example, to clear the links in the preceding symptom, run the following commands:

```
nvme disconnect -d nvme0
nvme disconnect -d nvme1
```

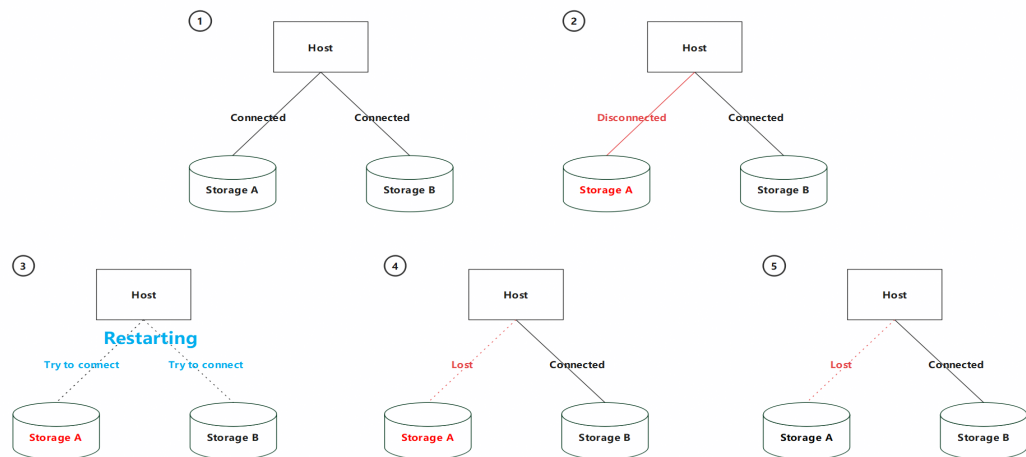
## 8.4.13 In the SAN HyperMetro Scenario, the Subpath of the Aggregated Disks Corresponding to the Mounted Volume Is Lost

### Symptom

The subpath of the aggregated disks corresponding to the mounted resource is lost.

### Root Cause Analysis

Figure 8-1 SAN HyperMetro subpath loss



As shown in [Figure 8-1](#), if the link between the host and the storage device is disconnected due to factors such as HBA/NIC exceptions, switch/network jitter, or storage array service port faults, the host restarts and triggers disk scanning again. In this case, the link to the faulty storage device is disconnected on the host. After the fault is rectified, the link information is lost after the host scans the disks again. As a result, the lost link will not be automatically restored.

To restore the link, you can drift over a pod to another host and enable CSI to automatically remount the link. To manually restore the lost link on the current host, perform the following steps:

### Solution or Workaround (iSCSI Protocol)

**Step 1** Run the following command to check whether the iSCSI node corresponding to the service IP address exists on the host. In the command, **192.168.1.100** indicates the service IP address. If the node exists, go to [Step 3](#). If the node does not exist, go to [Step 2](#).

```
iscsiadm -m node | grep 192.168.1.100
```

**Step 2** Run the following command to discover the iSCSI node:

```
iscsiadm -m discovery -t st -p 192.168.1.100
```

**Step 3** Run the following command to log in to the iSCSI node:

```
iscsiadm -m node -p 192.168.1.100 -l
```

**Step 4** Run the following command to query the iSCSI host ID based on the corresponding service IP address:

```
iscsiadm -m session -P3
```

**Step 5** Log in to DeviceManager, choose **Services > Host Groups > Hosts > Mapping**, and find the host LUN ID corresponding to the LUN.

**Step 6** Run the following disk scanning command to supplement the lost link. In the command, *host\_lun\_id* indicates the host LUN ID obtained in [Step 5](#), and *host\_no* indicates the host ID obtained in [Step 4](#).

```
echo "- - <host_lun_id>" > /sys/class/scsi_host/host<host_no>/scan
```

**Step 7** Run the following command to check whether the link is supplemented:

```
multipath -ll
```

----End

## Solution or Workaround (FC Protocol)

**Step 1** Run the following command to query all FC initiators on the current host:

```
cat /sys/class/fc_host/host*/port_name | awk 'BEGIN{FS="0x";ORS=" "}{print $2}'
```

**Step 2** Run the following command to query the host ID of the initiator whose path is lost. In the command, *port\_name* indicates the initiator name obtained in [Step 1](#).

```
for h in /sys/class/fc_host/host*; do echo $h: $(cat $h/port_name); done | grep <port_name>
```

**Step 3** Log in to DeviceManager, choose **Services > Host Groups > Hosts > Mapping**, and find the host LUN ID corresponding to the LUN.

**Step 4** Run the following disk scanning command to supplement the lost link. In the command, *host\_lun\_id* indicates the host LUN ID obtained in [Step 3](#), and *host\_no* indicates the host ID obtained in [Step 2](#).

```
echo "- - <host_lun_id>" > /sys/class/scsi_host/host<host_no>/scan
```

**Step 5** Run the following command to check whether the link has been supplemented:

```
multipath -ll
```

----End

## 8.5 Common Problems and Solutions for Interconnecting with the Tanzu Kubernetes Cluster

This section describes the common problems and solutions for interconnecting with the Tanzu Kubernetes cluster. Currently, the following problems occur during interconnection with the Tanzu Kubernetes cluster:

- A Pod cannot be created because the PSP permission is not created.
- The mount point of the host is different from that of the native Kubernetes. As a result, a volume fails to be mounted.
- The livenessprobe container port conflicts with the Tanzu vSphere port. As a result, the container restarts repeatedly.

## 8.5.1 A Pod Cannot Be Created Because the PSP Permission Is Not Created

### Symptom

When huawei-csi-controller and huawei-csi-node are created, only the Deployment and DaemonSet resources are successfully created, and no Pod is created for the controller and node.

### Root Cause Analysis

The service account used for creating resources does not have the "use" permission of the PSP policy.

### Solution or Workaround

**Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.

**Step 2** Run the **vi psp-use.yaml** command to create a file named **psp-use.yaml**

**Step 3** Configure the **psp-use.yaml** file.

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
 name: huawei-csi-psp-role
rules:
- apiGroups: ['policy']
 resources: ['podsecuritypolicies']
 verbs: ['use']

apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
 name: huawei-csi-psp-role-cfg
roleRef:
 kind: ClusterRole
 name: huawei-csi-psp-role
 apiGroup: rbac.authorization.k8s.io
subjects:
- kind: Group
 apiGroup: rbac.authorization.k8s.io
 name: system:serviceaccounts:huawei-csi
- kind: Group
 apiGroup: rbac.authorization.k8s.io
 name: system:serviceaccounts:default
```

**Step 4** Run the following command to create the PSP permission.

```
kubectl create -f psp-use.yaml
```

----End

## 8.5.2 Changing the Mount Point of a Host

### Symptom

A Pod fails to be created, and error message "mount point does not exist" is recorded in Huawei CSI logs.



## Root Cause Analysis

The native Kubernetes cluster in the **pods-dir** directory of huawei-csi-node is inconsistent with the Tanzu Kubernetes cluster.

## Solution or Workaround

- Step 1** Go to the **helm/esdk/** directory and run the **vi values.yaml** command to open the configuration file.

```
vi values.yaml
```

- Step 2** Change the value of **kubeletConfigDir** to the actual installation directory of kubelet.

```
Specify kubelet config dir path.
kubernetes and openshift is usually /var/lib/kubelet
Tanzu is usually /var/vcap/data/kubelet
kubeletConfigDir: /var/vcap/data/kubelet
```

----End

## 8.5.3 Changing the Default Port of the livenessprobe Container

### Symptom

The livenessprobe container of the huawei-csi-controller component keeps restarting.

## Root Cause Analysis

The default port (9808) of the livenessprobe container of huawei-csi-controller conflicts with the existing vSphere CSI port of Tanzu.

## Solution or Workaround

Change the default port of the livenessprobe container to an idle port.

- Step 1** Go to the **helm/esdk** directory and run the **vi values.yaml** command to open the configuration file.

```
vi values.yaml
```

- Step 2** Change the default value **9808** of **controller.livenessProbePort** to an idle port, for example, **9809**.

```
controller:
 livenessProbePort: 9809
```

- Step 3** Update Huawei CSI using Helm. For details, see [4.1.4.1 Upgrade Using Helm](#).

----End

## 8.5.4 Failed to Create an Ephemeral Volume

### Symptom

A [generic ephemeral volume](#) fails to be created, and the error message **PodSecurityPolicy: unable to admit pod: [spec.volumes[0]: Invalid value:**

**"ephemeral": ephemeral volumes are not allowed to be used spec.volumes[0]**  
is displayed.

## Root Cause Analysis

The current PSP policy does not contain the permission to use ephemeral volumes.

## Solution or Workaround

Add the permission to use ephemeral volumes to the default PSP **pks-privileged** and **pks-restricted**. The following is an example of modifying **pks-privileged**:

**Step 1** Use a remote access tool, such as PuTTY, to log in to any master node in the Kubernetes cluster through the management IP address.

**Step 2** Run the following command to modify the **pks-privileged** configuration.

```
kubectrl edit psp pks-privileged
```

**Step 3** Add **ephemeral** to **spec.volumes**. The following is an example.

```
Please edit the object below. Lines beginning with a '#' will be ignored,
and an empty file will abort the edit. If an error occurs while saving this file will be
reopened with the relevant failures.

apiVersion: policy/v1beta1
kind: PodSecurityPolicy
metadata:
 annotations:
 apparmor.security.beta.kubernetes.io/allowedProfileName: '*'
 seccomp.security.alpha.kubernetes.io/allowedProfileNames: '*'
 creationTimestamp: "2022-10-11T08:07:00Z"
 name: pks-privileged
 resourceVersion: "1227763"
 uid: 2f39c44a-2ce7-49fd-87ca-2c5dc3bfc0c6
spec:
 allowPrivilegeEscalation: true
 allowedCapabilities:
 - '*'
 supplementalGroups:
 rule: RunAsAny
 volumes:
 - glusterfs
 - hostPath
 - iscsi
 - nfs
 - persistentVolumeClaim
 - ephemeral
```

**Step 4** Run the following command to check whether the addition is successful.

```
kubectrl get psp pks-privileged -o yaml
```

----End

# 9 Command Parameter Description

## 9.1 Description of oceanctl Commands

### 9.1 Description of oceanctl Commands

#### Obtaining Help Information

- Obtain the oceanctl help information.  
`oceanctl --help`
- Check the oceanctl version.  
`oceanctl version`
- Specify the custom log file directory. The following example describes how to check the oceanctl version.  
`oceanctl version --log-dir=/path/to/custom`

#### Creating a Storage Backend

- Run the following command to obtain the help information about creating a backend.  
`oceanctl create backend -h`
- Run the following command to create a storage backend based on the specified yaml file.  
`oceanctl create backend -f /path/to/backend.yaml -i yaml`
- Run the following command to create a storage backend based on the specified json file. The **huawei-csi-configmap** file can be exported only in json format.  
`oceanctl create backend -f /path/to/configmap.json -i json`
- Run the following command to create a storage backend in the specified namespace.  
`oceanctl create backend -f /path/to/backend.yaml -i yaml -n <namespace>`
- Run the following command to create a storage backend and ignore the storage backend name verification, for example, uppercase letters and underscores (`_`). Do not run this command unless necessary.  
`oceanctl create backend -f /path/to/backend.yaml -i yaml --not-validate-name`
- Run the following command to create a storage backend and specify **provisioner**. **csi.oceanstor.com** is the driver name specified during installation. For details, see [Step 4](#).

 **NOTE**

This command is used only when a backend is created on the CCE or CCE Agile platform.

```
oceanctl create backend -f /path/to/backend.yaml -i yaml --provisioner=csi.oceanstor.com
```

## Querying a Storage Backend

- Run the following command to obtain the help information about querying a backend.  

```
oceanctl get backend -h
```
- Run the following command to query a single storage backend in the default namespace.  

```
oceanctl get backend <backend-name>
```
- Run the following command to query all storage backends in the specified namespace.  

```
oceanctl get backend -n <namespace>
```
- Run the following command to format the output. Currently, **json**, **yaml**, and **wide** are supported.  

```
oceanctl get backend <backend-name> -o json
```

## Updating a Storage Backend

- Run the following command to obtain the help information about updating a backend.  

```
oceanctl update backend -h
```
- Run the following command to update the specified storage backend in the default namespace.  

```
oceanctl update backend <backend-name> --password
```
- Run the following command to update a storage backend in the specified namespace.  

```
oceanctl update backend <backend-name> -n <namespace> --password
```

## Deleting a Storage Backend

- Run the following command to obtain the help information about deleting a backend.  

```
oceanctl delete backend -h
```
- Run the following command to delete the specified storage backend in the default namespace.  

```
oceanctl delete backend <backend-name>
```
- Run the following command to delete all storage backends in the default namespace.  

```
oceanctl delete backend --all
```
- Run the following command to delete a storage backend in the specified namespace.  

```
oceanctl delete backend <backend-name> -n <namespace>
```

## Creating a Storage Backend Certificate

- Run the following command to obtain the help information about querying a certificate.  

```
oceanctl create cert -h
```
- Run the following command to create a certificate for a single storage backend in the default namespace based on the specified .crt certificate file.

```
oceanctl create cert <name> -f /path/to/cert.crt -b <backend-name>
```

- Run the following command to create a certificate for a single storage backend in the specified namespace based on the specified .crt certificate file.  

```
oceanctl create cert <name> -f /path/to/cert.crt -b <backend-name> -n <namespace>
```
- Run the following command to create a certificate for a single storage backend in the specified namespace based on the specified .pem certificate file.  

```
oceanctl create cert <name> -f /path/to/cert.pem -b <backend-name> -n <namespace>
```

## Querying a Storage Backend Certificate

- Run the following command to obtain the help information about querying a certificate.  

```
oceanctl get cert -h
```
- Run the following command to query the certificate of a specified storage backend in the default namespace.  

```
oceanctl get cert -b <backend-name>
```
- Run the following command to query the certificate of a specified storage backend in the specified namespace.  

```
oceanctl get cert -b <backend-name> -n <namespace>
```

## Updating a Storage Backend Certificate

- Run the following command to obtain the help information about updating a certificate.  

```
oceanctl update cert -h
```
- Run the following command to update a certificate for a specified storage backend in the default namespace based on the specified .crt certificate file.  

```
oceanctl update cert -b <backend-name> -f /path/to/cert.crt
```
- Run the following command to update a certificate for a specified storage backend in the specified namespace based on the specified .crt certificate file.  

```
oceanctl update cert -b <backend-name> -n <namespace> -f /path/to/cert.crt
```
- Run the following command to update a certificate for a specified storage backend in the specified namespace based on the specified .pem certificate file.  

```
oceanctl update cert -b <backend-name> -n <namespace> -f /path/to/cert.pem
```

## Deleting a Storage Backend Certificate

- Run the following command to obtain the help information about deleting a certificate.  

```
oceanctl delete cert -h
```
- Run the following command to delete the certificate of a specified storage backend in the default namespace.  

```
oceanctl delete cert -b <backend-name>
```
- Run the following command to delete the certificate of a specified storage backend in the specified namespace.  

```
oceanctl delete cert -b <backend-name> -n <namespace>
```

# 10Appendix

- 10.1 Communication Matrix
- 10.2 Configuring Custom Permissions
- 10.3 Huawei CSI Resource Management
- 10.4 Kubernetes Feature Matrix

## 10.1 Communication Matrix

|                                  |                                             |                                      |                                      |                                      |
|----------------------------------|---------------------------------------------|--------------------------------------|--------------------------------------|--------------------------------------|
| Source Device                    | Host where CSI controller is located        | Host where CSI controller is located | Host where CSI node is located       | Kubernetes master node               |
| Source IP Address                | IP address of the source device             | IP address of the source device      | IP address of the source device      | IP address of the source device      |
| Source Port                      | 1024 to 65536                               | 1024 to 65536                        | 1024 to 65536                        | 1024 to 65536                        |
| Destination Device               | Storage device                              | Host where CSI controller is located | Host where CSI node is located       | Host where CSI controller is located |
| Destination IP Address           | Management IP address of the storage device | IP address of the destination device | IP address of the destination device | IP address of the destination device |
| Destination Port (for Listening) | 8088                                        | 9808                                 | 9800                                 | 4433                                 |
| Protocol                         | TCP                                         | TCP                                  | TCP                                  | TCP                                  |

|                                    |                                            |                                                                 |                                                           |                                                                           |
|------------------------------------|--------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------|
| <b>Port Description</b>            | Used to create, manage, and delete volumes | Used by Kubernetes to check the health status of CSI controller | Used by Kubernetes to check the health status of CSI node | Used to invoke webhook verification                                       |
| <b>Listening Port Configurable</b> | No                                         | No                                                              | No                                                        | Yes                                                                       |
| <b>Authentication Mode</b>         | User name and password                     | Certificate                                                     | Certificate                                               | Certificate                                                               |
| <b>Encryption Mode</b>             | TLS 1.3/TLS 1.2                            | TLS 1.3/TLS 1.2                                                 | TLS 1.3/TLS 1.2                                           | TLS 1.3/TLS 1.2                                                           |
| <b>Plane</b>                       | OM                                         | O&M plane                                                       | O&M plane                                                 | O&M plane                                                                 |
| <b>Special Scenario</b>            | None                                       | None                                                            | None                                                      | None                                                                      |
| <b>Remarks</b>                     | Enable some source ports.                  | -                                                               | -                                                         | For details about how to change the webhook port, see the CSI user guide. |

## 10.2 Configuring Custom Permissions

### User-defined Role Configurations

For different storage resources, refer to the following configurations:

- For NAS resources, configure the minimum permissions by referring to [Table 10-1](#).
- For SAN resources, configure the minimum permissions by referring to [Table 10-2](#).

#### NOTE

For details about how to configure permissions for user-defined roles, see [OceanStor Dorado 6000, Dorado 18000 Series Product Documentation](#).

**Table 10-1** Minimum permissions for NAS resources

| Permission Object | Parent Object        | Read/Write Permission | Function                   |
|-------------------|----------------------|-----------------------|----------------------------|
| workload_type     | file_storage_service | Read-only             | Queries the workload type. |

| Permission Object  | Parent Object               | Read/Write Permission | Function                                                                                                                |
|--------------------|-----------------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------|
| file_system        | file_storage_service        | Read and write        | Manages file systems.                                                                                                   |
| fs_snapshot        | file_storage_service        | Read and write        | Manages file system snapshots.                                                                                          |
| quota              | file_storage_service        | Read and write        | Manages file system quotas.                                                                                             |
| nfs_service        | file_storage_service        | Read-only             | Queries NFS services.                                                                                                   |
| share              | file_storage_service        | Read and write        | Manages NFS shares.                                                                                                     |
| dtree              | file_storage_service        | Read and write        | Manages dtrees.                                                                                                         |
| hyper_metro_pair   | hyper_metro                 | Read and write        | Creates file system HyperMetro pairs.                                                                                   |
| hyper_metro_domain | hyper_metro                 | Read-only             | Queries information about file system HyperMetro domains.                                                               |
| remote_device      | local_data_protection       | Read-only             | Queries remote device information.                                                                                      |
| storage_pool       | pool                        | Read-only             | Queries storage pool information.                                                                                       |
| smart_qos          | resource_performance_tuning | Read and write        | Manages SmartQoS policies.                                                                                              |
| system             | system                      | Read-only             | Queries storage device information (this object needs to be configured only when the owning group is the system group). |
| vstore             | vstore                      | Read-only             | Queries vStore information.                                                                                             |
| port               | network                     | Read-only             | Queries logical port information.                                                                                       |



**Table 10-2** Minimum permissions for SAN resources

| Permission Object | Parent Object               | Read/Write Permission | Function                                                                                                                |
|-------------------|-----------------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------|
| remote_device     | local_data_protection       | Read-only             | Queries remote device information.                                                                                      |
| hyper_clone       | local_data_protection       | Read and write        | Manages clone pairs.                                                                                                    |
| lun_snapshot      | local_data_protection       | Read and write        | Manages LUN snapshots.                                                                                                  |
| workload_type     | lun                         | Read-only             | Queries the workload type.                                                                                              |
| lun               | lun                         | Read and write        | Manages LUNs.                                                                                                           |
| host              | mapping_view                | Read and write        | Manages hosts.                                                                                                          |
| host_group        | mapping_view                | Read and write        | Manages host groups.                                                                                                    |
| initiator         | mapping_view                | Read and write        | Manages initiators.                                                                                                     |
| lun_group         | mapping_view                | Read and write        | Manages LUN groups.                                                                                                     |
| mapping_view      | mapping_view                | Read and write        | Manages mapping views.                                                                                                  |
| target            | mapping_view                | Read-only             | Queries iSCSI initiators.                                                                                               |
| port              | network                     | Read-only             | Queries logical ports.                                                                                                  |
| storage_pool      | pool                        | Read-only             | Queries storage pool information.                                                                                       |
| smart_qos         | resource_performance_tuning | Read and write        | Manages SmartQoS policies.                                                                                              |
| system            | system                      | Read-only             | Queries storage device information (this object needs to be configured only when the owning group is the system group). |
| vstore            | vstore                      | Read-only             | Queries vStore information.                                                                                             |
| hyper_metro_pair  | hyper_metro                 | Read and write        | Creates a block HyperMetro pair.                                                                                        |

| Permission Object  | Parent Object | Read/Write Permission | Function                                     |
|--------------------|---------------|-----------------------|----------------------------------------------|
| hyper_metro_domain | hyper_metro   | Read-only             | Queries block HyperMetro domain information. |

## 10.3 Huawei CSI Resource Management

This section lists the resource requests and limits used by each container of the Huawei CSI plug-in. For details about the unit, see [Resource units in Kubernetes](#).

**Table 10-3** Container resource requests and limits

| Pod Name              | Container Name             | CPU Request | CPU Limit | Memory Request | Memory Limit |
|-----------------------|----------------------------|-------------|-----------|----------------|--------------|
| huawei-csi-controller | huawei-csi-driver          | 50m         | 500m      | 128Mi          | 1Gi          |
|                       | storage-backend-sidecar    | 50m         | 300m      | 128Mi          | 512Mi        |
|                       | storage-backend-controller | 50m         | 300m      | 128Mi          | 512Mi        |
|                       | huawei-csi-extender        | 50m         | 300m      | 128Mi          | 512Mi        |
|                       | csi-attacher               | 50m         | 300m      | 128Mi          | 512Mi        |
|                       | csi-provisioner            | 50m         | 300m      | 128Mi          | 512Mi        |
|                       | csi-resize                 | 50m         | 300m      | 128Mi          | 512Mi        |
|                       | csi-snapshotter            | 50m         | 300m      | 128Mi          | 512Mi        |
|                       | snapshot-controller        | 50m         | 300m      | 128Mi          | 512Mi        |
|                       | liveness-probe             | 10m         | 100m      | 128Mi          | 128Mi        |
|                       | external-health-monitor    | 50m         | 300m      | 128Mi          | 512Mi        |
| huawei-csi-node       | huawei-csi-driver          | 50m         | 500m      | 128Mi          | 1Gi          |
|                       | csi-node-driver-registrar  | 50m         | 300m      | 128Mi          | 128Mi        |
|                       | liveness-probe             | 10m         | 100m      | 128Mi          | 128Mi        |

## Modifying Resource Requests and Limits

If you need to modify the resource requests and limits of a container, perform the following steps (in the following example, Helm is used to install Huawei CSI):

- Step 1** If Helm is used for installation, go to the `/helm/esdk/templates` directory. For manual deployment, the file to be modified is in the `/manual/esdk/deploy` directory. For details about the component package path, see [Table 4-1](#).
- Step 2** Modify the deployment template file.
- If the Pod name is **huawei-csi-controller**, modify the **huawei-csi-controller.yaml** file.
  - If the Pod name is **huawei-csi-node**, modify the **huawei-csi-node.yaml** file.

 **NOTE**

For details about Pod names, see [Table 10-3](#).

For example, to modify the resource request of the **huawei-csi-driver** container in the Pod named **huawei-csi-node**, run the following command to edit the configuration file and find the container whose **spec.template.spec.containers.name** is **huawei-csi-driver**. Modify resource requests and limits as required.

```
vi huawei-csi-node.yaml
```

Edit the following content.

```
containers
- name: huawei-csi-driver
 ...
 resources:
 limits:
 cpu: 500m
 memory: 1Gi
 requests:
 cpu: 50m
 memory: 128Mi
```

- Step 3** If Huawei CSI is not installed, the modification of resource requests and limits takes effect after Huawei CSI is installed by referring to [4.1.2.1.1 Installing Huawei CSI on Kubernetes, OpenShift, and Tanzu](#).
- Step 4** If Huawei CSI has been installed, the modification of resource requests and limits takes effect after Huawei CSI is updated by referring to [Upgrading Huawei CSI](#).

----End

## 10.4 Kubernetes Feature Matrix

This section describes the features of different Kubernetes versions supported by Huawei CSI.

**Table 10-4** Kubernetes versions and supported features

| Feature             | V1.16 | V1.17 | V1.18 | V1.19 | V1.20 | V1.21+ |
|---------------------|-------|-------|-------|-------|-------|--------|
| Static Provisioning | √     | √     | √     | √     | √     | √      |

| Feature                          | V1.16 | V1.17 | V1.18 | V1.19 | V1.20 | V1.21+ |
|----------------------------------|-------|-------|-------|-------|-------|--------|
| Dynamic Provisioning             | √     | √     | √     | √     | √     | √      |
| Manage Provisioning <sup>1</sup> | √     | √     | √     | √     | √     | √      |
| Expand Persistent Volume         | √     | √     | √     | √     | √     | √      |
| Create VolumeSnapshot            | x     | x     | x     | x     | √     | √      |
| Restore VolumeSnapshot           | x     | x     | x     | x     | √     | √      |
| Delete VolumeSnapshot            | x     | x     | x     | x     | √     | √      |
| Clone Persistent Volume          | x     | √     | √     | √     | √     | √      |
| Modify Volume <sup>2</sup>       | √     | √     | √     | √     | √     | √      |
| Raw Block Volume                 | √     | √     | √     | √     | √     | √      |
| Topology                         | √     | √     | √     | √     | √     | √      |
| Generic Ephemeral Inline Volumes | x     | x     | x     | x     | x     | √      |
| Volume Limits                    | x     | √     | √     | √     | √     | √      |
| FSGroup Support                  | x     | x     | x     | x     | √     | √      |

- Note 1: Manage Provisioning is a volume management feature customized by Huawei CSI. This feature allows existing storage resources to be managed by Kubernetes. You are not allowed to manage a storage resource for multiple times and concurrently delete or create a storage resource. When a storage resource is managed by multiple clusters, operations on the managed volume in a single cluster take effect only in the cluster and will not be synchronized to other clusters. Instead, you need to perform these operations on the managed volume in other clusters.
- Note 2: Modify Volume is a PVC change feature customized by Huawei CSI. This feature allows a common volume to be changed to a HyperMetro volume. To use this feature, ensure that the connected storage supports the volume HyperMetro feature.